

Tailoring Transition Metal Perovskite Oxides via Low-Temperature Hydrothermal Routes as Potential Candidates for Catalytic Applications

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To
My Mother & Motherland

Research publications, Symposiums and Conferences

1. **Kotha, V.;** Karajagi, I.; Ghosh, P. C.; Panchakarla, L. S. Potassium-Substituted LaMnO_3 as a Highly Active and Exceptionally Stable Electrocatalyst toward Bifunctional Oxygen Reduction and Oxygen Evolution Reactions. *ACS Appl. Energy Mater.* **2022**, 5 (6), 7297–7307. (*Chapter-4, Part-1)
2. Bhavani, B.; Chanda, N.; **Kotha, V.;** Reddy, G.; Basak, P.; Pal, U.; Giribabu, L.; Prasanthkumar, S. 1D Alignment of Co(II) Metalated Porphyrin-Napthalimide Based Self-Assembled Nanowires for Photocatalytic Hydrogen Evolution. *Nanoscale* **2022**, 14 (1), 140–146.
3. Nishad, H. S.; Gupta, S. P.; **Kotha, V.;** Patil, B. M.; Chakane, S. D.; Bute, M. G.; Gosavi, S. W.; Late, D. J.; Walke, P. S. Enhanced Van-Der Waals Separation in Hydrated Tungsten Oxide Nanoplates Enables Superior Pseudocapacitive Charge Storage. *J. Alloys Compd.* **2022**, 914, 165227.
4. **Kotha, V.;** Kumar, K.; Dayman, P.; Panchakarla, L. S. Doping with Chemically Hard Elements to Improve Photocatalytic Properties of ZnO Nanostructures. *J. Clust. Sci.* **2021**, (189), 1–8. (*Chapter-5)
5. Sugathan, V.; Ghosh, B.; Harikesh, P. C.; **Kotha, V.;** Vashishtha, P.; Salim, T.; Yella, A.; Mathews, N. Synthesis of Bismuth Sulphoiodide Thin Films from Single Precursor Solution. *Sol. Energy* **2021**, 230, 714–720.
6. Bera, A.; Pathak, S. S.; **Kotha, V.;** Prasad, B. L. V. Lamellar Bimetallic Thiolates: Synthesis, Characterization, and Their Utilization for the Preparation of Bimetallic Chalcogenide Nanocrystals through Mechanochemical Grinding. *Adv. Mater. Interfaces* **2021**, 8 (23), 2100898.
7. Mrinalini, M.; Suman Krishna, J. V.; Krishna, N. V.; **Kotha, V.;** Panchakarla, L. S.; Prasanthkumar, S.; Giribabu, L. Photobleaching of Triphenylamine–Phthalocyanine Entails Mixed Valence-State Triggered

- Self-Assembled Nanospheres. *J. Phys. Chem. C*. **2018**, 122 (34), 19946–19952.
8. **Kotha, V.**; Reddy P. L.; Maddu D.; Panchakarla, L. S. Importance of K substitution in LaMnO_3 towards high-performance selective reduction of α,β -unsaturated carbonyl compounds and nitroarenes. (*Submitted to ACS Nano Letters*).
 9. Nishad, H. S.; Gupta, S. P.; **Kotha, V.**; Magdum, V. V.; Gawade, V. V.; Patole, P. S.; Biradar, A. V.; Walke, P. S. Structural engineering of WO_3 nanoflowers via Na doping as a suitable cathode for aqueous zinc ion battery (*Submitted to Journal of Energy Storage*).
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Abstract

Perovskite oxides have been known for their multi-dimensional properties for decades. Among them, Lanthanum-based oxide perovskites have remained some of the most lucrative materials due to their significant properties like superconductivity, colossal magnetoresistance, and supercapacitance, exhibiting multiple utilities in multiferroic, and piezoelectric applications. However, their phase-controlled synthesis and ability to tune size and morphology for exploring properties have remained crucial questions to be addressed. To date, synthesis techniques of these materials mostly employ solid-state reaction conditions at very unconventional temperatures, with poor control of their morphology, phase, and yield. Despite their remarkable properties, the lack of a synthetic procedure to control size and morphology restricts their application in today's advanced technologies and main stream applications (catalysis), thus forming the need of the hour. This doctoral thesis aims to establish a low-temperature, hydrothermal solution-phase synthetic approach, where reasonable control over these compounds' size, composition, phase, and morphology could be facilitated. This dissertation shows our attempts to stabilize these perovskite phases at low temperatures using a solution-based synthetic approach. To facilitate the same, four materials were chosen, viz. LaCrO_3 (LCrO), LaMnO_3 (LMO), LaFeO_3 (LFO), and LaNiO_3 (LNO). Various optimization studies conducted on these materials facilitated not only a control on the phase; but also the morphology, size, crystallinity, and effective yield. In addition, a plausible mechanism for the nucleation of these materials has also been proposed. This was subsequently extrapolated successfully to synthesizing a few more submicron-sized materials like LCrO, $\text{La}_{1-x}\text{K}_x\text{MnO}_3$ (LKMO), and LFO, which proved that the synthetic protocol was generic and applicable to a wide diversity of similar materials. Apart from the classical hydrothermal method, which is static in nature, a modern dynamic hydrothermal method was introduced to enhance the kinetics of the formation of these Lanthanum-based Perovskite oxides (LPO) systems. Further, LKMO was synthesized in a one-dimensional fashion at low temperatures using the same chemical routes. This work was further extended to obtain truncated cubes of LKMO and double perovskites such as $\text{La}_{2-x}\text{K}_x\text{NiMnO}_6$, $\text{La}_{2-x}\text{K}_x\text{MnCrO}_6$, and $\text{La}_{2-x}\text{K}_x\text{CrFeO}_6$. Finally, the catalytic activity of the LKMO microcubes synthesized in these methods was studied in an electrochemical device and reduction of organic molecules. Apart from the perovskite oxides, ZnO nanoparticles were synthesized hydrothermally by doping with chemically hard elements, showing enhanced photocatalytic activity.

Keywords

Lanthanum-Based Oxide Perovskites, Catalysis, Manganite, Low-Temperature, Hydrothermal, Mechanism, Size, Crystallinity, Phase, Degrees of Freedom, Crystal Facets, Kinetics, Heterogeneous Nucleation, Morphology, Polarity, Chemical Hardness

Thesis Structure

Broadly, the research conducted in this doctoral thesis aims to establish a low-temperature, hydrothermal solution-phase synthetic approach for the synthesis of size-, phase-, composition- and morphology-controlled lanthanum-based oxide perovskites. This work is sectioned into the following chapters:

Chapter-1: Introduction and Literature Review gives an overall perspective on the importance of understanding the crystallization mechanism of lanthanum-based oxide perovskites; and how hydrothermal synthesis has been used to synthesize various binary, ternary, and quaternary metal oxides in controlled shape, composition, and phases.

Chapter-2: Static hydrothermal synthesis: controlled nucleation and growth talks about the conventional hydrothermal synthesis (SH-static hydrothermal) approach to generating lanthanum manganite. The key outcomes of these studies were the observations that solid-liquid diffusion governs the material formation, and the average size of the materials does not vary with time. Based on these results, it was imperative to study the factors that governed the kinetics of the reaction. We aimed to approach this by using the same- or similar- structure materials as precursor seeds, commonly known as substrate/seed-assisted synthesis. This method is well explored in synthesizing many materials such as graphene, TiO_2 , ZnO , GaN but has not been utilized to synthesize these bi- and tri- component systems. This clause has been explored in this chapter, and micro- and submicron cubes of LaCrO_3 , LaKMnO_3 (LKMO), LaFeO_3 have been synthesized. Another more controlled method to control the hydrothermal synthesis of LPOs is described in the next chapter.

Chapter-3: Dynamic hydrothermal synthesis: tailoring nucleation and growth deals with dynamic hydrothermal synthesis (facilitation of synthesis through continuous stirring) of perovskites to enhance the diffusion and simultaneous availability of the reagents. Thus, successfully synthesized LaCrO_3 , LaKMnO_3 (LKMO), LaFeO_3 phases and hollow cubes, truncated cubes, and nanorods of LKMO. Further, the electrical transport characteristics of LKMO nanorods have been discussed. A mechanism of nucleation has been proposed based on the observations. This was further extended to obtain double perovskites such as LaKNiMnO_6 , LaKMnCrO_6 , and LaKCrFeO_6 .

Chapter-4: Applications: study of electrocatalytic and organocatalytic activity of LKMO microcubes depicts, in the first part, the electrocatalytic applications of the above-synthesized LKMO microcubes in metal-air battery systems. The catalyst performed

remarkably well, with improved kinetics and activity for ORR (half-wave onset potential of 0.78 mV) and OER (onset potential at 10 mA cm⁻² current density is of 1.66 mV) in an alkaline environment. Additionally, as a two-electrode, this catalyst rendered the potential for use as an electrode in zinc-air batteries with unprecedented stability (no visible sign of degradation even after 1000 cycles; 100 hours). Subsequently, in the second part, the synthesized LKMO microcubes were used as organo-catalyst to evaluate catalytic efficiency in the reduction of α,β -unsaturated carbonyl compounds, and nitroarenes. The findings were compared to those obtained with La_{0.5}Sr_{0.5}MnO₃ (LSMO) and submicron LaMnO₃ (LMO), revealing that only LKMO exhibits enhanced catalytic activity against the selective reduction of α,β -unsaturated carbonyl compounds, and nitroarenes. LKMO demonstrated better catalytic conversion of 100% at 120 °C and 7 bar pressure, without any external activating agents. Systematic surface characterization demonstrated that LSMO covered microcubes with an inert SrO layer, while LKMO exposed catalytically active Mn sites on the surface. Due to its low cost, non-pyrophoric and catalytic properties, LKMO could be a possible substitute for precious metals in catalytic hydrogenation.

Chapter-5: Doping with chemically hard elements to improve photocatalytic properties of ZnO nanostructures illustrates simple and most abundant ZnO hydrothermally doped with divalent (Ni²⁺ and Sr²⁺) and trivalent (Cr³⁺ and Al³⁺) cations. The photocatalytic activity of undoped and doped ZnO nanostructures was tested towards the degradation of methyl orange (MO), methylene blue (MB), and congo red (CR). Al-ZnO exhibited a 91.25% removal rate compared to 40.88% for pure ZnO in MO degradation under ultra-violet (UV) radiation. The trend of photocatalytic activity of different photocatalysts (Al-ZnO > Sr-ZnO > Cr-ZnO > Ni-ZnO > ZnO) is identical to the trend of Pearson hardness values of Al³⁺ > Sr²⁺ > Cr³⁺ > Ni²⁺ > Zn²⁺ indicating the chemical hardness of the dopant plays a significant role in the photocatalytic dye degradation. These reactions also reveal that the dopant type affects the morphology of the ZnO nanostructures in given reaction conditions. We propose the polarity-induced mechanism to understand the changes in the morphology due to different dopants.

Chapter-6: Conclusion and Future Scope summarizes and concludes the scope of this current thesis. Describes the possible directions this thesis could open for future studies. Further, it also provides a foothold on novel discoveries that are possible through widening the horizons of this work.

Research highlights

- 1) A practically safe, green, economical, and energy-efficient protocol for the low-temperature, hydrothermal solution-based synthesis of potassium substituted cubic LMO perovskite materials has been described.
- 2) The influence of different reaction parameters, such as KOH concentration, precursor concentration, precursor ratio, and duration, on the regulation of size, morphology, and crystallinity of the LKMO structures has been examined.
- 3) A mechanism for the creation of transition metal-oxide perovskites has been proposed.
- 4) A generic strategy that could be applied to initiate better control on phase, morphology, and nucleation in similar La-based perovskites such as LaCrO_3 and LaFeO_3 has been developed.
- 5) Submicron cubes of LaCrO_3 , LKMO, and LFO have been synthesized by enhancing the secondary nucleation approach.
- 6) A dynamic hydrothermal method aided by stirring has been introduced to LPO systems.
- 7) In the DH method of synthesis, a shear-induced jamming phenomenon that aids nucleation has been proposed.
- 8) Hollow and truncated micron size LKMO cubes were synthesized.
- 9) Double perovskites of manganite were synthesized.
- 10) The dynamic hydrothermal method was used to obtain LKMO nanorods.
- 11) LKMO single nanorod FETs were fabricated and their electrical properties were investigated.
- 12) Electrocatalytic properties of the LKMO microcubes were explored by employing them as electrocatalysts in Zn-Air batteries.
- 13) As an organo-catalyst, LKMO microcubes promoted selective reduction of chalcones and nitroarenes with higher activity at nominal conditions.
- 14) The influence of dopant chemical hardness on photocatalytic dye degradation was investigated using ZnO nanoparticles.
- 15) A polarity-induced morphology tuning mechanism in ZnO nanoparticles is presented.

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Abbreviations

LPO	Lanthanum-based Perovskite Oxide
SH	Static Hydrothermal
DH	Dynamic Hydrothermal
TMSS	Topochemical Molten Salt Synthesis
LKMO	Lanthanum-Potassium-Manganese-Oxygen
STO	Strontium Titanate
La(OH)₃	Lanthanum Hydroxide
LCrO	Lanthanum Chromate
LFO	Lanthanum Ferrite
LNO	Lanthanum Nickelate
SEM	Scanning Electron Microscopy/Secondary Electron Microscopy
TEM	Transmission Electron Microscopy
HRTEM	High Resolution Transmission Electron Microscopy
SAED	Selected Area Electron Diffraction
IFFT	Inverse Fast Fourier Transformation
HAADF	High Angle Annular Dark Field
STEM	Scanning Tunnelling Electron Microscopy
XRD	X-ray Powder Diffraction
ND	Neutron Diffraction
MCE	Magnetocaloric effect
EDS/EDAX	Energy Dispersive Spectroscopy/Energy Dispersive X-ray Analysis
eBL	Electron Beam Lithography
VSM	Vibrating Sample Magnetometer
PPMS	Physical Property Measurement System
LSV	Linear Sweep Voltammetry
CV	Cyclic Voltammetry
BET	Braunneur Emmet Teller
XPS/ESCA	X-ray Photoelectron Spectroscopy/Electron Spectroscopy for Chemical Analysis
UPS	Ultraviolet Photoelectron Spectroscopy

CHAPTER-1:

Introduction and Literature Review

1.1 Introduction

Materials science played an essential role in the evolution of humans by forming a pivotal role in today's technological advances. The discovery of graphene ushered in a similar revolution for which new materials with multidimensional properties were required. For the synthesis of novel materials, new synthetic techniques must be developed. This is where a materials chemist's knowledge comes in handy. We need these chemists' unique vision to develop new synthetic techniques and improve existing ones.

1.2 Background on Perovskite:

1.2.1 Crystal structure:

Perovskite is one of the structural materials which has received tremendous attention for wide-ranging properties because of its flexible crystal structure. The word 'perovskite' was coined after Russian mineralogist Lev Aleksevich Perovskite in the 1840s.¹ It has a generic ABO_3 structure which can accommodate around 90% of the metals from the periodic table. The beauty of this is that it provides the flexibility for the metal ions to move around within the framework. The A and B cationic sites can be easily substituted because of which one can easily tune these materials' physical and chemical properties. Generally, the A-site cation is larger than the B-site cation, having 12-fold cuboctahedral coordination (**Figure 1.1(a)**). In addition to the simple atomic structure, the ideal cubic cell unit could have a specific cation distribution. The A cations are at the cube's corners, while the B cation is in the center and is coordinated with six anions in the cubic unit cell's face-centered positions (**Figure 1.1(b)**). In most cases, transition metal ion takes up the B-site, whereas lanthanides, alkali, and alkaline earth metals suit the A-site. Because of the partially filled d-orbitals, the B-site cations possess the flexibility to change their oxidation states by accepting or donating electrons.

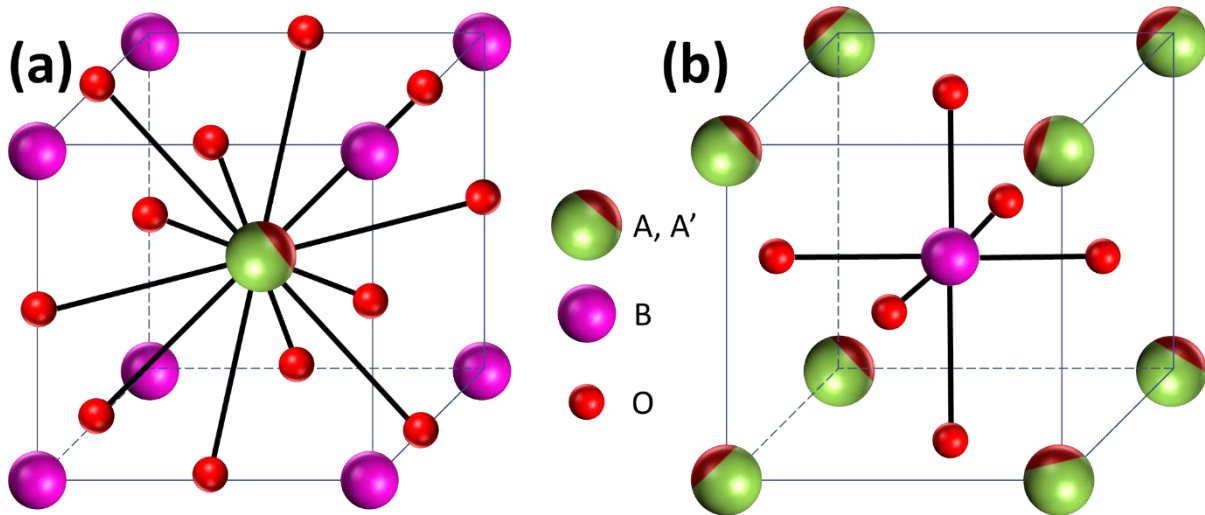


Figure 1.1 Structural representation of two different views of the $AA'BO_3$ ideal cubic perovskite (a) A/A'-atom at the centre and (b) A/A'-atom at the corner of the cubic unit cell.

In 1920, Goldsmith proposed some important features that relate the crystalline structure to chemical composition parameters as a result of his early exploration of the perovskite crystalline structure.^{1,2} The tolerance factor was one of the most fundamental ideas offered. The tolerance factor is a method that is commonly used to forecast the structural organisation and stability of a specific perovskite composition, both chemically and physically. This aspect is considered before deciding on the best processing approach for perovskite preparation.

Because strontium titanate oxide exhibits the atomic packing seen in **Figure 1.1(b)**, $SrTiO_3$ (STO) has the idealised cubic (fcc) structure. The atomic distribution of an ideal cubic structure is as follows, in conformity with the Wyckoff positions:

‘A’ cations are at 0.5, 0.5, 0.5 in XYZ coordinate system (**Figure 1.1(b)**), while

‘B’ cation is at 0, 0, 0 (**Figure 1.1(b)**), and

‘O’ anions are situated at 0.5, 0, 0; 0, 0.5, 0; and 0, 0, 0.5.

STO's idealised cubic structure has been altered in numerous ways. One of them is cation deficit at the A-site, which results in a ReO_3 type structure.² When this is followed by an octahedral rotation, another structural distortion leading to hcp type RhF_3 occurs. Aside from vacancies and tilts, minor atomic displacements from their ideal location also vary the final symmetry by a few tenths of an angstrom, resulting in significantly varied final symmetry across various materials. Because of the pure flexibility of its crystal structure, all of these different types of distortions may indeed be accommodated in a perovskite structure.¹⁻³ Aside

from the geometrical features, the electrical structure introduces a particular distortion known as the Jahn-Teller phenomenon.

1.2.2 Origin of Crystal Distortion:

The structural distortions on perovskite structure are caused by variances in ionic radii, as established by Goldschmidt's pioneering study.¹⁻³ The equation 1 is presented that geometrically connects the ionic radii (r_A , r_B , and r_O) for the perfect cubic cell with the lattice parameter a_0 . Where, r_A is the average ionic radii of A-site cations, r_B is the average ionic radii of B-site cations and r_O is the average ionic radii of oxygen atoms.

$$a_0 = \sqrt{2}(r_A + r_O) = 2(r_B + r_O) \dots \dots \dots (1)$$

The equation involving the unit cell length ratio is known as the Goldschmidt's tolerance factor t , and it is used to measure the amount of distortion obtained by a certain perovskite-structured molecule. Fundamentally, it accounts the ionic radii of the constituent atoms that meet the chemical stoichiometry and promotes pure ionic bonding between them; the mathematical expression for t is provided in Eq (2).

$$t = \frac{(r_A + r_O)}{\sqrt{2}(r_B + r_O)} \dots \dots \dots (2)$$

According to the above Eq. (2), an ideal perovskite cubic structure has a "t" value of one, which can be determined for SrTiO₃ with ionic radii of 1.44, 0.605, and 1.40. When the A ionic radius is somewhat smaller, the value of t is less than one, and the octahedral unit [BO₆] physically tilts forward to occupy the extra space. According to Eq. (2), the optimal value of grade of tolerance (t) to achieve cubic structure in perovskite compounds lies between 0.89 and 1.²⁻⁴

Among the existing, orthorhombic structured perovskites is most abundant and hugely explored for its diverse chemistry. This orthorhombic distortion arises because of the tilting of BO₆ octahedra in the formation stage itself. But noticing it is challenging because of the temperature raise which dilutes the tilt angle.⁵ Typically, an orthorhombic structure has unit cell twice in size that of cubic structure (**Figure 1.2(a)**). LMO like orthorhombic perovskites (where L = rare earth metals and M = 3d transition metal) shows high intrinsic distortion when their L³⁺ ionic size is nearly 1.11 Å. This can be reduced by increasing the average A-site ionic size to above 1.11 Å. Till date LTiO₃, LVO₃, LMnO₃, LFeO₃, and LNiO₃ orthorhombic perovskites are exhaustively studied by the crystallographers. Along with the structural factors, cooperative Jahn-Teller effect on the cation orbital ordering also studied.⁶

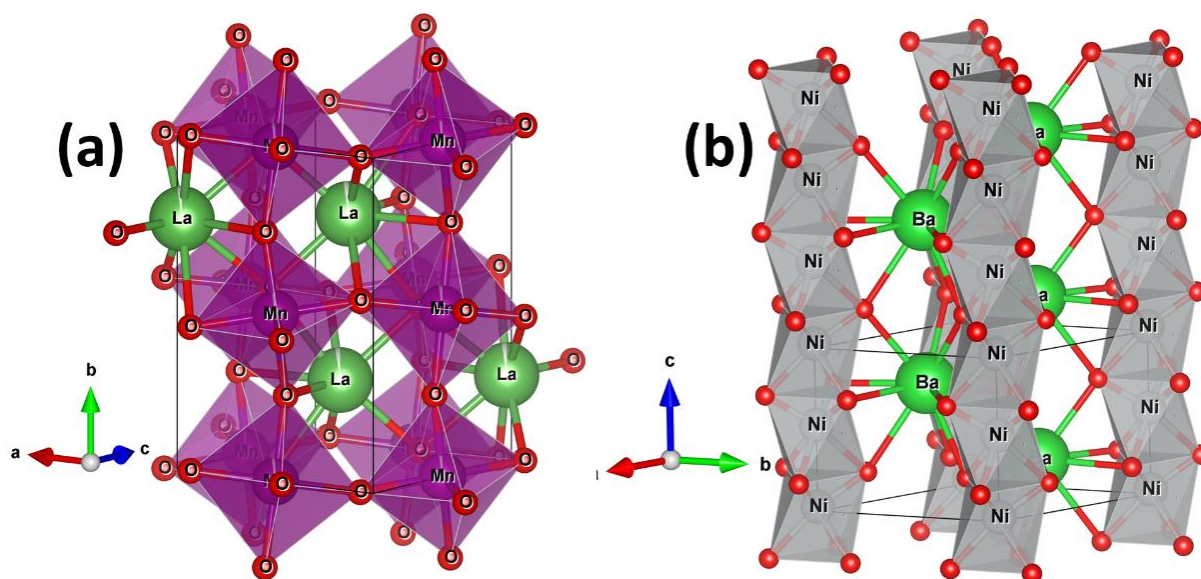


Figure 1.2 Schematic representation of (a) orthorhombic (LaMnO_3) and (b) face-sharing hexagonal (BaNiO_3) perovskite.

Another popularly studied systems are hexagonal structured perovskites. These compounds have tolerance factor, $t > 1$. This distortion takes place when one of the cations among A or B site becomes extremely large or small. Because of which the cubic structure distorts forming hexagonal as shown in **Figure 1.2(b)**. This structure is formed by closely packed layers of face sharing octahedra units. The stability of this is poor compared to the one which is formed by corner-sharing octahedra. But, some overcome this constraint by endorsing B-B bonding between BO_6 octahedra which is strong enough to endure the metal-metal repulsion.¹ Because of these stability constraints, hexagonal perovskites are uncommon. BaNiO_3 is an example of hexagonal perovskite which has $P6_3/\text{mm}$ space group. **Figure 1.2(b)** depicts a representative hexagonal perovskite unit cell structure. Here, the chains of face-sharing BO_6 octahedra are provided, oriented along the c-axis. Multiple structures can be obtained by further distortion as observed in cubic systems.

In this case, deformation does not have significant effect on the unit cell, whereas in a typical cubic cell any small amount of deformation leads to transformation into rhombohedral symmetry. As a result, the unit cell is likely to include at least one or two rhombohedral polyhedral units with angles of 60° or 90° . The anions, on the other hand, are displaced, resulting in the development of a huge unit cell with a 60° angle.

BaTiO_3 is a widely known tetragonal-structured perovskite exhibits a stable ferroelectric nature at ambient temperature. The TiO_6 octahedra in this structure are somewhat deformed (four Ti-O bond at $2.0 \text{ \AA}$, one at $1.86 \text{ \AA}$, and a longer one at $2.17 \text{ \AA}$). At $2.80 \text{ \AA}$, four oxygen

atoms are in cooperation with the barium atom, four at 2.88 Å, and four more at 2.83 Å. Other tetragonal perovskites (PbHfO₃, SrPbO₃, SrZrO₃, AgTaO₃, and others) have single-molecular cells and are isotypic with BaTiO₃. However, at high temperatures, majority of these materials exhibit tetragonal structure, making crystallographic investigations challenging. In general, a large number of perovskite-like materials undergo polymorphic changes. Some of these are significant in terms of physical features and uses. BaTiO₃ and KNbO₃ exhibit this kind of transformation from Rhombohedral to Orthorhombic to Tetragonal to Cubic under elevated temperatures.

Ferroelectric characteristics are found in low-temperature crystalline materials (cubic, tetragonal, and orthorhombic). In nature, all phase transitions are reversible, and all polymorphic forms contain a pseudo-cubic unit cell with $a_0 \sim 4\text{\AA}$. As a result, the polymorphism variation in the perovskite structure has an even greater influence on the crystal chemistry of this class of materials.⁴⁻⁶

1.2.3 Applications:

Since the discovery of the usefulness of perovskites materials over five decades ago, hundreds of research papers have been published explaining the physical and chemical characteristics of perovskites. These investigations have offered useful information about the principles of the chemical and physical components that contribute to structural aberrations in ABO₃ materials. The previous research also shows that there are still surprises in store for this specific set of compounds, particularly the perovskites with low tolerance factors, such as those shown in the phase stability diagram FeTiO₃-LiNbO₃-Pnma.⁷ A large number of elements in the periodic table are most likely found at A or B unit cell locations. This fact opens up a vast array of compounds with structural similarities and a wide range of characteristics. All of these functionalities are triggered by the existence of multiple degrees of freedom (**Figure 1.3**) in these materials, such as charge, spin, orbital, and lattice, and the coupling between them leads to exotic magnetism, superconductivity, and electricity.⁸ Colossal magnetoresistance,⁹⁻¹³ ferromagnetism (Sr₂FeRuO₆), Ferroelectricity (BaTiO₃), weak ferromagnetism (LaFeO₃), superconductivity (Ba_{0.6}K_{0.4}BiO₃),¹⁴ and high thermal conductivity (LaCoO₃) are among the most notable features. Insulating to metallic transitions are of particular interest in the design of devices for thermistor applications (LaCoO₃), fluorescence is applicable for laser devices (LaAlO₃:Nd), and transport properties have piqued the interest of researchers working on the development of high-temperature thermoelectric power devices (La₂CuO₄).^{1,5,7,9-15}

These materials have been studied for potential application in the preparation of solid electrolytes. Because of their superior performance, barium cerate (BaCeO_3) and barium zirconate (BaZrO_3) have been utilized in a variety of electrochemical devices. A solid electrolyte serves three important functions: (1) separates the anode from the cathode in the electrochemical cell (oxidizing and reducing sides), (2) if required, functions as an electronic insulator, enhancing the flow of electric current through an external circuit, and (3) it has a high ionic conduction coefficient, which is required to control the flow of electric current in the external circuit.¹⁶ Similarly, proton-conducting ceramics applications are classified based on two basic functions: (1) ability to produce electromotive force EMF upon the application of chemical potential and (2) ability to enhance electrochemical ion transport (oxygen or hydrogen) with an external power source. The use of an electrolyte which conducts cations mainly protons upon application of chemical potential finds a potential application in a device that can generate electrical energy.^{17,18} This electrochemical notion sparked the development of solid oxide fuel cell (SOFC) power sources. The SOFC is a type of energy conversion device that creates electricity through an electrochemical process that occurs at temperatures exceeding 800 °C. The mechanism to create electricity is the chemical reaction that occurs between the fuel (methane, hydrogen, natural gas) and the oxidizing agent (oxygen from air).^{16,17} The electrodes are the primary components accountable for the redox reaction in this specific electrochemical converter device; ion transport occurs in all cell parts (solid electrolyte and electrodes). In contrast to rechargeable battery systems, the SOFC does not need to be recharged and simply has to be continually supplied with a specific fuel to generate power.^{16–}

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Some perovskites have been used in a variety of applications, including photocatalytic processes. Various titanates and cobaltites were used to conduct considerable research on photocatalytic oxidation. In contrast to the easily reducible LaCoO_3 , titanates (SrTiO_3 and BaTiO_3) were shown to have a high photocatalytic action. Because certain rare earth elements (Nd^{3+} or Sm^{3+}) are inserted in the A site of the structure, perovskite materials have showed remarkable performance as hosts in laser systems. These are some of the primary components used in the manufacture of laser ion devices. Furthermore, certain perovskite oxides are highly resistive to electric field, making them ideal for the manufacture of dielectric materials that are poor electrical conductors and are commonly utilized for electrical insulation in power systems. Furthermore, these materials can be employed for energy storage, such as in capacitors. Typical conductor and semiconductor oxides include either B-ions with a net charge less than that of the stable B^{4+} , or B-ions with two distinct net charges. Materials such as LaVO_3 , LaTiO_3 , and

SrMoO₃ have been identified as excellent conductors or semiconductors. Sound generation, mechanical actuation as in piezoelectric materials as a result of dimensional change under a voltage source, transducers as in condensers, piezoelectric recorders, temperature detection as in pyroelectrics, and liquid crystals used in alphanumeric displays are all examples of dielectric perovskites' diverse applications.

Oxygen sensors are also made using perovskite oxide materials. The mechanism at work in these sensors is electrical conductivity in nature, which increases oxygen adsorption in the crystalline structure. The electrical conductivity of ABO₃ is related to the partial pressure of oxygen and the concentration of oxygen vacancies. SrTiO₃ is a compound with a high sensitivity to oxygen motion. SrSnO₃ is also a potential material for combustion monitoring sensors. Understanding the link between the composition of materials and crystalline structure is required for the creation of functional materials.^{16–22}

1.2.4 Synthesis:

The synthesis of perovskite oxide has received a lot of attention as design of variety of technological features and its development plays a critical role in attaining materials for diverse applications with unique physical morphology. Thin films, porous solids, and monodispersed powders of nanometric size are some of the recent breakthroughs. Furthermore, the desire for certain structural ordering displaying chains that improve unidirectional characteristics has encouraged several research groups to seek the fabrication of novel perovskite oxides with specified compositions.^{23,24} During the early 1940s, the standard solid-state reaction process (ceramic method) was widely utilized to generate perovskite oxide powders. For the fabrication of single crystals of different perovskite oxides, a comparable approach, the flux method, which is likewise based on solid-state reactivity, has been used.²⁴ Both techniques are sufficient for the synthesis of oxide compounds because they allow for the simple achievement of the required conditions owing to the reaction taking place in an air environment and at ambient pressure.²⁵ Homogeneous grinding and mixing are common processes in solid-state reaction techniques. The most often used solid reactants are oxygen and carbonate. These are blended in stoichiometric proportions prior to the application of a heat treatment at high temperatures (1000 °C) for extended periods of time (several hours or days). Grinding and heating cycles are repeated until the required pure phase is obtained. There is some uncertainty about the universal method's utility and ease of use, although meeting the prerequisites for atomic diffusion might be difficult. As a result, high temperatures are required to speed atomic

diffusion and avoid the creation of metastable phases. However, this has the drawback of increasing the volatility of reactants such as lead oxide (in the case of Pb^{2+} compounds such as PbTiO_3). Furthermore, the necessity to regrind the materials numerous times necessitates a significant amount of energy and time, as these factors have a considerable effect on obtaining the necessary product performance.^{23–25}

The solid-state metathesis (SSM) reaction technique has recently emerged as an effective processing strategy for the synthesis of a diverse variety of non-oxide materials. The procedure that takes place under thermodynamic equilibrium conditions is more efficient than the traditional solid-state reaction approach.^{23,26} The SSM reaction technique is a version of the SHS process in which chemical reactions occur at a quick pace. In a typical SSM reaction, an ionic exchange occurs between the reactants, resulting in thermodynamically stable perovskites. A substantial enthalpy shift happens along with the reaction, and high adiabatic reaction temperatures are attained. The solid-state metathesis process is carried out using alkaline precursors such as chalcogenide, silicide, or boride; the salt is then reacted with a metal halide in accordance with the general reaction as shown below.



where P = Li, Na, K, Mg, Ca, Sr, or Ba;

Q = B, Si, N, P, As, Sb, Bi, O, S, Se, or Te;

R = transition group, principal group or actinide metal, and

X = halogen.

One of the primary limitations of the techniques discussed above in terms of energy expenditure is the efficacy of transferring the heat necessary to achieve the chemical reactions under solid-state conditions. "Microwave irradiation" is one of the new processing approaches that has solved this problem. Heating is directly delivered to the solid reactants during microwave processing settings because it results from a molecular interaction between the electromagnetic field and some of the components forming the solid. This interaction changes according to parameters such as the dielectric and magnetic characteristics of the solid reactants, frequency and microwave power production, microwave permeability, and material size and density.²⁵ In compared to the traditional and SSM approaches, microwave heating enhances reaction kinetics by 10 to 1,000 times, leading in considerable variations in the crystalline structure and characteristics of the reaction products generated. Microwave processing has recently been extensively studied in the production of several perovskite-structured materials. The processing procedures for several single-phase perovskite materials synthesized using microwave irradiation can be found in reference²⁵.

Chemists working in material research are continuously looking for innovative ways to process materials that use less energy and are more ecologically friendly. The innovative approaches are extremely useful, allowing us to synthesize perovskite oxides under much gentler circumstances than the traditional solid-state procedure.^{27–29} Non-traditional ceramic materials, such as soft chemistry approaches, have a more homogenous distribution of ingredients, are purer, and may be created with a broad range of partial dopant replacements and forms (i.e. coatings and fibers) that are not truly attainable by normal procedures.³⁰ Soft chemical processing techniques also enable the creation of bulk monoliths without the usage of intermediary powders, which is accomplished by supercritical drying a precursor gel.³⁰

Coprecipitation and sol–gel methods have evolved as processing options for ceramic powders, including certain perovskite compounds, since the early 1960s. Certain innovative characteristics of these approaches are linked with the control of physical parameters such as size, aggregation state, and purity of the generated molecules. The operation of these technologies is based on concepts related to the mechanism of colloidal formation in aqueous fluids at extremely low temperatures. The beginning chemical reagents, i.e. salt solutions containing specified cation species, undergo chemical treatments at mild temperatures (100 °C) to generate stable colloid dispersions, or sols, when combined with an acid or basic solution in the aqueous "sol–gel" procedure. The particle size of the sols ranges from 1 nm to 1 μ m. The sols can be made from highly hydrolysable salt reagents such as ZrCl_4 or TiCl_4 , i.e. zirconate or titanate in the case of perovskite. The crystallization stage in "sol–gel" processing occurs at substantially lower calcination temperatures than those involved with the creation of the identical crystalline phase by the solid-state method. A preliminary dehydration stage is performed before to this procedure to separate the gel from the sol.^{30,31} Over the last three decades, enormous efforts have been made to discover novel alternative soft chemistry methods that allow us to lower the sol–gel crystallization temperature and manage the stoichiometry of the reaction products. One of the recently suggested processing techniques includes reacting a hydroxide reagent precursor, either NaOH or KOH, with $\text{Nb}(\text{OH})_5 \cdot x\text{H}_2\text{O}$. The dissolving of Nb_2O_5 in an HF solution was the first step in preparing the niobium precursor. The hydroxide powder combination, which contained stoichiometric proportions of the precursor reagents (sources of K and Nb), was heat-treated in air for 6 hours at temperatures ranging from 200 to 700 °C. The orthorhombic KNbO_3 perovskite phase was formed at 700 °C without the generation of any by-products, whereas the NaNbO_3 particles were created at temperatures as low as 500 °C.³¹

Approaches based on the synthesis of polymer precursors with organic solvent solutions, such as the Pechini technique and its alternative process polymerized complex method (PC). A chemical interaction between appropriate organometallic precursors and the liquid solvent results in the formation of a polymer comprising the cation precursors of the perovskite molecule. The polymer precursor is created by a low-temperature (200–400 °C) treatment that results in the compositional homogeneity of the components assigned to the polymer structure.^{32–34} Typically, amorphous polymer precursors are calcined to facilitate crystallization of the stable KNbO_3 ^{32,33} and BaTiO_3 ³⁴ ceramic phases. Furthermore, polymer precursors can be engineered in terms of components and composition to increase their combustion performance. A combustion approach has been developed based on this idea. As a novel soft chemical pathway, a combustion process has been developed based on this idea. This technology is useful in industry since it allows for the processing of huge batches of powder. Furthermore, the creation of nanocrystalline oxide ceramics powders occurs in a relatively short period at a lower calcination temperature. The particles generated have the maximum degree of purity as well as properties such as narrow particle size distribution and a greater surface area, which improves sintering ability.^{33,34} The exothermic breakdown of a fuel-oxidant precursor is the basis for the combustion process. The reaction favors the development of a fine monodispersed powder with a perovskite structure, or a partly degraded precursor with significant carbon traces. The final outcomes are determined by the precursor and the fuel-to-oxidant ratio utilized to perform the combustion. During the decomposition, the organic component breakdown allows for a quick increase in temperature paired with gas generation, resulting in particle coalescence and short diffusion paths. As a result, a foamed porous aggregate formed by agglomerated pure-phase nanoparticles may be generated at low temperatures.^{33,34}

To generate perovskite oxides, several processing approaches involving non-equilibrium reaction conditions are also of industrial relevance. The glycothermal method is a unique process that involves crystallization of oxide particles, such as KNbO_3 , and is carried out at supercritical pressures, depending on the organic solvent used. The organic solvent capacity at supercritical circumstances is similar to that of conventional polar liquids, but it has greater transport capabilities (diffusivity, viscosity, and thermal conductivity, among others).³⁵ In terms of repeatability and environmental concerns, this technology is dependable. However, the expensive cost of organometallic reagent precursors and the limited kinds of organic solvents are some drawbacks of this approach. Hydrothermal processing, a comparable technology that uses just water as the solvent, has been widely employed for more than three

decades to produce a wide range of perovskite materials. In the mid-twentieth century, this method was investigated in the engineering disciplines of crystal formation and metal leaching. Until now, hydrothermal technology has extensively covered a variety of multidisciplinary domains of materials science. Hydrothermal media have lately been used in the field of solid-state chemistry to prepare a wide range of ceramic compounds. Because of the influence of a combination of factors, such as solvent media, temperature, and pressure, on the ionic reaction equilibrium, this method provides an efficient reaction environment for synthesizing perovskite powders. The traditional hydrothermal (CH) technique is an effective approach for enhancing the crystallization of micro/nanometric morphology regulated and crystal growth-oriented particles. Furthermore, this approach is dependent on the solubility of inorganic salts in water under varying temperature and pressure circumstances. The vapor pressure is another important component that has a significant impact on boosting the heterogeneous reactions in this process.

1.3 Hydrothermal Synthesis:

The term "hydrothermal" is derived from the etymological root of the Greek word "hydrous," which means "water," and "thermal," which means "heat." The word "hydrothermal" derives from geology. Sir Roderick Murchinson, a Scottish geologist, presented this name to the Scientific Society in the mid-eighteenth century. In practical terms, he stated that the development of mineral species on Earth's crust is caused by the reaction of water subjected to high temperatures and pressures. Another acceptable technical definition by the Scientific Society refers to the word "hydrothermal" as any heterogeneous chemical reaction that occurs in the presence of a solvent medium. At temperatures over 25 °C and pressures above 0.1 MPa, the reactions take place inside a hermetically sealed vessel system. It makes no difference whether the solvent is aqueous or non-aqueous under hydrothermal conditions. Under hydrothermal circumstances, the crystallization of solid phases occurs at autogenous pressure, which is produced by the saturated vapor pressure of the fluid at the prescribed temperature and composition of the hydrothermal solution. In this regard, gentle operating conditions, such as temperatures of treatment below 350 °C and pressures less than 50 MPa, are favored in industrial and commercial processing.^{36,37} The autoclave vessel's lining material strength is the processing characteristic that facilitates the shift from moderate to severe reaction conditions during a hydrothermal treatment. The liner material may corrode quickly under harsh treatment conditions in highly concentrated acidic or basic solutions.

Progress in experimental work study in this field of examination has allowed for a better knowledge of how chemical reactions created in hydrothermal fluids behave. Several oxide and non-oxide materials were crystallized with proper adjustment of the experimental conditions during the hydrothermal treatment (T 200 °C and P 1.5 MPa). Recent scientific and technical advances have made hydrothermal synthesis more cost-effective. One example is the synthesis of ceramic particles, which may be done in a single step by combining sophisticated pressure reactor technology with processing approaches described for a variety of inorganic substances.³⁸

1.4 Thermodynamic modelling of the hydrothermal production of perovskite oxide:

The major preoccupation of the chemical community in the early 1990s was the development of monodispersed perovskite oxide tiny particles with regulated shape. Hydrothermal processing is one chemical approach that offers a suitable environment for the manufacture of a large range of ceramic materials. Hydrothermal reaction enables dynamic interaction of dissolved species of various elements by tuning processing parameters such as solvent, temperature and pressure which impact mobility and interaction ability of solvated ions. The reactions lead to the formation of crystalline oxide of single or multicomponent phases. The wide range of precursor chemical reactants (such as water, soluble salts, hydroxides, and oxides, among others) enables the preparation of solvents suitable for use as hydrothermal media. This processing parameter encourages a wide range of reaction paths in order to crystallize a solid in a specific hydrothermal reaction system. In general, phase-pure perovskite oxides with precise stoichiometry, particle size, and shape may be hydrothermally generated in a single step using low-cost chemicals at temperatures and pressures five times lower than those required for standard processing procedures.

Previously, the practical basics were developed from the experience gathered over the previous five decades. To fully exploit the novelty of hydrothermal technology for the processing of inorganic compounds, some critical key processing factors must be considered, such as the selection of a suitable precursor system (highly reactive and cost-effective) required for the optimization of the chemical equilibrium that enhances the crystallization of the desired phase. The success of the chosen experimental strategy may be assessed by adjusting the chemical precursor concentrations and mixing ratios, the pH of the hydrothermal media, the

temperature, and the pressure level.^{39,40} Because of the multiple factors involved, this method is relatively difficult and time-consuming. As a result, since the early 1990s, a technique based on thermodynamic modelling to mimic hydrothermal reactions has been developed to estimate the efficacy of the precursor hydrothermal system.⁴¹ To simulate the chemical processes, OLI Systems Inc. (USA) created an algorithm that includes the thermodynamic foundation. The model can generate thermochemical data for a wide range of perovskite compounds with excellent consistency and precision, including single-phase BaTiO_3 , SrTiO_3 , CaTiO_3 , PbTiO_3 , SrZrO_3 , and certain chosen solid solutions of $\text{Ba}_{1-x}\text{Sr}_x\text{TiO}_3$ and $\text{PbZr}_{1-x}\text{Ti}_x\text{O}_3$.^{42–50} Furthermore, these investigations have yielded useful information about the behavior of solutions under varying pressure and temperature circumstances. Solubility, stability, and yield product quantity, as well as reaction processes, are among the most studied properties under hydrothermal settings.

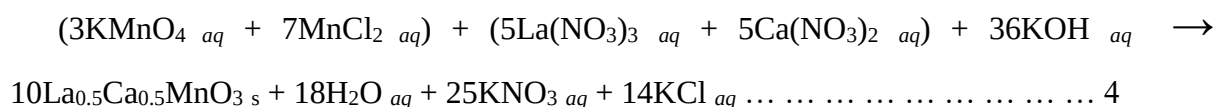
1.5 Lanthanum-based transition metal perovskite oxides (LPOs):

Much research has been done in recent years to synthesize perovskite-type LaCrO_3 and LaMnO_3 (LCrO and LMO) under moderate hydrothermal conditions. Several studies have been published on the synthesis of nanoparticles under hydrothermal conditions. Zheng et al. investigated the effect of solvent alkalinity using KOH concentrations ranging from 2 to 16 M.⁵¹ Another parameter studied in this study was the molar ratio of $\text{KOH/Cr} = 10\text{--}80$, and the hydrothermal treatments were carried out at 260 °C for 7 days. When the chosen B-cation exhibits amphoteric behavior, as in the case of Cr^{3+} , the authors discovered that a highly concentrated alkaline solvent is necessary. As a result, 8 M of KOH was required in the reaction media to produce pure LCrO.

Hydrothermally, a number of solid solutions with perovskite-type structures have been produced. The orthorhombic structured $\text{La}_{1-x}\text{Sr}_x\text{CrO}_3$ (LSCrO) synthesis was carried out at an early stage employing several reaction media to prepare the co-precipitated gel precursor. Triton as non-ionic surfactant and NH_4OH as co-precipitation medium were used to make perovskite LSCrO powders; the pH of the beginning aqueous media was 8. The common ion effect induced by the metal precursor, as well as the steric barrier caused by the presence of surfactant micelles, influenced the reaction rate associated with this procedure.⁵² The dissolution–crystallization method was also used to synthesize LSCrO fine particles in a medium with a NaOH concentration of 0.5 M. Even at the concentration chosen, the solvent promotes homogenous nucleation, resulting in the creation of scattered particles. The

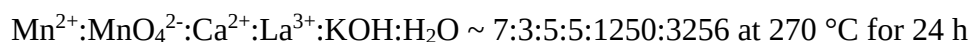
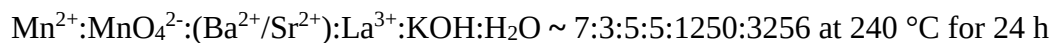
hydrothermal solvent chemically interacts with the solid species in the vessel, resulting in solid dissolution and ionic saturation of the aqueous solvent medium. As a result, the LCrO and LSCrO powders precipitate spontaneously.⁵³ These powders were also synthesized in a KOH solution reaction media. The KOH solution was combined in two phases in this technique. KOH was first combined with CrCl₃ to form Cr(OH)₃, and then used to modify the alkalinity of the reaction medium after the sources of Sr and La were added. Furthermore, because Cr³⁺ is an amphoteric element, a greater alkalinity is required for the manufacture of LSCrO. In fact, the optimal alkalinity range for crystallization of LSCrO solid solutions is 5 to 8 M.⁵⁴ The production of rare-earth and yttrium orthochromite perovskites followed a similar chemical route. Because a rare earth element partially replaces one of the key elements in the perovskite structure, a strong alkaline solution (i.e. 10–12 M KOH) must be employed as a precursor solvent.⁵⁵

Under hydrothermal circumstances, several forms of lanthanum manganite solid solutions have been produced. LBM nanowires (La_{0.5}Ba_{0.5}MnO₃) with a cubic perovskite structure were synthesized at 270 °C for 25 hours with a net [OH⁻] = 10 M using La(NO₃)₃, Ba(OH)₂, KMnO₄, and MnCl₄ as precursors.⁵⁶ Similarly, a solid solution with orthorhombic structure was obtained under comparable circumstances of synthesis of LCM (La_{0.5}Ca_{0.5}MnO₃). The reaction related with the examined hydrothermal processing is as follows⁵⁷:



Perovskite LSMO (La_{1-x}Sr_xMnO_{3+δ}) were recently produced at 150 °C for 20 hours by increasing the suspension (hydrothermal medium) pH to 9 using an ammonia solution. Based on the findings of exploratory runs, a systematic investigation was carried out with the goal of obtaining the appropriate chemical compositions of manganite solid solutions.⁵⁸ The manufacture of this specific perovskite chemical is facilitated by a method of dendritic nucleation. This fact was proven using crystalline phase evolution investigations of the reaction product generated after each treatment, which was carried out in a 4 M NaOH hydrothermal medium. At temperatures below 220 °C, the SLM solid solution with hexagonal perovskite structure and particle shape of hexagonal platelets was created. The mechanism of embryo nucleation, on the other hand, differs when the reaction temperature is 240 °C. Near this temperature, the LSMO particles were created by epitaxial nucleation, which occurs preferentially at the hexagonal platelet's edges. As a result, under the hydrothermal conditions studied, the formation of "tree-like" LCM dendrites is favored.⁵⁹ Spooren et al. examined the

synthesis of LCMO, LSMO, and LBMO solid solutions using a KOH reaction medium in recent years; the use of alternative solvents was also researched.⁶⁰ However, the generation of reaction by-products was not prevented in any of the novel solvents studied. The hydrothermal synthesis of LBMO, LSMO, and LCMO in KOH solvent solution was accomplished for various molar compositional ratios of:



Several authors are now engaged in the investigation of perovskite solid solutions.^{61,62} The research was aimed at developing a facile synthesis approach for LCSMO ($\text{La}_{1-x}\text{Ca}_x\text{Sr}_x\text{MnO}_3$) solid solutions with orthorhombic structure and cuboidal shape. For 72 hours, the synthesis was carried out at 240 °C. The traditional hydrothermal and microwave-assisted hydrothermal methods were used to produce perovskite LAMO ($\text{La}_{1-x}\text{Ag}_x\text{MnO}_3$) using a reaction medium saturated with NH_4OH solution to adjust the solution pH to 8.5.⁶³ In addition, orthorhombic perovskite manganates YCMO (YMnO_3 and $\text{Y}_{1-x}\text{Ca}_x\text{MnO}_3$, $x = 0.07, 0.55, 0.65$) were synthesised in a single step using a moderate hydrothermal synthesis. To stabilize the single phase of YCMO, the optimal concentration of the alkaline solvent medium used to perform the hydrothermal crystallization was 23 M.⁶⁴ The pH of the solvent solution is an important factor in the crystallization of PSMO and NSMO manganites.

The optimal KOH molar concentration for achieving equilibrium between dissolution crystallization events is between 5.0 and 10 M. As a result, solid solutions PSMO ($\text{Pr}_{0.5}\text{Sr}_{0.5}\text{MnO}_3$), NSMO ($\text{Nd}_{0.5}\text{Sr}_{0.5}\text{MnO}_3$), and alkali-earth manganese oxides such as 4H- SrMnO_3 and 2H- BaMnO_3 were synthesized using KOH at the above-mentioned concentrations.⁶⁵ In the case of perovskite RMnO_3 ($\text{R} = \text{Dy, Ho, La, Pr, Nd, Tb, and Bi}$), the high alkalinity of the solutions is a frequent characteristic of the mild hydrothermal synthesis. This high alkalinity creates a crucial condition that has a significant impact on the crystallization and composition of the RMnO_3 product during the hydrothermal synthesis.^{66,67} The high alkalinity of the reaction solution is also an ideal condition that effects the crystallization of perovskite manganites significantly. Thus, to maintain the essential alkalinity for the hydrothermal process, a strong solvent such as KOH must be used.⁶⁸

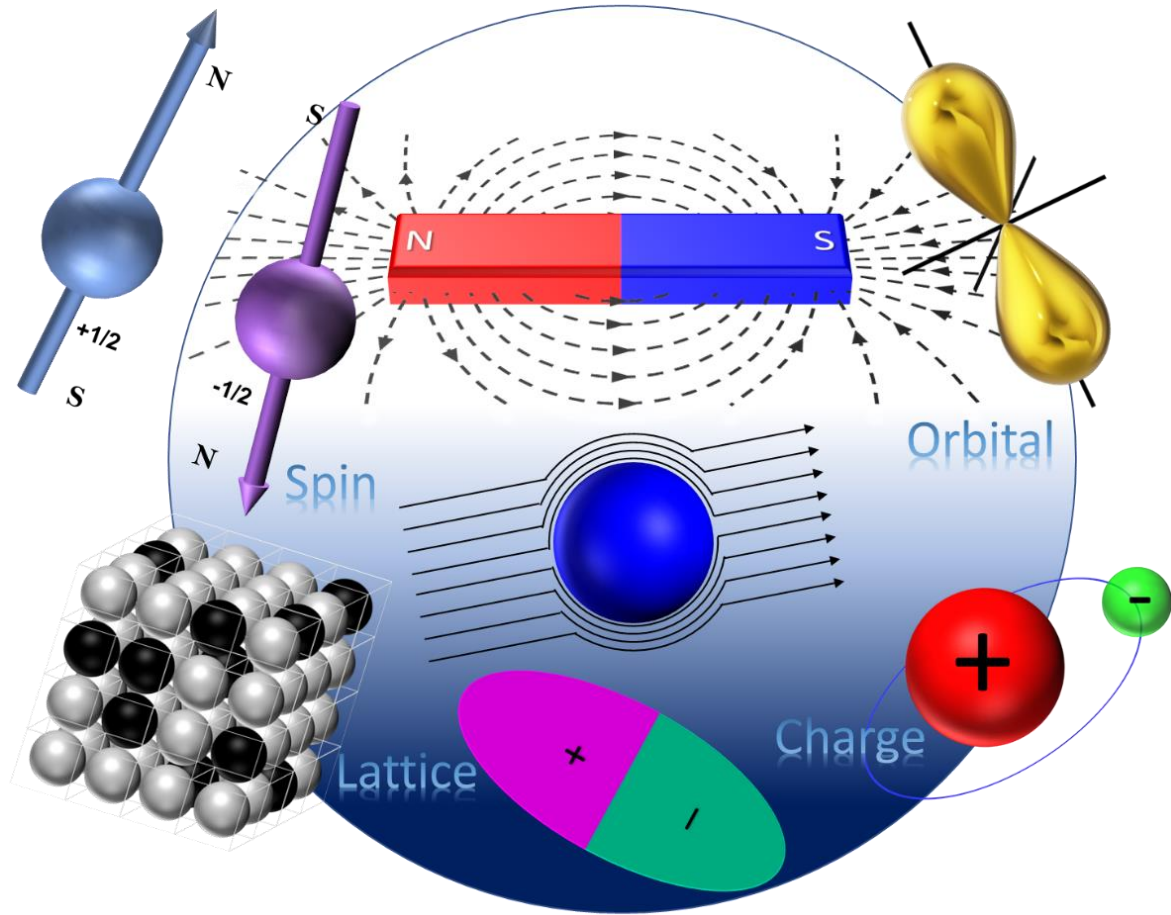


Figure 1.3 A graphical representation of degrees of freedom (D.O.F.).

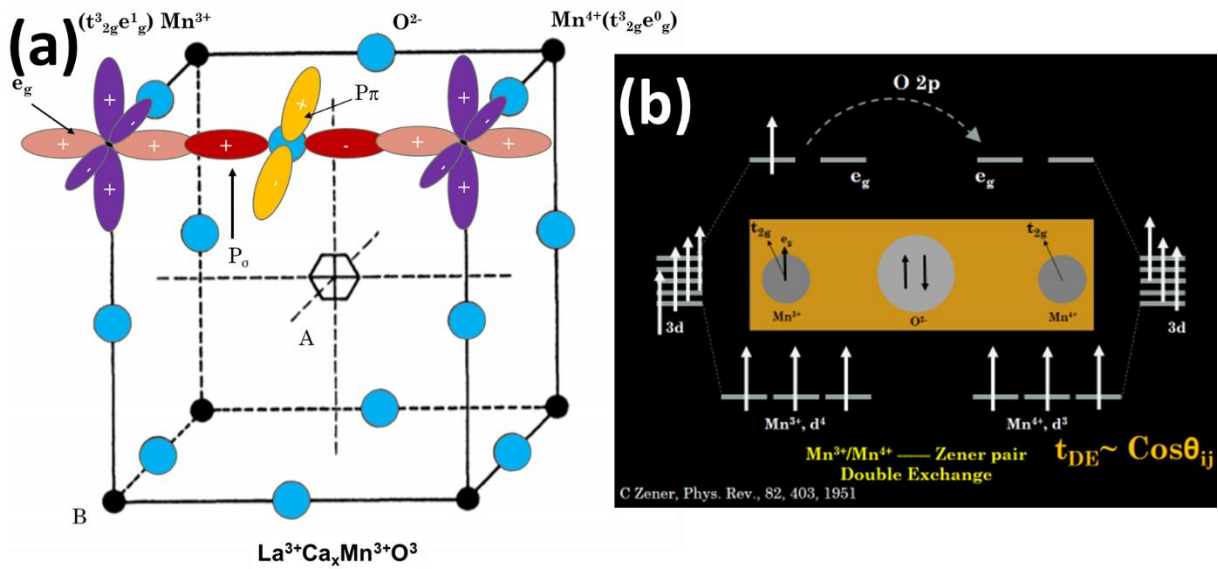


Figure 1.4 Pictorial representation of double exchange phenomena in LCMO (a) (b).⁶⁹

In these perovskite manganites, the $\text{Mn}^{+3}/\text{Mn}^{+4}$ ratio leads to the transition from an insulator to a metallic property (IMT) and further to the magnetic phase transition of the material at temperatures above ambient.⁷⁰ Instead of divalent alkaline earth metals, substituting half of the La^{+3} with monovalent alkali metal ions in these rare earth manganites has the potential to achieve equal amount of hole doping; since the hole density becomes twice that of divalent ion doping for a similar amount of monovalent dopant used.⁷¹ As well increases the exchange integral, θ_{ij} allowing for double exchange and a more ferromagnetic material (**Figure 1.4**). However, the synthesis of such monovalent alkali ion-doped materials raised huge practical concerns due to their high rate of evaporation.⁷² This resulted in very poor control being exerted on the composition of the final product. This further paved the way for the use of alkali halide fluxes for controlled crystal growth and was tested in $(\text{LaX})\text{MnO}_3$ type materials where X could be appropriate monovalent alkali ions such as Na or K.⁷³ This use of fluxes enhanced the rate of diffusion in these solid-state reactions, thus facilitating the crystallization of the final product at comparatively lower reaction temperatures of 700 °C – 900 °C. The perovskites synthesized by this technique; namely. LaNaMnO_3 and LaKMnO_3 showed a ferromagnetic transition T_c above ambient temperature.⁽⁷⁴⁾

Knowledge of crystal growth mechanisms and how different reaction parameters affect their morphology is important to effectively synthesize the material with its desired properties for the desired activity.⁷⁵ This has been the reason why there have been countless attempts by researchers around the world to effectively synthesize materials with a morphology that exhibits a corresponding activity. Solid-phase chemistry has largely dominated the synthesis of these transition-metal perovskite materials to date because of their simplicity, feasibility, and convenience.⁷⁶ The traditional mode of synthesis of these wondrous materials has been solid-state chemistry at reaction temperatures ranging from 600 °C to 1,000 °C.^(71,77–79) However, precise control of the local stoichiometry in low-dimensional ternary or quaternary component systems such as these lanthanum-based materials is very difficult at high temperatures.⁷⁷

Surprisingly, very limited attention has been paid in understanding the mechanism of formation of these wondrous CMR materials such as lanthanum-based oxide perovskites such as LaKMnO_3 (LKMO). On the other hand, the high reaction temperatures involved could probably have been the reason why it has hitherto been difficult to achieve different morphologies such as cube, rod, wire, and truncated shapes in these lanthanum-based transition metal perovskite materials. Furthermore, there are no satisfactory explanations for how the

different reaction parameters affect the nucleation process or the resulting size and morphology of the materials.⁽⁷⁵⁾

Solution-phase synthesis has been largely overlooked, particularly in the domain of synthesis of these types of nanomaterials. With the ability to better control nucleation and morphology even at high reaction temperatures, solution-phase chemistry is an extremely lucrative option for the controlled synthesis of nanomaterials, where controlling the size and morphology is crucial to exhibit the desired magnetic and electrical properties. Lanthanum transition metal-based perovskites, LnTmO_3 or LnATmO_3 alkaline earth metal doped/substituted materials ($\text{Ln}=\text{La}$; $\text{Tm}=\text{Cr, Mn, Fe, Co, Ni \& Al}$; $\text{A}=\text{Na, K, Ba, Sr}$) have not been explored by solution-based chemistry. In complex systems such as these involving more than two components, simple kinetic products with AO or BO phases dominate as well as the desired complex thermodynamically stable products with ABO_3 structures; within the same given reaction time.⁽⁸⁰⁾ In such scenarios, more important and desirable than reaction time or feasibility is the precise control of the nucleation step; so that the synthesis is better channelled. Solution chemistry offers this precise control and fidelity in synthesis than any other synthesis chemistry has shown to be inadequate.

Hydrothermal treatment has been a widely used technique for performing solution-phase synthesis since times immemorial.⁽⁸¹⁾ This is because of the advantages it offers such as: a small reaction volume, access to high temperature and pressure conditions in a safe manner, and the option that comes closest to scalability. In addition, the process is generally safe, environmentally friendly, energy efficient, and inexpensive.

The aim of this current work was to address the crystal phase transitions through synthetic techniques that have not yet been explored and reported at lower temperatures to the best of our knowledge; and to investigate how different reaction parameters affect the morphology of these materials. LaKMnO_3 perovskite oxide microcubes with excellent ferromagnetic and electrical properties were successfully synthesised, characterised, and applied using a viable, safe, energy-efficient, environmentally friendly, low-cost, low-temperature solution-phase hydrothermal synthesis. In addition, it has been thoroughly studied how various reaction parameters such as alkali concentration (KOH), precursor concentration (La), precursor metal ratios (La/Mn), and reaction time play important roles in determining the size, morphology, and nucleation processes of the formation of these materials. A plausible mechanism of formation of this LKMO phase has been proposed, along with a clear elucidation of how altering the mechanism of nucleation from homogeneous to heterogeneous can help in achieving morphological control over the nature of materials synthesized. Interestingly, it was

also proven that this strategy is generic and could be applied to the controlled synthesis of other similar perovskite materials such as LaCrO_3 and LaFeO_3 .

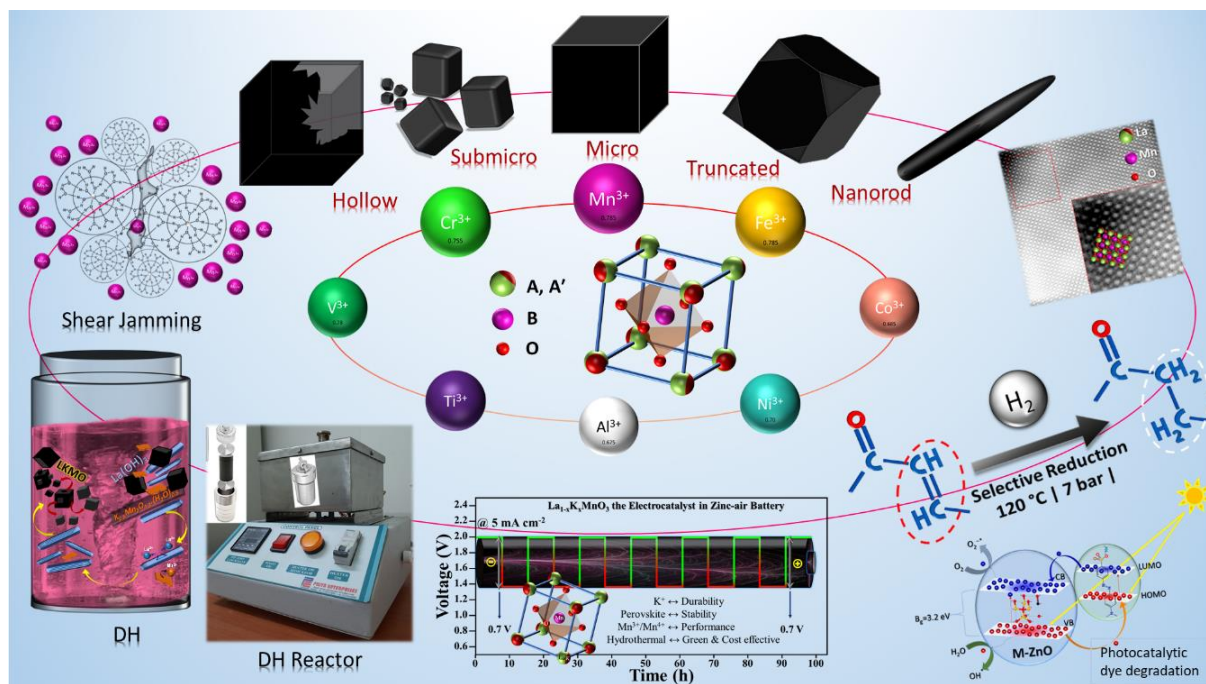


Figure 1.5 An overall graphical representation of this thesis; showing various LKMO morphologies that we have synthesised thus far, nucleation mechanism studied, catalytic applications studied.

Aside from using a precursor for secondary nucleation or heterogeneous nucleation, in-situ stirring proves to improve the kinetics over normal static hydrothermal processes. This is known as the dynamic hydrothermal (DH) process. The DH method has shown several improvements, including improved kinetics, lower reaction temperatures, new morphologies, and controlled sizes, and it also allows us to understand the effect of stirring on nucleation. Truncated cubes and one-dimensional structures such as nanorods could be synthesized using this DH method, which could be useful for catalysis and for the transport studies. In addition, the DH method was successful in synthesising double perovskites such as $\text{La}_{2-x}\text{K}_x\text{BMnO}_3$ ($B = \text{Cr, Fe, Ni}$). Because of the presence of two transition metals, which are responsible for the overall chemical property of the catalyst, these have demonstrated improved catalytic activity. This is because the 3d orbital of the B-site transition metals dominates the valance band and conduction band of perovskite materials. The lack of theoretical/thermodynamic models for lanthanum-based ternary systems is a challenge in this field. With many advancements, solution-based synthesis has proven to be a viable controlled method of synthesis. The LKMO microcubes produced have demonstrated high stability and bifunctional electrocatalytic

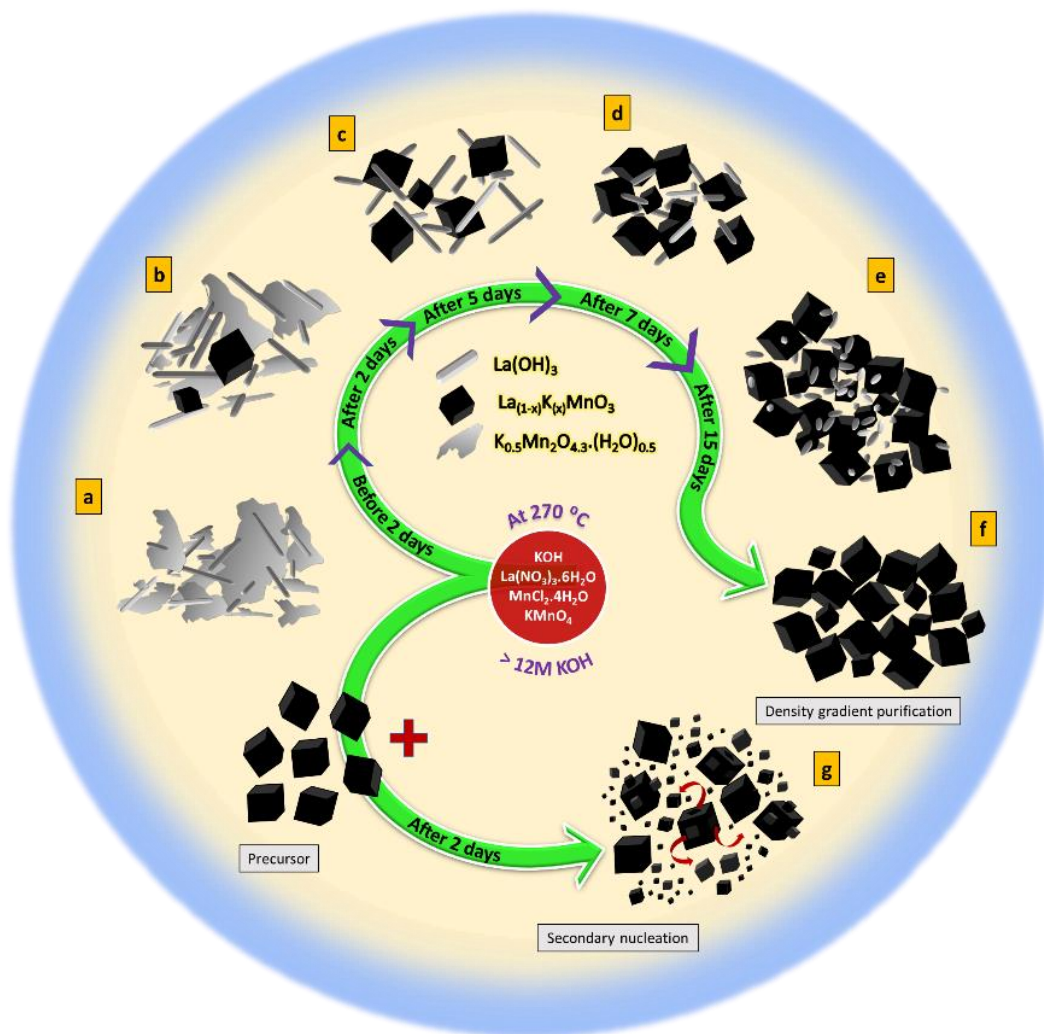
activity. Also demonstrated improved activity and selectivity in the reduction of pharmaceutically important organic molecules. With controlled synthesis, a pseudo-cubic phase LKMO material with enhanced magnetic properties was obtained, which would have been difficult to obtain under normal conditions. This high-symmetry phase has the potential to exhibit a strong magnetocaloric effect, MCE. In their two-probe device IV-characteristics, the LKMO nanorods demonstrated Poole-Frenkel type defect/trap assisted electrical conductivity.

The LKMO microcubes electrocatalytic activity has been discussed in zinc-air battery. Also, the catalytic reduction of organic molecules has been discussed. Apart from the perovskite oxides hydrothermal synthesis could synthesize photocatalytically ZnO nanoparticles doped with chemically hard elements.

Finally, all of these materials have been tested and proven to have excellent electrical and magnetic properties. This low temperature solution-phase hydrothermal synthesis opens up enormous possibilities for the preparation of a wide range of complex systems with precise control over nucleation, phase, size, crystallinity, and morphology; this, in turn, broadens the horizons for studying one-dimensional surfaces and interfaces to reveal unique physical, electrical, mechanical, and electrochemical properties.

CHAPTER-2:

Static hydrothermal synthesis: controlled nucleation and growth



This chapter describes the impact of static hydrothermal synthesis on obtaining different phases (LaCrO₃, LKMO, and LFO) and morphologies (micro, nano, cubes) of lanthanum-based oxide perovskites.

2.1 Abstract

Lanthanum-based perovskite oxides (LPOs) with desired facets and small sizes hold great promise for a wide range of heterogeneous catalytic applications. They are frequently formed in basic hydrothermal conditions. The synthesis typically yields several micron-sized crystals, with a total reaction time ranging from 5 to 14 days. The nucleation and growth of LPOs are poorly understood, both experimentally and theoretically. In this study, we report on the synthesis of La_{1-x}K_xMnO₃ (LKMO) using a low-temperature (265 °C) hydrothermal technique and investigate the growth mechanism. When reaction parameters (reaction temperature, KOH concentration, precursor amount, and time) are optimised, selectively faceted microcubes with sizes ranging from 3 to 30 microns are produced. The nucleation of LKMO is affected by both the reaction temperature and the KOH concentration. The nucleation of LKMO is affected by both the reaction temperature and the KOH concentration. The amount of K⁺ incorporation in LMO changes as the KOH concentration in the reaction changes, as do the crystal structure and magnetic properties. The reaction produces K_{0.5}Mn₂O_{4.3}.(H₂O)_{0.5} at a lower KOH concentration (8 M). The above phase dissolves as the KOH concentration increases, while La(OH)₃ crystallises. The inverse relationship between the solubility of the byproducts La(OH)₃ and K_{0.5}Mn₂O_{4.3}.(H₂O)_{0.5} during the reaction limits the simultaneous availability of La and Mn ions in the solutions for burst nucleation. The current study demonstrates that the crystallization neither follows the classical LaMer model (nucleation followed by diffusion-controlled growth) nor two-step Watzky and Finke model (nucleation and growth are temporally overlapped). We propose a mechanism called temporally separated crystallization (nucleation and growth), in which the nucleation and growth of each (set of) crystal/s formed in the reaction are temporally separated. Time-dependent studies indicate that the reaction goes through a prolonged primary nucleation step, followed by rapid crystal growth and secondary nucleation. Furthermore, the addition of the pre-synthesized cubes promoted secondary/heterogeneous nucleation, reducing the overall reaction time from 7 days to 2 days and yielding submicron cubes. The hydrothermal synthesis is extended further to yield microcubes and submicron cubes (25-450 nm) of LaCrO₃ and (≈180 nm) LaFeO₃. The

current method could aid in the synthesis of nanocrystals of other lanthanum-based perovskite oxides and the investigation of their size and facet-dependent properties.

2.2 Introduction

Lanthanum-based transition metal perovskite oxides (LPOs) have remained an extremely important category of materials due to their significant properties such as superconductivity⁸² (SC), colossal magnetoresistance^{83,84} (CMR), catalysis^{85–89}, supercapacitance,^{90,91} and electrode for solid oxide fuel cells^{92,93}. Their unique inherent phenomena such as charge, spin, and characteristic orbital ordering have compelled the recent past to witness a tremendous surge in the research of doped lanthanum-manganese oxide-based (LMO) perovskite materials.⁹⁴ The structural stability of perovskites with respect to the oxidation state of the metal, substitution with other elements, and oxygen non-stoichiometry are unrivalled. In the ABO_3 structure, the transition metal site (B) is usually catalytically active. The catalytic ability of perovskites includes oxidation, reduction, redox, combustion, eliminating volatile organic compounds, to name a few.⁸⁵ Perovskites with desired planes exposed are also considered as substitutes for noble metals for heterogeneous catalysis.^{95,96} However, the use of perovskites to its full potential in catalytic reactions is limited due to the inability to make them in low-dimensions with desired facets exposed.⁷⁵

The conventional way of synthesizing perovskite oxides has been either through solid-state chemistry employing high reaction temperatures ($> 800\text{ }^{\circ}\text{C}$) with several intermediate grindings^{71,77–79} or through flux-based methods using alkali halides/hydroxides.^{74,97–100} However, controlling the size, morphology, and uniformity of the crystals with smooth atomic surfaces is difficult through high-temperature methods, where nucleation is faster than growth. Solution-based methods are more energy-efficient and known to have better control over nucleation and growth as well as over morphology.^{101,102} Atomic flat surfaces can be achieved in solution-based synthesis, which is a result of slow nucleation compared to growth.¹⁰¹ It also offers control over doping and substitution of perovskites oxides.¹⁰²

The hydrothermal methods have been applied to synthesize various ceramic oxides in the last two decades. This technique is energy-efficient and allows reactions at high concentrations of acids and bases.^{38,102,103} The ionic mobility in this technique is controlled by the dynamic interaction of various parameters such as type of solvent, temperature, and pressure.¹⁰³ Hydrothermal method is the only solution-based method that has been shown some success to synthesize lanthanum-based perovskite oxides (LPOs).⁸¹ This technique has been

used for the synthesis of divalent cations (such as Ba^{104,105}, Sr^{106–109}, and Ca¹⁰⁶) substituted LMO and also shown the morphology tuning. However, this technique could not be extended to the monovalent substituted LMO or other LnTmO₃ perovskites or their substituted analogue where Ln=lanthanides; Tm=Cr, Mn, Fe, Co, Ni & Al. Hydrothermal synthesis of LnCrO₃ (Ln=La, Pr, Nd, Sm, Er, Tm, Yb, Lu), LaFeO₃ usually produces several (1-30) micron-sized crystals with a reaction duration of 3-14 days.^{110–113} Thus, the synthesis inhibits to use size and facet dependent catalytic properties of LnTmO₃. Hence, use of these materials in polarity dependent activities including facet dependent functional electronic components, spatial parting of photogenerated electrons and holes, gas sensors and photocatalysts are limited.⁷⁵ Moreover, there has been no satisfactory explanation on how the different reaction parameters affect the nucleation process or no thermodynamic models that could predict the crystal growth kinetics¹⁰³. Further, the inability to identify temporal size evolutions in the growth stage suggest that the nucleation of LaTmO₃ in hydrothermal conditions does not follow the classical LaMer model^{114,115}, which indicates burst nucleation trailed by diffusion-controlled growth of surviving nuclei. Employing high concentration of bases for crystallizing LaTmO₃ could corrode the reaction vessel; thus, it is also essential to develop hydrothermal synthesis technique below 280 °C so that Teflon can be used as a reaction vessel. Therefore, there is a need to understand these systems' nucleation and growth to synthesize small crystals under mild conditions further.

The recent past has compelled to witness a tremendous surge in the research of doped lanthanum-manganese oxide-based (LMO) perovskite materials for their unique inherent degrees of freedom.⁹⁴ The divalent doping changes the oxidation state of manganese from Mn⁺³ to Mn⁺⁴. The appropriate Mn⁺³/Mn⁺⁴ ratio leads to an insulator to metallic transition (IMT) and simultaneously to the ferromagnetic transition below a transition temperature.⁷⁰ On the other hand, substituting monovalent alkali metal such as Na⁺ or K⁺ at the La⁺³ sites in LaMnO₃ generates two equivalent of hole doping per unit formula.⁷¹ It has been believed that during the synthesis of manganite, the K⁺ is weakly attached to the surface of MnO₆ octahedra, which is replaced by the La³⁺ in the subsequent growth of another unit cell on the surface.¹¹⁶ We have realized something different is happening. As per the literature, there are plenty of reports concerning the Mn³⁺/Mn⁴⁺ mixed-valence, cation vacancies, or oxygen hyper-stoichiometry such as intrinsic complexity in lanthanum manganites.¹¹⁷ Thus, to stabilize the perovskite phase of lanthanum manganite at low temperatures, the presence of K⁺ is helping, leading to the incorporation of it in the A-site.

In this work, we have synthesized a relatively large cation, K^+ , substituted $LaMnO_3$ microcubes through a low-temperature hydrothermal synthesis. We systematically explored different reaction conditions (including temperature, alkali (KOH) concentration, precursor (La) concentration, precursor metal ratios (La/Mn), and reaction time) to understand the nucleation and growth of lanthanum-based perovskite oxides by taking LKMO as a model system. The plausible mechanism of formation of this LKMO phase has been proposed. We have extended this method to synthesize $LaCrO_3$ and $LaFeO_3$ microcubes. We have observed a decrease in average particles size with increasing reaction duration, revealing the domination of secondary nucleation at extended times. Which suggest that crystallization of these systems does not follow existing nucleation and growth models (classical LaMer model or Watzky and Finke model) completely. The formation of side products ($La(OH)_3$ and $K_{0.5}Mn_2O_{4.3} \cdot (H_2O)_{0.5}$) during the reaction limit simultaneous availability of La and Mn ions in the solutions for burst nucleation. We propose a plausible step-wise crystallization mechanism, where different crystallization events are temporally separated. Secondary or heterogeneous nucleation has already been thoroughly explored to decrease the kinetic barrier of the nucleation process.¹⁰² Still, this phenomenon has not been extended to lanthanum-based perovskite oxides. Thus, adding pre-synthesized cubes of the same materials or $SrTiO_3$ nanoparticles at the beginning of reactions is explored to understand the effect of secondary/heterogeneous nucleation on particles sizes of the end product. We have observed, the formation of well faceted submicron crystals of $LaKMnO_3$, $LaCrO_3$, $LaFeO_3$ and decreased overall reaction time from 7 to 2 days. Further, we have also studied the magnetic and electrical properties of LKMO microcubes obtained at two different KOH concentrations. The phase analysis reveals that this method of synthesis is a more controlled one to obtain higher symmetry (pseudo cubic) manganite.

2.3 Existing issues

No thermodynamic models are available to study the thermodynamics and kinetics of lanthanum-based ternary systems. As a result, there is a scarcity of information about these systems.

Because much of their synthesis was application-oriented and required enormous activation energies, solid-state synthesis methods were preferred till date.

Without the knowledge to control crystallisation, it is impossible to probe size, shape, and facet dependent properties.

2.4 Objectives & Action plan

1. Obtaining the perovskite phase for lanthanum manganite, chromite, and ferrite using the static hydrothermal (SH) method.
 - i. To accomplish this, we must optimise the mineralizer (KOH) concentration.
 - ii. The concentration and ratio of the precursors must be optimized.
 - iii. Reaction time, temperature, and knowledge of stirring and sonication times are required.
2. Reduce the size of the single-crystal microcubes of these materials to submicron levels.
 - i. This necessitates understanding of the nucleation and growth processes of these materials.
 - ii. It is necessary to closely monitor the reaction process as the reaction parameters change.
 - iii. We have devised some techniques to overcome the existing barriers resulting from the above knowledge.
3. Fine-tune the reaction conditions further to mold the morphology of these materials and obtain nanorods.
 - i. Use of surfactants to control the morphology in the nucleation.
 - ii. Using seeds to direct the crystallization into anisotropic growth.

2.5 Materials and Methods

2.5.1 Materials

All metal salts used in the experiments, viz. $\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$, $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$, $\text{Cr}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, $\text{Fe}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$, $\text{Sr}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, $\text{Ti}(\text{OC}_3\text{H}_7)_4$ (TiIP) were purchased from Merck (AR 99%) and employed as such in the experiments performed. Potassium salts KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$ were purchased from Merck (AR 99%) and used as such. KOH was purchased from Merck (AR 99%) and used as such. Ethanol (Emparta, Merck, ACS) and Milli-Q water were used for the post-reaction workup. No other purification was performed on any of the chemicals. While weighing, hygroscopic materials ($\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$) and pyrophoric chemicals (KMnO_4) were handled with care to ensure minimal exposure to the atmosphere. All of the hydrothermal reactions described in this chapter were carried out in a 25 ml Teflon autoclave container.

2.5.2 Synthesis

2.5.2.1 Preparation of LKMO microcubes:

The perovskite $\text{La}_{1-x}\text{K}_x\text{MnO}_3$ (LKMO) microcubes were synthesized using a low-temperature hydrothermal method, which is similar to the approach adopted in the literature.^{106,108,118} In a typical synthesis, $\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$, $\text{MnCl}_2 \cdot \text{H}_2\text{O}$, and KMnO_4 are taken into a beaker containing 2-5 ml of Milli Q water in a molar ratio of 1:0.7:0.3 (under variation of the equivalents from 1 to 8 millimoles). These reagents are dissolved in water by stirring until a homogenous solution is formed. The above solution is transferred into a 25 ml Teflon-based autoclave. Subsequently, 10 ml aqueous KOH is added into the reaction mixture such that the resultant final concentration of KOH varies from 9 to 18 M and is allowed to undergo stirring for 20-24 hours. The hydrothermal reaction is performed by heating the autoclave at 265 °C under autogenous pressure for different periods of time ranging from 20 hours to 15 days (**Table 2.1**). After completion of the reaction, the supernatant is decanted, and the resultant solid precipitate is washed thoroughly with MilliQ water and ethanol until the pH is found neutral. The product is dried at 70 °C for 12h. The reaction yields LKMO microcubes as a major product and $\text{La}(\text{OH})_3$ nanorods as a minor product. These lighter minor products have been separated from the heavier product in the workup process itself by using the concept of density gradation.

2.5.2.2 Separation of $\text{La}(\text{OH})_3$ nanorods from LKMO

The $\text{La}(\text{OH})_3$ rods (a byproduct of the above reaction) were separated by either of the two following methods:

2.5.2.2.1 Treatment with oxidant

The as-synthesized compound containing LKMO microcubes and $\text{La}(\text{OH})_3$ nanorods (80 mg) is transferred into a 25 ml Teflon autoclave having 15 ml H_2O_2 solution (30% w/v) under stirring. After allowing the mixture to stir overnight, it was subjected to hydrothermal treatment at 180 °C for 12 hours to dissolve the $\text{La}(\text{OH})_3$ nanorods. The resultant precipitate was washed several times with water to eliminate the excess oxidant and other unreacted materials and dried at 70 °C in ambient conditions.

2.5.2.2.2 Density-based separation of La(OH)₃ nanorods from the LKMO microcubes

The difference in densities was used to separate the LKMO microcubes from the La(OH)₃ rods. 100 mg of as-prepared solid powder (containing a mixture of LKMO microcubes and La(OH)₃ rods) is sonicated for a few minutes before letting it settle naturally for 5 minutes. The heavier LKMO microcubes sank to the bottom, leaving the relatively lighter La(OH)₃ rods. The greyish-white supernatant formed by these rods was decanted from the heavy solid precipitate containing LKMO microcubes. This procedure was repeated until a clear transparent supernatant was obtained, indicating that La(OH)₃ rods had been completely separated from the LKMO microcubes. This was then dispersed in ethanol to separate any remaining La(OH)₃ rods and kept for drying. This separation's obtained phase purity was then tested using PXRD and SEM characterizations. Compared to the above H₂O₂ method (**Figure 2.2(c)**), this density-based separation (**Figure 2.2(b)**) method is much safer and easier.

2.5.2.3 Preparation of LaCrO₃ microcubes

In a similar hydrothermal condition, we have also synthesized LaCrO₃ microcubes starting from La(NO₃)₃·6H₂O, Cr(NO₃)₃·9H₂O, and K₂Cr₂O₇ are taken in a molar ratio of 0.5:0.7:0.3 (of 1 millimoles equivalent). A precaution that was taken particularly to prevent Chromium from getting oxidized in the atmospheric air; was to ensure as little stirring as possible, thus preventing too much subjection to the atmosphere. An aqueous KOH solution was added into the above mixture under stirring until the resultant final concentration of solution reached 9 M. This was finally hydrothermally treated at 265 °C for 5 days. After completion of the reaction, the supernatant is decanted, and the resultant solid precipitate is washed thoroughly with MilliQ water and ethanol until the pH is found neutral. The product is dried 70 °C for 12h. The reaction yields LaCrO₃ microcubes as the major product.

2.5.2.4 Preparation of LaFeO₃ microcubes

In a similar hydrothermal condition, we have also synthesized LaFeO₃ microcubes starting from La(NO₃)₃·6H₂O and Fe(NO₃)₃·9H₂O are taken in a molar ratio of 1:1 (of 1 millimoles equivalent) into a 25 ml Teflon container. An aqueous KOH solution was added into the above mixture under stirring until the resultant final concentration of solution reached 12 M. This was finally hydrothermally treated at 265 °C for 5 days. After completion of the reaction, the supernatant is decanted, and the resultant solid precipitate is washed thoroughly with MilliQ

water and ethanol until the pH is found neutral. The product is dried 70 °C for 12h. The reaction yields LaFeO_3 microcubes as the major product.

2.5.2.5 Synthesis of LKMO submicron cubes: using LKMO microcubes precursor

At first, 20mg of the LKMO microcubes was taken into a 25ml Teflon-based autoclave containing 2-3 ml Milli Q water. Into this $\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$, $\text{MnCl}_2 \cdot \text{H}_2\text{O}$, and KMnO_4 are taken in a molar ratio of 1:0.7:0.3 (of 5 millimoles equivalent) and stirred until a homogenous solution formed. This reaction mixture was then subjected to the addition of aqueous KOH until the resultant final concentration of solution reached 18 M and then allowed to stir for 24 hours. This was then hydrothermally treated at 265 °C for 2 days. After completion of the reaction, the supernatant is decanted, and the resultant solid precipitate is washed thoroughly with MilliQ water and ethanol until the pH is found neutral. The product is dried 70 °C for 12h.

2.5.2.6 Synthesis of LaCrO_3 submicron cubes: using LaCrO_3 microcubes precursor

Initially, 20-30mg of thoroughly cleaned LaCrO_3 microcubes, which were synthesized in the first batch, were taken into a clean 25ml Teflon container. Into which 2-3ml Milli Q water was added and sonicated for 20-30 mins for nice dispersion. Into this $\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$, $\text{Cr}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, and $\text{K}_2\text{Cr}_2\text{O}_7$ are taken in a molar ratio of 0.5:0.7:0.3 (of 1 millimoles equivalent) and stirred for 10 mins. Then KOH was added into the above mixture under stirring until the resultant final concentration of solution reached 18 M. This was hydrothermally treated at 265 °C for 2-7 days. The obtained product was multiply washed with Milli Q water until the pH became neutral. Finally, one round of workup was done with ethanol and kept for drying at 70 °C overnight.

2.5.2.7 Synthesis of SrTiO_3 nanoparticles

By means of solvothermal reaction, we have synthesized STO nanoparticles using three different molar ratios of Ti ions in $\text{Ti}(\text{OC}_3\text{H}_7)_4$ and Sr ions in $\text{Sr}(\text{NO}_3)_2$ as starting materials. The Sr/Ti atomic ratios were taken as 1:1, 1:1.5, and 1.5:1. After dissolving 1.47.9 μL of $\text{Ti}(\text{OC}_3\text{H}_7)_4$ in 5 mL of 2-propanol, 105.7 mg $\text{Sr}(\text{NO}_3)_2$ and 5 mL of 2 M KOH aqueous solution were slowly added while stirring. Then this solution was transferred into an autoclave. Then in another autoclave, we had varied the amount of $\text{Sr}(\text{NO}_3)_2$, which is 221.84 μL , except that every other thing was the same as the previous one. Thirdly, we had changed the quantity of

Ti(OC₃H₇)₄ to 158.58 μ L while keeping others as same. Then all the three autoclaves were kept under oven at 200 °C for 15 h.¹¹⁹ After cooling, and the samples were centrifuged, washed with Milli-Q water repeatedly until the pH came down to neutral, and then dried in an oven at 60 °C overnight.

2.5.2.8 Synthesis of LKMO submicron cubes: using STO nanoparticles as precursor

20mg of the thoroughly washed STO nanoparticles sample was taken into a 25ml Teflon-based autoclave containing 2-3 ml Milli Q water. Into this La(NO₃)₃.6H₂O, MnCl₂.H₂O, and KMnO₄ are taken in a molar ratio of 1:0.7:0.3 (of 5 millimoles equivalent). This solution was stirred and sonicated until a homogenous solution formed. This reaction mixture was then subjected to the addition of aqueous KOH until the resultant final concentration of solution reached 18 M and then allowed to stir for 24 hours. This was then hydrothermally treated at 265 °C for 2 days. After completion of the reaction, the supernatant is decanted, and the resultant solid precipitate is washed thoroughly with MilliQ water and ethanol until the pH is found neutral. The product is dried 70 °C for 12h.

2.5.2.9 Synthesis of LaFeO₃ submicron cubes: using STO nanoparticles as a precursor

Initially, 20-30mg of thoroughly cleaned STO nanoparticles were taken into a clean 25ml Teflon container. Into which 2-3ml Milli Q water was added and sonicated for 20-30 mins for nice dispersion. Into this La(NO₃)₃.6H₂O, Fe(NO₃)₃.9H₂O are taken in a molar ratio of 1:1 (of 1 millimoles equivalent) and stirred for 10 mins. Then KOH was added into the above mixture under stirring until the resultant final concentration of solution reached 12 M. This was finally hydrothermally treated at 265 °C for 2 days. After completion of the reaction, the supernatant is decanted, and the resultant solid precipitate is washed thoroughly with MilliQ water and ethanol until the pH is found neutral. The product is dried 70 °C for 12h. The reaction yields LaFeO₃ microcubes as the major product.

2.5.3 Characterization

Components present in the synthesized materials were characterized through Powder X-ray Diffraction (PXRD) using a Panalytical Empyrean diffractometer system with CuK α (λ =1.5418 Å) with large beta-filter Nickel to filter the K β lines. Samples for the XRD analysis were prepared by packing onto the depression of a quartz slide. Morphology of the materials

was studied through Field Emission Scanning Electron Microscopy (FESEM, *Carl Zeiss*). Elemental composition and distribution in the materials were studied by Energy Dispersive Spectroscopy (EDS) mapping. The sample for SEM and EDS analysis was prepared by taking less than 1 mg of the powder product, crushing mildly to ensure the absence of lumps, and sticking it to carbon tape on the SEM sample stub. Further, nitrogen was blown on the surface to ensure that no loose powders got into the polepiece; imaged and analyzed without coating. Magnetoresistance and resistivity of the synthesized materials were studied by the 4-probe resistivity measurement method. Around 2.5 g of the pure LKMO microcubes sample was compressed into a pellet through the hot press method at 800 °C. Piezoelectric response of the synthesized materials was tested on the Asylum/Oxford MFP3D origin Atomic Force Microscopy (AFM) machine in Piezo Force Microscopy (PFM) contact mode. The individual sample cubes were mounted onto an elastomeric matrix for a better hold for an unwavering and firm contact of the former with the PFM tip. The piezoelectric response of the synthesized materials was measured by applying DC bias in the range of -5V to +5V; and measuring triplicates on five different locations on each cube on a 5 μm X 5 μm area. Temperature-dependent magnetization of the peroxide-washed sample of LKMO was measured using the Quantum Design, USA PPMS Vibrating Sample Magnetometer (VSM) at 300 Oe with field and zero-field cooling (FC and ZFC). For qualitative and quantitative elemental information X-ray photoelectron spectroscopy (XPS) was carried out using Kratos Analytical, UK (AXIS Supra). Monochromatic Al K-alpha of 75 W was used as the x-ray source. The survey and high-resolution scans were done at 160 eV and 20 eV pass energy with ≈ 2 eV and ≈ 0.5 eV resolution, respectively. Charge corrections were performed with respect to the C 1s peak at 284.8 eV. Ultraviolet Photoelectron Spectroscopy (UPS) was done using the same above AXIS Supra, Kratos Analytical, UK facility. TEM, HRTEM, ED, and STEM_EDS characterizations are performed with a FEI Tecnai G2, F30 microscope operating at 300 kV (resolution: point 2 Å and line 1 Å) equipped with a EDAX Octane ELITE T70, BF/DF STEM-HAADF detector. For TEM analysis the powder samples dispersed in a 1 ml Eppendorf tube using iPA solvent and ultrasonically dispersed for 0.5 to 1 hr. The fine dispersed TEM sample looks transparent. One to two drops of this dispersion deposited on the dark side of a carbon coated copper grid balanced in specialized swiss self-closing antimagnetic SS tweezer under IR-lamp.

2.6 Results and Discussion

2.6.1 Effect of temperature

The hydrothermal synthesis to make perovskite oxides in the literature adopted temperatures between 100 °C to 374 °C.⁽⁸⁰⁾ However, the reactions carried out in this work at temperatures below 240 °C yielded only $\text{La}(\text{OH})_3$ nanorods and $\text{K}_{0.5}\text{Mn}_2\text{O}_{4.3}(\text{H}_2\text{O})_{0.5}$ sheets of birnessite-such as phase. The formation of the birnessite-such as phase was also observed in the literature.⁽⁶⁰⁾ The formation of the LKMO phase is observed only when the reaction temperature is above 260 °C.

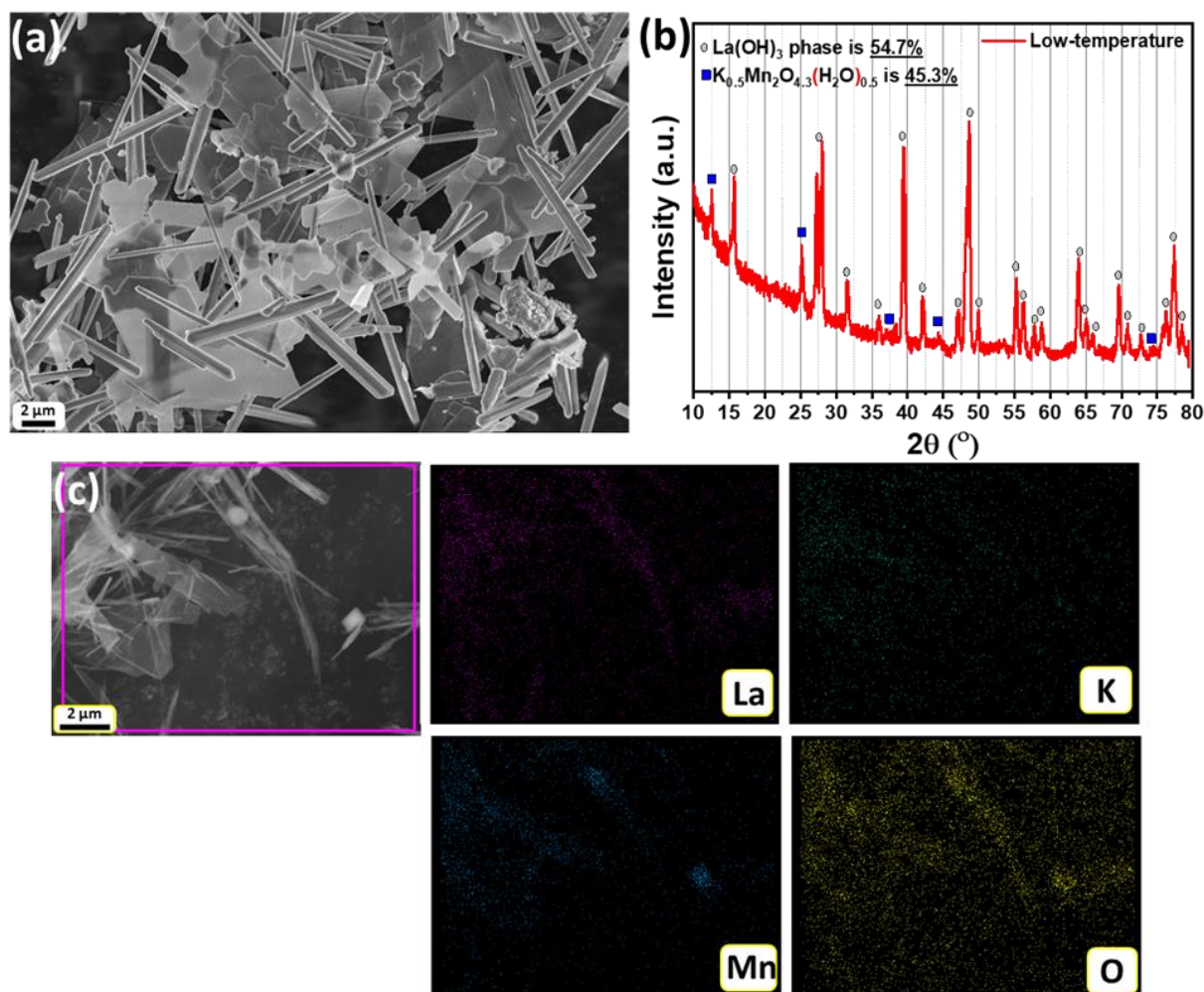


Figure 2.1 (a) SEM image and (b) XRD pattern and (c) EDS mapping indicating the elemental distribution of by-products obtained at 240 °C which are $\text{La}(\text{OH})_3$ rods and $\text{K}_{0.5}\text{Mn}_2\text{O}_{4.3}(\text{H}_2\text{O})_{0.5}$ sheets of birnessite-like phase.

Figure 2.1 shows the SEM images of the different products other than that indicative of the LKMO phase obtained. Hence, following this observation, all the hydrothermal

syntheses carried out further in this work (**Table 2.1**) were at a fixed temperature of 265 °C. The reaction temperature close to the hydrothermal reactor is also monitored by an external thermometer, and this temperature varies between 264-266 °C during the reaction.

2.6.2 Morphological and structural characterization of LKMO microcubes

The SEM image is shown in **Figure 2.2(a)** of as-synthesized LKMO microcubes. The sizes of LKMO microcubes range from 2.0 to 23.0 μm with an average size of 11.0 μm . Size distribution of the cubes is presented as a histogram in **Figure 2.2(d)**. (By tuning the reaction conditions, we can vary the size of the cubes from 0.5 to 30 μm .) The above reaction also yields $\text{La}(\text{OH})_3$ nanorods along with LKMO microcubes, as can be seen from **Figure 2.2(a)**.

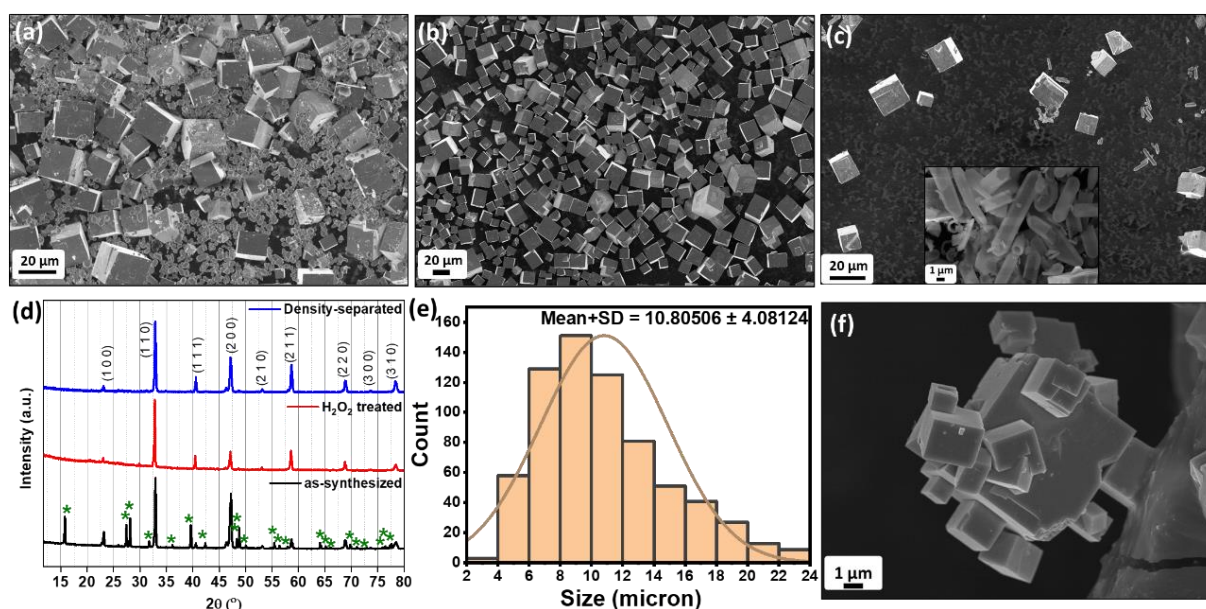


Figure 2.2 (a) Typical SEM image of as-synthesized LKMO sample, (b) the size-distribution histogram with an average size of the LKMO cubes ≈ 11 microns, (c) SEM image of LKMO sample after separating the by-products via density-separation method, (d) SEM image of LKMO sample after separating the by-products via H_2O_2 treatment inset showing the supernatant containing degraded by-products, (e) XRD pattern of the LKMO microcubes obtained via hydrothermal treatment at 265 °C for 7 days with $[\text{KOH}] = 18 \text{ M}$ at 5mmol [precursor] and $R_{\text{La}/\text{Mn}} = 1.0$, (f) growth of smaller cubes on the externally added LKMO cubes as a precursor.

Post-synthesis separation of the $\text{La}(\text{OH})_3$ nanorods from the LKMO microcubes is achieved by density-based separation as well as by H_2O_2 treatment. **Figure 2.2(b and d)** show SEM images of LKMO microcubes after efficient removal of $\text{La}(\text{OH})_3$ nanorods using density-based separation and H_2O_2 treatment, respectively. Powder XRD analysis of the as-prepared samples reveals the formation of crystalline LKMO and $\text{La}(\text{OH})_3$ phases (**Figure 2.2(d)**). XRD patterns

recorded after the separation techniques indicated the absence of $\text{La}(\text{OH})_3$ phase (**Figure 2.2(d)**). As shown in, it is clear that the obtained LKMO shows rhombohedral R-3cH crystal structure; the prominent XRD peaks at 33.12° , 40.77° , 47.4° , and 58.87° can be assigned to the (110), (111), (200), and (211) facets of the standard LKMO crystal, respectively. Among the two separation techniques, the density-based technique is easy to implement and effectively separates $\text{La}(\text{OH})_3$ nanorods from LKMO microcubes, which is evident from both the SEM image (**Figure 2.2(b)**) and XRD patterns (**Figure 2.2(d)**).

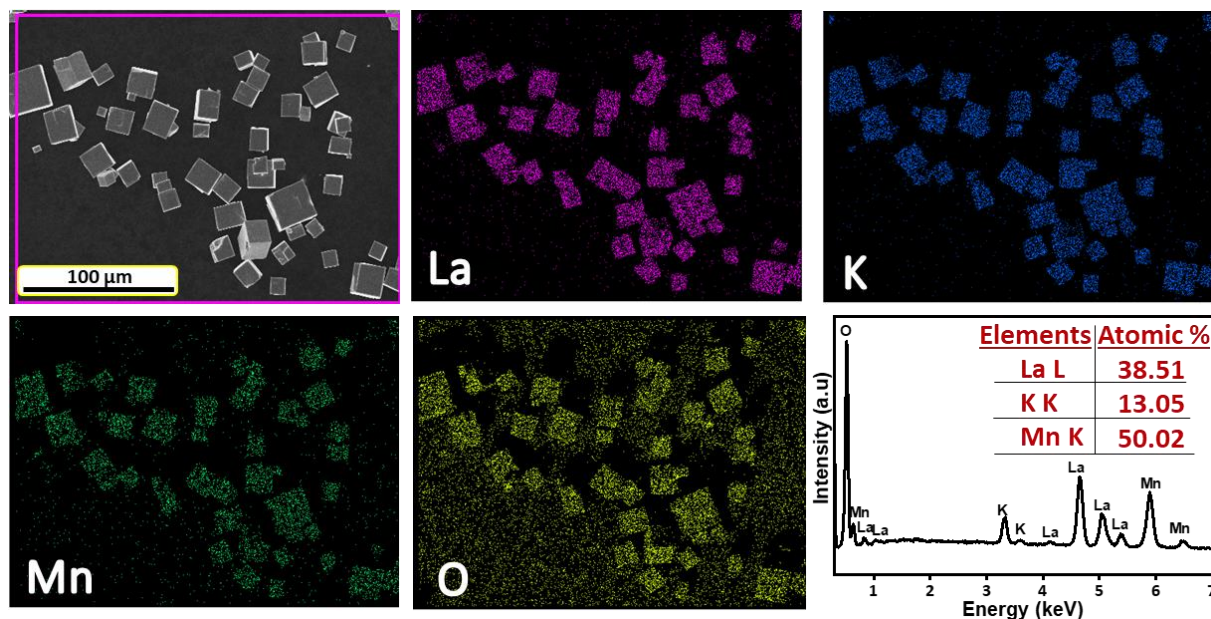


Figure 2.3 EDS elemental mapping and EDS spectrum indicating the elemental composition of the LKMO microcubes after removal of the $\text{La}(\text{OH})_3$ nanorods by density-based separation

The SEM micrograph entirely accords to the images of the EDS mapping (**Figure 2.3**) of La, K, Mn, and O atoms. All the atoms are homogeneously dispersed on the microcubes. The semi-quantitative EDS analysis (**Figure 2.13**) on microcubes shows the ratio of La:K:Mn as 0.77:0.26:1.0. The stoichiometry of the microcubes obtained in these experiments is calculated based on their individual atomic percentages. As there can be some trace K^+ from the KOH left out on the surface of the cube thus we considered Mn as 100% stoichiometry and the difference between Mn% and the La% as the K% in the respective samples. The excess presence of oxygen in the data could be attributed to external atmospheric sources. The yield and amount of K^+ substitution in the microcubes were found to be variant depending on the reaction conditions (described below). Though K^+ is known to loosely bind the lattice because of phase stabilization at low temperatures, it's incorporated here for the phase to form.

The initial experiments have produced LKMO microcubes with only a limited yield. The yield is improved by optimizing the reaction condition such as reaction temperature, KOH concentration, precursor concentration, precursor ratio, and reaction time. The experimental **Table 2.1** portrays a compilation of all the reaction parameters investigated along with a summary of their resultant yield and morphological data.

Table 2.1 The details of the various experiments carried out to estimate the effect of various parameters; viz. K (mineralizer (KOH) concentration in M(mol/L)), P (precursor (La, Mn) concentration in mM), R (precursor ratio $R_{La/Mn}$) and T (time taken for the reaction in days) at 265 °C. The base value $K=18M$, $P=5M$, $R=1$ & $T=7days$ is the best-optimized values that correspond to the master entry in this table

<u>Sample</u>	[KOH] (mol/L)	[Precursor] in M (mol/L)	$R_{La/Mn}$	Time in days	XRD	SEM			
						LKMO		La(OH) ₃	
					Phase %	Cube size (μ m)	Length (μ m)	Dia (μ m)	Rod Aspect Ratio
K0	0	5	1	7	---	---	---	---	---
K2	2	5	1	7	---	---	---	---	---
K5	5	5	1	7	---	---	1.94	0.32	5.27
K7	7	5	1	7	7.2	0.68	1.65	0.45	3.50
K8	8	5	1	7	13.1	5.5	1.83	0.57	3.25
K9	9	5	1	7	26	7.87	2.07	0.83	2.91
K12	12	5	1	7	64.1*	9.98	4.66	0.62	6.93
K16	16	5	1	7	31.8	11.00	2.49	1.62	1.44
K18P5	18	5	1	7	99.8	28.96	2.17	1.25	1.65
P8	18	8	1	7	93.4	14.81	2.15	1.08	1.87
P1	18	1	1	7	62	4.76	6.07	1.37	4.50
P1R0.67	18	1	0.67	7	79	12.81	---	---	---
T1	18	1	1	1	48.9*	1.60	3.94	0.98	3.99
T2	18	1	1	2	50.4	4.63	5.47	1.49	4.14
T3	18	1	1	3	54.3	4.80	5.77	1.01	6.16
T4	18	1	1	4	69.9*	5.01	4.29	0.96	4.62
T5	18	1	1	5	74.5*	5.30	4.92	1.17	4.21
T6	18	1	1	6	100**	5.31	7.67	1.14	6.81

Figure 2.4(a) is the TEM image of an LKMO microcube, which exhibits the rhombohedral symmetry identified from selected area electron diffraction (SAED) shown in **Figure 2.4(b)**. Based on these findings, it is possible to conclude that the hydrothermal method used in this study produces a more crystalline LKMO phase than the solid-state method used in the literature.¹²⁰

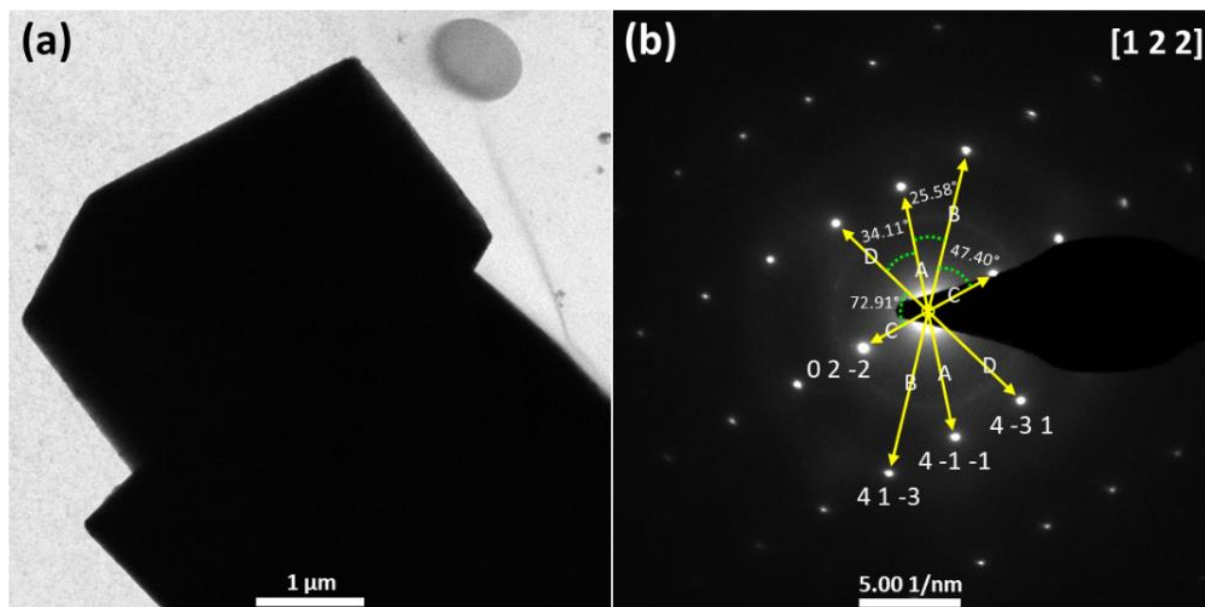


Figure 2.4 (a) Typical TEM image of as-synthesized LKMO microcube and (b) SAED pattern along the [1 2 2] zone-axis, synthesized via hydrothermal treatment at 265 °C for 7 days with 18 m KOH at 5 mmol precursor concentration.

X-ray photoelectron spectroscopy (XPS) was carried out to investigate the chemical state of hydrothermally synthesized LKMO microcubes, as shown in **Figure 2.5**. The survey spectrum (**Figure 2.5(a)**) represents the presence of La, K, Mn, and O elements. **Figure 2.5(b)** shows the La 3d high-resolution spectrum with peak deconvolutions into core and satellite peaks. The corresponding binding energies (BE) are in good agreement with the reported values.^{121,122} Both the spin-orbit split 3d_{5/2} at 833.61 eV and 3d_{3/2} at 850.31 eV levels exhibit roughly comparable intensity double-peak structures.

The spin-orbit splitting between La 3d_{5/2} and La 3d_{3/2} is 16.7 eV. The additional multiplet split of the individual peak is due to electron transfer from the O valence band to the empty La 4f orbital.^{123,124} These shake-up satellite peaks are observed at 836 eV and 838 eV for La 3d_{5/2}; 852.29 eV and 854.85 eV for La 3d_{3/2} respectively. The peak at 847.88 eV is known to be the Auger peak.¹²¹ The presence of K can be confirmed from the K 2s peak at 376.6 eV, as illustrated in **Figure 2.5(c)**.^{125,126} Besides, the peaks at 293.57 eV and 295.51 eV can be

attributed to K 2p_{3/2} and 2p_{1/2} core levels of the K 2p spectrum shown in **Figure 2.5(d)** indicates the existence of potassium in the perovskite lattice.

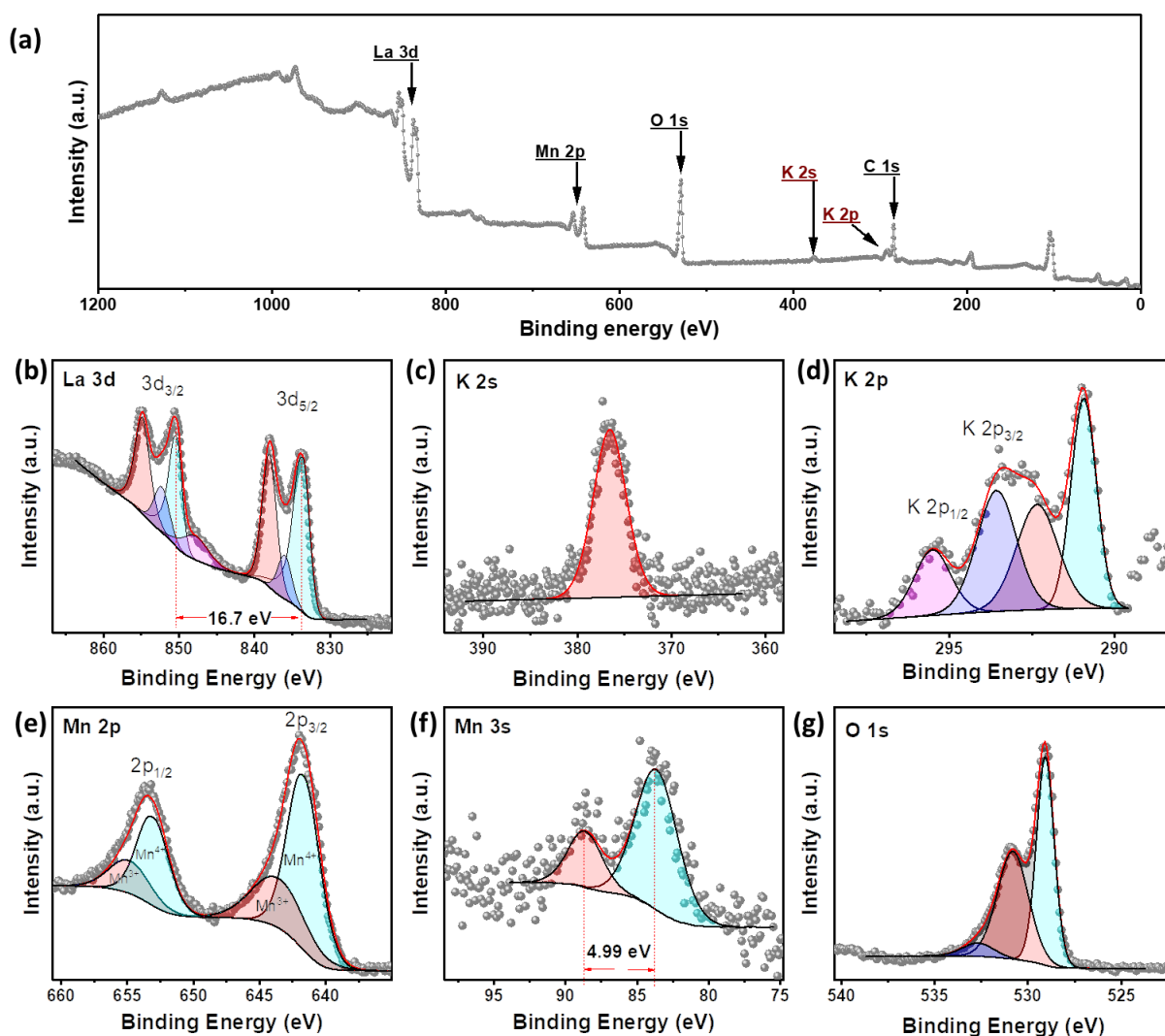


Figure 2.5 XPS spectra of LKMO microcubes synthesized via hydrothermal treatment at 265 °C for 7 days with 16 m KOH at 5 mmol precursor concentration. (a) Survey spectrum, and high-resolution spectra of (b) La 3d, (c) K 2s, (d) K 2p, (e) Mn 2p, (f) Mn 3s and (g) O 1s contributions. The narrow-scan spectra were fitted into individual components (turquoise, blue, red, and pink areas) using the combination of Gaussian and Lorentzian algorithm with Shirley background algorithm.

Mn 2p core-level spectrum (**Figure 2.5(e)**) shows spin-orbit split 2p_{3/2} and 2p_{1/2} peaks centered around 641.89 eV and 653.39 eV. Mn 2p_{3/2} exhibits two components with maxima at 641.64 eV and 643.51 eV indicating Mn³⁺ and Mn⁴⁺ respectively, and Mn 2p_{1/2} exhibits two components with maxima at 653.04 eV and 654.79 eV corresponding to Mn³⁺ and Mn⁴⁺ respectively (as shown in **Figure 2.5(e)**). From the area of the components of Mn 2p_{3/2}, the amount of Mn³⁺ is around 68%, and Mn⁴⁺ is 32%. For accurate quantification of Mn average

valence, the use of multiplet splitting values of the Mn 3s is preferential.¹²⁷ Mn 3s core level exhibits exchange split due to the interaction of 3s hole (generated in the X-ray photoelectron emission) and 3d electrons of the Mn.¹²⁸ In **Figure 2.5(f)**, we see splitting of Mn 3s into 83.64 eV and 88.63 eV peaks having 4.99 eV separation. This indicates that the formal oxidation state of Mn is +3.5 as per ref.¹²⁹ Because Mn⁴⁺ has a higher oxidation capacity than Mn³⁺, the increased Mn⁴⁺ concentration facilitates the disproportionation reaction of intermediate HO₂⁻ to generate a four-electron route, boosting ORR stability. Electrocatalytic activity is relevant to the surface oxygen species in addition to the B-site metal species and valence state of the perovskite oxides. The O 1s high-resolution spectrum was deconvoluted into three components using Shirley-type background shape, as can be seen from **Figure 2.5(g)**. The 529.1 eV component belongs to the lattice oxide resembling the B.E. of Mn-O bonding. The adjacent higher B.E. 530.8 eV component can be correlated with the La/K-O bonding. As the earlier lower B.E. component is intense than the later component, signifying the surface of the material/electrocatalyst is of Mn-O bonding environment.^{128,130} The third higher B.E. component could be ascribed to surface/defect-related oxygen or surface hydration.

2.6.3 Effect of alkali (KOH) concentration:

The concentration of KOH employed in the synthesis played a major role in the formation of different phases-morphologies-topology of LKMO. At concentrations (< 4 M), the reaction yielded only agglomerated particles of manganese oxide **Figure 2.6(a and b)**. Between 4-7 M concentrations of KOH, reaction yielded LaMn₂O₅ and La(OH)₃ nanorods (SEM images in **Figure 2.6(c-f)** and XRD profile in **Figure 2.7(c-f)**. Further, as the KOH concentration increased up to 8-9 M, the commencement of the formation of the LKMO phase was observed (in 7 days) (**Figure 2.6(e & f)**). However, the yield of the LKMO phase is small in comparison with that of La(OH)₃ even after 7 days. At this concentration, LaMn₂O₅ is not observed in the reaction product. The yield of the LKMO phase was found to be higher in the range of 12-18 M KOH concentration (see **Figure 2.6(i-k)**). The overall yield of the LKMO phase after 7 days was 80 %. The yield and crystallinity from the SEM and XRD show 18 M is better compared to the rest of the lower KOH concentrations. The 12 M shows drastic difference in surface morphology and the phase which can be seen from the SEM and XRD. Increasing KOH concentration beyond 18 M decreased the yield of the LKMO phase (see **Figure 2.6(l)**). The above observations indicate that with increasing concentrations of KOH, the availability of Mn ions increases in the solution, whereas La ions decrease (formation of crystalline La(OH)₃ is

favourable). Thus, nucleation and dissolution of $\text{La}(\text{OH})_3$ at higher concentrations of KOH is the rate-limiting step for the effective synthesis of LKMO.

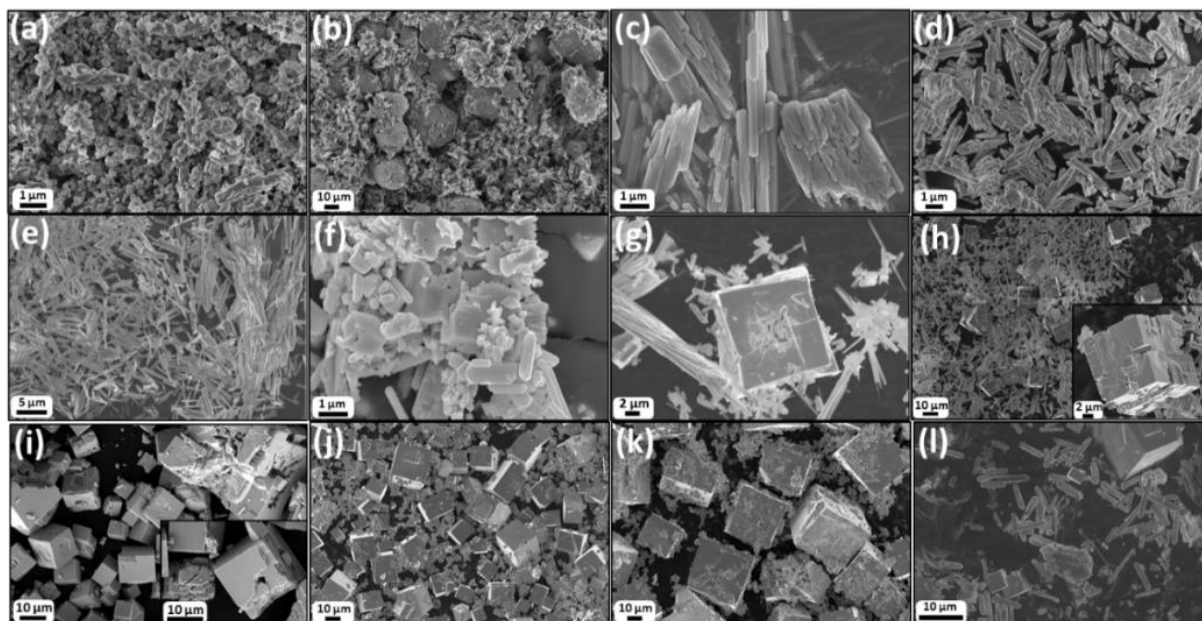


Figure 2.6(a-l) SEM images of the LKMO samples synthesized via hydrothermal process at 265 °C for 7days ($[\text{La}, \text{Mn}] = 5.0\text{M}$), ($R_{\text{La}/\text{Mn}} = 1$) with varying KOH concentrations: (a) 0 mol/L (K0), (b) 2 mol/L (K2), (c) 4 mol/L (K4), (d) 5 mol/L (K5), (e) 6 mol/L (K6), (f) 7 mol/L (K7), (g) 8 mol/L (K8), (h) 9 mol/L (K9), (i) 12 mol/L (K12), (j) 16 mol/L (K16), (k) 18 mol/L (K18P5), and (l) 21 mol/L (K21).

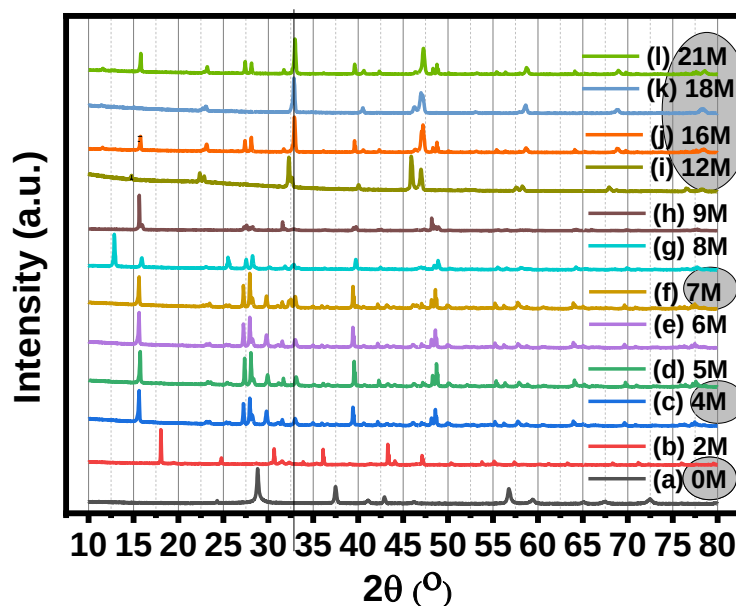


Figure 2.7 XRD patterns of the LKMO samples synthesized via hydrothermal process at 265 °C for 7days ($[\text{La}, \text{Mn}] = 5.0\text{M}$), ($R_{\text{La}/\text{Mn}} = 1$) with varying KOH concentrations: (a) 0 mol/L (K0), (b) 2 mol/L (K2), (c) 4 mol/L (K4), (d) 5 mol/L (K5), (e) 6 mol/L (K6), (f) 7 mol/L (K7), (g) 8 mol/L (K8), (h) 9 mol/L (K9), (i) 12 mol/L (K12), (j) 16 mol/L (K16), (k) 18 mol/L (K18P5), and (l) 21 mol/L (K21).

The concentration of KOH also affects the potassium incorporation inside the LMO as well as the phase and morphology of the LKMO cubes. EDS analysis on microcubes obtained at 12 M KOH (**Figure 2.13**) shows the ratio of La:K:Mn is $\approx 0.91:0.21:1.0$. The LKMO adopted a tetragonal structure (**Figure 2.8(d)**). The microcubes have shown porous in nature at the center of the cube. At 18 M KOH, the ratio of La:K:Mn in microcubes is $\approx 0.77:0.26:1.0$ (**Figure 2.13**). At this concentration, LKMO adopted a rhombohedral structure. The XRD patterns show merging of the (110) and (104) planes (corresponding to rhombohedral structure) into one leading to a more cubic structure (**Figure 2.7(k)**). The crystallinity of microcubes is improved both in XRD patterns as well as in SEM images. SEM images show a solid core with no porous nature. The two phases have shown different magnetic properties, as described later in the **section 2.8**.

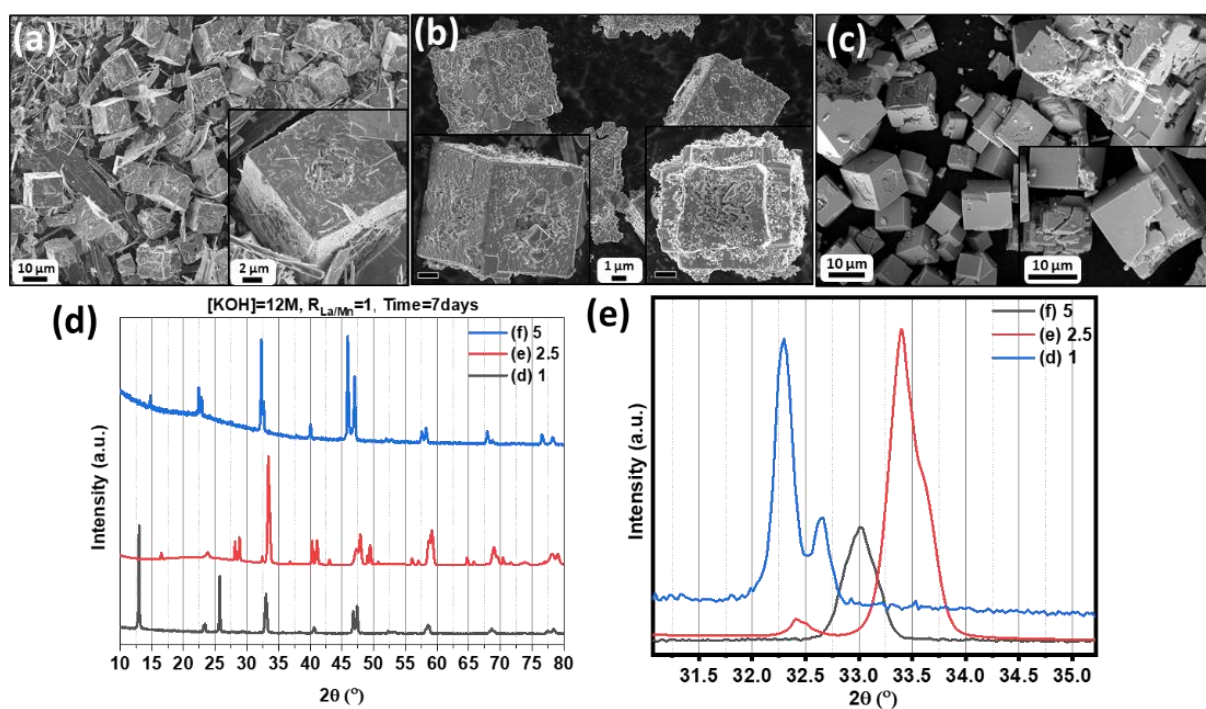


Figure 2.8 a-c SEM images of the LKMO samples synthesized via hydrothermal process at 265 °C for 7 days ($R_{La/Mn} = 1.0$) at 12 M KOH, with various $[La, Mn]$: (a) 1 (K12P1), (b) 5 (K12P2.5), (c) 8 (K12P5), and the (d & e) XRD pattern of K12P1, K12P2.5, & K12P5.

2.6.4 Effect of precursor concentration

Reaction kinetics generally increases with an increase in the concentration of the reagents in a reaction. Hence, keeping all the other parameters constant (at 18 M KOH), the reaction is conducted at three different precursor concentrations (1 mmol, 5 mmol, and 8 mmol) in order

to study its effect on the reaction time (Table 1; P1, K18P5 & P8). The reaction yielded $\text{La}(\text{OH})_3$ nanorods and $\text{K}_{0.5}\text{Mn}_2\text{O}_{4.3} \cdot (\text{H}_2\text{O})_{0.5}$ nanosheets after 1 day. When the reaction is conducted for 7 days, the yield of the phase and size of the LKMO microcubes (more than 90% yield and average size: 29 microns) are improved.

SEM images and XRD profiles of the products are depicted in **Figure 2.9**. Where we can see the length and diameter of $\text{La}(\text{OH})_3$ rods produced at 8 mmol decreased after 7 days compared to 1 mmol.

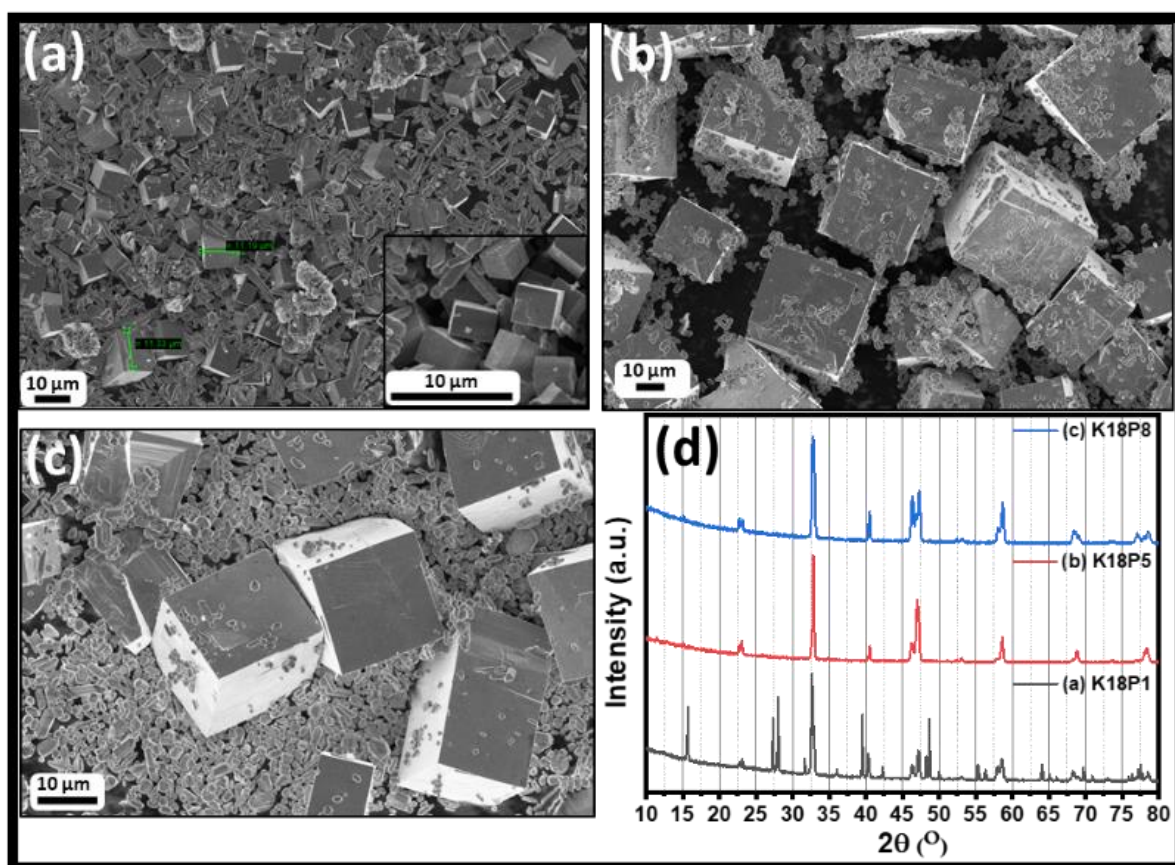


Figure 2.9 (a-c) SEM images of the LKMO samples synthesized via hydrothermal process at 265 °C for 7days ($R_{\text{La}/\text{Mn}} = 1.0$) at 18 M KOH, with various [La, Mn] : (a) P1, (b) P5, (c) P8, and the (d) XRD patterns of P1, P5, & P8.

From **Figure 2.10**, we see the cubes are growing, consuming the rods as the length of the rods is decreasing with the size of the cubes. Even at 9 M KOH, better cubes can be obtained by increasing the precursor concentration to 8mmol. This shows the importance of the precursor concentration. Even though the KOH is not sufficiently present, but higher precursor concentration allows the LKMO phase to form.

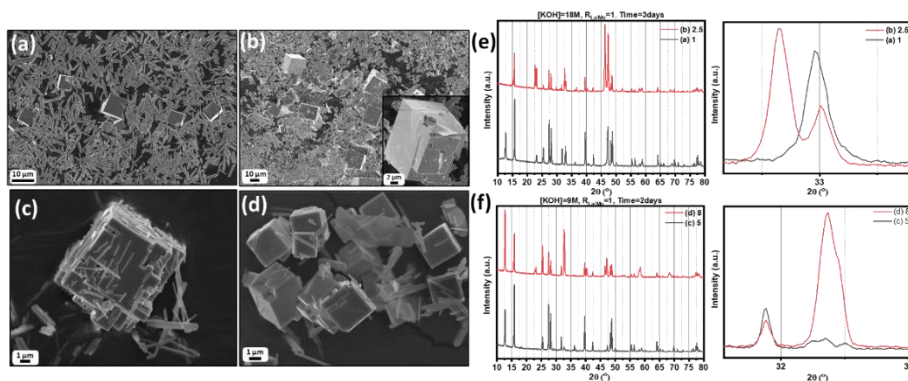


Figure 2.10 SEM images of the LKMO samples hydrothermal synthesized at 265 °C ($R_{La/Mn} = 1.0$), with various [La, Mn], reaction time & KOH concentrations: (a) (P1T3), (b) (P2.5T3), (c) (K9P5T2), (d) (K9P8T2), and Overall and 32°-33° focused (e) XRD pattern of P1T3, P2.5T3, & (f) XRD pattern of K9P5T2, K9P8T2.

From **Figure 2.8(a-c)**, at 12 M KOH, we see the cubes forming are hollow, and this hollowness again can be seen of different structures, which are dependent on the precursor concentration. In 5 mmol case, the holes are deep, but at 1mmol case, they are on the surface looking and amorphous in appearance. The hollowness of the forming LKMO cubes, from 1mmol to 2.5 mmol and 5 mmol at KOH of 12 M, we see the cracks are different, and the surface is getting smoother. Whereas at the higher-end above 12 M (**Figure 2.9**) the cubes are more solid.

2.6.5 Effect of precursor ratio

The molar ratio of the precursor (La:Mn) played a pivotal role in the formation of the LKMO phase. The amount of K^+ incorporation is also affected by this ratio. At La: Mn is 1:1, the XRD pattern in **Figure 2.11(c)** of 18KP1 shows the difference between (110) and (104) indicative of lower symmetry (having La:K:Mn as 0.97:0.03:1).

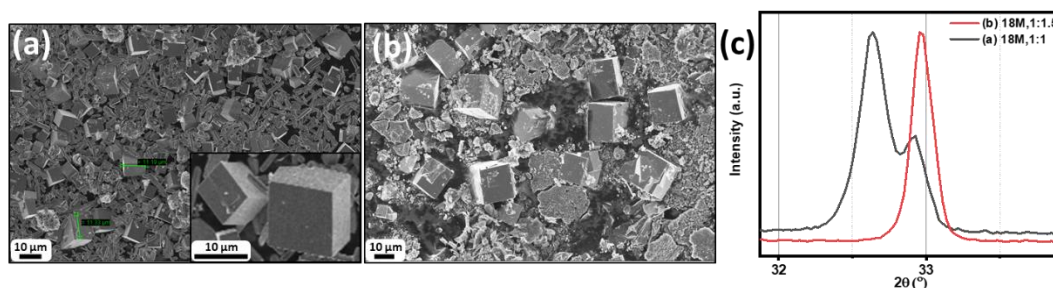


Figure 2.11 a and b SEM images of the LKMO samples synthesized via hydrothermal process at 265 °C for 7days ($[La, Mn] = 1.0M$) & 18 M KOH with various $R_{La/Mn}$ 1 (P1) and (b) 0.67 (P1R0.67), and (c) Typical XRD patterns at around $2\theta = 32^\circ - 34^\circ$.

These two peaks merge as K^+ is more in the crystal. When La:Mn changes from 1:1 to 1:1.5 [(Table 2.1; entries P1 & P1R0.67) is depicted in Figure 2.11 & Figure 2.12.], the amount of K^+ inside the LKMO increased to 18% leading to the reduction of cooperative Jahn-Teller distortion in LMO. Thus, the crystal adopts higher symmetry. A similar observation is also reported in the literature in the flux method synthesis.⁽¹³¹⁾ As 18 m is more favourable for the LKMO cubes formation thus the 1:1.5 has bigger cubes ≈ 3 times more than the 1:1. Indicating faster growth and also quick nucleation leading to more yield.

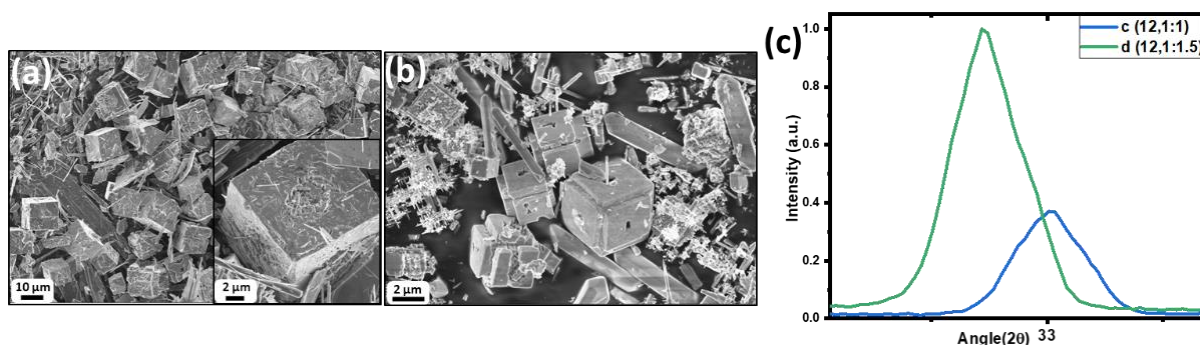


Figure 2.12 a,b SEM images of the LKMO samples synthesized via hydrothermal process at 265 °C for 7days ([La,Mn]=1.0M) & 12 m KOH with various $R_{La/Mn}$: 1 (K12P1) and (b) 0.67 (K12P1R0.67) (c) Typical XRD patterns at around $2\theta = 32^\circ - 34^\circ$.

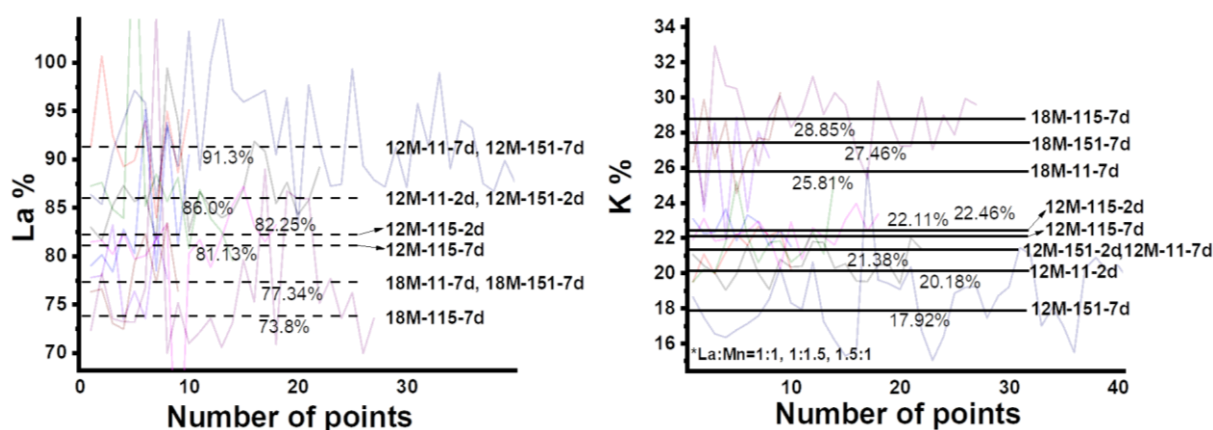


Figure 2.13 Atomic% analysis of La and K in LKMO at different KOH concentration (12 m & 18 m), precursor ratio (1:1, 1:1.5, & 1.5:1), and reaction time (2d & 7d) conditions from several EDS points.

2.7 Mechanisms:

2.7.1 Nucleation:

Under hydrothermal conditions, nucleation rate and subsequent particle growth depend on relative supersaturation, which is influenced by processing parameters such as the concentration of the precursor feedstock, mixing and temperature. Thus, adjusting these synthesis conditions effects the final size distribution of particles. However, synthesizing complex oxides such as ABO_3 through hydrothermal route; especially when lanthanides are involved; suffers from simultaneous side reactions such as the formation of oxides and hydroxides of individual elements ($A(OH)_x/AO_y$ and BO_z) because of the stronger interaction with oxygen. Thus, the choice of mineralizers, their concentrations and temperature play a major role to obtain single-phase perovskites oxides.

In our hydrothermal synthesis, base concentration, precursor amount, temperature, and time played an important role in LKMO synthesis. Crystallization of $La_{1-x}K_xMnO_3$ instead of $LaMnO_3$ in the hydrothermal conditions could be a result of difficulty in crystallizing Jahn-Teller active $LaMnO_3$. The incorporating potassium decrease cooperative Jahn-Teller distortion and helping in nucleation. In the synthesis procedure of LKMO, although LKMO is a stable thermodynamic phase, kinetics products are dominated under basic hydrothermal conditions. At low pH ($[KOH] < 8M$), $LaMn_2O_5$ phase is dominated in the reaction, as seen from **Figure 2.6(d)**, while above 8M $[KOH]$ $LaMn_2O_5$ phase is dissolved in the reaction, but La is precipitated as $La(OH)_3$ rods, as seen from **Figure 2.6(h)**. The inverse solubility relation of Mn and La by-products with pH leads to difficulty in crystallizing the LKMO phase. With increasing pH at elevated temperatures, the solubility of $La(OH)_3$ is slightly improved and leads to the formation of the LKMO phase. However, nucleation is very slow as we have observed crystallization has been observed only after one day (at $[KOH]$ 18 M at 265 °C). The size of the crystals is on an average of 5 microns (sizes range from 2-9 microns). With increasing time from 1 to 7 days, we have observed the formation of new nuclei (the number of LKMO cubes are increased) and increased dissolution of $La(OH)_3$ nanorods (length of $La(OH)_3$ rods are decreased). This can be seen from **Figure 2.14**. We have not observed smaller cubes that are below 1 micron in this process, rather observed only increasing in the number of cubes with time.

The above observation, temporally separated crystallization, differs from the classical nucleation and growth model, where nucleation happens in a single step and follows diffusion-

controlled crystal growth. In this case, a number of nuclei are the same during the process, but size increases with time. Accordingly, one should observe the size evolution of particles with time. Alternatively, Watzky and Finke developed a model to explain the temporal overlap of nucleation and growth.^{115,132} According to this model, nucleation is a slow continuous process (slow nucleation is a consequence of reaction limited) and growth is a fast-autocatalytic process. Accordingly, to this, within a given reaction process, nucleation and growth are predominantly overlapped events. This model has explained the kinetics of different nanoparticles.^{132–135}

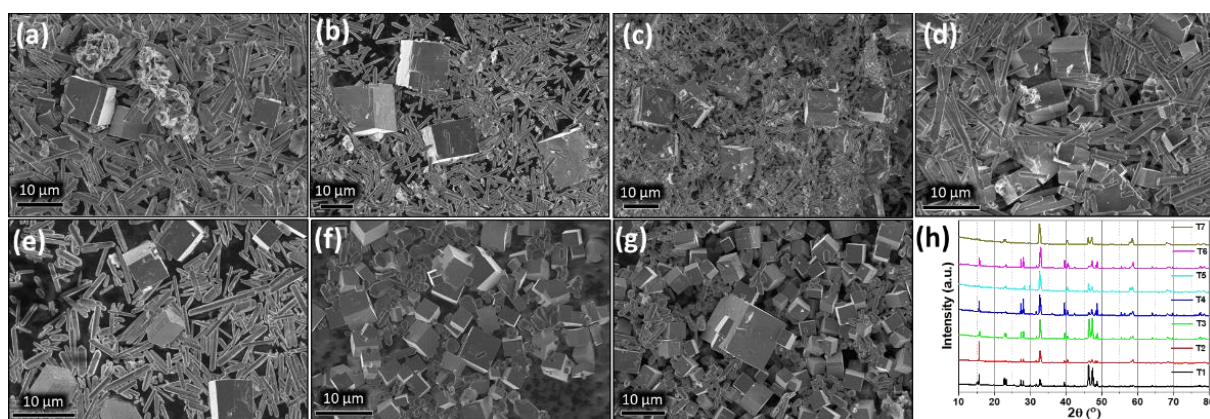


Figure 2.14 a-g SEM images of the LKMO samples synthesized via hydrothermal process at 265 °C, 18 M KOH, ([La,Mn]=1.0M), ($R_{La/Mn}=1$) with varying reaction time: (a) 1day (T1), (b) 2days (T2), (c) 3days (T3), (d) 4days (T4), (e) 5days (T5), (f) 6days (T6), (g) 7days (T7), and (h) XRD pattern of as-synthesized LKMO sample with time variation.

However, this model does not explain the temporally separated crystallization events observed in the present case, where different events of crystallization (nucleation and growth) have minimal overlap.

In our case, a high concentration of Mn in the solution (in the dissolved state) and low solubility of $\text{La}(\text{OH})_3$ leads to unavailability of both the cations simultaneously at one place, thus making it difficult in achieving supersaturation. For nucleation to happen, there is a need for a relative difference in the level of supersaturation; which could be either homogeneous globally or heterogeneous locally. However, the finite probability of local supersaturation leads to slow nucleation. The freshly formed nuclei have high surface energy; thus, they might dissolve back into the solution leading to difficulty in observing the nucleation at shorter durations. But, when the size of crystals crosses the critical nuclear size, the crystal grows with high rates (consuming already dissolved La and Mn) to decrease the surface energy, leading to the formation of 2 to 9 micron cubes. At this stage, the availability of dissolved La and Mn in

the solution helps for the growth of already formed nuclei than the initiation of another nucleation step (as nucleation is a slow process than the growth). Now the solution is undersaturated, and rate of dissolution of $\text{La}(\text{OH})_3$ rods are higher than LKMO cubes (at micron size crystals) thus leading to the continuous dissolution of $\text{La}(\text{OH})_3$ rods into the solution. Freshly dissolved $\text{La}(\text{OH})_3$ can create the condition for another local supersaturation and leads to another nucleation step, thus increase the number of cubes (nuclei) with time. The already formed LKMO cubes slowly increase their size during the reaction, but the growth of newly formed small crystals is faster. This temporally separated nucleation and growth of each (set of) crystal(s) formed in the reaction could be a possible reason for not observing lanthanum-based perovskite oxides at nanosizes.

Similar conditions have been used to synthesize LaCrO_3 and LaFeO_3 in the usual hydrothermal conditions. In both cases, we got microcubes, as can be seen from **Figure 2.16(a)** and **Figure 2.16(e)**, respectively. The elemental distribution for the same can be seen from the EDS mapping in **Figure 2.17(a)** and **Figure 2.17(b)**, respectively. In the literature, it has been reported that similar conditions have been used to get chromite and ferrite where the particles sizes are above 1 microns.^{110,112,113,136} Synthesizing them at low dimensions (sizes less than 1-micron) is found to be challenging.

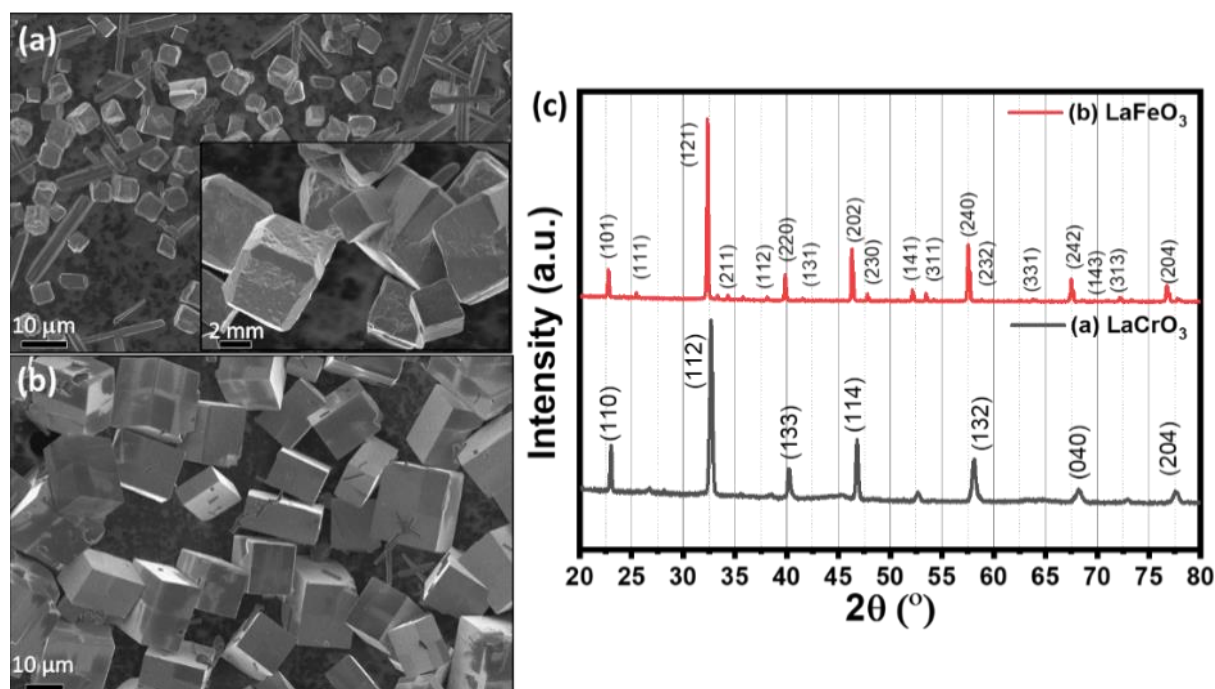


Figure 2.15 SEM image of (a) LaCrO_3 and (b) LaFeO_3 microcubes synthesized hydrothermally, (c) XRD pattern of LaCrO_3 and LaFeO_3 microcubes after density-gradient separation of byproducts.

As the above experiments reveal the domination of secondary nucleation at extended times, we have attempted to synthesize perovskites in the presence of pre-synthesized cubes.

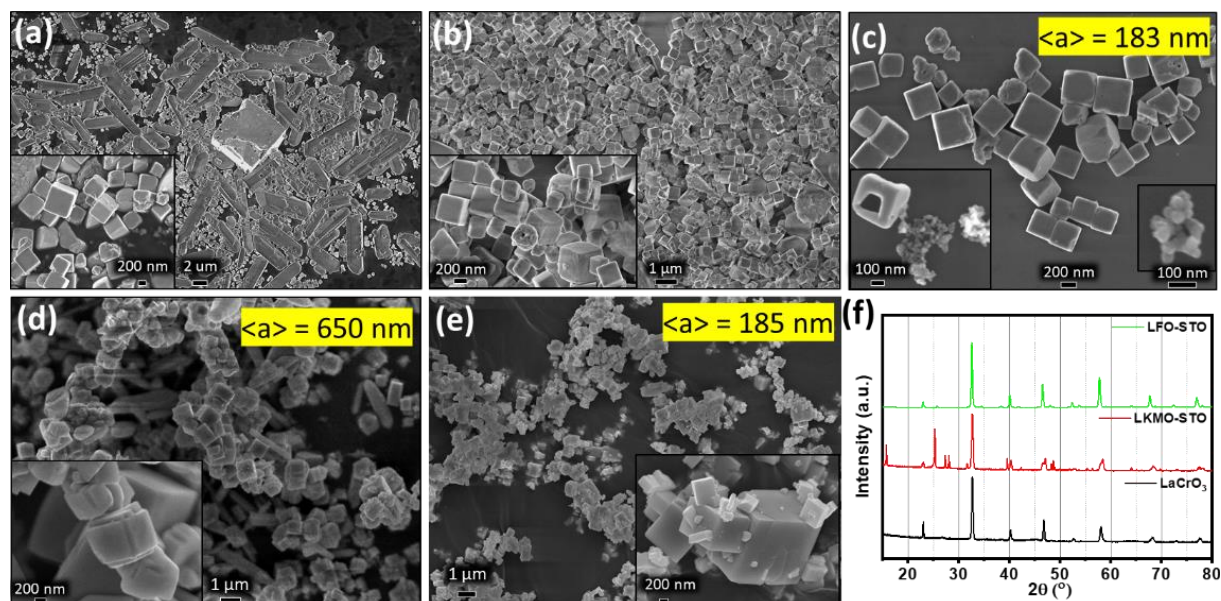


Figure 2.16 SEM image of (a) as-synthesized LaCrO_3 submicron cubes by secondary nucleation by adding pre-synthesized microcubes of the same, inset showing even smaller cubes of LaCrO_3 , (b & c) density-gradient separated LaCrO_3 submicron cubes synthesized hydrothermally, (d) LaKMnO_3 synthesized by using the STO nanoparticles (e) LFO submicron cubes synthesized by using the STO nanoparticles; inset showing the lower magnification of the same, and (g) XRD patterns of LaCrO_3 , LKMO-STO, and LFO-STO submicron cubes

Figure 2.2(f) shows SEM images of LKMO synthesized when in the presence of pre-synthesized cubes. The small LKMO cubes around 1 micron in size can be clearly seen from the image. The image also shows the formation of small cubes on top of the big cubes suggesting the secondary nucleation process. The overall reaction time has come down to 2 days (to complete the 80%). On the other hand, hydrothermal condition using SrTiO_3 nanoparticles (inset in **Figure 2.16(c)**) as seeds (average size of 170 nm) yielded LKMO smaller cubes (Fig. **Figure 2.16(d)**) with an average size of 650 nm. Similarly, LaFeO_3 submicron cubes (**Figure 2.16(f)**) with an average size of 180 nm were synthesized using STO nanoparticles as the precursor seeds. In the case of LaCrO_3 , hydrothermal conditions using pre-synthesized LaCrO_3 as a precursor have yielded small LaCrO_3 cubes (**Figure 2.16(b)**) with sizes ranging from 80 nm - 450 nm. The above observation clearly demonstrates the importance of secondary/heterogeneous nucleation in the synthesis of small cubes.

In the case of secondary or heterogeneous nucleation, the decrease in overall reaction time and the observation of small cubes could be explained as follows. The kinetic barrier for nucleation decreases in the presence of seeds, resulting in a faster nucleation process, and

nucleation occurs at multiple locations. As a result, the consumption of dissolved La in the solution is greater for both nucleation and growth, resulting in a rapid undersaturation of the reaction solution. The dissolution of $\text{La}(\text{OH})_3$ rods into solution increases due to the common ion effect (Le Chatelier's principle). The freshly dissolved La in the solution is distributed among all the formed nuclei, lowering the average increment of size of cubes but a large number of nuclei; thus, small cubes are observed at the end of the reaction. In the meantime, newly formed cubes can serve as new nucleation centres. As a result, the entire process is supported by faster kinetics, which reduces the overall reaction time as previously discussed.

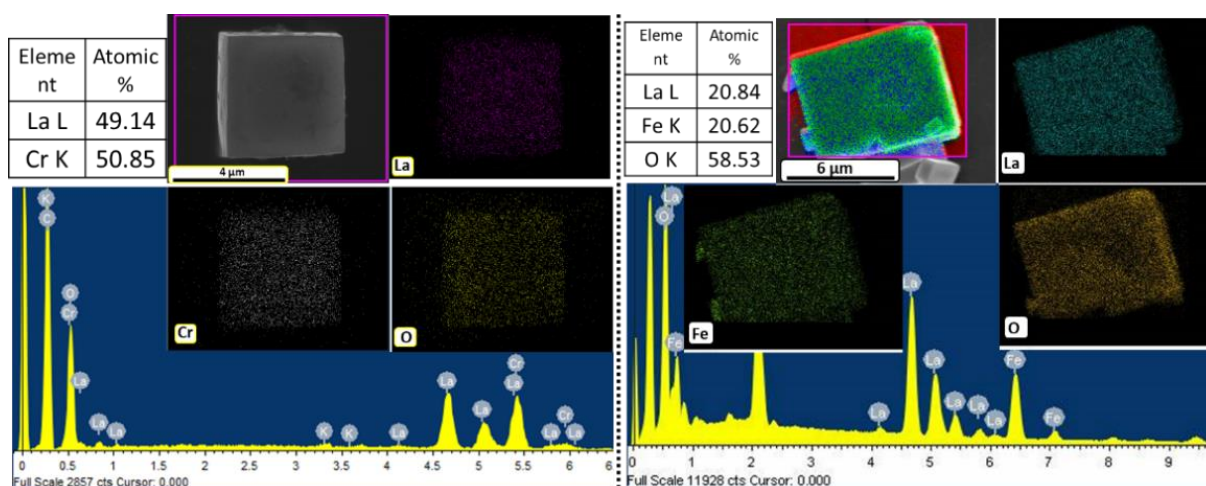


Figure 2.17 EDS and mapping of density-separated (a) LaCrO_3 and (b) LaFeO_3 microcubes indicating the elemental composition and distribution

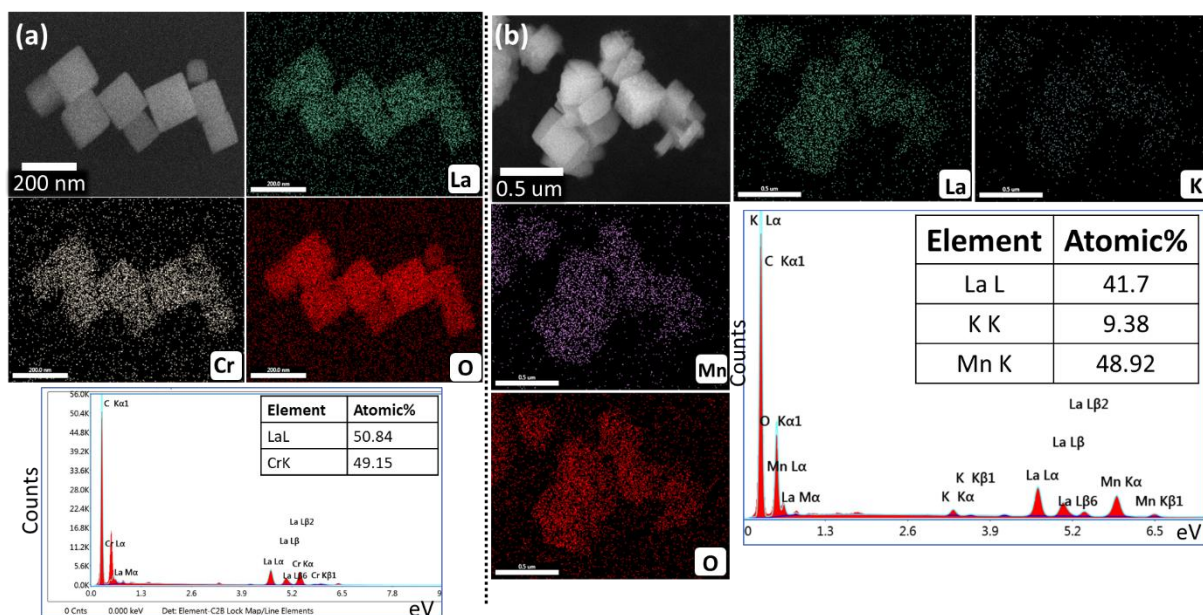


Figure 2.18 EDS spectrum and mapping of density-separated (a) LaCrO_3 and (b) LaKMnO_3 submicron cubes indicating the elemental composition and distribution

2.8 Magnetic properties measurement:

We have measured the magnetic properties of the LKMO cubes sample, which are synthesized at different KOH concentrations. **Figure 2.19** displays the magnetization-temperature (M-T) curves, zero fields cooled (ZFC), and field cooled (FC) measured for the LKMO sample synthesized at 12 m and at 18 m KOH concentration under 300 Oe field. The inset of each, **Figure 2.19**, shows the dM/dT -T curve, determined from the M-T curves, whose minimum gives the Curie temperature T_c (ferromagnetic to paramagnetic phase transition temperature). The red curve from **Figure 2.19(a)** shows the ZFC, and the black curve shows the FC of the 12 m LKMO sample with the Curie transition temperature around 277K. **Figure 2.19(c)** shows the ZFC and FC of the 18 m sample with the Curie transition temperature around 220K. **Figure 2.19(b)** and **(d)** show the field variations of magnetization (M-H) done at different temperatures between 2K to 300K for 12 m and 18 m, respectively. We can see from **Figure 2.19(d)** that the material has the saturation magnetization, M_s that is far from saturation.

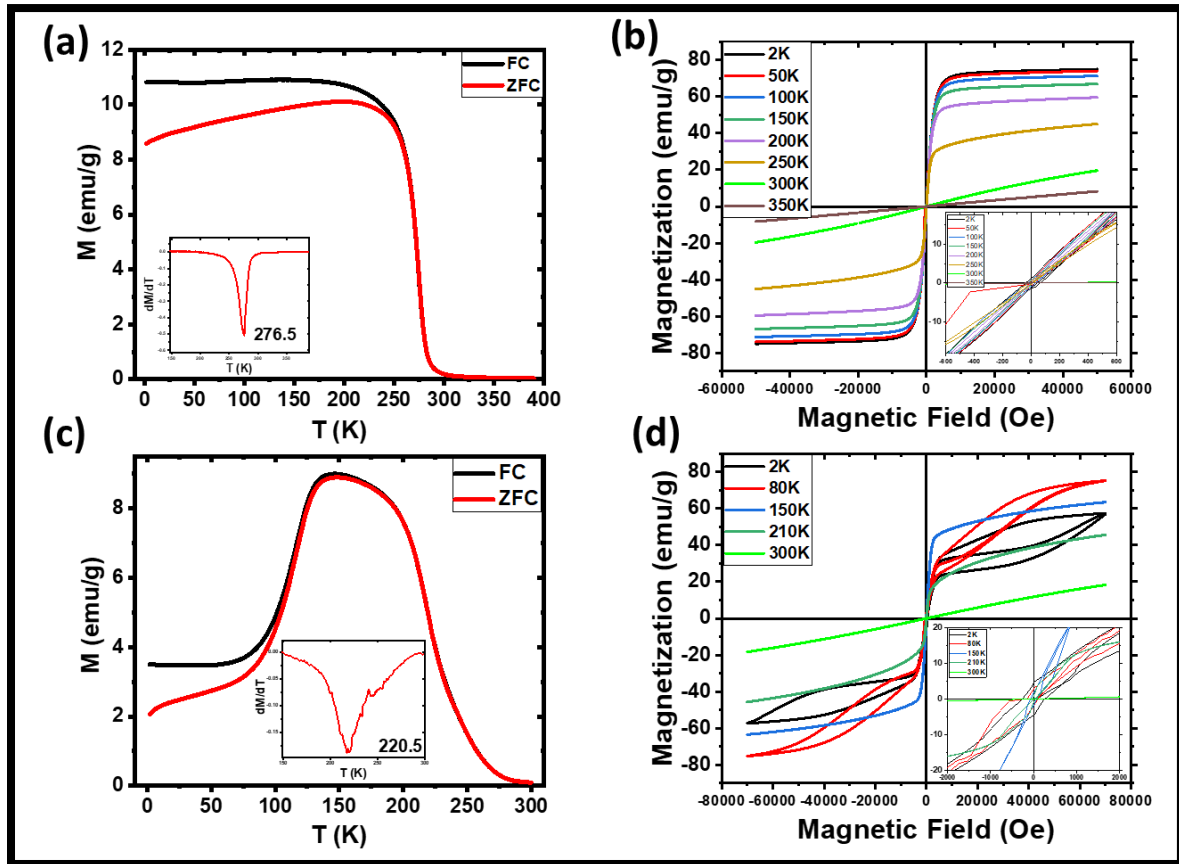


Figure 2.19 Temperature-dependent magnetization of LKMO samples (a) 12 m KOH & (c) 18 m KOH obtained at 300 Oe. The inset shows the first-order derivative of the respective zero-field cooling magnetization with

respect to temperature. (b) & (d) are the field-dependent magnetization of LKMO samples synthesized at 12 m and 18 m KOH, respectively, at different temperatures varying from 2K to 300K.

This shows that these materials exhibited a paramagnetic behaviour at and above ambient temperatures and experienced no magnetic saturation up to 50 KOe. However, an anomaly was observed in the hysteresis loops at the high magnetic field range indicative of a metamagnetic transition⁽¹³⁷⁾ where it was observed that the application of a critical magnetic field melted the charge-ordered, CO state and induced an irreversible ferromagnetic, FM ordering; thereby showing a steep increase in the magnetic moment. This metamagnetic transition was related to the field-induced melting of the strain-induced at the phase boundary between canted anti-ferromagnetic cAFM and FM clusters. It was reversible upon retracing the same magnetization path on subsequent sweeping of the field. The 12 m and 18 m samples show the coercivity, H_c 263 Oe and 40 Oe respectively at 2K shown in the inset of **Figure 2.19(b)** and **(d)**. 12 m case seems to have better exchange integral, θ_{ij} suitable for double exchange of unpaired 3d e_g electron along $Mn^{3+}-O^{2-}-Mn^{4+}$ leading to ferromagnetic phase and higher T_c . In contrast, 18 m case shows superexchange along $Mn^{4+}-O^{2-}-Mn^{4+}$ leading to canted antiferromagnetic phase (hysteresis loop at low-temperature under high field as shown in **Figure 2.19(d)**). One plausible explanation could be the variation of K substitution. In the 12 m cases, 22% K substitution was observed, whereas this has increased to 28% in the 18 m cases. By controlling the reaction parameters, the K quantity of the LKMO material and thus the magnetic properties can be altered.

2.9 Phase transformation:

Rietveld refinement was carried out to understand the crystallographic or structural information of the LKMO microcube samples. **Figure 2.20(a)** below depicts the refinement of an 18 m-LKMO XRD pattern indicating Pnma space group with $a=5.43 \text{ \AA}$, $b=7.72 \text{ \AA}$, and $c=5.45 \text{ \AA}$ lattice parameters. While the 12 m-LKMO (**Figure 2.20(b)**) shows to be of I4/mmm space group which is hexagonal (higher in symmetry than orthorhombic). To investigate the effect of potassium, we synthesized pure $LaMnO_3$ (LMO) using Molten Salt Synthesis (MSS) method, as described in the auxiliary section.

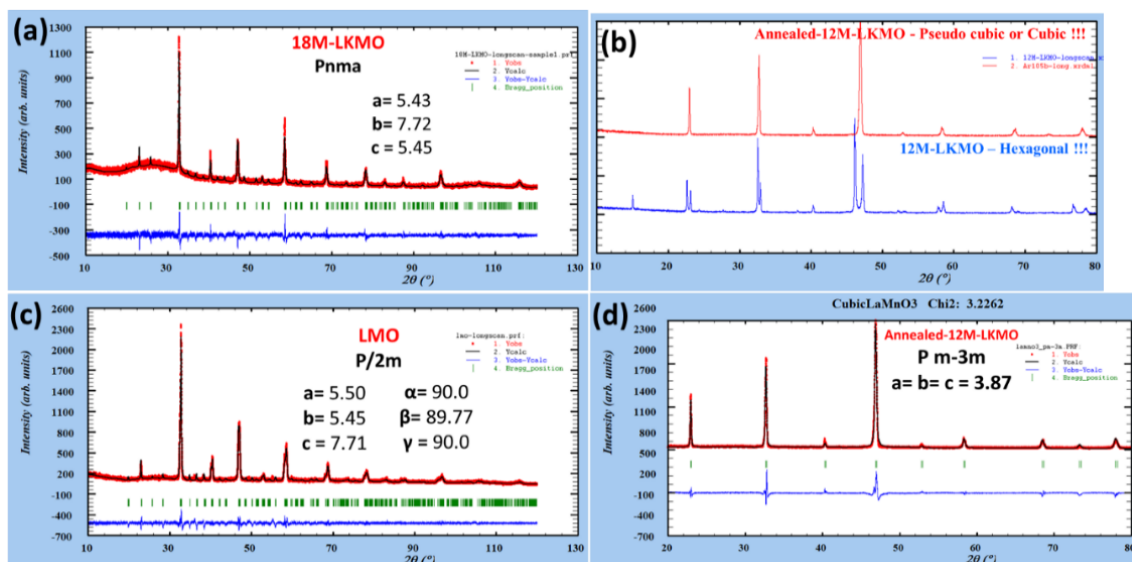


Figure 2.20 Rietveld refinement of (a) 18 M-LKMO, (b) 12 M-LKMO*, (c) LMO, and (d) annealed 12 M-LKMO; indicating the observed (Yobs), calculated (Ycal), and difference (Yobs-Ycal). (*Neutron diffraction (ND) will provide accurate refinement of both as-prepared and annealed 12 M LKMO.)

As shown in **Figure 2.20(c)**, we refined the XRD pattern of LMO, which indicates a P/2m space group, i.e., monoclinic. This indicates that potassium substitution improves the crystal symmetry significantly. We then annealed the hydrothermally synthesized 12 M and 18 M LKMO microcube samples at 900 °C temperature from 12 to 24 hrs to better understand the stability of the K as well as to verify structural transformation and magnetic property enhancement. The SEM images (**Figure 2.21**) show no morphological changes, whereas the XRD results are exciting. The phase of 18 M-LKMO is relatively unchanged, whereas 12 M-LKMO has evolved into higher symmetry. The lattice parameters are shown in **Figure 2.20(d)**.

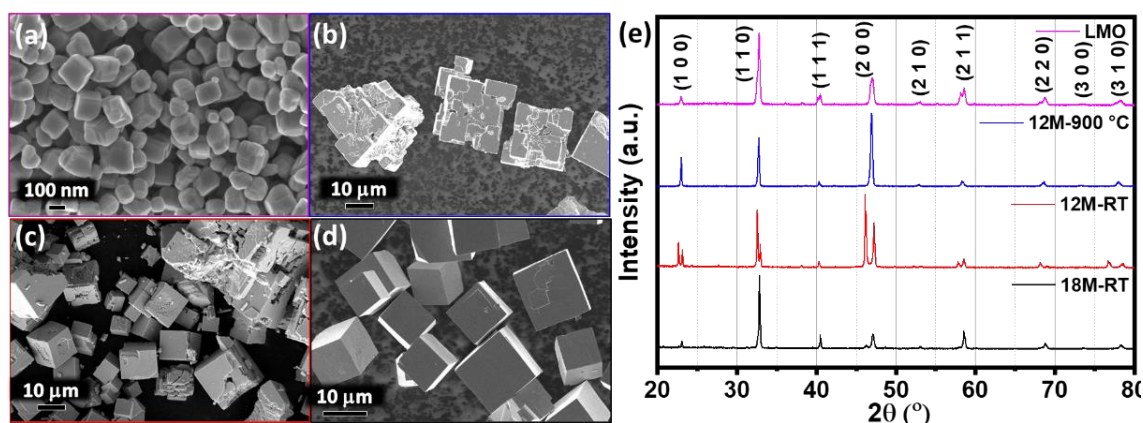


Figure 2.21 SEM micrographs of (a) MSS-synthesized LMO nanoparticles, (b) 900 °C annealed 12 M-LKMO microcubes, (c) as-synthesized 12 M-LKMO microcubes, (d) as-synthesized 18 M-LKMO microcubes, and (e) XRD patterns of LMO, 18 M, 12 M, and annealed 12 M LKMO.

This annealed 12 m system goes to a much higher symmetry pseudo-cubic $Pm\bar{3}m$ with lattice parameter 3.87 Å. The position of K appeared to be off in both the as-synthesized and annealed 12 m LKMO microcubes. Neutron diffraction can provide us with more accurate structural parameters for the same. When the magnetic M-T measurement (**Figure 2.23**) was performed on annealed 12 m, we discovered a significant increase in curie temperature to 330.8 K from 276.8 K (**Figure 2.19**).

The unaltered presence of K (27.34% in 18 m and 21.74% in 12 m) from the EDS spectrum after annealing the sample to and above 850 °C to 1000 °C for 12hrs to 24hrs (**Figure 2.22**), indicates high potential application in the solid-oxide fuel cells (SOFCs), magnetocaloric (MCE), and thermochromic applications.

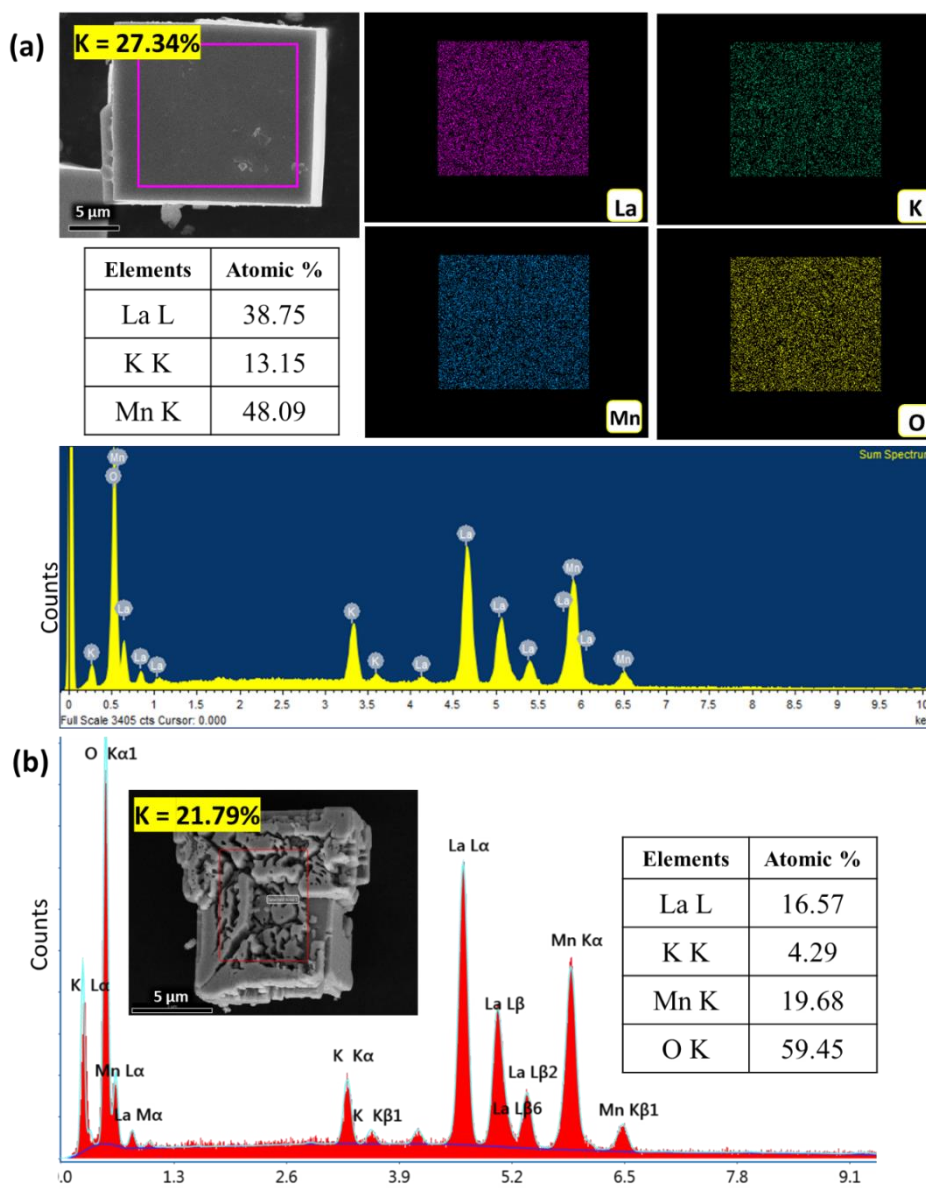


Figure 2.22 EDS elemental quantification of (a) 18 m and (b) 12 m LKMO microcubes after annealed at 900 °C.

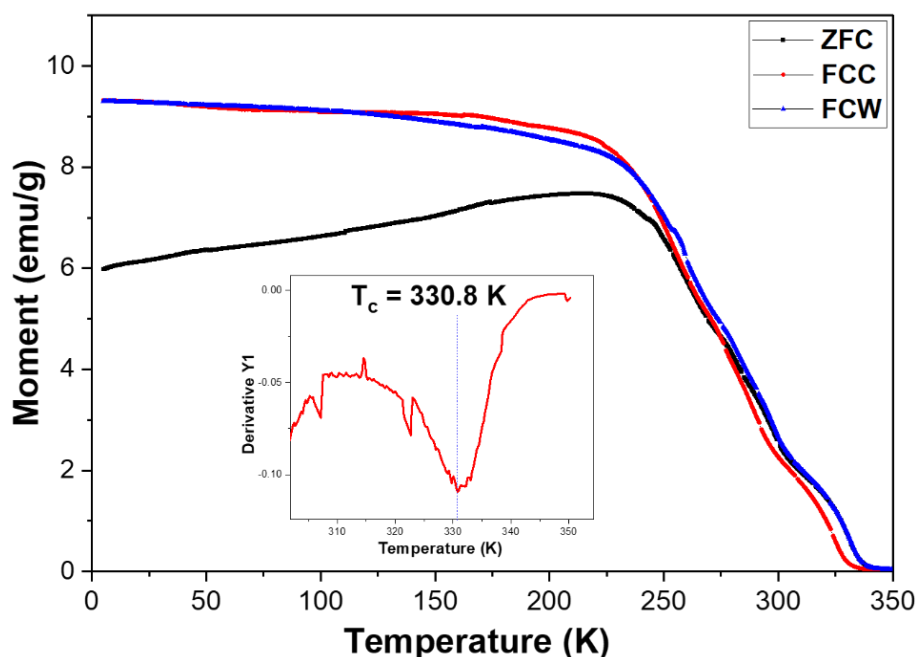


Figure 2.23 Magnetization-Temperature curves (ZFC, FCC, FCW) of annealed 12 M LKMO microcube sample, with the first-derivative of ZFC indicating the T_c shown in the inset.

2.10 LKMO synthesis as a generic growth strategy:

As discussed in the previous sections of this work, LKMO microcubes were synthesized through a low-temperature, hydrothermal solution-phase synthetic approach; and successfully optimized to produce not only the desired morphology and phase but also an appreciable yield; by tailoring various reaction parameters. However, the efficiency of this synthesis would cease to be as helpful and significant if it were not generic, viz., applicable to other Lanthanum-containing systems as well. In order to prove this, an attempt to synthesize other Lanthanum-containing systems was tried, employing Chromium (Cr) and Iron (Fe) in place of Manganese (Mn). In the literature, people have synthesized LaCrO_3 ^(51,111) and LaFeO_3 ^(138,139) by a hydrothermal method where 5 to 14 days takes for the reaction completion. These LaCrO_3 and LaFeO_3 microcubes were synthesized using a similar procedure as that for LaKMnO_3 , with a high KOH concentration of 9 M and 12 M, respectively, precursor concentration of 5 mM, and a typical reaction time of 5 to 7 days for 90 % completion. **Figure 2.15a** and **b** show the SEM images, **Figure 2.15c** XRD profiles, and **Figure 2.17a** and **Figure 2.17b** shows the EDS mapping of these samples. The LaFeO_3 microcubes were of the size range of 5 – 10 μm , while that of the LaCrO_3 was around 1 – 2 μm . A noteworthy fact was that; in the combined presence of La and Cr, the former tended to become La_2O_3 in the solution phase, thus causing the LaCrO

phase to retard; which made it mandatory to use Cr slightly in higher quantities than La⁽⁵¹⁾. This also meant that the structural morphology could show some changes in comparison with the usual reported ones synthesized at markedly lower KOH concentrations of 3 – 6 M. Such as the LKMO microcubes, the latter showed a very good magnetic momentum; as is visible from **Figure 2.24a**. An interesting fact was that in contrast to the originally reported products^(54,78,111), the ones synthesized in this work showed both T_c and T_n (as is depicted in **Figure 2.24a**). The possible reason for this phenomenon is not currently within the scope of this work.

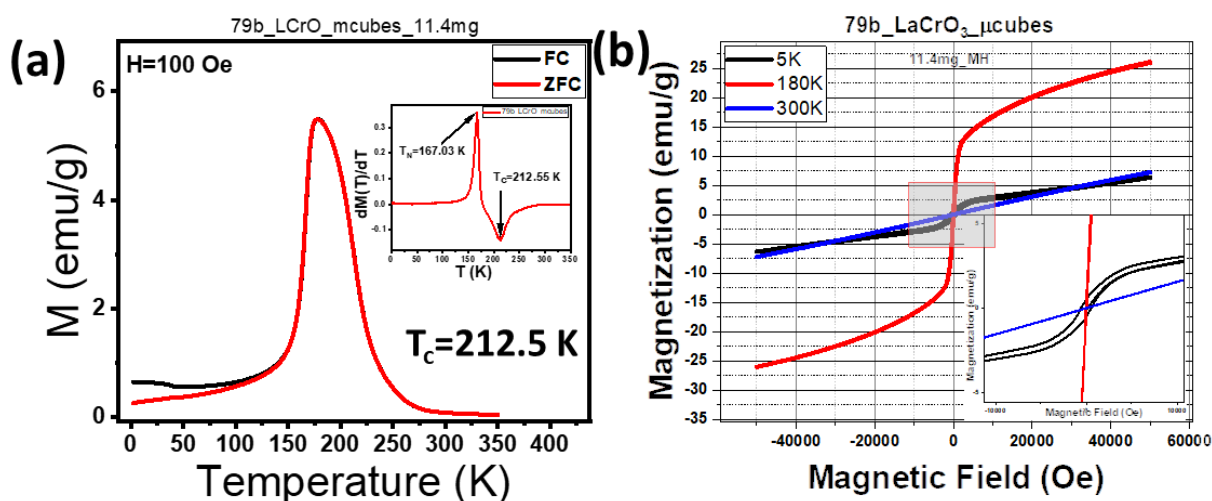


Figure 2.24 (a) MT and (b) MH studies of LaCrO₃ microcubes at 100 Oe field and 5K, 180K and 300K temperatures, respectively.

2.11 Effect of doping:

Tuning the percentage of the A-site dopants was demonstrated to change the magnetic and electrical resistivity phenomenally of the material⁽¹⁴⁰⁾, where the highest ferromagnetic (FM) phase was shown to be possible with 33% Calcium doping in LCMO; deviation from which percentage caused the material to show a decrease in ferromagnetism and quantitatively undergo various phase changes such as AFM, canted AFM, charge-ordered and so on. This study provided the basis for understanding the importance of the percentage of K-substitution at the A-site in LMO and determining how crucially this parameter played a role in deciding the magnetic phase of the resultant LMO. With this knowledge, an attempt to synthesize LKMOs with varying amounts of K substitution was tried; by varying the potassium source in the synthesis, i.e., the KOH concentration. It was observed that in a typical synthesis, while 1

– 5 mM precursors with 12 – 21 M KOH formed LKMO microcubes containing 13 – 20 % potassium, concentrations of precursors above 5 mM and KOH below 12 M lead to microcubes with a distinctly lower potassium substitution. This could be due to the capability of a monovalent cation to drive the oxidation of two Mn ions from the (+3) oxidation state to the (+4) one within the same time as would be taken by its divalent counterpart; which would ultimately result in a quicker achievement of the spin locking between the adjacent Zener pairs and initiate the double exchange transfer integral to go below unity. Thus, ultimately, it could be successfully demonstrated in this work that the highest possible ferromagnetic phase of LMO could be achieved just by a meagre amount of K-substitution. **Figure 2.19** shows the inverse dependence of T_c on the degree of K-substitution. This evidently shows that the materials exhibit metallicity and ferromagnetism below this temperature (T_c) and insulation above it, the very property which makes them an ideal CMR material. It, however remains to understand further the different kinds of magnetic and electrical transitions that could be achieved by varying the degree of K-substitution.

2.12 LKMO microcube growth

Several systematic studies were carried out by varying the reaction parameters such as temperature, precursor concentration, KOH concentration, reaction time, and pre-synthesis stirring time in order to understand the growth pattern of these LKMO microcubes. On average, large cubes of a size of around 15 μm were formed, which were characterized accordingly by a series of analytical techniques. All these investigations assisted in realizing the importance of the kinetically stable intermediates formed in the due course of the reaction.

As discussed in the previous section, hydrothermal synthesis is usually governed by the classical theory of nucleation and growth. Individual ions undergo homogeneous nucleation before reaching their activation energies, subsequently after which they grow into bigger particles with time. However, the growth mechanism of the LKMO microcubes in this current work has been estimated to follow a deviated path from the classical growth mechanism, which to the best of our knowledge, has not been reported yet.

A close look into the synthesis of LaMnO_3 gave the knowledge that while the Lanthanum formed the kinetically more stable $\text{La}(\text{OH})_3$ nanorods, Manganese formed the potassium-substituted $\text{K}_{0.5}\text{Mn}_2\text{O}_{4.3}(\text{H}_2\text{O})_{0.5}$ (Manganese oxide with intercalated water) sheets. It was estimated that the lanthanum precursor mediated $\text{La}(\text{OH})_3$ nanorods, under the constant hydrostatic environment, agglomerated together, excluding the manganese precursor

molecules from their localized system. This led to the deposition and subsequent diffusion of the manganese into this agglomeration to further form the materials. Thus, the outer surface of the La-agglomerate acted as a primary nucleation site which oriented manganese diffusion in the direction of the cube formation. It is noteworthy that this very diffusion, which has been proven to date to be occurring only in solid-phase reactions at temperatures above 600 °C, could be achieved in this hydrothermal solution-phase synthesis protocol at a temperature as low as 265 °C. Further, this La-cluster seemed to undergo a structural reconstruction as the diffusion of the Mn ions occurred deeper and deeper into the core. Thus, the phase of the particles kept getting better with the size remaining constant; as the reaction proceeded with time, thus ultimately yielding the LKMO microcubes. Here, a remarkable clause for consideration could be the use of higher reagent (precursor) concentrations in order to increase the rate of diffusion.

Another interesting parameter here was found to be the role of KOH in LKMO formation. KOH in this current protocol had been more active in terms of a mineralizer than an active reagent. Identified as a ‘double-edged sword,’ a higher concentration of KOH has been demonstrated to decrease the activation energy, which in turn increased the phase formation. However, the capping effect shown by the alkali on the active faces of the nuclei blocked further nucleation and growth of the material, subsequently decreasing the yield (as is shown in **Figure 2.6**). When this background was applied to the current work, it was understood that increasing the concentration of the precursors in the solution phase needed to be potentiated with a decreased KOH concentration as well in order to obtain a high-yielding synthesis of LKMO.

Interestingly, growth of LKMO cubes of sizes varying from 0.6 – 1.0 μm was observed when pre-synthesized LKMO cubes of smaller sizes were used as a nucleation seed, proving the fact that the heterogeneous nucleation that had occurred followed the classical theory of nucleation and growth (**Figure 2.2(f)**). This could be because the growth rate of the material was so rapid that the formed nuclei hit the saturation limit of growth of the LKMO in usual cases; but once there was the availability of a pre-synthesized LKMO seed, the surface energy became so low, allowing smaller particles to easily settle on the surface and nucleate; further yielding larger microcubes.

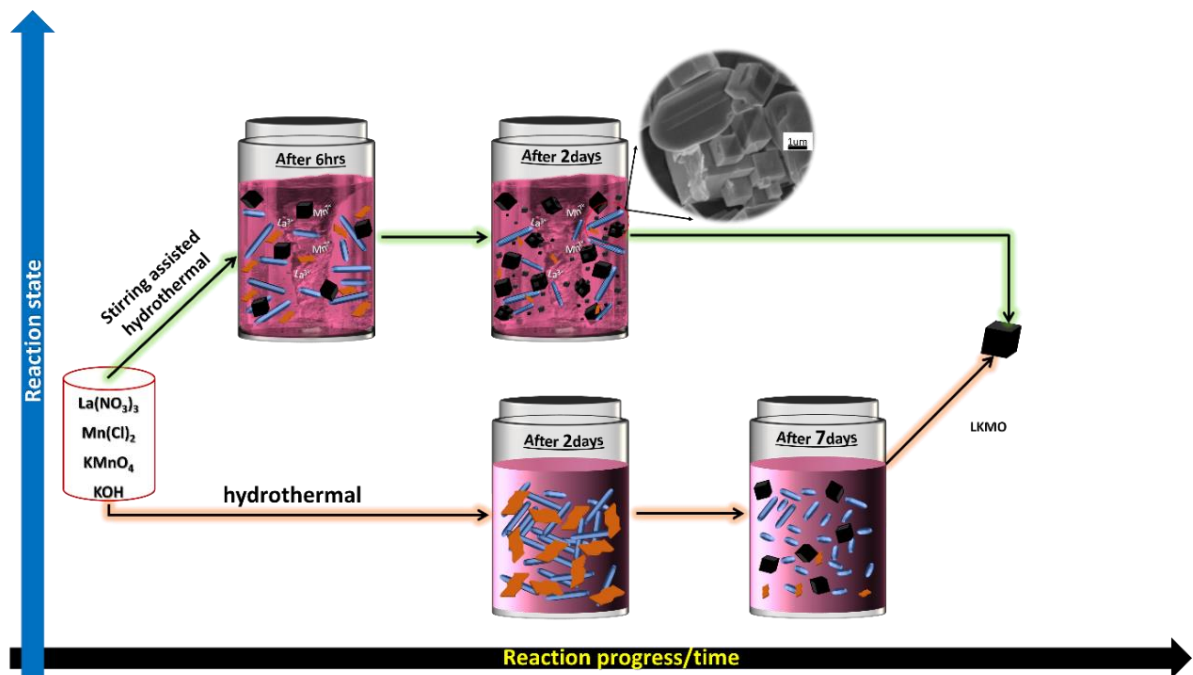
Once the preliminary techniques such as SEM, EDS and XRD were employed to throw light on the morphology, crystallinity, phase, size, and chemical composition of the materials, a few characterizations were performed to further gain insight into their spectral, magnetic, and electrical properties.

2.13 Conclusion

In summary, $\text{La}_{1-x}\text{K}_x\text{MnO}_3$ (LKMO) microcubes of the size range 6 – 30 μm have been successfully synthesized through a low-temperature solution-phase hydrothermal synthesis; and characterized to exhibit various phenomenal electrical, magnetic and electrochemical properties (discussed in **Chapter 4**). It was found that the yield and morphology of the LKMO microcubes were largely dependent on the reaction parameters such as KOH concentration, time, temperature, precursor concentration, and precursor ratio. The highest yield and accurate cubic morphology of the LKMO were obtained by a hydrothermal synthesis at 265 °C involving a precursor concentration of 5 mM, a ratio of 1, KOH concentration of 18 M, within a period of 7-14 days. Moreover, a mechanism for the formation of these LKMO microcubes has been proposed, and their stoichiometry; calculated. These materials, owing to their efficient exchange interaction between Mn^{+3} and Mn^{+4} demonstrated by the presence of Curie and Neel temperature, allow these materials to exhibit exceptional capability in magnetoresistance, supercapacitance, and oxygen reduction ability. A controlled synthesis of high-symmetry (pseudo-cubic) phase of manganite (A12 M-LKMO) was achieved. Furthermore, the synthetic hydrothermal strategy described in this chapter is a scale-up efficient one; and generic, which could be easily extended to the synthesis of other lanthanum-based systems as well.

CHAPTER-3:

Dynamic hydrothermal synthesis: tailoring nucleation and growth



This chapter describes the effect of stirring to overcome the limitations of static hydrothermal synthesis of the same LPOs.

3.1 Abstract

LPOs are extremely durable. They require a large amount of activation energy to synthesise. The major limitations of their synthesis via static hydrothermal are the unavailability of active reagents due to reagent trapping in the form of respective by-products. Because the byproducts dissolve slowly, the reaction can take up to 10 to 15 days. To enhance the reaction kinetics, stirring assisted hydrothermal, also known as dynamic hydrothermal (DH), method was performed. Hydrothermal synthesis is typically performed in an oven (below 200 °C) or furnace (above 200 °C), where stirring is not possible. As a result, a stirring-assisted reactor must be custom-built to perform reactions at low temperatures, specifically 250-300 °C. The idea behind this is that these products are mostly diffusion-limited, so to increase this, we need to create extra turbulence in the reaction environment, which results in an improvement of the reaction conditions such that the product that takes 1-2 weeks now takes 1-2 days. Based on the observations, an internal mechanism (shear jamming) of product formation has been hypothesised. We also got truncated cubes, nanorods, and hollow cubes of the same (LKMO) materials, also some double perovskites.

3.2 Introduction

Previously, the effect of stirring was studied in organic and some inorganic reactions. However, no conclusive evidence exists to show how stirring affects the rate of the reaction. The role of stirring in materials synthesis has not been explored earlier. It is unclear in the literature whether stirring aids in nucleation. There are few literature reports on the synthesis of materials using the dynamic hydrothermal method. Stirring ensures uniform mixing and better control over the final product (**Figure 3.1**). Vediappan K. et al. have studied the effect of stirrer rotation frequency on the synthesis of nano C-LiFePO₄.¹⁴¹ The frequency of agitation dictated particle size distribution, particle agglomeration, self-assembly, morphology, and crystallinity of the products. According to their findings, the rotation frequency has an effect on the nucleation-growth kinetics. The existing material synthesis methods are divided into two broad categories: solid-state and solution-state (hydrothermal). The solid-state method produces small spherical primary particles, while the hydrothermal method produces primary

particles of various shapes and sizes. The addition of stirring to these methods will improve control over the size and morphologies of the primary particles. These primary particles also influence the formation of secondary particles, which determines the material properties. Sallem F. et al. 2017's report demonstrates how the mechanical stirring reduces the required reaction time and evolves the morphology of titanate nanotubes into nanoribbons and multiwall TNTs.¹⁴²

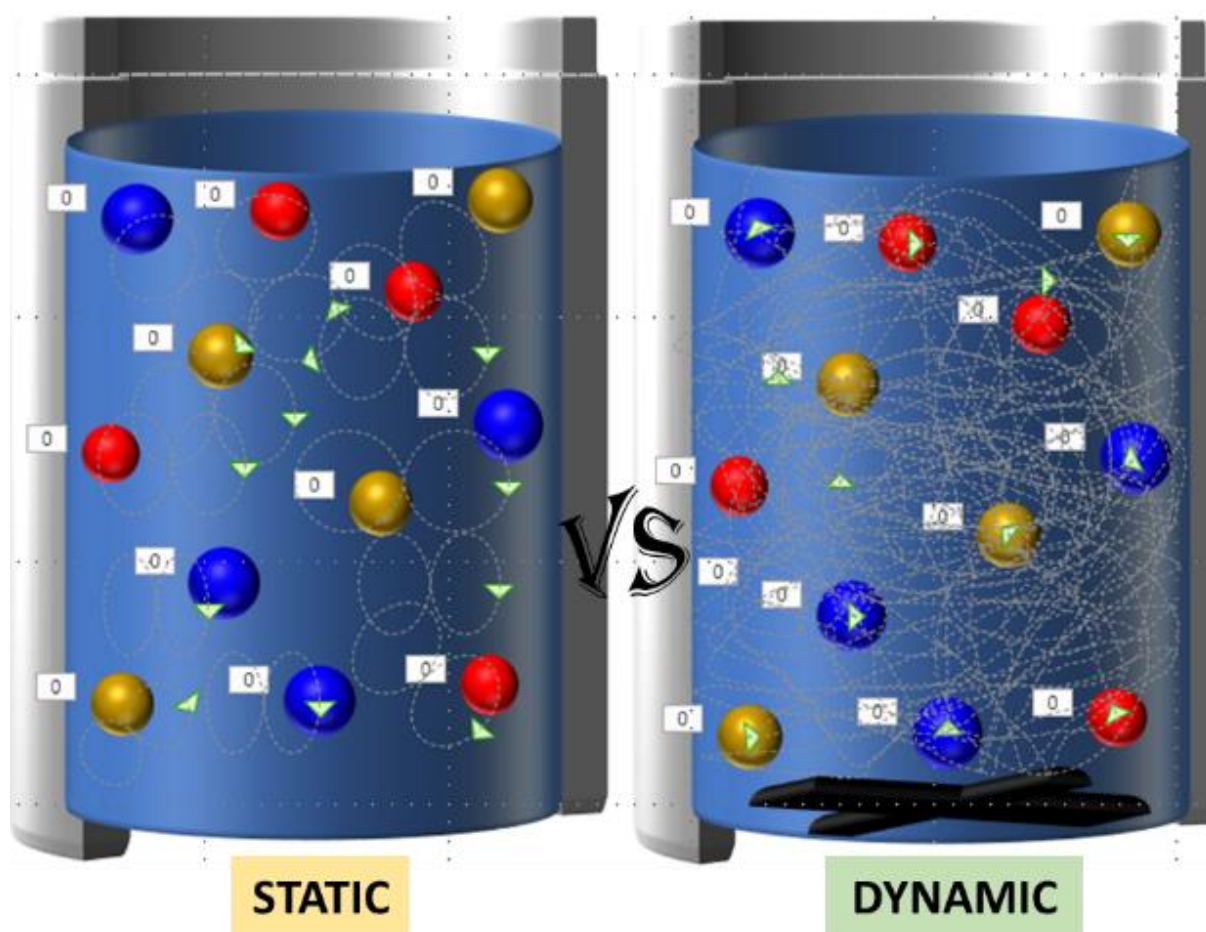


Figure 3.1 Schematic representation; cross-sectional view of static vs dynamic hydrothermal autoclave where red, blue, and gold coloured spheres are reagent ions, magnetic bead is shown in black color. The grey lines indicate that the ionic movement is very much localized for static case. Whereas it is comprehensive for dynamic case.

This demonstrates that the stirring increased the contact area between the particles and the unreacted surrounding solution, thereby increasing the kinetics (**Figure 3.1**). This same group investigated the effect of stirring type, stirring rate, and stirring time variation on the morphology of the resulting titanate nanotubes. Papa A. et al. investigated the effect of stirring

mode on the yield and the crystallinity of the P25 nanotubes.¹⁴³ According to Gao J. et al. (2015), continuous stirring in the dynamic hydrothermal method promotes surface recrystallization and the formation of well-dispersed hollow BaTiO₃ nanoparticles.¹⁴⁴ According to them, stirring appears to be a vital parameter that is rarely discussed in the literature.

Based on this understanding of the literature, the stirring assisted hydrothermal or dynamic hydrothermal synthesis method could open up new possibilities. Among many modern approaches, the addition of sonication or microwave treatment in conjunction with the classical static hydrothermal (SH) suffers from a lack of control over the morphology. These add-ons are extremely energetic and are only available on small scales. Long reaction times, harsh alkaline conditions, and low nucleation efficiencies characterise solution phase chemistry. As a result, the addition of stirring to the SH is the most significant advance that this field can anticipate. Theoretically, stirring can accelerate nucleation, but this has only been demonstrated in bulk systems such as polystyrene beads or a colloidal system. However, we are demonstrating for the first time in a complex LPO. It is proposed that shear-induced jamming can improve the nucleation process; otherwise, it is a very high energy penalty step/process, particularly when lanthanides are involved.

So, in this chapter, we study the stirring assisted solution-based synthesis of various lanthanum-based transition metal perovskites at low temperatures. It was explored how stirring can enhance nucleation and growth compared to traditional hydrothermal synthesis. We chose our LKMO material synthesis as a model for this purpose. We achieved 90% completion with micron-sized LKMO crystals in two days, whereas this normally takes weeks via SH due to the slow dissolution of byproducts (La(OH)₃). Stirring (shear) appears to influence not only the rate of dissolution but also how La(OH)₃ dissolves, i.e. which planes dissolve first or rapidly. Without stirring, dissolution of these rods occurs in the [0 0 1] direction, resulting in rod shortening. Stirring appears to have little effect on length, but it does change diameter, implying that dissolution occurs via the (110) plane.

The same DH method was used to obtain LaCrO₃ and LaFeO₃ perovskites in significantly shorter duration. Along with the above-mentioned normal perovskites, we were also succeeded in synthesizing La_{1-x}K_xMnBO₆ (B = Cr, Fe, and Ni) like double perovskites. We demonstrated the synthesis of {1 1 1} truncated LKMO microcubes and hollow microcubes, demonstrating the influence of stirring on the morphology of the material. Stirring has a significant impact on dimensionality, allowing us to obtain nanorods of LKMO perovskite using the DH method. Stirring-assisted synthesis, we believe, opens up a wide range of possibilities for creating

variety of complex systems at low temperatures and exerting control over their nucleation and morphologies, and thus the resulting properties.

3.3 Existing issues

The static hydrothermal method has no control over any other parameters besides the external temperature. This includes the inability to improve the reaction's dynamics. The science of the dynamic hydrothermal method is poorly understood.

3.4 Objectives & Action plan

1. Obtain perovskite phases of lanthanum chromite, manganite, and ferrite.
2. Understand the influence of stirring on nucleation by observing the resulting phase and morphology.
3. Obtain double perovskite phases of manganite with other transition metals.
4. Tune the morphology to obtain single-crystal nanorods of these materials.
 - i. To accomplish this, we must optimise the mineralizer (KOH) concentration.
 - ii. The concentration and ratio of the precursor must be optimized.
 - iii. Reaction time, temperature, and knowledge of stirring and sonication times are required.
5. Use these nanorods to fabricate two and four probe devices and characterize their electrical properties.

3.5 Materials and Methods

3.5.1 Materials

3.5.1.1 Reagents

All the metal salts involved, viz. $\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$, $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$, $\text{Cr}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, $\text{Fe}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$, $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, and $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ were purchased from Merck (AR 99%) and employed as such in the experiments performed. Potassium salts KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$ were purchased from Merck (AR 99%) and used as such. KOH was purchased from Merck (AR 99%) and used as such. For the post-reaction workup, ethanol (Emparta, Merck, ACS) and Milli-Q water were used. No further purification was carried out for any chemical.

Hygroscopic materials ($\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$) and pyrophoric chemicals (KMnO_4) were carefully handled while weighing, ensuring minimum exposure to the atmosphere. All the hydrothermal reactions referred to in this work were performed in a 25 ml Teflon autoclave container.

3.5.1.2 Equipment

Figure 3.2 is showing a custom-made dynamic hydrothermal reactor. It contains;

- Heating chamber (aluminium block) with surrounding glass wool
- Heating controlling unit (PID, heating indicator)
- Two electric heating rods (10 mm diameter, 65 mm length)
- Thermocouple (J-type (for low temperature accuracy))
- Magnetic stirrer
- Rotation controller
- A couple of switches (main and heating)

We used one of the 25ml capacity autoclave as mold to make the base at the centre in an aluminium block. At the back side two holes for the heating rods and one hole for thermocouple made. The entire unit is based and surrounded with glass wool for insulation. Single step PID controller with heating indicator is used to set the program and observe the operation. The heating chamber is situation on top of the heating controller unit in between the magnetic stirrer is present. On the left side of the heating controller a knob to control the stirring rate is present. A Teflon coated magnetic bead stirs the reaction mixture from inside.



Figure 3.2 The setup for the dynamic hydrothermal synthesis of LKMO microcubes

3.5.2 Synthesis

3.5.2.1 Preparation of LKMO microcubes, truncated microcubes and nanorods

In **Chapter 2**, the best optimised K18R11P5T7 procedure for the synthesis of LKMO microcube sample was chosen. In a typical synthesis, $\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$, $\text{MnCl}_2 \cdot \text{H}_2\text{O}$, and KMnO_4 are taken into a beaker containing 2-5 ml of Milli Q water in a molar ratio of 1:0.7:0.3 (5 millimoles). These reagents are dissolved in water by stirring until the formation of a homogenous solution. The above solution is transferred into a 25 ml Teflon-based autoclave. Subsequently, KOH is added into the reaction mixture such that the resultant final concentration of KOH becomes 18 M and is allowed to undergo stirring for 10-20 mins. The dynamic hydrothermal reaction is performed at 800 to 1100 rpm by heating the autoclave at 265 °C under autogenous pressure for different periods of time ranging from 6 hours to 2 days. After completion of the reaction, the supernatant is decanted, and the resultant solid precipitate is washed thoroughly with MilliQ water and ethanol until the pH is found neutral. The product is dried 70 °C for 12h.

High-indexed {111} truncated microcubes of the LKMO phase were obtained by adjusting the precursor concentration to 2.5 mM while keeping the other reaction parameters constant. Nanorods of the LKMO phase were obtained by adjusting the KOH concentration to 12 M while keeping the other reaction parameters constant.

3.5.2.2 Preparation of LaCrO_3 and LaFeO_3 microcubes

To synthesis of LCrO and LFO in DH the procedures optimized in **Chapter 2** for the respective phases (Section 2.5.2.3 and 2.5.2.4) were chosen. The prepared mixtures were hydrothermally treated at 265 °C for 2-3 days under 800rpm stirring. After completion of the reaction, the supernatant is decanted, and the resultant solid precipitate is washed thoroughly with MilliQ water and ethanol until the pH is found neutral. The product is dried at 70 °C for 12h. The reaction yields pure phase of LCrO and LFO microcubes.

3.5.2.3 Synthesis of $\text{La}_{2-x}\text{K}_x\text{MnBO}_6$ (B = Cr, Fe, and Ni) double-perovskite microcubes

Double-perovskites with the general formula $\text{La}_{1-x}\text{K}_x\text{MnBO}_6$ (B = Cr, Fe, and Ni) were synthesised in a single direct step. A 25ml Teflon container was taken with an 80 percent filling capacity. Into this $\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ salt with $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}:\text{KMnO}_4$ (0.7:0.3) salt and

($\text{Cr}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, or $\text{Fe}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$, or $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$) were added in a molar ratio La:Mn:B to be 2:1:1. Then 2 to 3 ml MilliQ was added and thoroughly mixed the precursor salts for 15-20 minutes, until a homogeneous solution was obtained. Finally, the calculated amount of KOH was added to achieve the necessary 18 M solution. This allowed for 10 more mins to be stirred before being hydrothermally treated at 265 °C for 2-3 days at 800rpm stirring. After the reaction is finished, the supernatant is decanted, and the solid precipitate is thoroughly washed with MilliQ water and ethanol until the pH is neutral. The product is dried for 12 hours at 70 °C.

3.5.2.4 Synthesis of hollow cubes of LMKO

Purely washed 30 to 50 mg of pre-synthesized LKMO microcubes were placed in a 25 ml Teflon container with 2 to 3 ml MilliQ and sonicated for 10 to 15 minutes to ensure good dispersion. Then, with constant stirring, NaOH was added until the final concentration reached 18 M. This was allowed to stir for one whole night and then hydrothermally treated at 265 °C 800 rpm for one 1day. After decanting the clear upper solution, the final compound appeared brown to black. SEM and EDS were used to characterise this.

3.6 Results and Discussion

The synthesis of high activation energy demanding LPO like robust materials poses risk financially and facility wise. Generally, hydrothermal synthesis is known to be the solo existing solution-based technique where one can achieve the required energetics as well as the flexibility to tune the features of the product. Combination of temperature with internal pressure is the key to it. But to synthesize LKMO in SH, 270 °C temperature with 80% filling factor wasn't sufficient. When 18 M KOH was used in combination with above temperature and pressure then only we could get LKMO phase as seen in the previous **Chapter 2**. Because, KOH is a mineralizer and it decreases the huge activation energy required to obtain LPOs. But Still, the reaction was taking more than 5 to 7 days of time.

Apart from the temperature and mineralizer concentration none other parameters could exert any control over the reaction thermodynamics in case of SH method. The understanding of nucleation mechanism and thus the limiting step (dissolution of the byproducts) demands accommodation of an external flexible reaction parameter. Among the existing sonication or microwave assisted modern hydrothermal, which brings tremendous amount of uncontrolled

energy, agitation/stirring could be the cheapest, easiest and more controlled parameter. Fusion of stirring with SH can be called as dynamic hydrothermal (DH). Popularly top assisted external stirring reactors are known but these are very costly and regular maintenance is required because of vacuum locks. When it comes to the regular hydrothermal reaction it is done in the oven (below 200 °C) or furnace (above 200 °C) where one cannot do stirring. Thus, we have designed our own dynamic hydrothermal reactor which is shown in the **Figure 3.2** with the help of Priya enterprises. With this one can conduct DH to do reactions at low temperatures, especially at 250-300 °C. The synthesis conditions that one can achieve are ranging up to 300 °C temperature and upto 1500 rpm stirring. As per the gained understating LKMO is mainly diffusion-limited, so to enhance we need to create extra turbulence in the reaction environment. Also, to understand the effect of shear during the reaction synthesis of LKMO phase is used as model example.

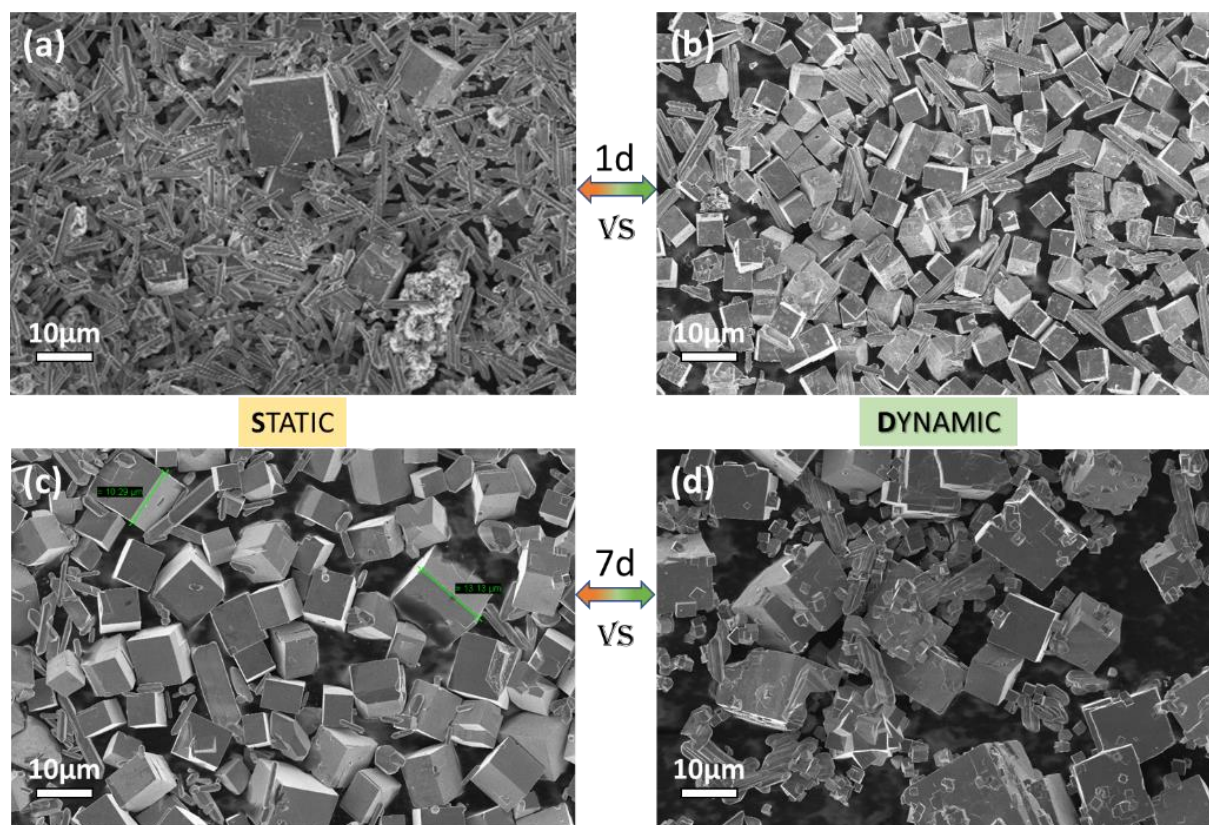


Figure 3.3 Comparison of SEM micrographs of LKMO microcubes obtained by static and dynamic hydrothermal at 1 and 7 days.

Under the same conditions of precursors (5 mM), KOH (18 M), ratio (1:1), and temperature (265 °C), shear (850 rpm) is applied to understand how it affects the formation of the LKMO phase. This delivered assorted results. First of all, the kinetics have increased so vastly that

within 6 hours of time LKMO microcubes forms (which took minimum 1 day to observe without stirring the reaction) (Check the **TOC** of this chapter). and the reaction which usually takes 1 to 2 weeks is completing in 1 to 2 days. From the above **Figure 3.3** we can say high yield at lesser time and size reduction of LKMO is easily possible by DH.

Except the uniformity of the reagents people really didn't explore much about the effect of stirring on the product formation. Several observations are made under shear-induced conditions. Under stirring, there is formation of grooves on $\text{La}(\text{OH})_3$ rods, which (might) suggest the dissolution of $\text{La}(\text{OH})_3$ is high (**Figure 3.4(b)**). In without stirring reaction conditions, the size of $\text{La}(\text{OH})_3$ rods is reduced (dissolving 001 planes first), while under stirring, 110 planes of $\text{La}(\text{OH})_3$ rods are dissolved. This could be due to the formation of micro/nano wavy patterns in the solution during the shear. As the relative motion of solution with respect to $\text{La}(\text{OH})_3$, 3 ribbons are different, which could affect $\text{La}(\text{OH})_3$ ribbons such that where $\text{La}(\text{OH})_3$ ribbons faces/interacts with the crest portions of sheared produced waves, dissolve fast.

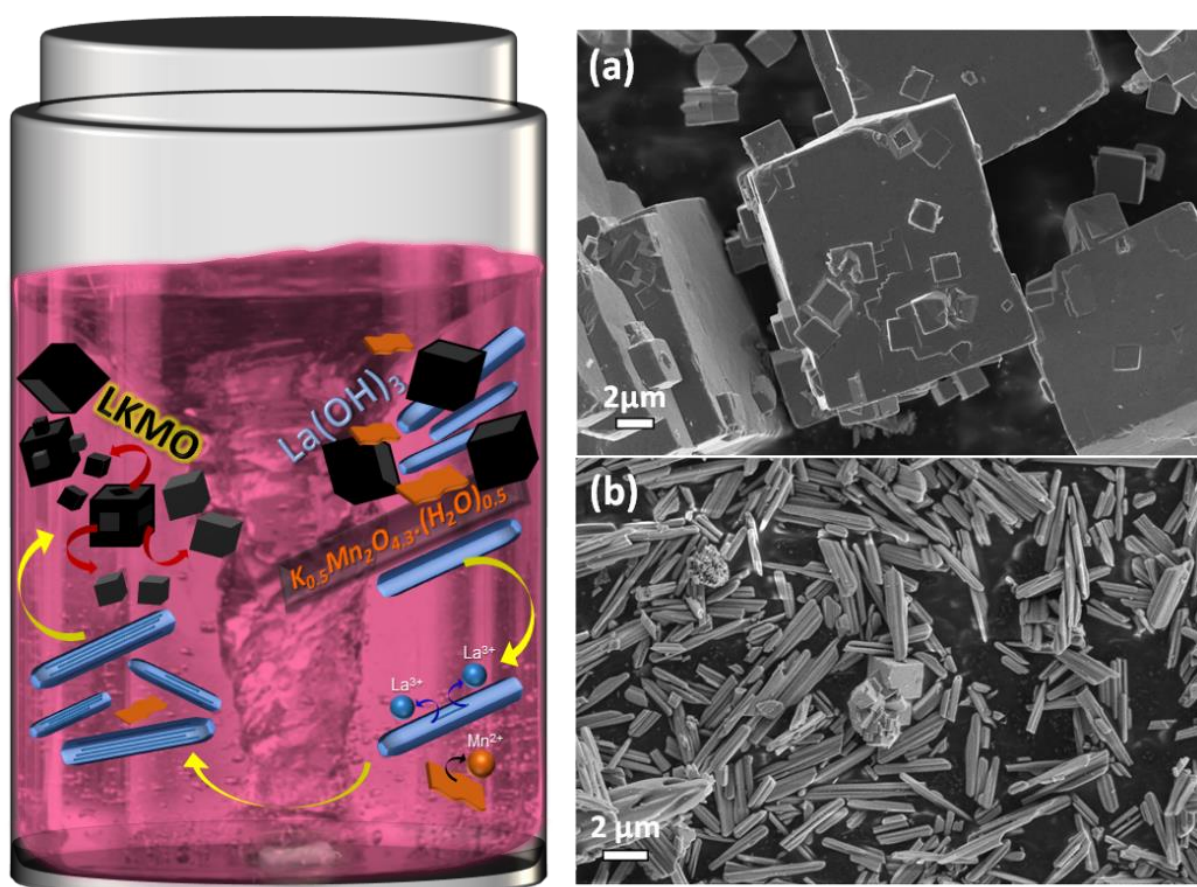


Figure 3.4 Schematic representation of formation of LKMO in DH; SEM micrograph showing formation of (a) smaller cubes from bigger cubes, (b) dissolution of $\text{La}(\text{OH})_3$ rods forming grooves.

In the initial times (6 hrs) uniform sized cubes ($\approx <7>$ microns) (single-step nucleation) formed and after a prolonged time size distribution from 300 nm to 10 microns observed (**Figure 3.4(a)**). This could be due to, under stirring conditions, small cubes are formed on top of big cubes which are sheared off from the big cubes during the reaction. This is quite possible because stirring breaks the gold nanorods into particles.¹⁴⁵ This reduces the probability of multi-step nucleation (secondary nucleation) rather enhances the occurrence of heterogeneous nucleation. In other words, reaction shifts from multi-step nucleation to heterogeneous nucleation, which is energetically more favourable. Big cubes have rough surfaces which could serve as nucleation sites for hetero-generous nucleation. SEM images show marks in big cubes after removing small cubes from them.

As per Fletcher et. al. a crystal formation passes through four stages.¹⁴⁶ These stages are shown in **Figure 3.5** left hand side. The transition of these stages there occur two rate limiting steps. First one is the diffusion process which happens at the beginning where the dissolved ions starts to aggregate into clusters. The second one is the organization of prenucleating clusters and embryos which is the formation of well-defined lattice. Teng et. al. published different internal ionic arrangements during the crystal nucleation. Here the stability of each arrangement is defined by the ratio of surface energy gain to the solution chemical potential. Once an arrangement crosses the critical nucleus size then it becomes stable and then growth commences (**Figure 3.5**).

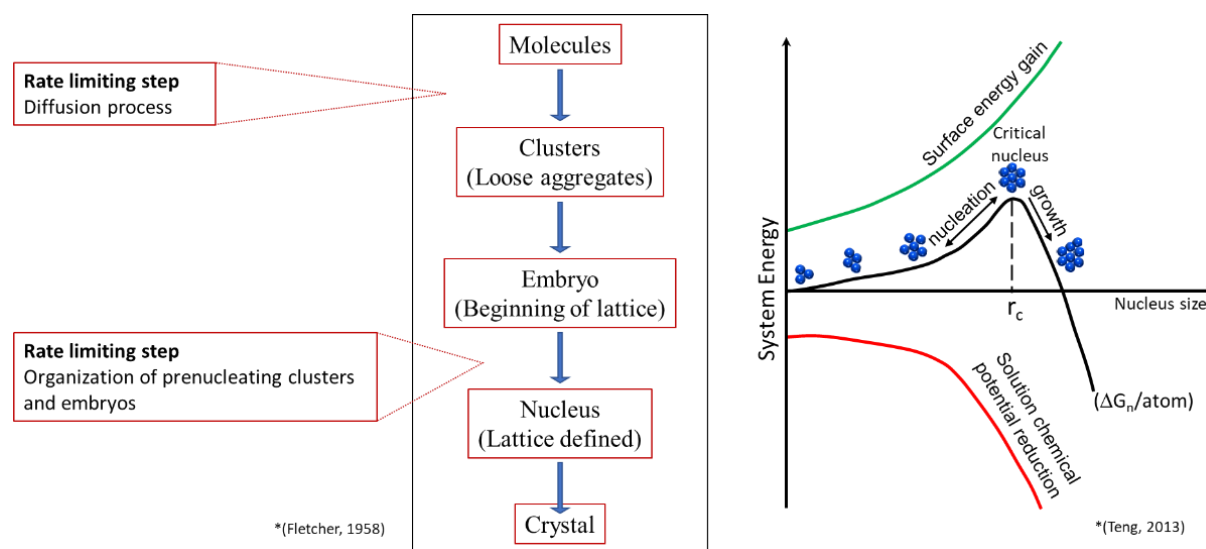


Figure 3.5 (Left side) Different stages of crystal formation¹⁴⁶ and Internal organisation of ions in crystal nucleation¹⁴⁷ (right side)

La^{3+} is known to form the hydrodynamic sphere with the hydroxide and the water. The entire diameter of which is of the order of some nanometres (**Figure 3.6**). This is true when La dissolves into the solution. In case of SH there is enough Mn^{x+} ion concentration and poor La^{3+} ion concentration in the solution, at the prenucleation stage.

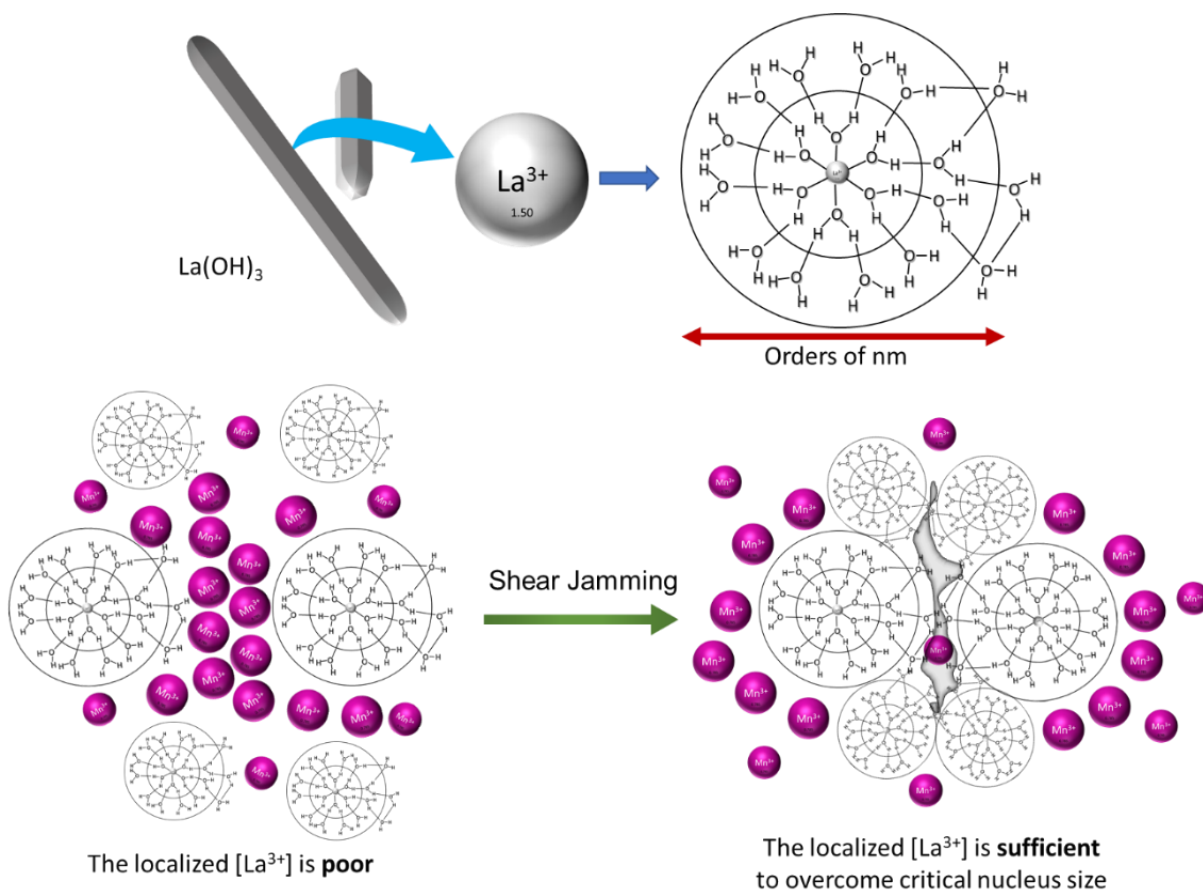


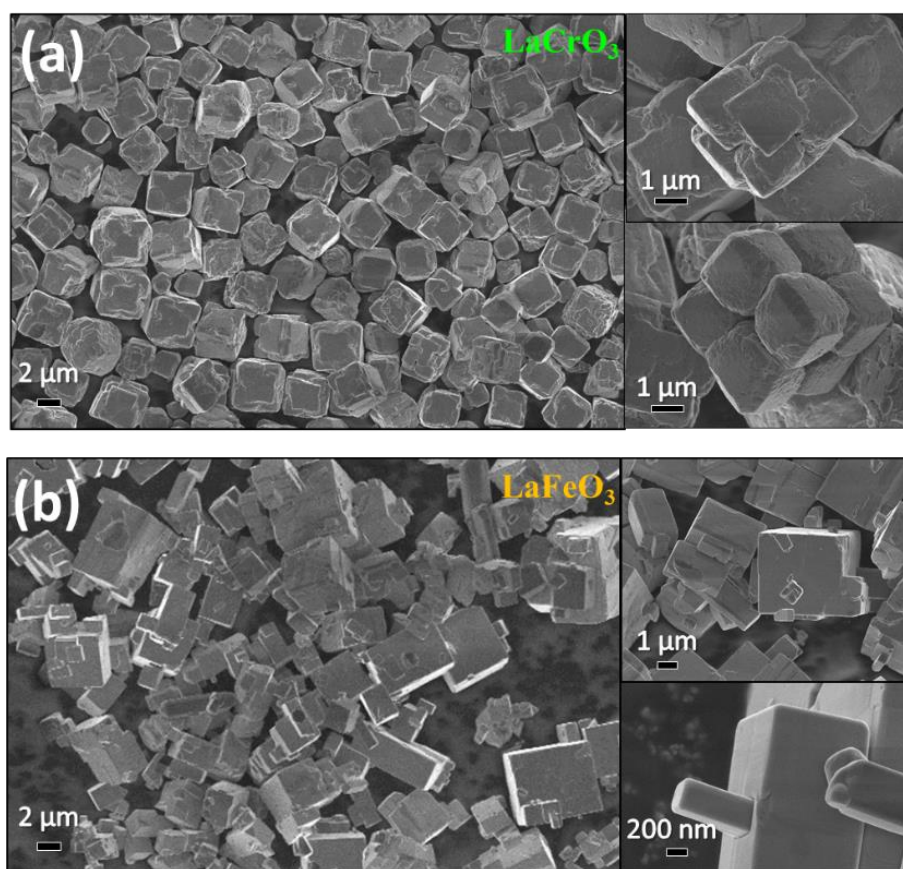
Figure 3.6 Schematic representation of the shear jamming situation proposed in DH.

The reaction kinetics can be improved by increasing the temperature or also can be increased by adding the extra precursor. But even then, locally La^{3+} ion concentration is low due to the formation of $\text{La}(\text{OH})_3$ rods (discussed in **section 2.7**). In this case, two La^{3+} ions with its hydrodynamic spheres are physically separated by the Mn ions and the probability of having these two near is very less. So, whenever there is a nucleation, it happens locally and the nuclei size is extremely small. It might not cross the critical nuclei size thus it dissolves back as ions to the solution.

In DH method the stirring can instigate a situation called shear jamming. This concept of particle shear induced jamming is a consequence of dynamical structural transition.^{148,149} If it is happening for the 100nm particles still it can happen for 1 to 2 nm particle. There is no

logic/reason why it can't happen in LKMO case. So, now if this kind of jamming materializes at some places, might not be everywhere; wherever it happens the amount of La^{3+} ions becomes high at that place (as shown in **Figure 3.6**). If this happens the formed nuclei can quickly cross the critical nuclei size after which it won't dissolve back. This nucleus is extremely energetic thus it rapidly consumes the rest of the available La and Mn ions in the immediately vicinity and grows bigger. Because of this situation there won't be any further saturation to allow subsequent nuclei to form. For second nucleation once again, shear jamming has to facilitate.

This is the proposal for the phenomena taking place at the atomic scale as per our observation, which explains the improvement of the reaction rate of the synthesis of LKMO microcubes. The same DH method used to synthesize LaCrO_3 and LaFeO_3 LPOs. **Figure 3.7** represents the morphology and phase of the same. Here also DH shown superior results compared to SH. The obtained XRD patterns indicate formation of pure phase without any secondary or byproducts in both the cases. Thus, it clarifies that the stirring brings complete homogeneity of solution also enhances the reaction rate as within 2 days both the reactions reached their completion. These usually takes 5 to 7 days in case of SH method.



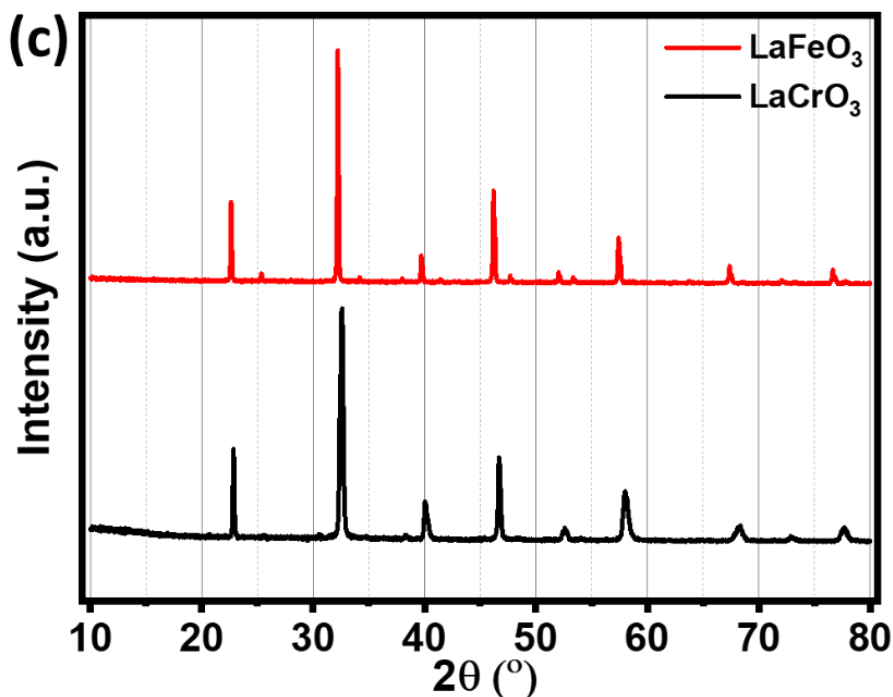


Figure 3.7 SEM images of (a) LaCrO_3 and (b) LaFeO_3 sample synthesized by dynamic hydrothermal method and (c) corresponding XRD patterns.

Hollow cubes

The synthesized LKMO microcubes, when subjected to concentrated NaOH or KOH, resulted in internal corrosion/dissolution leading to hollow cubes. This explains the LKMO cubes has the tendency to dissolve back and form the respective byproducts. This formation of hollow cubes or dissolution of the LKMO microcubes shows DH is better than SH. Because, in the case of SH the precursor concentration is not uniformly distributed and thus there will be a place where the formed cubes will dissolve back where as in DH case we have more uniformity and thus the cubes stay stable.

By looking at the SEM images (**Figure 3.8**), the extent of the corrosive nature of the base seems very deep. Compared to the surface, edges and corners of the cubes are more prone to corrosion. Thus, these works as the active centres through which the base starts to dissolve the Mn and La from the cube and corrodes inside. The EDS spectrum and mapping shows that the perovskite phase at the shell level still maintained.

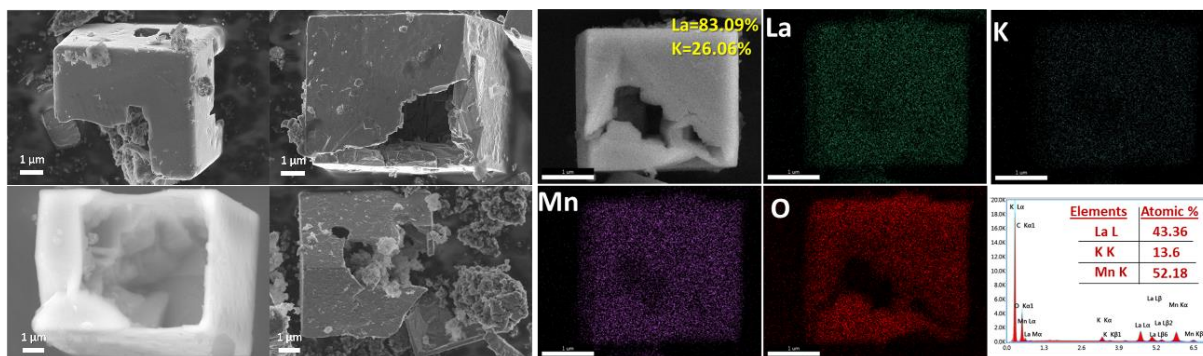


Figure 3.8 SEM micrographs and EDS spectrum-mapping of LKMO hollow cubes.

Facilitate in catalysis

Catalysis is very specific to the surface of the catalyst material. The atomic arrangements in each facet are exclusive thus different facets hold specific energetics and thus catalytic activity. When LKMO is prepared in DH at reduced precursor concentration (< 2.5 mM) resulted high-index $\{1\ 1\ 1\}$ faceted/truncated microcubes (**Figure 3.9**). This shows the capability to alter the morphology thus expose different facets of LPOs which was quite challenging to achieve in SH method.

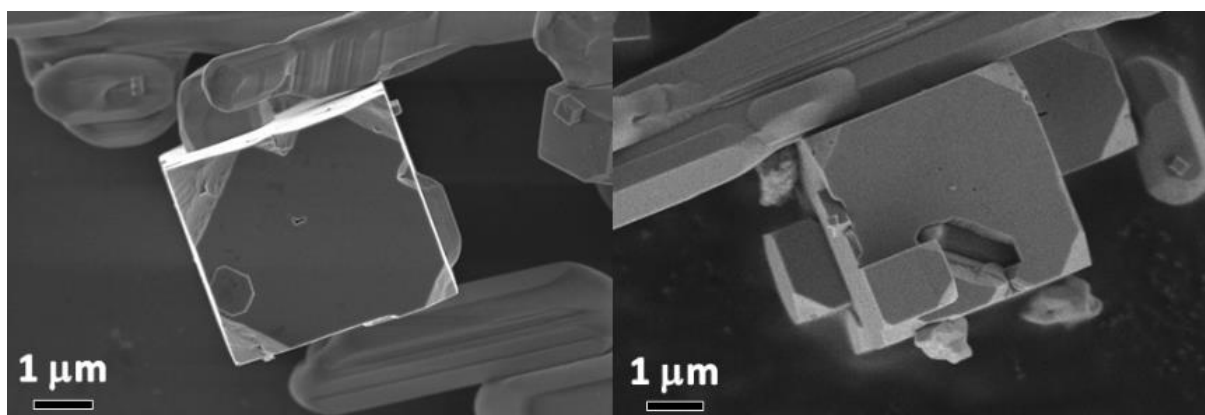


Figure 3.9 SEM images of high-index $\{111\}$ truncated LKMO microcubes obtained by DH method.

Catalytic activity in LPO systems usually dictated by the B-site atom. The covalency of Mn-O bonds plays significant role in designing catalyst. A-site substitution of K can modify the valence state of the B-site Mn and thus the filling of the d sates involved. As upper edge of the valency band and lower edge of the conduction band are defined by the oxygen 2p states and transition metal 3d states. Thus, the electronic properties of Mn-O bonds define the activity and stability of the catalyst. Having established DH as a better alternative to the more conventional SH method we further explored its potential in synthesis a series of La₂-

$x\text{K}_x\text{MnBO}_6$ ($B = \text{Cr}, \text{Fe}, \text{and Ni}$) double perovskites. Because of the presence of second transition metal with different 3d state filling can work in conjunction with the Mn resulting enhanced catalytic activity. If we look at the SEM images in **Figure 3.10a** of $\text{La}_{2-x}\text{K}_x\text{MnNiO}_6$ (LKMNO) we see truncated microcubes similar to the above truncated LKMO microcubes. Elemental distribution and composition can be found from the **Figure 3.10c and d** EDS mapping and spectrum. The relatively equal atomic percentage presence of both the Mn and Ni can be clearly seen here indicating the double perovskite. Here Mn brings the potential of oxygen reduction while Ni is known for oxygen evolution thus making the LKMNO to be a bifunctional electrocatalyst.

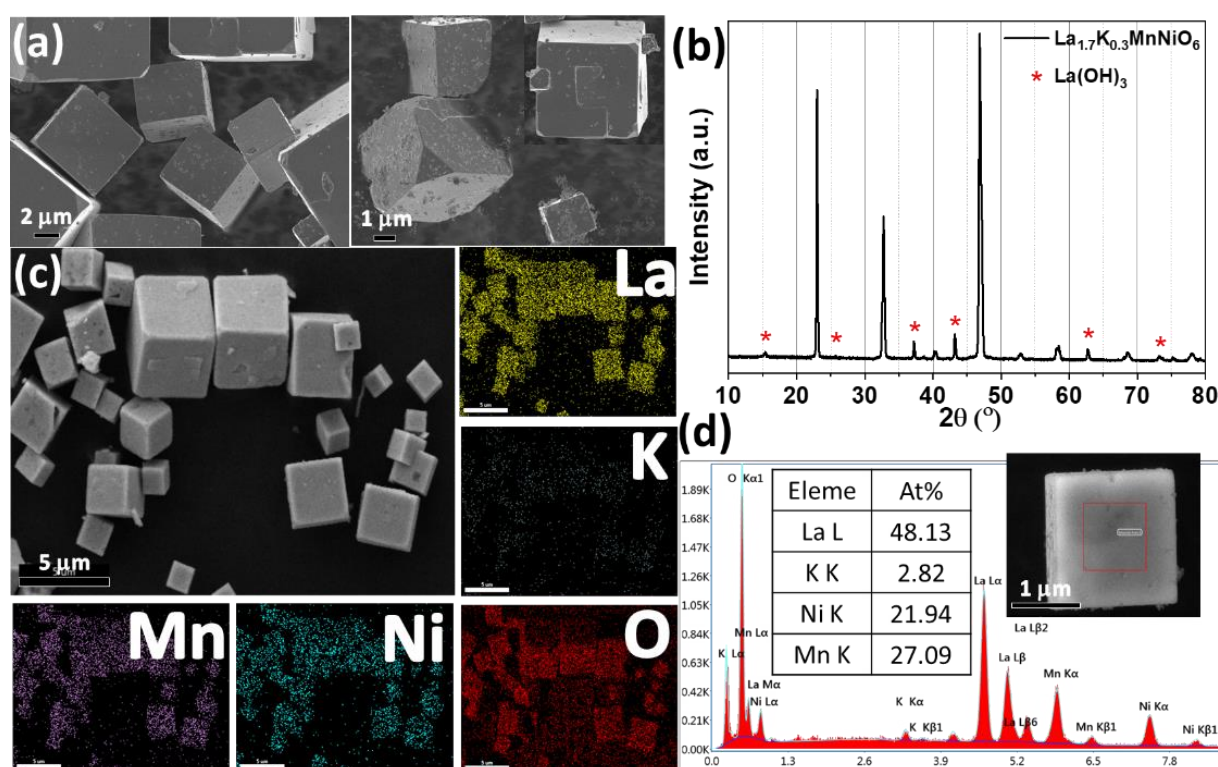


Figure 3.10 (a) SEM micrographs indicating the {111} faceted LKMNO microcubes, (b) XRD pattern, (c) EDS mapping, and (d) EDS spectrum of LKMNO.

Similarly, one can obtain $\text{La}_{2-x}\text{K}_x\text{MnCrO}_6$ and $\text{La}_{2-x}\text{K}_x\text{MnFeO}_6$ double perovskites as shown in below **Figure 3.11(a and b)**. Both the cases uniform distribution of both the transition metals on the cubes can be seen.

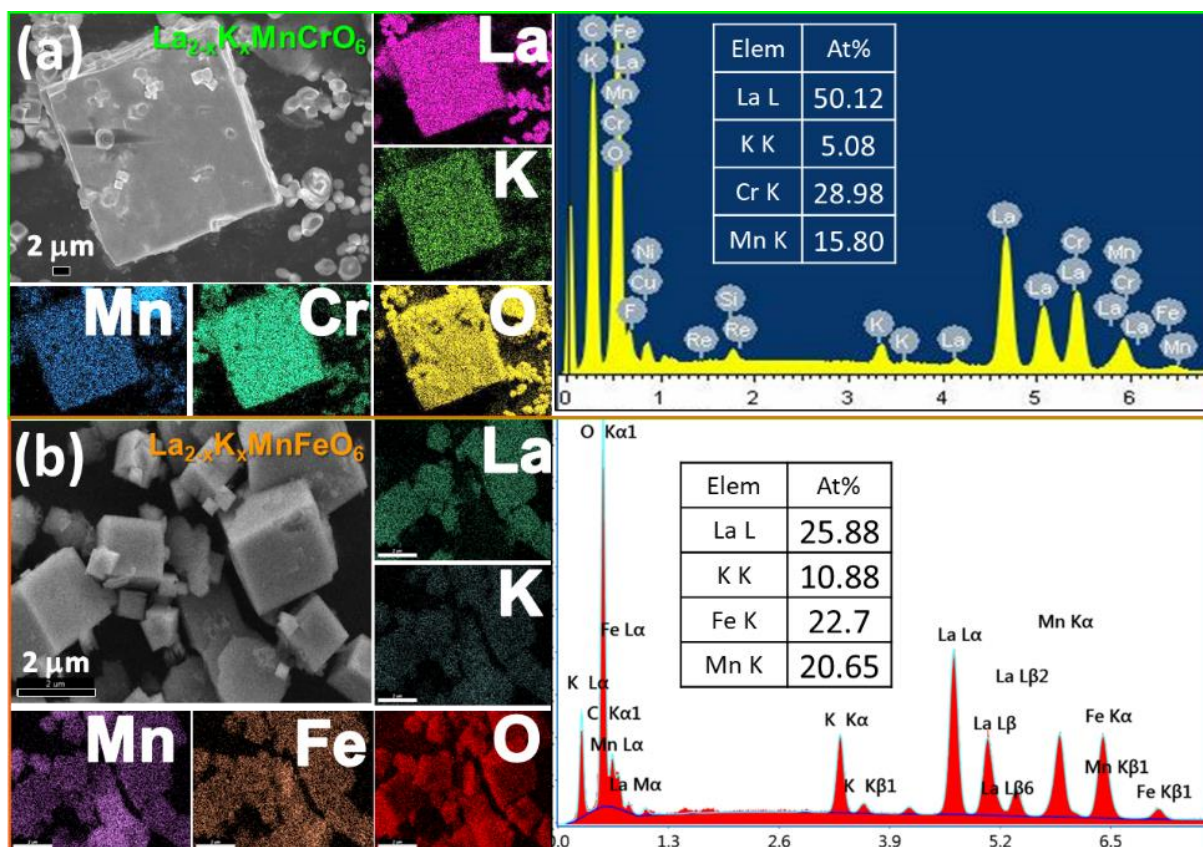


Figure 3.11 EDS mapping of (a) LKMCrO and (b) LKMFO double perovskites.

Electrical characterization of LKMO nanorods

The reaction environment can bring serious changes to the resulting product morphologies. Lowering KOH (12 M) concentration can bring one-dimensional structures compared instead of cubes when crushed LKMO microcubes used as seeds in DH. The careful observation to the electron microscope images indicates the growth of these nanorods is screw dislocation assisted growth which is one of the prevalent growth mechanisms (**Figure 3.12**). In this process microns of length LKMO nanorods obtained within 2 days of time. By closely looking at the SEM images in **Figure 3.12** infers every nanorod is growing from a core and they themselves start to protrude new nanorods. This might be because of the presence of topological defects which are acting as active centres.

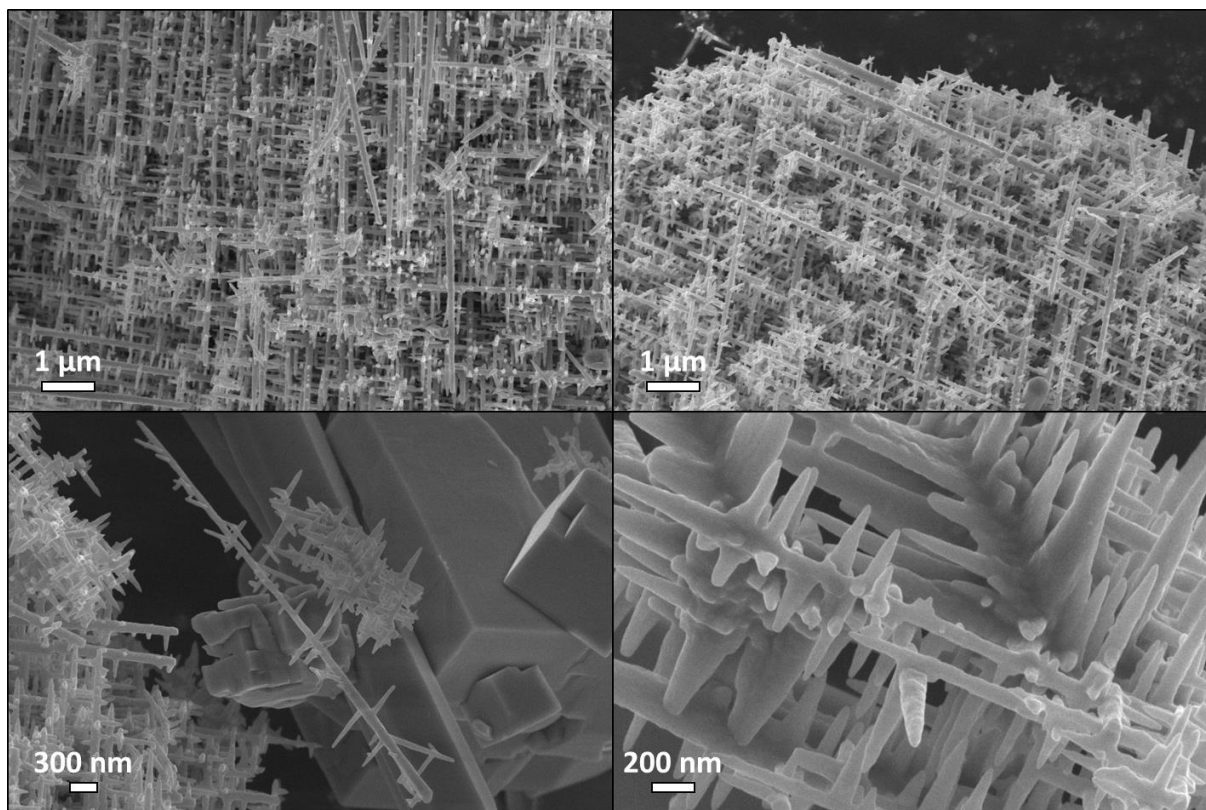


Figure 3.12 SEM micrographs of LKMO nanorods obtained in DH at 12 M KOH.

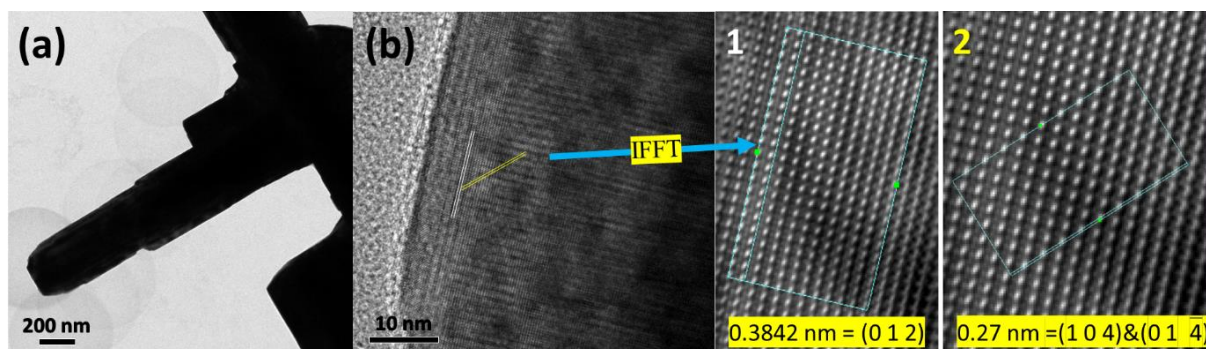


Figure 3.13 (a) TEM image of LKMO nanorod, (b) HRTEM image showing the lattice fringes where 1 and 2 are IFFT images showing the lattice d-spacing and corresponding hkl plane.

A detailed TEM characterization has been carried out to understand the composition and structure of these nanorods. The TEM image in **Figure 3.13a** depicts the screw dislocation assisted growth of these nanorods. Analysis of the HRTEM image in **Figure 3.13b** indicates the surface of these nanorods oriented in $[0\ 1\ 2]$ direction. This is clearly measured from the Inverse Fast Fourier Transformation (IFFT) using the Digital Micrograph software. In (1) this the spacing between two bright spots measured (as shown in **Figure 3.13b(1)**) to be 0.38 nm.

This is corresponding with the (0 1 2) plane of LKMO phase. The **Figure 3.13b(2)** measures another d-spacing of 0.27 nm which corresponds with the (0 1 $\bar{4}$) plane of LKMO phase. The Selected Area Electron Diffraction (SAED) analysis in **Figure 3.14a** also shows presence of (0 1 2) and (0 1 $\bar{4}$) along with (0 0 6) and (0 2 $\bar{2}$). More importantly the SAED pattern along with the HRTEM image proves that the LKMO nanorod is single crystalline in nature and it doesn't have any observable internal defects. In addition to the main intense Bragg reflections, which are indexed to a rhombohedral perovskite structure, there are superlattice reflections.

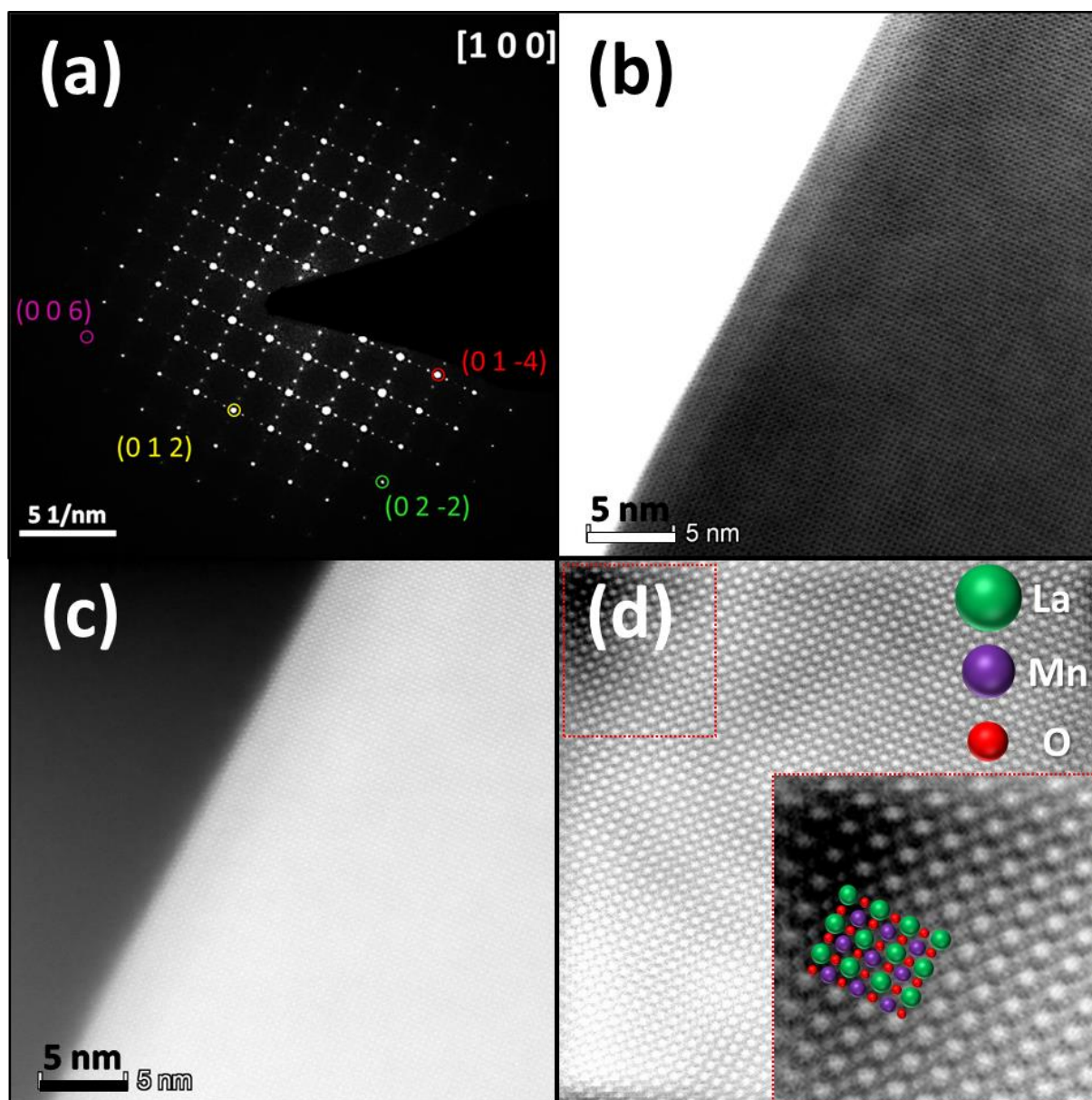


Figure 3.14 (a) Indexed SAED along [1 0 0] zone axis, (b) STEM-BF, (c) STEM-HAADF image of LKMO nanorod, and (d) IFFT of bright field STEM image showing a model overlay of the La and Mn columns of the LKMO nanorod in the magnified red rectangle.

The HAADF-STEM imaging differentiates high atomic number element with brighter contrast because of larger electron scattering angle due to higher electrostatic interactions amongst the nucleus and electron beam. **Figure 3.14(a)** and **(b)** show high-resolution STEM images in both BF and DF, respectively, confirming good uniformity of $\text{La}_{1-x}\text{K}_x\text{MnO}_3$ at the atomic scale. The dark spots in **Figure 3.14(b)** preferably represents the heavy element La column. For a better representation IFFT was collected from the BF-STEM (**Figure 3.14(c)**) as shown in **Figure 3.14(d)**. Here the bigger spots correspond to La-atomic columns. smaller spots correspond to Mn-atomic columns and in between them oxygen atomic-column present, as represented in the model in the inset of **Figure 3.14(d)**. Here green atoms represent La, purple represents Mn, and red is O. There was no clear Z-contrast to distinguish K atomic columns.

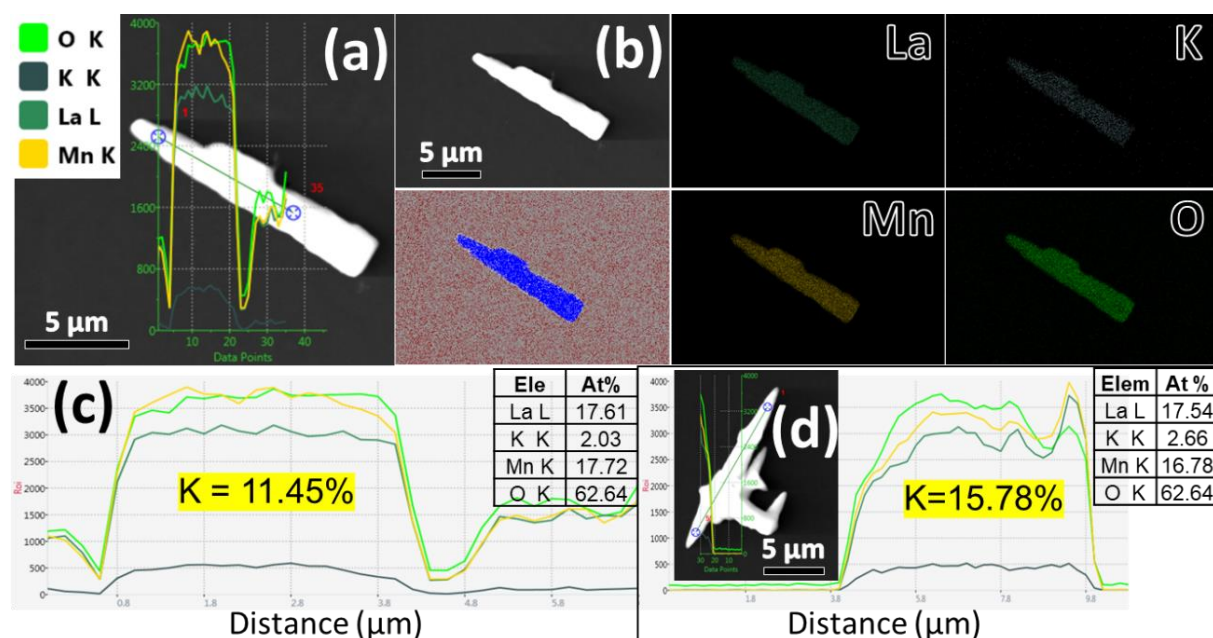


Figure 3.15 (a) STEM image, (b) EDS mapping, (c) Line mapping along the length of LKMO nanorod, and (d) Second line mapping case of LKMO nanorod for K%.

Further, the elemental identification and location are examined from the chemical mapping using EDS in STEM (**Figure 3.15(a)** and **(d)**) which is a powerful technique at atomic resolution. The EDS spectrum indicates the K in the LKMO nanorods is averaged around $\approx 13\%$ (**Figure 3.15(c)** and **(d)**). **Figure 3.15(b)** displays homogeneous distribution of La, K, Mn, and O elements throughout the nanorod. Thus, these LKMO nanorods could show better electrical properties.

Fabrication of two probe devices from LKMO nanorods

To measure the electrical properties of the LKMO single nanorod devices electron beam lithography was used to fabricate the two probe devices on the single LKMO nanorods. At first, Si/SiO₂ substrates of 1 by 1cm were taken and washed thoroughly with acetone-iPa-MilliQ. Then, first-level marker lithography was done on these substrate chips. These were then spin-coated with the IPA dispersed LKMO nanorods sample. This process was optimized to isolate individual LKMO nanorods. Then the location of the LKMO rods was marked. Then based on their location, the second-level mask was written using KLayout and Clewin 4 softwares. To find out the suitable metal contact KPFM was used to calculate the work function which came to be 4.7 eV. Based on this we choose adhesive 2 nm Cr layer on top of the Si/SiO₂, on top of which 80 nm Pt and 100 nm Au was sputter coated. As EL9 and PMMA e-resists were used the metal deposited device was left in acetone for lift-off over the night. The obtained device was etched at it's back to remove the SiO₂ using mild HF-buffer. This allows to apply the gate bias to the formed device.

Two probe measurements were done on these devices by using the instrument called Proxima. Here we have studied the 2-terminal IV, IDVD, and IDVG. We obtained the current, carrier concentration, carrier mobility (μ), carrier type information.

$$\mu = \frac{g_m * L * \ln(\frac{2t}{r})}{2\pi\epsilon_0\epsilon * V_{sd}} \dots \dots \dots 5$$

From the above Eq. (5)¹⁵⁰ and **Figure 3.16**, g_m is transconductance, V_{sd} is source-drain voltage, L (383 nm) is the channel length of the nanorod FET, t (100 nm) thickness of the silicon oxide layer, r (86 nm) the nanorod radius and ϵ (3.9) the dielectric constant of silicon.

IV characteristics obtained from the LKMO nanorod SEM micrograph are shown in **Figure 3.16 (a & b)**. The device has a channel length of 383 nm, and 173 nm is the diameter at the middle of the channel. Despite being thick, the device exhibited inverse-‘S’ such as IV characteristics which are typical of silicon nanowires. Previous studies explained this behavior using either quantum mechanical effects^{152,153} or the non-linearity of the metal silicon contacts¹⁵⁴. Ponomareva et al. have theoretically studied the electronic properties of silicon nanowires. They have found that unsaturated bonds at the silicon nanowire surface play a key role in determining the electrical characteristics of a silicon nanowire.¹⁵⁴

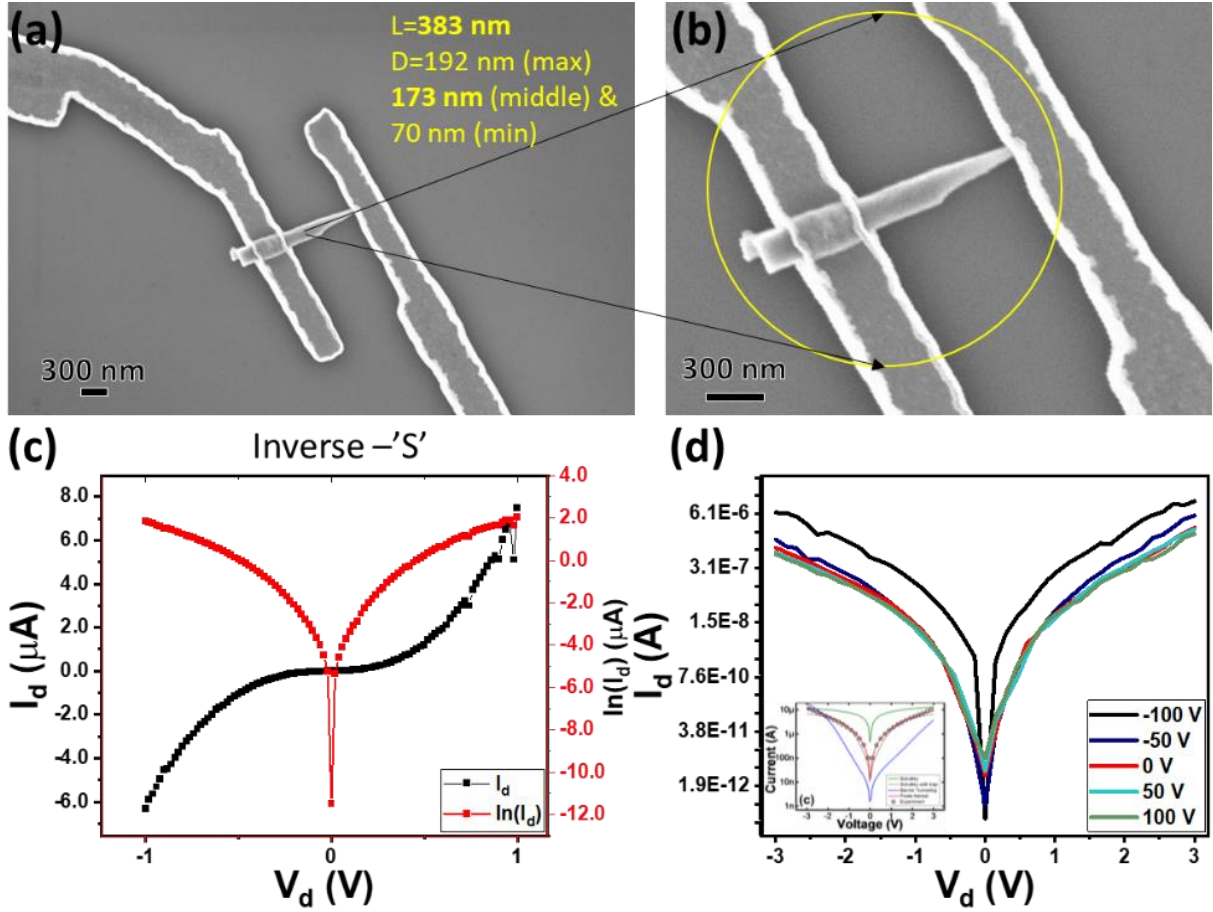


Figure 3.16 (a) SEM image of two probe IV devices of LKMO single nanorod, (b) enlarged representatively, (c) 2-terminal IV or drain current and absolute values, and (d) variation of drain current w.r.t. the drain voltage, inset showing comparison of theoretical and experimental fitting for Schottky model shown by S. Acharya. (2018).¹⁵¹

The electric transport characteristics of LKMO nanorods were carried out as shown in **Figure 3.16(c & d)**. By looking at which the LKMO nanorods are showing p-type behavior. Because in a typical semiconductor such as silicon transistors, the pMOSFETs show conduction in negative gate bias whereas stays off in the positive gate bias.

... .. 6

B is a constant,

E is electric field,

ϕ_t is zero field ionization energy,

k_B is the Boltzmann constant,

T is absolute temperature,

e is the electronic charge,

ϵ_0 is the permittivity of the free space,

K is the optical dielectric constant

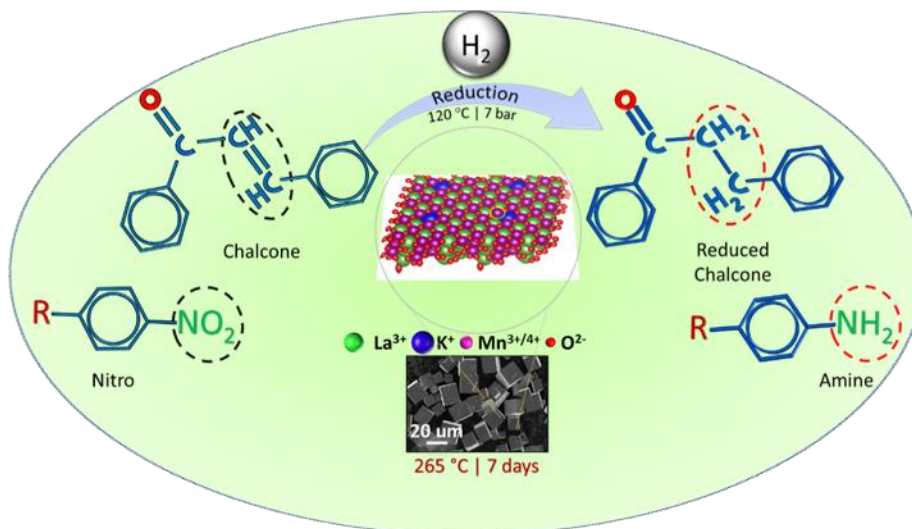
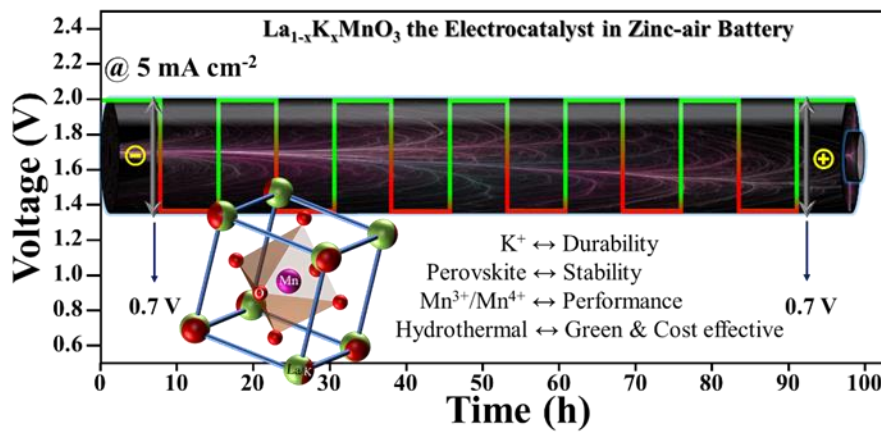
From the literature, the electrical transport properties of La(Fe, Ga)O_3 perovskite oxide nanowires study shows the possible conduction mechanism.¹⁵⁵ Similar to this, our LKMO rods also show inverse-S such as IV characteristics and a slight trend of increase in current with negative gate bias. Thus, we assume that the carrier transport in our nanorod follows the bulk-limited conduction mechanism where electron jumps from trap sites to conduction band that is known as Poole-Frenkel conduction Eq. (6) instead of interface-limited Schottky emission.

3.7 Conclusion

We have improved the kinetics of synthesis of these LPOs by introducing the stirring. We decreased the overall reaction time from 7 days to 2 days. Proposed shear jamming taking place in the reaction under stirring conditions. This phenomenon has been shown to reflect the grooves on the La(OH)_3 rods and enhanced kinetics of LKMO formation. We have obtained nanorods, high-index $\{1\ 1\ 1\}$ faceted microcubes, and hollow cubes of the LKMO sample. These can be used as selective catalysts pertaining to their specific facets' exposure. The DH method can be used to obtain bifunctional catalysts like double perovskites. We fabricated the single rod devices of LKMO nanorods and studied their electrical properties. The results indicate that the nanorod is of p-type material and shows the Poole-Frenkel conduction mechanism, which is due to the cation vacancies, which probably originates due to potassium (cation) doping. In future, one can try to synthesize the nanorods of other materials such as cobaltite, nickelate and aluminate by this method.

CHAPTER-4:

Applications: study of electrocatalytic and organocatalytic activity of LKMO microcubes



Part-1: Importance of K Substitution in LaMnO₃ to Achieve Highly Active and Exceptionally Stable Bifunctional Electrocatalyst Towards Oxygen Reduction and Oxygen Evolution Reactions

4.1 Abstract

Efficient, low-cost, and long-lasting electrocatalysts that accelerate sluggish oxygen evolution reaction (OER) and oxygen reduction reaction (ORR) kinetics are highly desired for a variety of applications, including metal-air batteries and fuel cells, but they are uncommon. Such bifunctional catalysts primarily suffer either from low-catalytic activity or degradation during prolonged cycling. Here, we have used the low-temperature hydrothermally synthesized (as described in **Chapter 2, Section 2.5.2**) potassium substituted LaMnO₃ microcubes as a perovskite oxide-based electrocatalyst. In an alkaline environment, the catalyst performs admirably, with improved kinetics and activity for ORR (half-wave onset potential of 0.78 V) and OER (onset potential at 10 mA cm⁻² current density is of 1.66 V). Furthermore, as a two-electrode, this catalyst renders the potential for use as an electrode in zinc-air batteries with unprecedented stability (no visible sign of degradation even after 1000 cycles of charge-discharge).

4.2 Introduction

Rechargeable zinc-air batteries have drawn much attentiveness in energy conversion and storage applications due to their high theoretical energy densities, eco-friendliness, low cost, and safety.¹⁵⁶ The output performance of zinc-air battery enormously relies on the reaction rates of anodic oxygen evolution reaction (OER) and cathodic oxygen reduction reaction (ORR). However, slower kinetics and complicated multi-electron transfer processes of OER and ORR have posed many scientific challenges due to their resulting higher overpotentials.^{157–160} To date, the conventional electrocatalysts to boost OER or ORR rates are based on precious metals like Pt, Ru, and Ir. However, their utilization in practical applications is critically hindered by poor durability, scarce resource, and prohibitive cost.^{161,162} Thus, the precious metals are being replaced with alternatives with comparable electrocatalytic activity to reduce

the cost. Therefore, developing high-performance and stable electrocatalysts is highly desirable yet remarkably challenging for OER and ORR of zinc-air battery.

Transition metal oxides have been proposed as alternative materials for replacing precious metals. Most high-performing bifunctional electrodes based on metal oxides such as Co₃O₄ and MnO₂ suffer from low conductivity. Thus, adding conducting carbon-based materials is a usual practice. However, carbon gets oxidized during electrochemical cycles and destroys the conductive carbon network.^{163–165} The resulting carbonaceous precipitates clog pores of the electrodes and increase impedance, thus reducing performance and cell failure.^{166,167} In this regard, perovskite oxides have been considered promising efficient bifunctional electrocatalyst candidates due to their high intrinsic activity, chemical and structural stability, and compositional flexibility.^{168–171} In particular, LaMnO₃ (LMO) has shown better catalytic activity and stability compared to many non-noble transition metal-based electrocatalysts, as reported by Hayashi M. et.al.^{172,173} Yabuuchi, N. et al. reported the unusual catalytic activity of LMO due to the presence of ideal d-orbital occupancy and optimum Mn-O bond.¹⁷⁴ Doping +2 cations in perovskite oxides, La_xA_{1-x}MO₃ (M=Co, Ni, Mn; A=Ca, Sr, Ba), are valuable for improving the conductivity, thus eliminating the addition of external conducting agents.^{174–176} However, +2 cation substitution often terminates the surface with the AO layer due to charge neutrality and buries the catalytic MO layer inside the crystal.¹⁷⁷ Thus, the catalyst is not able to perform to its full potential. On the other hand, substitution with +1 cation can solve this problem,¹⁷⁸ but such studies have not been performed yet. Further, incorporating +1 cations inside perovskite shows synthetic challenges via high temperature solid-state synthesis¹³¹ (due to the significant evaporation rate of alkali metals during the reaction), and alternative low-temperature solution-based synthesis is yet to be explored. Thus, establishing low temperature solution-based synthesis, which can offer better control over nucleation and morphology will open plethora of opportunities to study and design the robust electrocatalysts.

Herein, we successfully synthesized catalytically active potassium substituted LaMnO₃ perovskite oxide microcubes via a low-temperature, energy-efficient solution-phase hydrothermal method. The K⁺ substitution increases the average A-site ionic radii, thus decreasing the crystal structure towards a higher symmetry.⁷² Further, the K⁺ substitution also improves exchange interaction between the manganese ions and the effective transfer integral of the, e_g electrons.^{179,180} The electrocatalytic activity and durability of designed electrocatalysts are examined by electrochemical studies in a three-electrode system and half-cell testing studies in zinc-air battery.

4.3 Materials and Methods

4.3.1 Materials

The 20 wt% Pt/C catalysts were procured from Johnson Matthey (UK), and Nafion (5 wt%) solution was purchased from DuPont. **Section 2.5.1.** lists the materials required for the synthesis of the LKMO catalyst.

4.3.2 Synthesis of LKMO microcubes:

As discussed in **Chapter 2**, the LKMO microcubes were synthesized by taking (**Figure 4.1**) La(NO₃)₃.6H₂O, MnCl₂.H₂O, and KMnO₄ into a 25 ml Teflon container in a molar ratio of 1:0.7:0.3 (of 5 mmol equivalents). These reagents are dissolved in 4 ml of Milli-Q water by stirring until the formation of a homogenous solution. Subsequently, KOH is added into the reaction mixture such that the resultant final concentration of the solution becomes 18 M and is allowed to undergo stirring for 24 hours. The hydrothermal reaction is performed by heating the autoclave at 265 °C under autogenous pressure for a period of 7 days. After completing the reaction, the supernatant is decanted, and the resultant solid precipitate is washed thoroughly with Milli-Q water and ethanol until the pH is found neutral. The product is dried at 70 °C for 12 hrs. The reaction produces LKMO microcubes as a major product and La(OH)₃ rods as a minor byproduct. La(OH)₃ rods can be separated from the denser product in the workup process as described in the **section 2.5.2.2.2.**

LaMnO₃ (LMO) and La_{0.5}Sr_{0.5}MnO₃ (LSMO) microcubes have been synthesized using the same above low-temperature hydrothermal method, which is similar to the approach adopted in the literature.^{106,109,181} For detailed information **section 4.8.2** and **4.8.3** can be referred.

Importance of K Substitution in LaMnO_3 to Achieve Highly Active and Exceptionally Stable Bifunctional Electrocatalyst Towards Oxygen Reduction and Oxygen Evolution Reactions

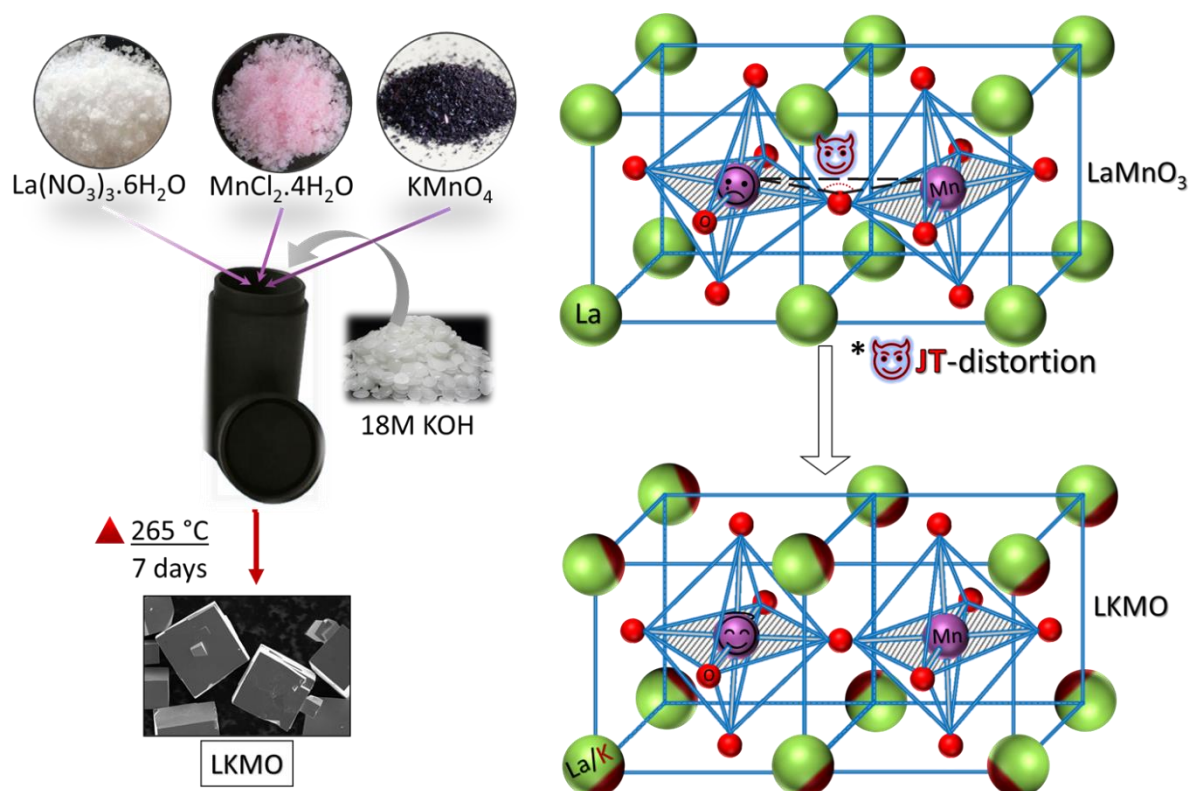


Figure 4.1 Left side depicts the hydrothermal synthesis of LKMO microcubes, and the right side shows the drop in JT-distortion with potassium substitution

4.3.3 Electrochemical characterization:

The electrochemical studies were carried out using a three-electrode electrochemical cell setup in 0.1 M KOH solution for the synthesized electrocatalysts using a bipotentiostat. The catalyst coated glassy carbon was used as a working electrode after polishing with alumina powder of 0.5 microns. A graphite rod was used as a counter electrode. The catalyst of 3 mg was mixed and sonicated well with 120 μl of water, 80 μl of IPA, and 30 μl of Nafion (5 wt%) solutions to obtain the slurry. The requisite quantity of slurry was drop casted onto the working electrode. The silver-silver chloride (Ag/AgCl) electrode filled with 3 M KCl was used as a reference electrode, and all the potential values mentioned in the chapter were converted to RHE values. The linear sweep voltammetry studies were performed to evaluate the electrocatalytic ORR and OER activity of the synthesized electrocatalysts in 0.1 M KOH with oxygen saturated solution at a scan rate of 5 mV s^{-1} . The ORR was performed between the sweep potential of 0.2 V to 1.2 V vs RHE and 1 V to 2 V vs RHE for OER. The ORR and OER studies were compared at 1600 rpm. The ORR kinetics was investigated using the Koutecky-Levich plots with the following equation:

$$\frac{1}{I} = \frac{1}{I_k} + \frac{1}{I_d} = \frac{1}{nFAkC^0} - \frac{1}{0.62nFAD_{O_2}^{2/3}v^{-1/6}C^0\omega^{1/2}} \dots \dots \dots (1)$$

$$I_k = \frac{I * I_d}{I_d - I} \dots \dots \dots (2)$$

A is geometric electrode area (cm²), k is the reaction rate constant, C⁰ is O₂ saturated concentration in 0.1 M KOH solution, D_{O₂} is the oxygen diffusion coefficient, ω is the rotation rate (radian s⁻¹), v is solvents kinetic viscosity, n is the involved number of electrons in the reaction, I_d is the diffusion current density, F is Faraday constant, I_k is the kinetic current density, and I is measured current density.

4.3.4 Zinc-air battery fabrication and testing:

A zinc-air battery of 2.5 cm² active area was fabricated by using an acrylic cell setup consisting of zinc anode, air cathode with 8 M ZnO + 6 M KOH as electrolyte. The zinc electrode was developed by electro-deposition of zinc on the stainless-steel mesh of 4 cm² of electrode surface area. The 12 mg cm⁻² of catalyst loading was maintained on the air electrode of the battery. The performance and cycle life of the battery was tested using a battery cycler (Arbin instrument LBT21084). The charge-discharge cycles of the battery with 360 s per cycle period were performed by maintaining the airflow rate of 40 ml min⁻¹ at the cathode during discharge, and the voltage polarization were observed over the life cycle test.

4.4 Electrochemical measurements

Electrochemical characterizations were carried out to assess the catalytic activity of synthesized LMO, LSMO, LKMO, and commercial Pt/C and IrO₂ catalysts. The cyclic voltammograms in N₂ purged solution show the pure capacitive behaviour for the LKMO catalyst. The prominent ORR peak was observed at 0.8 V in the O₂ purged 0.1 M KOH solution (**Figure 4.2**). The LSVs were further recorded for the different catalysts at 1600 rpm to analyse the ORR kinetics, as shown in **Figure 4.3**. The ORR onset potential for LMO, LSMO, LKMO, and Pt/C was found to be 0.75 V, 0.83 V, 0.86 V, and 0.93 V, and their corresponding half-wave potentials were found to be 0.58 V, 0.72 V, 0.78 V, and 0.85 V, respectively.

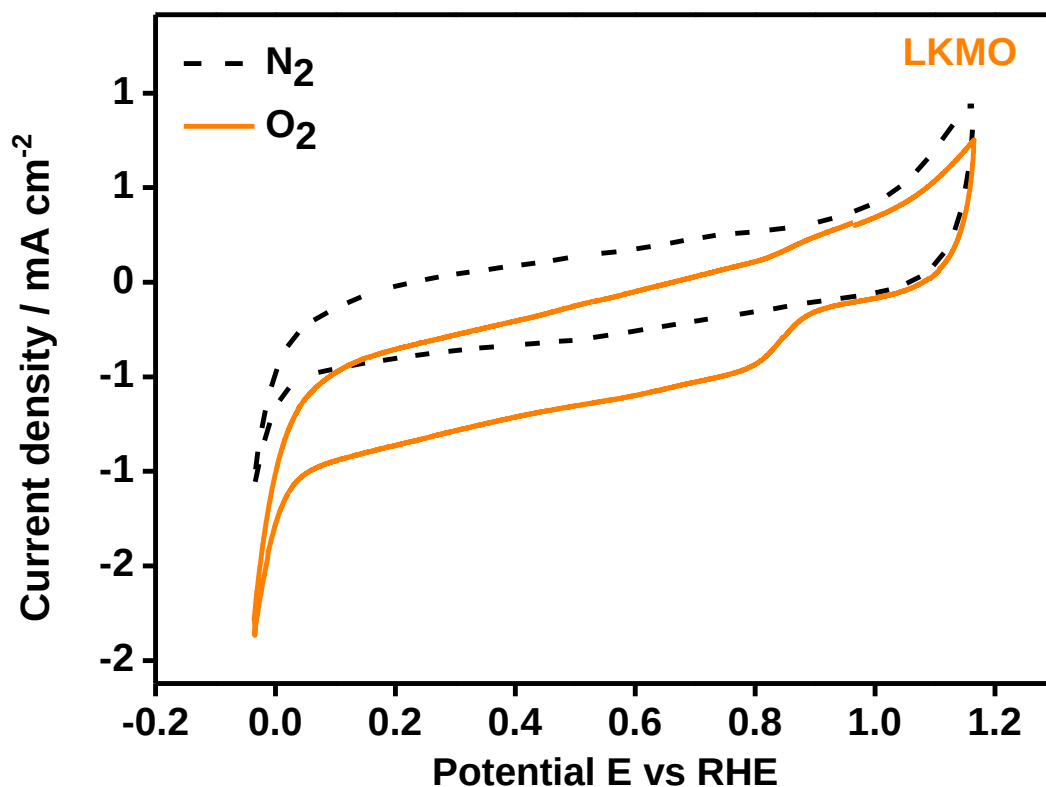


Figure 4.2 Cyclic voltammogram of LKMO at N₂ and O₂ saturated 0.1 M KOH electrolyte.

The synthesized LKMO catalyst showed a higher limiting current density and higher positive shift in the onset potential for ORR compared to many previously reported perovskites in alkaline medium. Tafel slopes from LSV curves studied the kinetics of ORR catalytic activity. The LKMO electrocatalyst exhibited a lower Tafel slope of 84 mV dec⁻¹ than LSMO (97 mV dec⁻¹), LMO (121 mV dec⁻¹), and is very close to Pt/C (59 mV dec⁻¹) (**Figure 4.3(b)**). The lower the Tafel slope value indicates the faster reaction kinetics and higher intrinsic ORR activity.

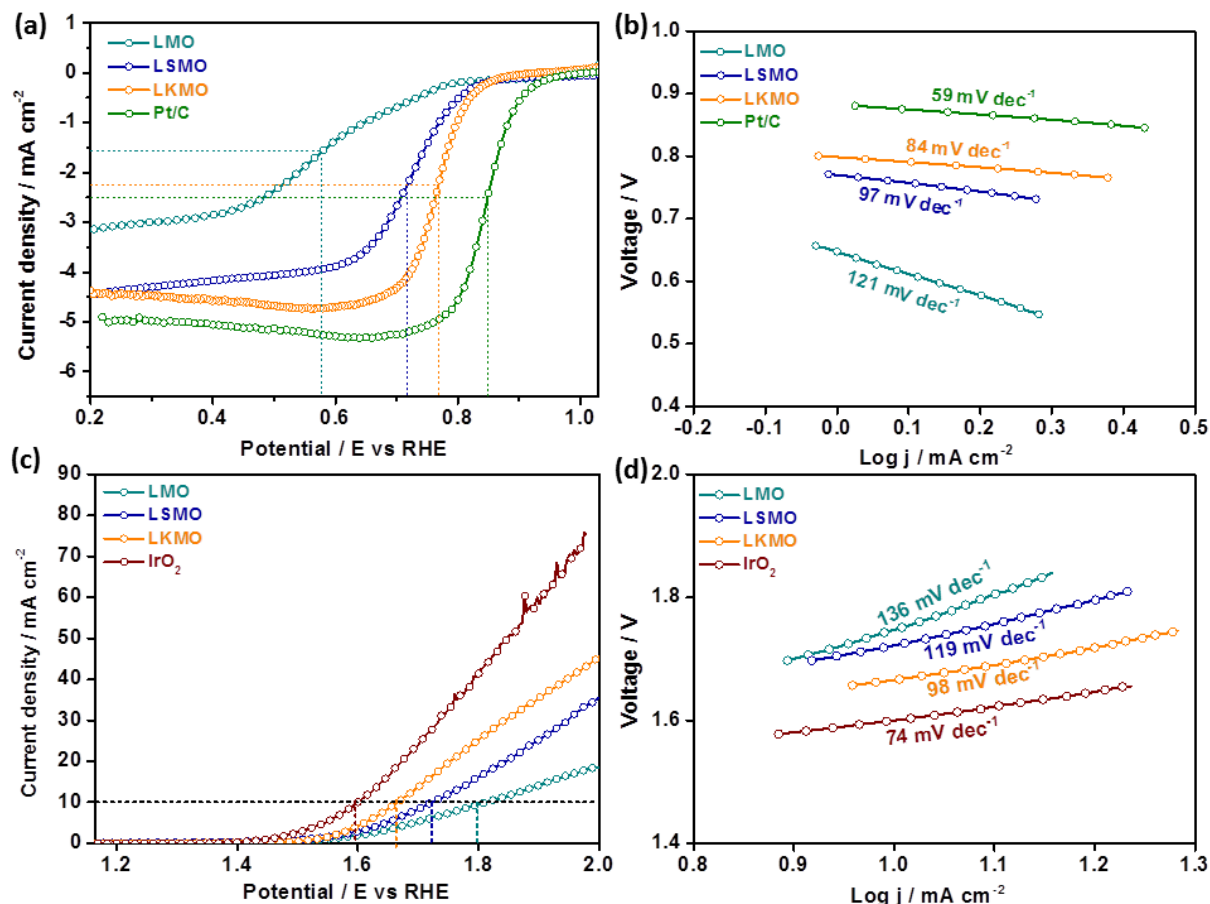


Figure 4.3 (a) ORR polarization curves of LMO, LSMO, LKMO, and Pt/C (20%) catalyst in 0.1 M KOH oxygen saturated at 1600rpm (b) Tafel slope plots for ORR (c) OER polarization curves of LMO, LSMO, LKMO and IrO₂ at 1600rpm. (d) Tafel slopes plots for OER.

The OER activities of synthesized catalysts were evaluated by LSV at 1600 rpm and compared to the state-of-art noble electrocatalysts (IrO₂), as shown in **Figure 4.3(c)**. The obtained potential at a current density of 10 mA cm⁻² was found to be 1.79 V, 1.72 V, 1.66 V, and 1.59 V for LMO, LSMO, LKMO, and IrO₂, respectively. The earlier onset potential and smaller overpotential indicate the higher OER catalytic activity of synthesized LKMO electrocatalysts. The measured catalytic kinetics for OER shows that the LKMO catalyst had a lower Tafel slope of 98 mV dec⁻¹ than LSMO (119 mV dec⁻¹), LMO (136 mV dec⁻¹) and is comparable to IrO₂ (74 mV dec⁻¹) as shown in **Figure 4.3(d)**. The LKMO catalyst exhibited a better or comparable performance to many reported perovskite and transition metals-based OER electrocatalysts (Table 4.1).

Importance of K Substitution in LaMnO₃ to Achieve Highly Active and Exceptionally Stable Bifunctional Electrocatalyst Towards Oxygen Reduction and Oxygen Evolution Reactions

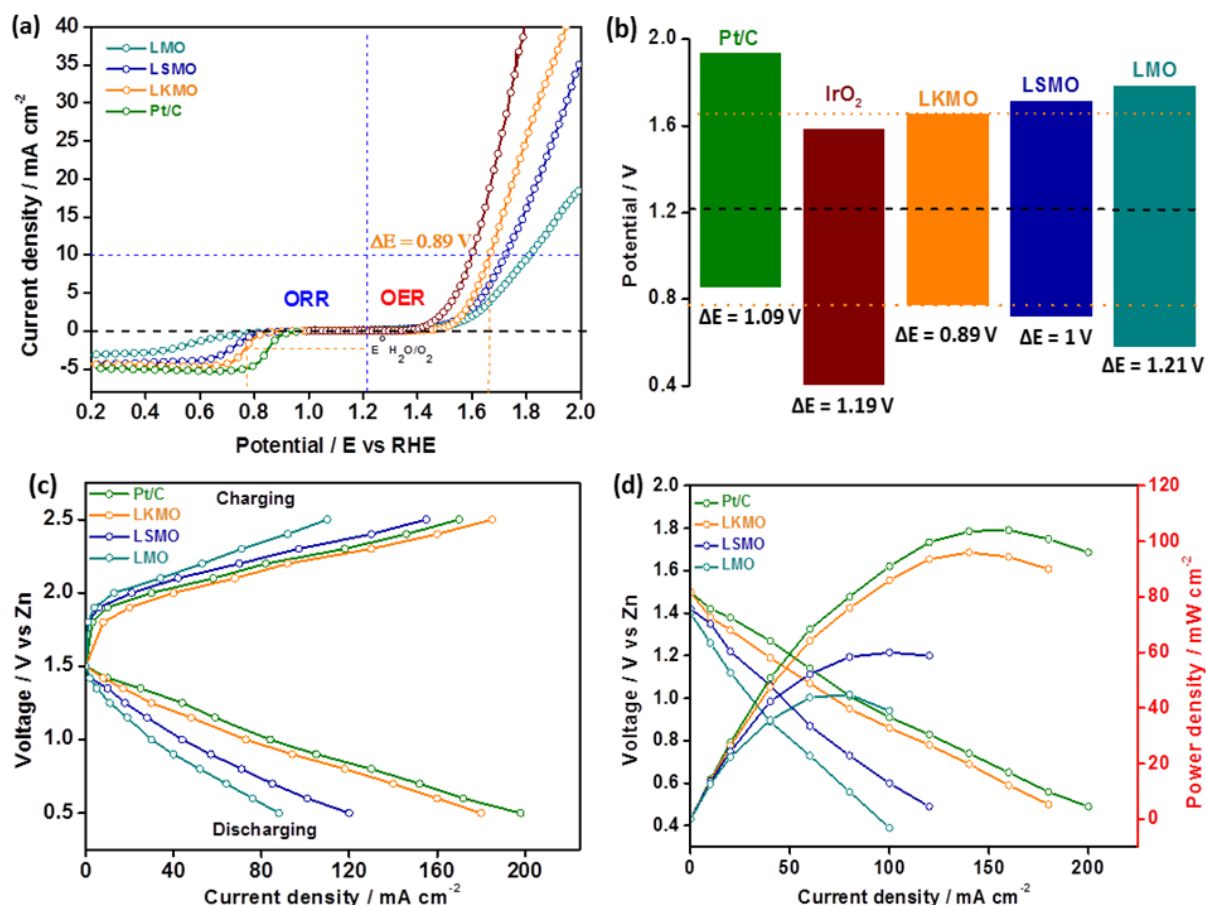


Figure 4.4 (a) LSV curve in the range of potential from 0.2 to 2 V, ΔE ($E_{j=10} - E_{1/2}$) is a measure of OER and ORR bifunctionality. (b) Bifunctionality measurement from LSV for OER and ORR potential to achieve a current density of 10 mA cm⁻² and 3 mA cm⁻², respectively. (c) Polarization curves for charge-discharge of zinc-air battery. (d) Power density curve for LKMO, LSMO, LMO, and Pt/C (20%) as electrocatalyst at air electrode of zinc-air battery.

The bi-functionality of synthesized electrocatalysts was evaluated by measuring the difference in the OER and ORR ($\Delta E = E_{j=10}$, OER - $E_{1/2}$, ORR) potentials at the air cathode of zinc-air battery (**Figure 4.4(a)**). The lower value of ΔE identified the higher bifunctional activity of the catalyst as potential characteristics of zinc-air battery electrocatalysts. The obtained ΔE value for the LMO, LSMO, LKMO, Pt/C, and IrO₂ was 1.21 ± 0.03 V, 1.0 ± 0.03 V, 0.89 ± 0.02 V, 1.09 ± 0.03 V, and 1.19 ± 0.02 V, respectively.

The OER and ORR performance of synthesized electrocatalysts are compared with the state-of-art and other recently reported bifunctional electrocatalysts (Table 4.1), further confirming the excellent OER and ORR activity of the LKMO catalyst. The obtained overpotentials for the developed electrocatalysts for OER (@10 mA cm⁻²) and ORR (from onset potentials) were calculated for LMO, LSMO, LKMO, Pt/C and IrO₂ to be (0.39 V, 0.48

V), (0.35 V, 0.40 V), (0.33 V, 0.37 V), (0.43 V, 0.30 V) and (0.26 V, 0.59 V), respectively (Figure 4.4(b)).

Figure 4.4(c and d) show the voltage profiles and power density curves of zinc-air battery during charge-discharge utilizing LMO, LSMO, LKMO, and Pt/C as electrocatalysts. The LKMO cathode exhibited a maximum current density of $\approx 200 \text{ mA cm}^{-2}$ at 2.5 V, which is higher than LMO ($\approx 120 \text{ mA cm}^{-2}$), LSMO ($\approx 160 \text{ mA cm}^{-2}$), and Pt/C ($\approx 188 \text{ mA cm}^{-2}$) during the charging process. Similarly, the catalyst LKMO achieved a high discharge current density of $\approx 190 \text{ mA cm}^{-2}$ at 0.5 V compared to LMO ($\approx 90 \text{ mA cm}^{-2}$) LSMO ($\approx 120 \text{ mA cm}^{-2}$) and it is very close to Pt/C ($\approx 200 \text{ mA cm}^{-2}$). The maximum peak power density for LMO, LSMO, and LKMO was found to be $\approx 45 \text{ mW cm}^{-2}$, $\approx 58 \text{ mW cm}^{-2}$ and $\approx 98 \text{ mW cm}^{-2}$, respectively and it is comparable to benchmark Pt/C (104 mW cm^{-2}). It is evident that the synthesized catalyst (LKMO) has higher performance compared to reported values.^{182–184}

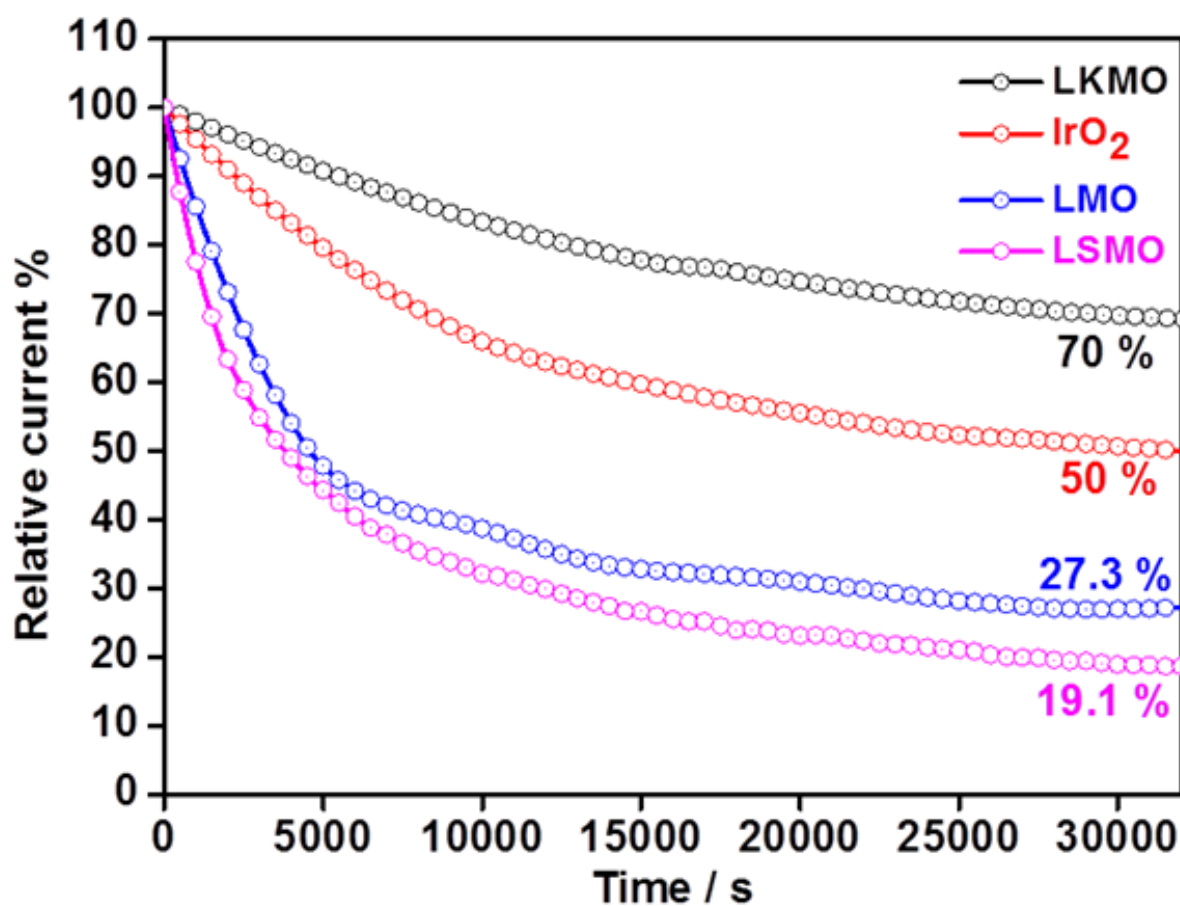


Figure 4.5 Chronoamperometric response of LMO, LSMO, LKMO, and IrO₂ at 1.5 V vs. RHE

Importance of K Substitution in LaMnO₃ to Achieve Highly Active and Exceptionally Stable Bifunctional Electrocatalyst Towards Oxygen Reduction and Oxygen Evolution Reactions

The superior performance of LKMO as an electrocatalyst compared to the existing LMO and LSMO can be attributed to the charge of the dopant, monovalent K than divalent Sr. +2 cation substitution often terminates the surface with the AO layer due to charge neutrality and buries the catalytic MO layer inside the crystal.

The stability of synthesized electrocatalysts was assessed by chronoamperometric measurements, as shown in **Figure 4.5**. The catalyst LMO, LSMO, LKMO, and IrO₂ maintain 27.3 %, 19.1 %, 70 %, and 50 %, respectively, of its initial current density after continuous operation for 8 h at 1.5 V vs RHE. The higher stability of the catalyst can be ascribed to the size of the potassium (1.64 Å), which is slightly higher than the divalent cations, thus stabilizing the Goldsmith tolerance factor, which defines the overall perovskite structure stability. Also, the higher concentration of Mn⁴⁺ decreases the Jahn-Teller distortion associated with the Mn³⁺ ions.⁷²

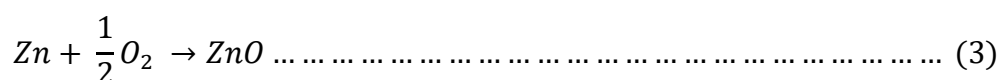
Table 4.1. The onset potentials of different catalysts are tabulated here.

Catalyst	ORR			OER			ΔE	Ref
	E onset (V)	E _{1/2} (V)	Tafel (mV dec ⁻¹)	E onset (V)	E@10 mAc m ⁻²	Tafel (mV dec ⁻¹)		
LKMO	0.86 ±0.02	0.78 ±0.04	84	1.56 ±0.04	1.66 ±0.05	98	0.89 ±0.02	This Work
LSMO	0.83 ±0.03	0.72 ±0.06	97	1.58 ±0.06	1.72 ±0.05	119	1.0 ±0.03	This Work
LMO	0.75 ±0.03	0.58 ±0.05	121	1.62 ±0.04	1.79 ±0.05	136	1.21 ±0.03	This Work
LSCM-80	0.89	0.71	83		1.78	133	1.07	185
BSCF-80	0.78	0.64			1.62	68.6	0.99	186
La _{1.5} Sr _{0.5} Ni _{0.5} Mn _{0.5} Ni _{0.5} Ru _{0.5} O ₆ /C		0.73			1.66		0.92	187

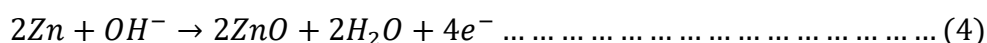
Importance of K Substitution in LaMnO₃ to Achieve Highly Active and Exceptionally Stable Bifunctional Electrocatalyst Towards Oxygen Reduction and Oxygen Evolution Reactions

LaMn _{0.7} Co _{0.3} O ₃	0.88	0.73	110		1.82	151	1.09	188
LaNi _{0.85} Mg _{0.15} O ₃	0.82	0.69	105		1.68	95	0.99	189
Ba _{0.5} Sr _{0.5} Co _{0.8} Fe _{0.2} O _{3-δ}					1.74	94		190
LaMnO ₃	0.78	0.62	110.2		>2	154	>1.3	191
PrBa _{0.85} Ca _{0.15} MnFeO _{5+δ} / CB		0.77			1.64		0.87	192
LaNiO ₃ NR/rGO		0.62			1.70		1.07	193
Nd _{1.5} Ba _{1.5} CoFeMnO _{9-d}		0.69			1.58		0.89	194
La _{1.7} Sr _{0.3} Co _{0.5} Ni _{0.5} O _{4+δ}	0.69	0.56			1.82	125		195
IrO₂				1.49 ±0.04	1.59 ±0.05	74	1.19 ±0.02	This Work
Pt/C	0.93 ±0.01	0.85 ±0.02	59	1.66 ±0.05	1.94 ±0.06	173	1.09 ±0.03	This Work

A zinc-air battery typically consists of an anode (Zinc-plate), a cathode (air), a separator in between them and the electrolyte, as shown in **Figure 4.6**. As an electrolyte, 6 M KOH and additives like Zn(Ac)₂ or ZnO are mixed into a mixture to facilitate the reversible conversion of zinc anode in zinc-air batteries. Electrodeposited zinc on SS mesh or plate forms anode,¹⁹² and catalyst coated on SS mesh forms the cathode of Zn-air battery. The air electrode was developed by coating the electrocatalyst on SS mesh with a highly hydrophobic PTFE for oxygen permeability and to avoid air blockage by an aqueous electrolyte.¹⁹⁶ The electrochemical reactions of zinc-air battery are as follows



During discharge, the Zn electrode forms the Zn(OH₄)²⁻ by reacting with OH⁻ ions as intermediate, further decomposing to ZnO.



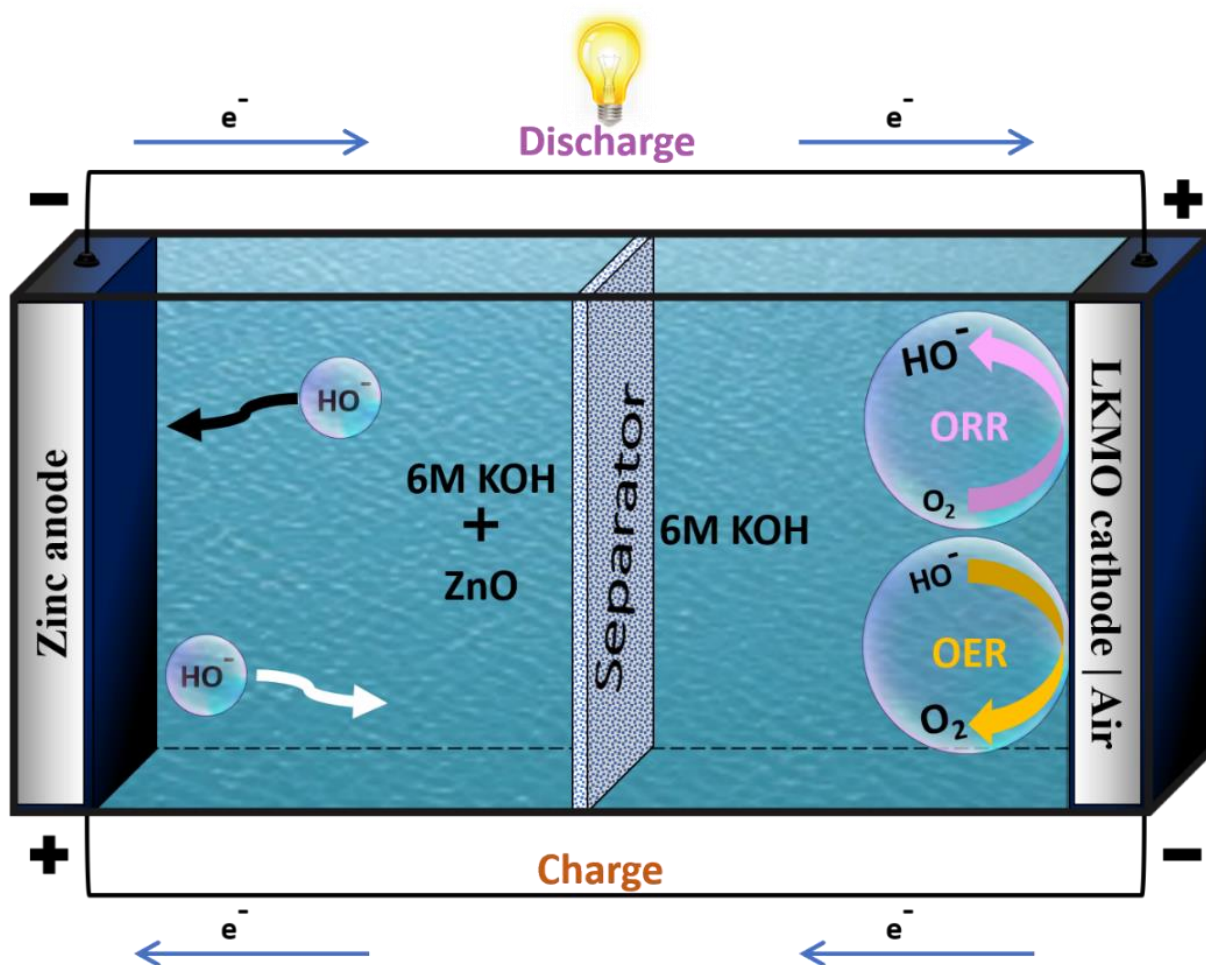


Figure 4.6 Schematic illustration of the working of the Zinc-air battery.

The utilization of bifunctional electrocatalyst in the rechargeable zinc-air battery is highly desirable. The use of LKMO electrocatalysts in the assembled rechargeable zinc-air battery exhibited a high open circuit potential of ≈ 1.42 V, indicating its good catalytic activity.

Importance of K Substitution in LaMnO₃ to Achieve Highly Active and Exceptionally Stable Bifunctional Electrocatalyst Towards Oxygen Reduction and Oxygen Evolution Reactions

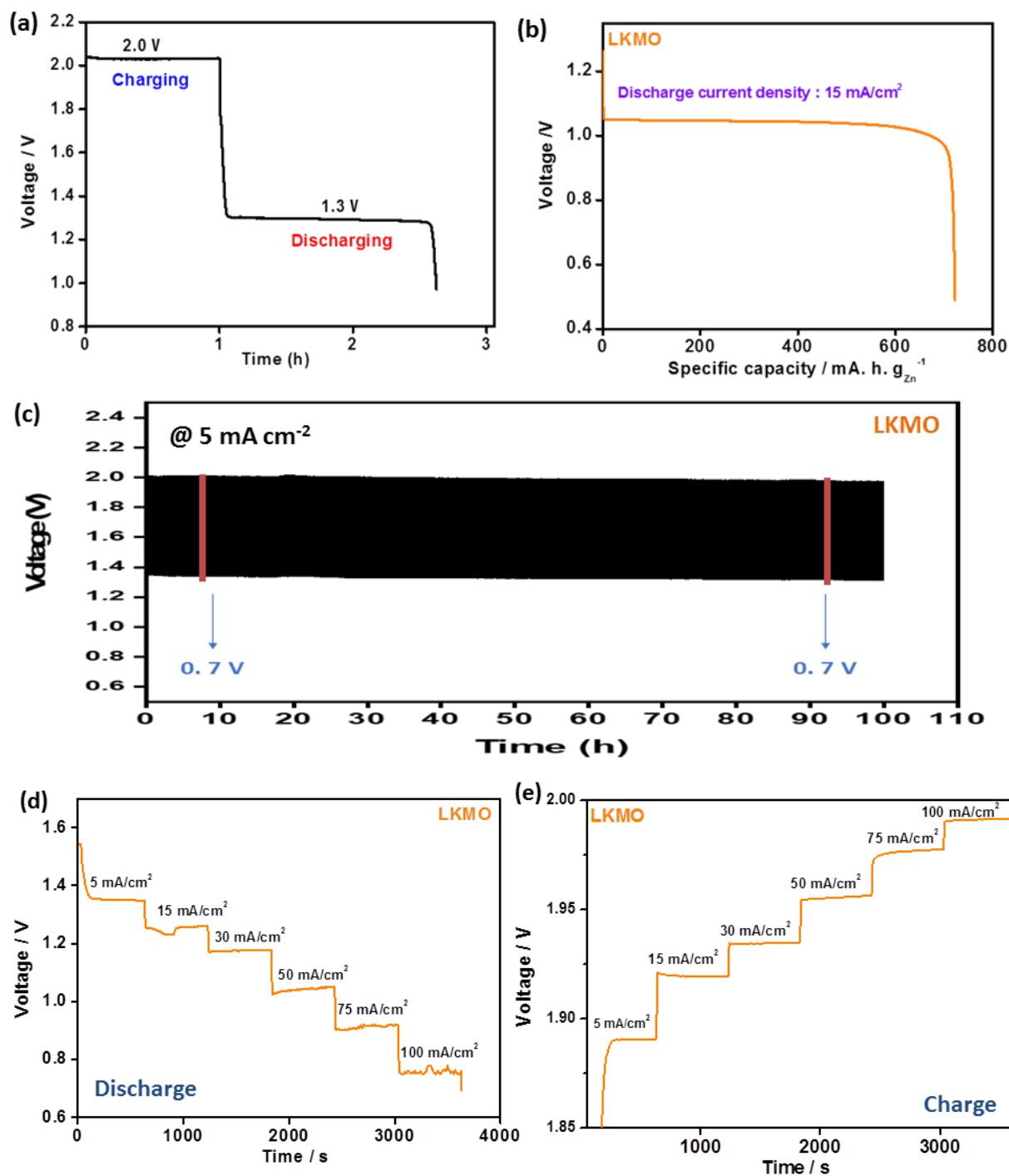


Figure 4.7 (a) Charge-discharge curve for zinc-air battery with LKMO electrocatalyst. (b) Specific discharge capacity of the battery at 15 mA cm⁻² current density with LKMO catalyst (c) Cycling of battery with LKMO at 5 mA cm⁻² (d) Discharging of battery with various current densities starting from 5 mA cm⁻² to 100 mA cm⁻². (e) Charging of battery with different current densities starting from 5 mA cm⁻² to 100 mA cm⁻².

Importance of K Substitution in LaMnO_3 to Achieve Highly Active and Exceptionally Stable Bifunctional Electrocatalyst Towards Oxygen Reduction and Oxygen Evolution Reactions

Table 4.2. The stability of different perovskite catalysts is listed here.

Electro-catalyst	Mass loading (mg/cm)	Anode Electrolyte	Voltage polarization	Charge-Discharge condition	Stability	Reference
$\text{LaNiO}_3/\text{N-CNT}$	0.72	Zn plate 6M KOH	~ 1.2 V at a current density of 20 mA cm^{-2}	600s per cycle period with a current density of $\sim 17.6 \text{ mA cm}^{-2}$	75 cycles; Voltage increased from 0.1 to 0.2 V at the end of the cycling	197
$\text{CoO}/\text{N-CNT} + \text{NiFe LDH}$	1.0/5.0	Zn foil, $0.2\text{M Zn}(\text{AC})_2 + 6 \text{ M KOH}$	~ 0.69 V at 10 mA cm^{-2} of current density	200s per cycle period with a current density of $\sim 10 \text{ mA cm}^{-2}$	60cycles; negligible change in the voltage at the end of cycling	198
$\text{LaNiO}_3/\text{rGO}$	-	Zn plate 6 M KOH	~ 0.98 V at a current density of 25 mA cm^{-2}	3600 s per cycle period with a current density of $\sim 25 \text{ mA cm}^{-2}$	97 cycles; negligible change in the voltage at the end of cycling	193
$\text{La}_{0.7}(\text{Ba}_{0.5}\text{Sr}_{0.5})_{0.3}\text{Co}_{0.8}\text{Fe}_{0.2}\text{O}_{3-\delta}/\text{C}$	-	Zn plate 6M KOH	-	-	100 cycles; negligible change in the voltage	170
$\text{BaTiO}_{3-\delta}/\text{C}$	-	-	-	-	5000 s; stable Charge-discharge Cycles	199

Importance of K Substitution in LaMnO₃ to Achieve Highly Active and Exceptionally Stable Bifunctional Electrocatalyst Towards Oxygen Reduction and Oxygen Evolution Reactions

NiCo ₂ S ₄ /N-CNT	1.0	Zn foil 0.2 M ZnCl ₂ + 6 M KOH	~0.63 V at 10 mA cm ⁻² of current density	400 s per cycle period at ~10 mA cm ⁻² of current density	150 cycles; very less increase in the potential (0.06 V) at the end of cycles	200
Co ₃ O ₄ -doped Co/CoFe	-	0.2 M ZnCl ₂ + 6M KOH	0.9 V at a current density of 5 mA cm ⁻²	1300 s per cycle period with ~5 mA cm ⁻² of current density	180 cycles; very stable during cycling	197
N-Co/CNF- 300-10	-	Zn foil, 0.2 M Zn(AC) ₂ + 6 M KOH	0.77 V at a current density of 10 mA cm ⁻²	15 min per cycle period with ~10 mA cm ⁻² of current density	100 cycles; voltage increased to 0.12 V at the end of cycles	201
Mn-S-C	12	Zinc electrodeposited SS mesh ZnO + 6 M KOH	0.75 V at a current density of 10 mA cm ⁻²	200 s per cycle period with a current density of ~10 mA cm ⁻²	100 cycles; negligible voltage change at the end of cycles	202
LKMO	12	Zinc electro- deposited SS mesh ZnO + 6 M KOH	0.7 V at 5 mA cm⁻² of current density	360 s per cycle period with ~5 mA cm⁻² of current density	1000 cycles; 100 hrs, no change in the voltage at the end of cycles	This work

Figure 4.7(a) display the charge-discharge curve by utilizing LKMO as electrocatalysts in the air electrode of the zinc-air battery. The specific capacity of the battery was measured at a current density of 15 mA cm^{-2} , and it was found to be $742 \text{ mA} \cdot \text{h} \cdot \text{g}_{\text{Zn}}^{-1}$ as shown in **Figure 4.7(b)**. The cycling performance of zinc-air battery with LKMO as electrocatalyst is shown in **Figure 4.7(c)** at 5 mA cm^{-2} of charge and discharge current density. The voltage polarization was proposed for the 1000 cycles of charge-discharge with LKMO catalyst. The voltages at initial cycles were found to be $\sim 1.3 \text{ V}$ and $\sim 2 \text{ V}$ for discharge and charge, respectively, and the corresponding voltaic efficiency was found to be 65% for the initial cycles of charge-discharge. However, the voltage gap for 1000 cycles of continuous charge-discharge remained the same. The LKMO electrocatalysts in zinc-air battery shows the excellent performance with no change in voltage polarization of charge-discharge and lowest change in round trip peak overpotential. **Figure 4.7(c & d)** shows the various discharge and charge of zinc-air battery with different current densities using LKMO electrocatalysts starting from 5 mA cm^{-2} to 100 mA cm^{-2} . The performance of battery with synthesized LKMO catalysts was compared as shown in **Table 4.2** with the reported metal-based electrocatalysts.

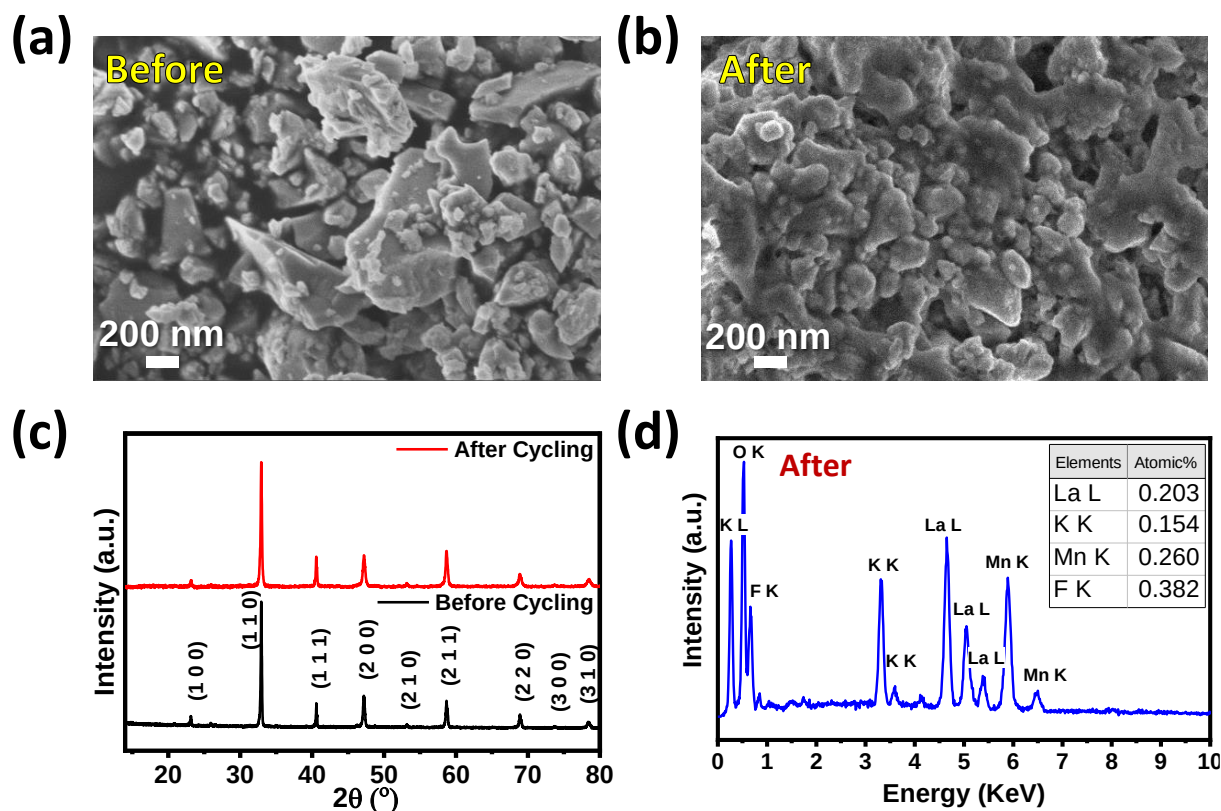


Figure 4.8 SEM micrographs (a) before, (b) after cycling, (c) XRD pattern of the catalyst before and after cycling, and (d) EDS spectrum-quantification of LKMO after cycling.

XRD, FESEM, and EDS studies were further carried out to investigate the morphology, structural stability, and composition of the cycled LKMO catalyst. SEM micrographs in **Figure 4.9(a and b)** indicate no significant change in the morphology of the used LKMO catalyst compared to the pristine one. The XRD patterns of pristine and cycled electrodes are shown in **Figure 4.9(c)** demonstrating no change in the peak shape and intensity of the cycled LKMO compared to the pristine one. It can be found from the EDS (**Figure 4.9(d)**) the presence of La and Mn in the post LKMO was consistent with that of the electrode before cycling. The excess K is due to cycling in the 6 M KOH electrolyte. The presence of F can be attributed to the Nafion (binder). These results confirm the structure, stability, and composition of the catalyst prepared.

4.5 Conclusion:

In summary, we have synthesized potassium substituted LaMnO₃ microcubes via the hydrothermal method at low temperatures. The electrochemical and full cell performance of LaSrMnO₃, LaMnO₃, and LaKMnO₃ were evaluated. Owing to the presence of potassium in the perovskite structure, the improvement in the catalytic activity and excellent stability for OER and ORR compared to many reported perovskites and transitional metal-based electrocatalysts was observed. Potassium substitution helps form more Mn³⁺-Mn⁴⁺ Zener pairs and creates abundant oxygen vacancies. The superior stability over 1000 charge-discharge cycles and lower round-trip peak potential was achieved by utilizing LKMO as electrocatalysts in the zinc-air battery. The robust structure and chemical inertness of synthesized LKMO electrocatalysts has attributed excellent catalytic activity compared to LMO and LSMO perovskite-based catalysts. We believe this work will offer a new aspect to design electrocatalysts for large-scale applications in rechargeable zinc-air batteries.

Part-2: Importance of K substitution in LaMnO₃ towards high-performance selective reduction of α,β -unsaturated carbonyl compounds

4.6 Abstract

Manganites based on lanthanum and their substituted analogs have attracted considerable interest due to their multifunctional properties, including colossal magnetoresistance, magnetocaloric effects, and catalytic oxidation. However, its application to significant heterogeneous catalytic hydrogenation reactions has not been thoroughly investigated. Primarily, these uses have focused mostly on precious metals and Raney nickel. This chapter is all about investigating the catalytic property of the synthesized micron-scale K substituted LaMnO₃ (LKMO) under hydrothermal conditions. These findings are compared to those obtained with LaSrMnO₃ (LSMO) and submicron LaMnO₃, revealing that only LKMO exhibits enhanced catalytic activity against the selective reduction of α,β -unsaturated carbonyl compounds, and nitroarenes. LKMO demonstrated better catalytic conversion of 100% at 120 °C and 7 bar pressure, without any external activating agents. Systematic surface characterization demonstrates that LSMO covers microcubes with an inert SrO layer, while LKMO exposes catalytically active Mn sites on the surface. Due to its low cost, non-pyrophoric and catalytic properties, LKMO could be a possible substitute for precious metals in catalytic hydrogenation. These principles could also be extended to other types of heterogeneous catalysis, and catalytic activity could be further improved by developing synthesis techniques to make them on a nanometer scale.

4.7 Introduction

Controlling the surface to expose catalytically active sites is vital for heterogeneous catalysis.²⁰³ Thus, synthesizing materials that could reveal active facets has gained interest over the last decade. In the pharmaceuticals and fine chemical industry, chemoselective hydrogenation of α,β -unsaturated carbonyl compounds are important but challenging.^{204–208} Chemoselective hydrogenation with traditional catalysts (Pt, Pd, Ru, Rh, Ir, or Raney-Ni) is extremely difficult.^{209–213} Additionally, conventional catalysts have a number of drawbacks,

including the usage of expensive metals, high flammability, and a need for dry reaction conditions. Thus, it is highly beneficial to produce a cost-effective, easy-to-handle catalyst. Some studies have shown that supported nanoparticles based on 3d metals such as CoO_x²¹⁴, CoNi²¹⁵ have shown some success in the temperature range of 110–130 °C and H₂ pressure of 20-30 atm.

Lanthanum-based 3d metal perovskite oxides (LMOs) have been one of the most viable categories of materials for decades due to significant properties such as superconductivity⁸², colossal magnetoresistance^{83,84} and supercapacitor^{90,91}. The catalytic activity of LMOs and their substitutional analogues are mainly concentrated on oxidative reactions as well as some NO_x reductions.^{216–218} They have not been studied towards the reduction of α,β -unsaturated carbonyl compounds. In particular, Sr substituted LaMnO₃ (LSMO) have been shown to be a suitable catalyst towards the oxidation of CO²¹⁹, CH₄^{220,221}, and soot²²². The improved activity is attributed to the high lattice oxygen mobility in substituted LMO.²²³ Compared to the divalent cations such as Sr²⁺ and Ca²⁺, the substitution of La³⁺ by monovalent cation such as K⁺ leads to the formation of more Zener pairs (Mn⁺³/Mn⁺⁴) and more stable LMO. This leads to better electronic structure yielding better thermal, magnetic and electrochemical performance.⁷¹ Especially in the case of K⁺, the size of it is very close to the La³⁺ makes it substitute easily and result in high T_c at lesser substitution.¹²⁰ This chapter discusses a study of catalytic activity LKMO microcubes in the reduction of α,β -unsaturated molecules. We compare the catalytic activity of LKMO with other catalysts such as LaMnO₃ and LaSrMnO₃. We argue better performance of LKMO compared to LSMO (with a similar Mn⁴⁺ to Mn³⁺ ratio) could be assigned to the availability of more Mn sites on the surface of the LKMO. The structural stability of perovskites is superior compared to many other classes of materials, which makes them suitable/ideal for catalytic applications. However, a perovskite metal oxide is not known to show high catalytic hydrogenation performance. Designing such materials would greatly help for the future of catalytic hydrogenation with high stability.

4.8 Materials and Methods

4.8.1 Materials and characterization

The hydrothermally synthesized and purified 18 μm LKMO microcubes, as discussed in **Chapter 2**, are used. Purification of organic compounds were performed by column chromatography using silica gel (60-120 and 100-200 mesh) and thin layer chromatography

was performed on Merck pre-coated silica gel 60-F254 plates. ¹H and ¹³C NMR studies were conducted on a Jeol Spectrospin spectrometer at 400 MHz and 100 MHz, 500 MHz and 100 MHz respectively. Chemical shifts were reported in ppm relative to the residual CDCl₃; coupling constants (*J*) were reported in Hertz. Components present in the synthesized materials were characterized through Powder X-ray Diffraction (PXRD) using a Panalytical Empyrean diffractometer system with CuK α ($\lambda=1.5418$ Å) with large beta-filter Nickel to filter the K β lines. Samples for the XRD analysis were prepared by packing onto the depression of a glass slide. Morphology of the materials was studied through Field Emission Scanning Electron Microscopy (FESEM, *Carl Zeiss*). Elemental composition and distribution in the materials were studied by Energy Dispersive Spectroscopy (EDS) mapping. The sample for SEM and EDS analysis was prepared by taking less than 1 mg of the powder product, crushing mildly to ensure the absence of lumps and sticking it to carbon tape on the SEM sample-stub, and analyzed without coating. XPS spectra were recorded on a Kratos AXIS 165 with a dual anode (Mg and Al) apparatus using the Mg K α anode.

4.8.2 Synthesis of LSMO microcubes:

The La_{0.5}Sr_{0.5}MnO₃ (LSMO) microcubes have been synthesized using the low-temperature hydrothermal method, which is similar to the approach adopted in the literature.^{106,108,118} In a typical synthesis, La(NO₃)₃.6H₂O, Sr(NO₃)₂.4H₂O, MnCl₂.H₂O and KMnO₄ are taken into a beaker containing 2-5 ml of Milli-Q water in a molar ratio of 1:1:1.3:0.7. These reagents are dissolved in water by stirring until the formation of a homogenous solution. The above solution is transferred into a 25 ml Teflon based autoclave with a filling capacity of 80%. Subsequently, 10 ml aqueous KOH is added into the reaction mixture such that the resultant final concentration of KOH to 12 M and allowed to undergo stirring for 20-24 hours. The hydrothermal reaction is performed by heating the autoclave at 265 °C under autogenous pressure for a different period of times ranging from 32 hours. After completing the reaction, the supernatant is decanted, and the resultant solid precipitate is washed thoroughly with Milli-Q water and ethanol until the pH was found neutral. The product is dried at 70 °C for 12 hrs.

4.8.3 Synthesis of LMO sub-microcubes:

The LaMnO₃ (LMO) microcubes have been synthesized using the molten salt synthesis technique as explained in auxiliary section.²²⁴ In a typical synthesis, equal amount of

Figure 4.9a depicts a FESEM micrograph of the LMO submicron cubes obtained by the molten salt method as described in the literature.²²⁴ The submicron cubes range in size from 30 to 156 nm, with an average of 88 nm. In **Figure 4.9b**, depicts a FEG-SEM micrograph of LSMO particles prepared by hydrothermal treatment. The typical sizes vary between 1-10 μm . The SEM images reveal the porous nature of LSMO. In **Figure 4.9c**, we show a FEG-SEM image of LKMO micron size cubes prepared by basic (18 M KOH) hydrothermal treatment of $\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ with $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$ and KMnO_4 at 265 $^\circ\text{C}$ for 7 days. The typical sizes range from 4.5 to 27.5 μm , with an average size of 12.7 μm . The XRD patterns (**Figure 4.9d**) of the LMO, LSMO, and LKMO could be indexed to the ICSD collection code 082224. SEM and powder XRD analysis of the as-prepared samples of LKMO reveals the formation of crystalline LKMO and $\text{La}(\text{OH})_3$ phases. SEM and XRD patterns recorded after the density-based separation techniques indicated the absence of $\text{La}(\text{OH})_3$ phase (**Figure 4.9b**, **1c**, and **1d**).

Figure 4.10 shows the EDS elemental mapping and EDS spectrum of the LKMO microcubes after removal of the $\text{La}(\text{OH})_3$ rods by density-based separation. The SEM micrograph entirely accords to the images of the EDS mapping of La, K, Mn, and O atoms. All the atoms are homogeneously dispersed on the microcubes. The semi-quantitative EDS analysis on microcubes shows the ratio of La:K:Mn as 0.74:0.27:1.0.

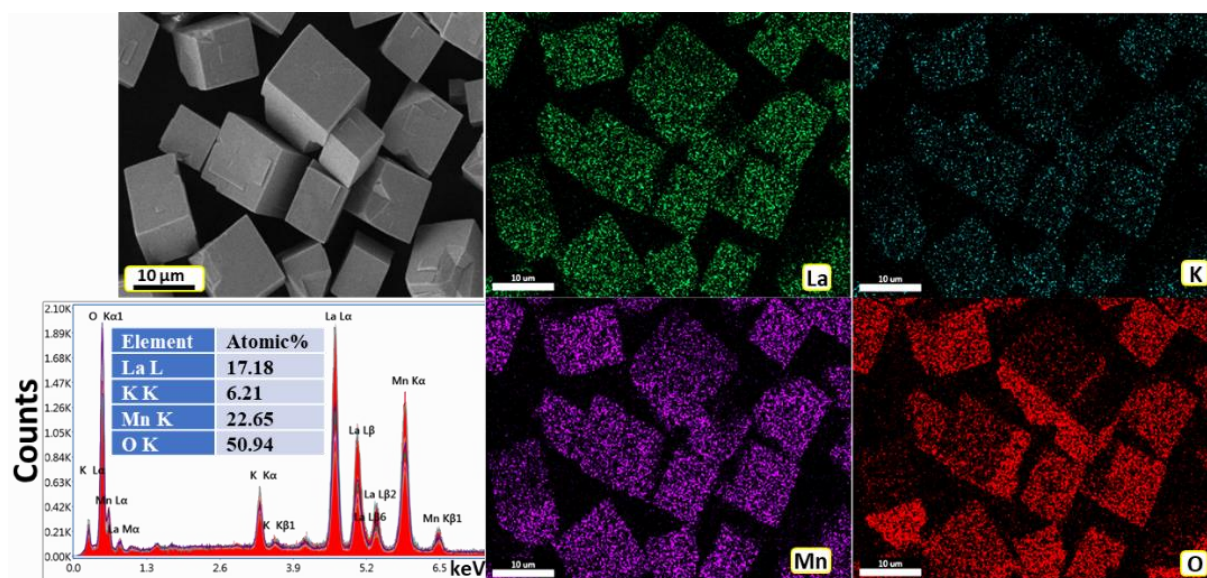


Figure 4.10 EDS elemental mapping and EDS spectrum of the LKMO microcubes after removal of the $\text{La}(\text{OH})_3$ nanorods by density-based separation.

Importance of K substitution in LaMnO_3 towards high-performance selective reduction of α,β -unsaturated carbonyl compounds

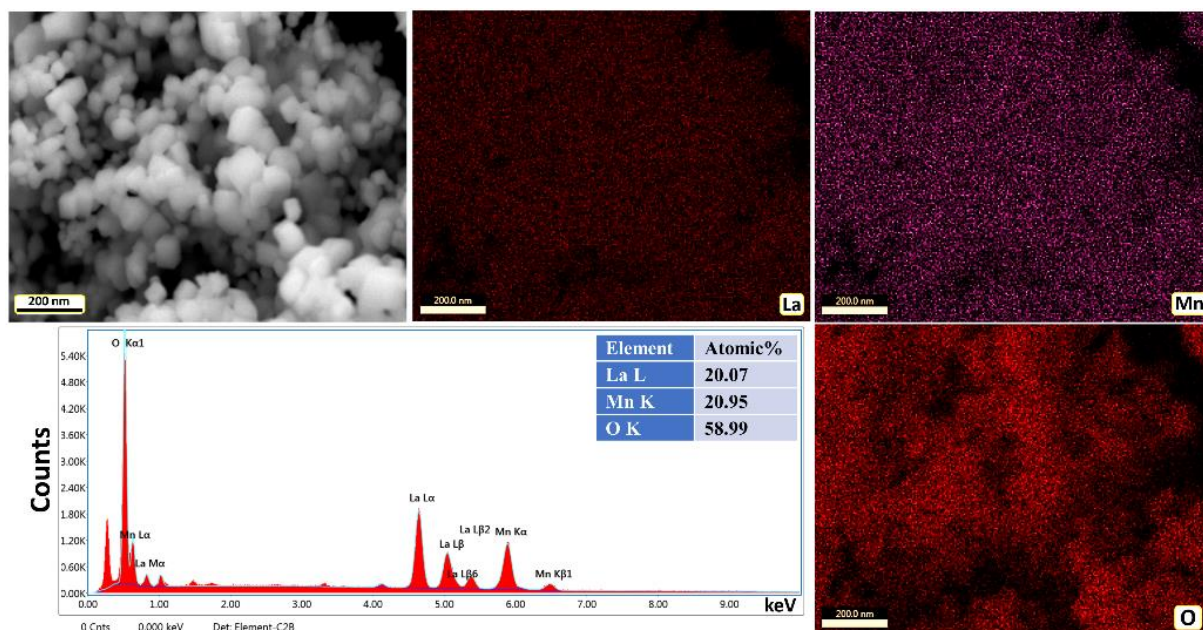


Figure 4.11 EDS elemental mapping and EDS spectrum of the LMO submicron-cubes. The SEM micrograph entirely accords to the images of the EDS mapping of La, Mn and O atoms. All the atoms are homogeneously dispersed on the microcubes.

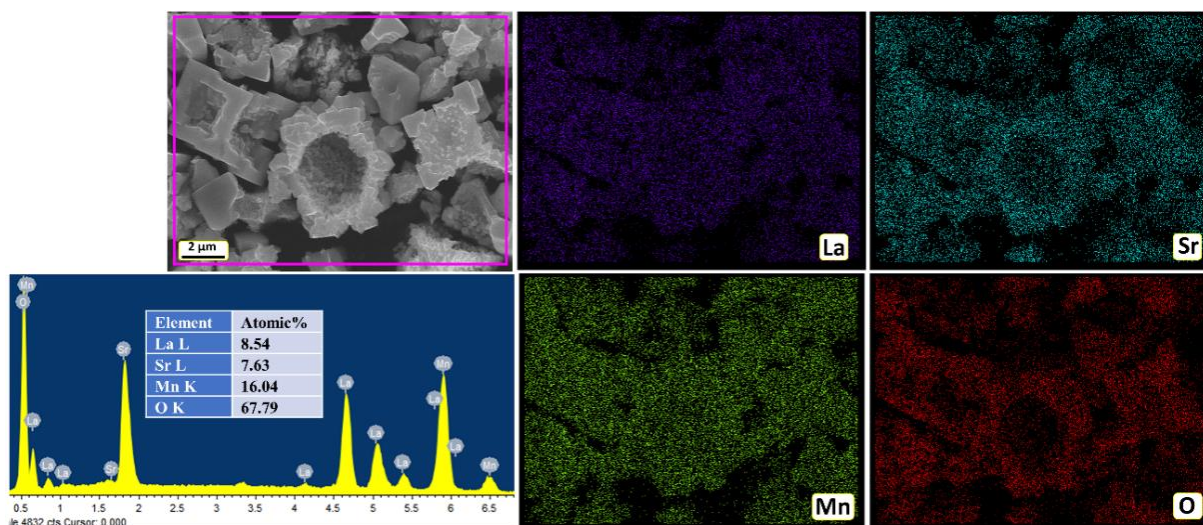


Figure 4.12 EDS elemental mapping and EDS spectrum of the LSMO particles. The SEM micrograph entirely accords to the images of the EDS mapping of La, Sr, Mn and O atoms.

EDS mapping and spectra are collected on several LKMO cubes (more than 25), and the composition ratio among different cubes varies only marginally. The EDS mapping and spectra collected on LSMO are shown in **Figure 4.12**. EDS analysis on LSMO cubes shows the ratio of La:Sr:Mn as 0.53:0.48:1.0. This can be averaged to $\text{La}_{0.5}\text{Sr}_{0.5}$. Similarly, EDS analysis on LMO shows a La:Mn ratio of 1:1 (**Figure 4.11**).

4.9.1 General procedure for hydrogenation

In a stainless steel reactor with 100 mL capacity was charged with α,β -unsaturated carbonyl compound (0.5 mmol), Catalyst (30 mg) and toluene (6 mL), reactor was evacuated to remove the air and purged with hydrogen (3 times at 5 atm) and finally pressurized with 7 atm of H₂. Then the reaction kept for stirring at 120 °C for 12 h and progress of the reaction monitored by TLC. After the reaction was complete, autoclave was cooled to room temperature and the hydrogen was released carefully. The crude reaction mixture was purified by chromatography on silica (eluent: hexane/ethyl acetate = 10:1) to afford desired saturated carbonyl compounds.

4.10 Catalytic activity

Catalysis is a surface phenomenon, and perovskites are known to show different compositions in bulk and at the surface. This variation becomes more prominent with doping. When divalent cations such as Sr²⁺ are introduced inside the LMO, SrO generally terminates the outer layer due to the electrostatic charge balance.¹⁷⁷ SrO can further convert to SrCO₃ in aerobic conditions. However, the SrO layer is not catalytically active towards hydrogenation. To understand the surface composition, we have analysed LMO, LSMO, and LKMO using XPS. Survey scans of the samples are shown in **Figure 4.13**. The patterns were charge corrected with amorphous carbon at 284.7 eV for all the samples. Peaks were fitted with mixture of Gauss and Lorentz function with 0.3 blend. In the case of 2p (Mn) and 3d (Sr) spectrum peak area ratios (1:2 for p and 2:3 for d-subshell) were considered for fitting the components as per their spin-orbit splitting j values. The high-resolution Mn 2p spectra are shown in **Figure 4.13**. It has contributions from Mn³⁺ and Mn⁴⁺ but accurate quantification is challenging due to close overlap among the multiplets of the 2p_{3/2} for both the oxidation states.^{228,229} By looking at the binding energy of Mn 2p_{3/2} the oxidation state of Mn in all the samples is in between +3 and +4. The Mn 2p_{3/2} peak is centred at 640 eV for all the samples (**Figure 4.13**) with FWHM of 2 eV in LMO. The binding energy of Mn 2p_{3/2} shifts to higher binding energy in the case of doped samples (K and Sr), and its FWHM increases compared to LMO due to the contribution of Mn⁴⁺ in doped samples.

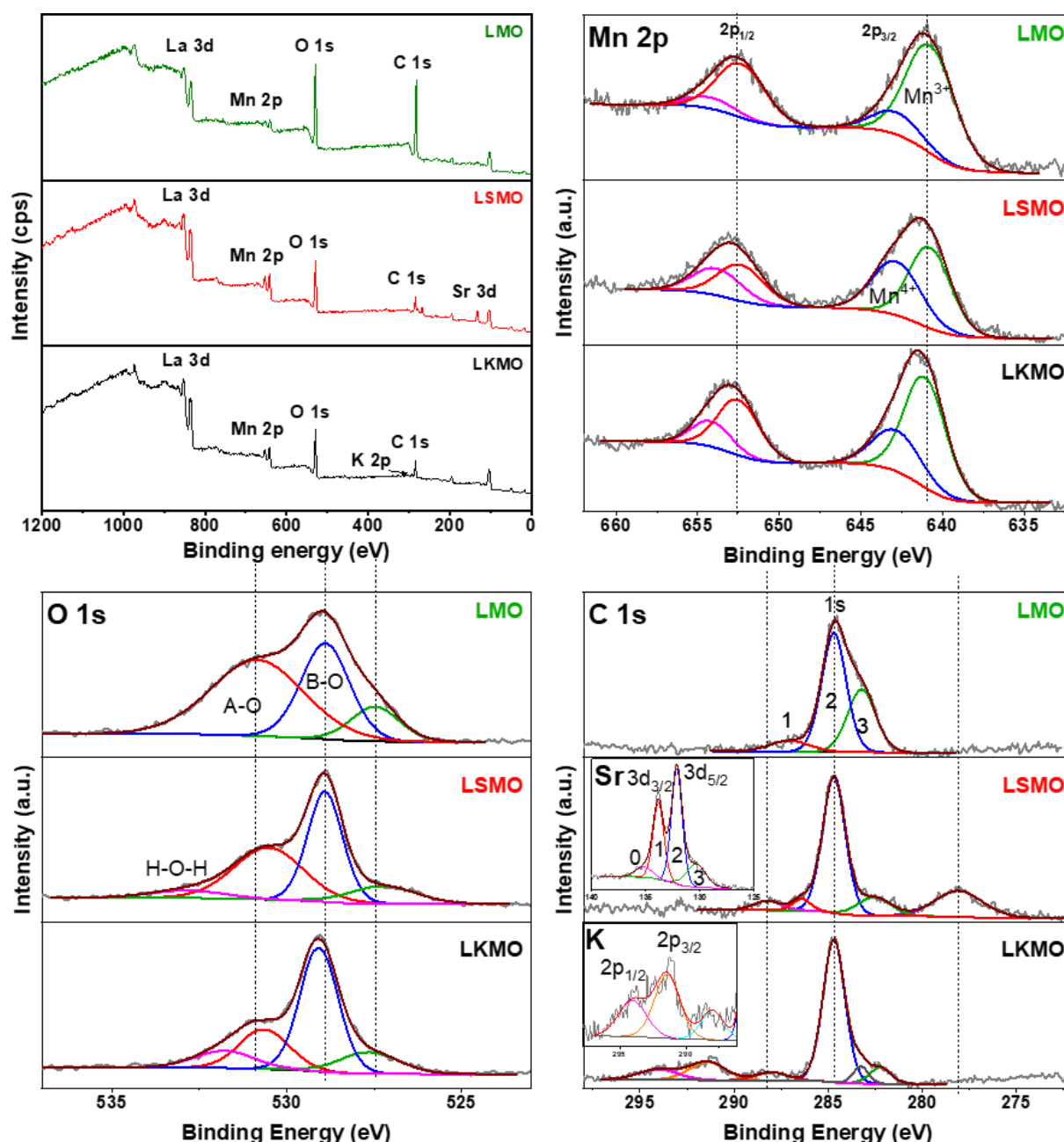


Figure 4.13 XPS survey scan of LMO, LSMO, and LKMO samples and their high-resolution scans for the peaks Mn 2p, O 1s, and C 1s; insets show K 2p (LKMO) and Sr 3d (LSMO)).

The O 1s peak generally has three contributions(**Figure 4.13**). The lowest binding energy component in each sample is correspond to the perovskite bulk or oxygen bonded to the core Mn (B-O). The highest binding energy of smaller area percent comes due to moisture. The peak around 530 eV are less straightforward. Some literature assigned it to SrO precipitates at the surface of LSMO while this can be originated from surface SrCO₃, Sr(OH)₂ also the oxygen bonded with A-site cations (perovskite terminating with A-O layer).²³⁰ From the O 1s

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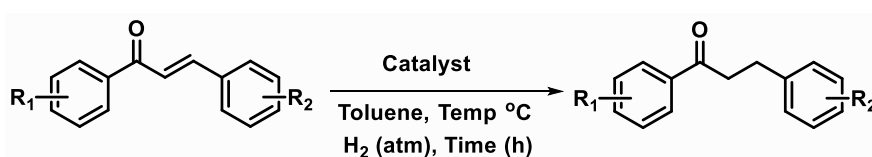
spectrum; compared to LSMO, LKMO shows less dopant concentration at the surface and exposes more Mn at the surface, which is catalytically more active.

The formation of SrCO₃ can be clearly seen from the **Figure 4.13** Sr 3d spectrum in the inset. In case of LSMO in addition to the lattice Sr 3d (peak 1 and 2) additional Sr (peak 0) present. This was deconvoluted keeping equal FWHM values for the Sr 3d 3/2 and 5/2 components with 1.7 eV peak separation. Also, the corresponding C1s spectrum shows peak around 288.5 eV attributing to the carbon atoms of carbonate species. Thus LKMO can commence its catalytic activity for longer cycles without any degradation in activity promising to be superior catalyst than LSMO.

Table 4.3 Binding energy (B.E.) and relative concentration of Mn³⁺, Mn⁴⁺, and oxygen species from the deconvolution of Mn 2p and O 1s XPS peaks of LMO, LSMO, LKMO.

Sample/Element	Mn				O			
LKMO	3+	4+	3+	4+		B-O	A-O	H-O-H
Line	2p _{3/2}	2p _{3/2}	2p _{1/2}	2p _{1/2}	1s	1s	1s	1s
B.E. (eV)	641.02	642.75	652.46	654.11	527.61	529.06	530.62	531.78
Area (%)	46.54	20.13	23.27	10.06	14.10	52.60	21.91	11.39
LSMO								
B.E. (eV)	640.72	642.76	652.29	653.77	527.24	528.90	530.52	532.86
Area (%)	37.57	29.10	18.78	14.55	12.73	42.03	39.15	6.08
LMO								
B.E. (eV)	640.69	642.79	652.29	654.18	527.47	528.89	530.83	
Area (%)	55.29	11.37	27.65	5.69	12.50	34.69	52.81	

Table 4.4 Optimization of LKMO for hydrogenation of α,β -unsaturated carbonyl compounds.^a



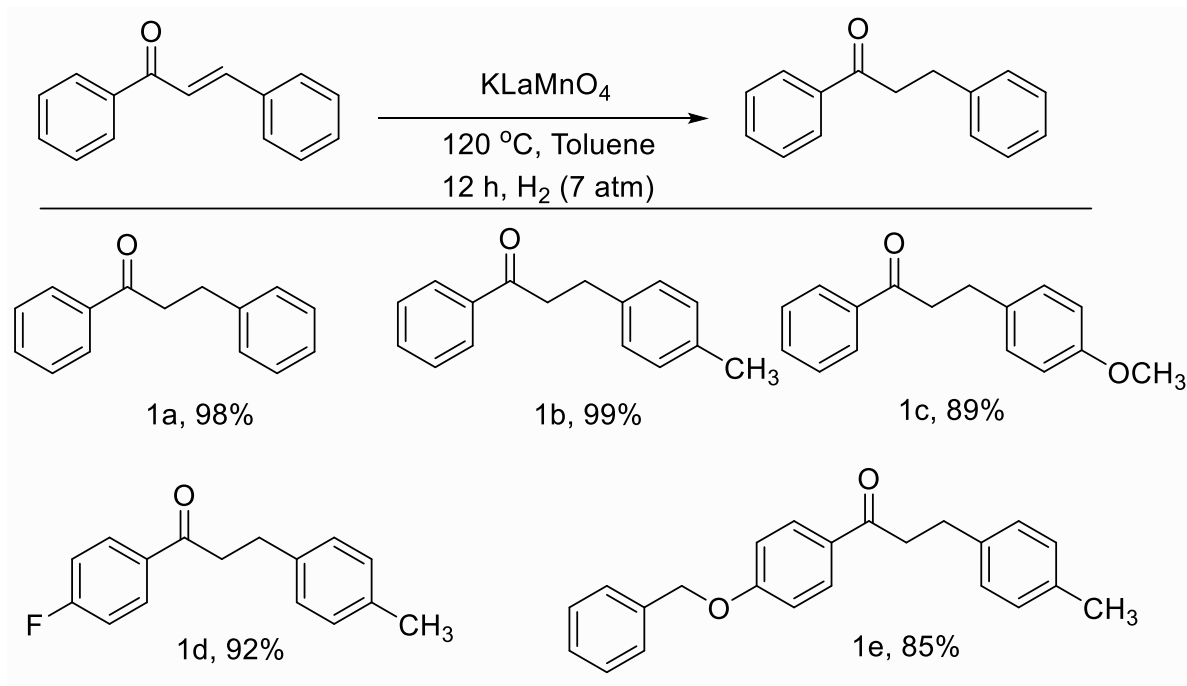
S.No	Catalyst	Yield (Conversion%) ^b
1	LMO submicron cubes	4.2
2	La_{0.77}K_{0.23}MnO₃ microcubes	100
3	La_{0.5}Sr_{0.5}MnO₃ particles	58.8

^a Reaction conditions: Temperature (120 °C), Time (12 h), α,β -unsaturated carbonyl compound (0.5 mmol), Catalyst (30 mg) and Toluene (5 mL); ^b GC-MS yields

Next, to understand the catalytic activity of the prepared catalysts, a model hydrogenation reaction was performed on α,β -unsaturated carbonyl compound as shown in **Table 4.4**. The initial studies were performed with simple LMO and were found to be less active. Further doping of divalent Sr^{2+} and monovalent K^+ subsequently enhanced the catalytic activity towards the hydrogenation of α,β -unsaturated carbonyl compound. These experimental findings are clearly correlating with our understanding, as discussed earlier of these systems (see **Table 4.4**).

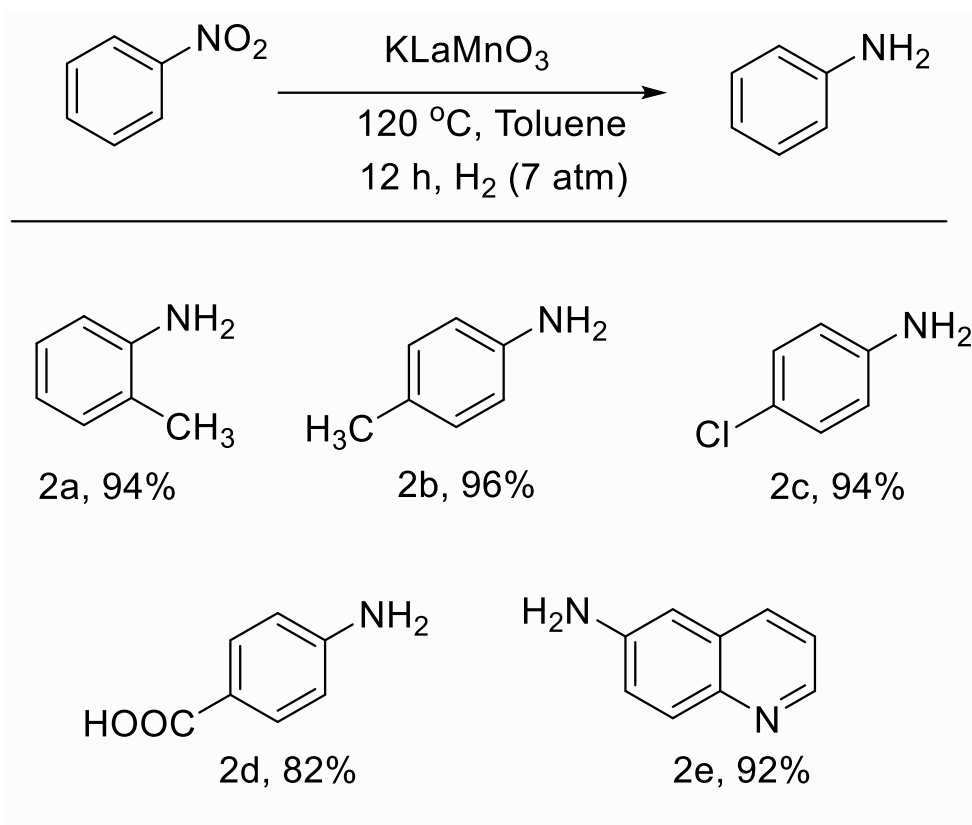
Further, after optimizing the reaction conditions for the hydrogenation of α,β -unsaturated carbonyl compounds, we have examined the substrate scope of the LKMO with functionalized α,β -unsaturated carbonyl compounds. From **Table 4.5**, it is clear that the catalyst has functional group tolerance towards halo, carbonyl, and benzyl groups (**1d** and **1e**). Along with the functional group tolerance, the catalyst has shown good to an excellent conversation with all substrates in **Table 4.5**. Next, to understand the wide scope of the catalyst, we have extended the hydrogenation method for the reduction of nitroarenes (**Table 4.6**). The *ortho* and *para*-substituted nitroarenes were converted to corresponding anilines (**2a** and **2b**) with excellent yields. *p*-Chloro nitrobenzene also could be converted to the corresponding aniline without any dehalogenation side product (**2c**).

Table 4.5 Substrate scope of the LKMO for hydrogenation of α,β -unsaturated carbonyl compounds.



Reaction conditions: α,β -unsaturated carbonyl compound (0.5 mmol), Catalyst (30 mg), Toluene (5 mL).

Table 4.6 Catalytic hydrogenation of nitroarenes to corresponding amines.



Importance of K substitution in LaMnO₃ towards high-performance selective reduction of α,β -unsaturated carbonyl compounds

Reaction conditions: Nitroarenes (0.5 mmol), Catalyst (30 mg), Toluene (5 mL).

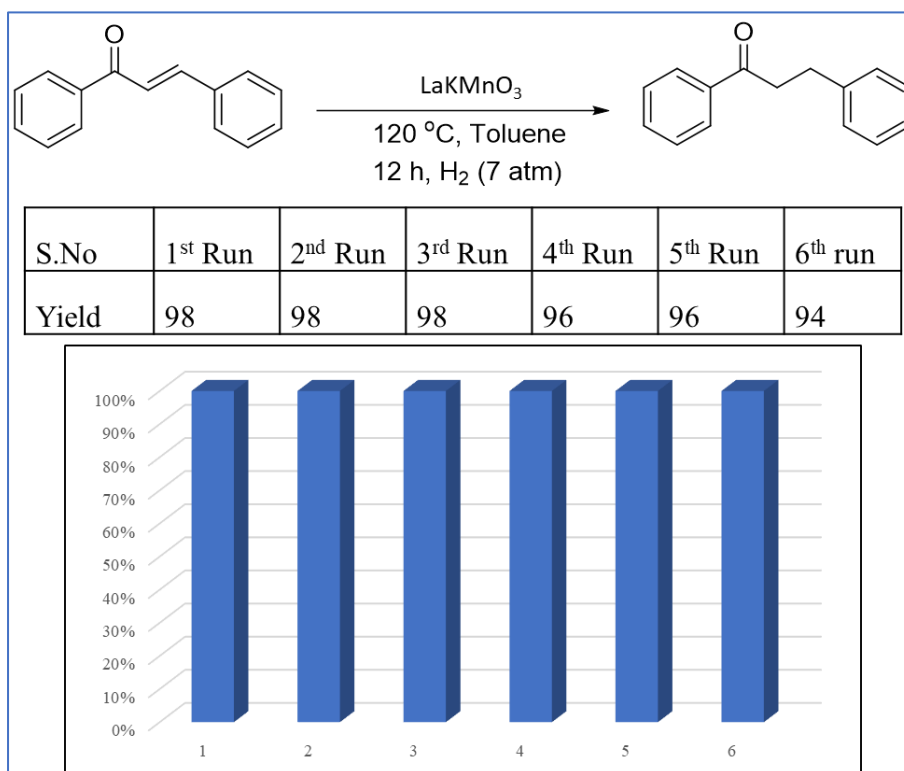


Figure 4.14 Plot describing the recyclability of the LKMO catalyst.

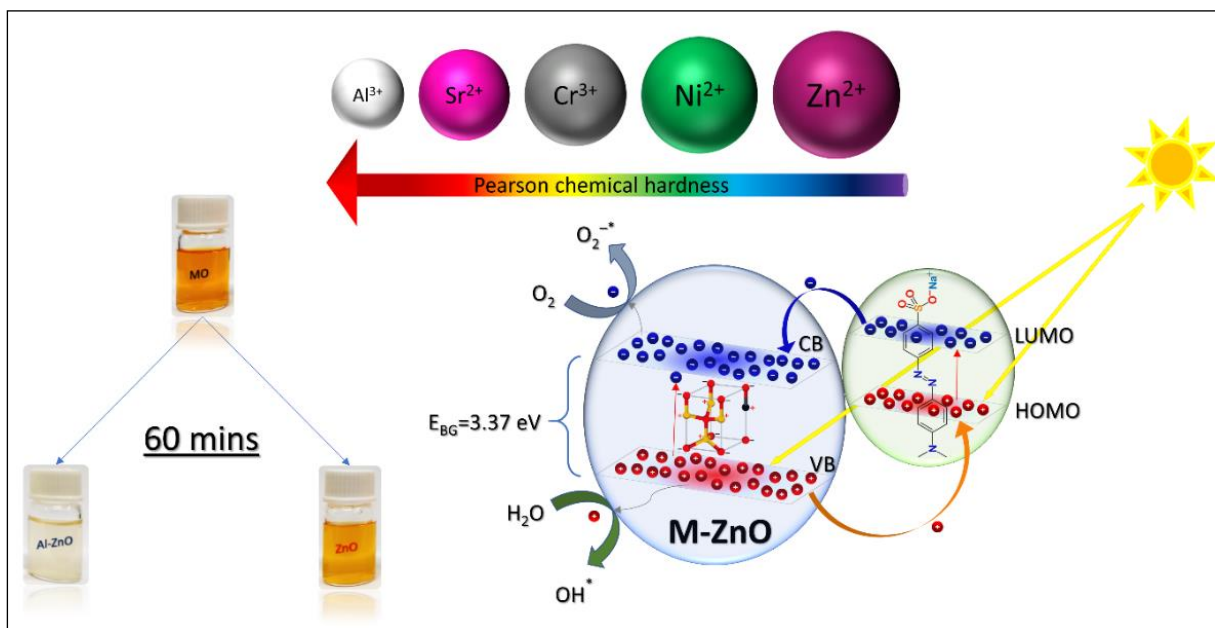
4-nitrobenzoic acid and 6-nitroquinoline (bicyclic heteroaromatic compound) also could be converted to the corresponding amines in excellent yields. To understand the catalyst stability and recyclability for the hydrogenation process, the catalyst has been reused up to five successive runs without any appreciable loss in its catalytic activity (**Figure 4.14**).

4.11 Conclusion

Finally, we demonstrated the catalytic activity of LKMO in the reduction of chalcones and nitroarenes. This hydrothermally synthesised LKMO could be a viable candidate to replace noble metals in catalysis. These LKMO microcubes have a remarkable activity of 98 percent conversion in the reduction of α,β -unsaturated compounds, which is far superior to existing materials. Despite having a smaller surface area, the micron size cubes have shown a high level of activity. We anticipate that this will lead to a better understanding of the fundamental principle underlying the formation/synthesis/crystallization mechanism of trimetallic-complex systems. Leading the research to investigate the synthesis and catalysis of other ternary systems with various morphologies for activity and selectivity.

CHAPTER-5:

Doping with chemically hard elements to improve photocatalytic properties of ZnO nanostructures



5.1 Abstract

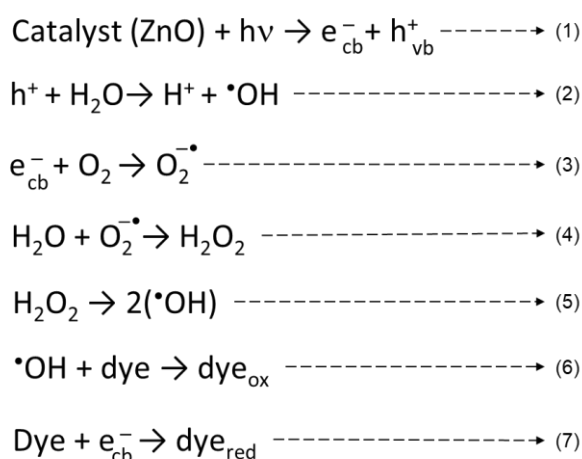
Here we report improved photocatalytic activity of ZnO nanostructures through judicious choice of the dopant, particularly by choosing the dopant with high chemical hardness. We have doped ZnO nanostructures with divalent (Ni^{2+} and Sr^{2+}) and trivalent (Cr^{3+} and Al^{3+}) using hydrothermal conditions. The photocatalytic activity of undoped and doped ZnO nanostructures was tested towards the degradation of methyl orange (MO), methylene blue (MB), and congo red (CR). The Al-ZnO exhibited a 91.25% removal rate compared to 40.88% for pure ZnO in MO degradation under ultra-violet (UV) radiation. The trend of photocatalytic activity of different photocatalysts ($\text{Al-ZnO} > \text{Sr-ZnO} > \text{Cr-ZnO} > \text{Ni-ZnO} > \text{ZnO}$) is similar to the trend of Pearson hardness values of $\text{Al}^{3+} > \text{Sr}^{2+} > \text{Cr}^{3+} > \text{Ni}^{2+} > \text{Zn}^{2+}$. These reactions also reveal that the dopant type affects the morphology of the ZnO nanostructures in given reaction conditions. We propose the polarity-induced mechanism to understand the changes in the morphology due to different dopants.

5.2 Introduction

Photocatalytic degradation of dyes using semiconductor nanoparticles is an efficient and cost-effective strategy to degrade environmentally concerning organic pollutants.^{231–240} However, strategies to improve the photocatalytic properties of existing catalytic need to be developed. Tuning the catalyst to improve photocatalytic activity is important for air purification and wastewater treatment. The widely investigated photocatalysts for these purposes are TiO_2 and ZnO due to their non-toxic nature, abundant availability, high reactivity. However, they absorb light in the near UV region, only <5% of our solar spectrum.^{241–243} Several strategies have been adopted so far for improving the photocatalytic activity, including doping with other elements such as Al^{244,245}, Eu²⁴⁶, Mg^{247,248}, Mn^{249,250}, Fe²⁵¹, Ce^{252,253}, Y²⁵⁴, Sr^{255,256}, La²⁵⁷, Ta²⁵⁸, Cu²⁵⁹, Nd²⁶⁰, Bi²⁶¹, Hf²⁶², Sn²⁶³, Sm²⁶⁴, Pd²⁶⁵, Se²⁶⁶, Ag²⁶⁷, and co-doping such as Co-Ni²⁶⁸. Dopants tune the properties of photocatalyst in several ways; some metals decrease the band gap²⁵⁴ and thus accommodate the visible part of the solar spectrum to be absorbed. Some dopants introduce trap states into the crystal that helps to trap the excited electrons-holes and diffuse them to the surface.^{246,269,270} Some dopants also tune the size and morphology of the photocatalyst, making it into a thin flakes-such as structure where the bulk generated e^- - h^+ pairs are easily transported to the surface.²⁶³ Also, in some cases, the presence

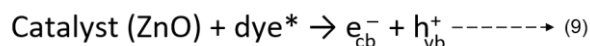
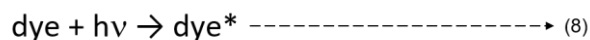
of dopants makes the surface of the photocatalyst rough and porous allowing more adsorption of the dyes and organic pollutants.^{250,267} However, judiciously choosing a dopant by looking into their chemical nature to improve the photocatalytic properties are less explored especially correlating between photocatalytic activity and dopant chemical hardness. As the adsorption of chemically hard species such as oxygen and water species on photocatalyst is involved in the dye degradation mechanism, systematic studying of the chemical hardness effect on the dye degradation process is crucial.

Two mechanisms have been put forward in the literature to explain the dye degradation, namely, a) photocatalytic dye degradation and b) dye-sensitization mechanism.^{231,233,235} In the earlier case, the dispersed photocatalyst absorbs the radiation and generates electrons and holes.^{233,235,271–275} These photogenerated charge carriers reduce and oxidize oxygen and water species that are adsorbed on the photocatalyst surface, resulting in active superoxide radical anion and hydroxide radicals, which react with the dye and degrade it (See **Scheme 1**).^{245,248,253} In the dye-sensitization mechanism, the dye absorbs light in the visible region (around 465 nm in MO) and gets excited. The photoexcited dye transfers the electron from its lowest unoccupied molecular orbital (LUMO) to the photocatalyst's conduction band (CB). At the same time, one unit of hole migrates from dye to catalyst as the fermi level of the MO is above that of the ZnO (photocatalyst). The remaining process in the mechanism follows similar steps as of the earlier mechanism. Under UV irradiation, ZnO-based photocatalysts follow photocatalytic dye degradation. In contrast, the dye-sensitization mechanism is applicable in visible light.



Scheme 1 Steps representing the dye degradation mechanism.

The dye sensitization mechanism also follows the similar above steps, except the dye gets excited by exposure to UV light, and the excited dye reacts with the catalyst (as shown in the below equations 8 & 9).



As ZnO follows the photocatalytic dye degradation mechanism, adsorption of oxygen and water species on photocatalyst is essential. This chemically hard oxygen and water can interact more with a photocatalyst if the photocatalyst contains more chemically hard dopants such as Al^{3+} or Sr^{2+} . Here, we report doping of ZnO nanostructures with different divalent (Ni^{2+} , Sr^{2+}) and trivalent cations (Cr^{3+} and Al^{3+}) in hydrothermal conditions. Even though ZnO nanostructures are doped with different cations in the literature,^{245,276–278}, the correlation of dopant hardness on photocatalytic activity has not been studied earlier. We have also investigated the optical and morphological changes in the doped ZnO nanostructures. The mechanism of morphology change due to different types of cations is proposed. These doped nanostructures are further tested for photocatalytic activity under UV and artificial sunlight towards the most common organic dyes such as MO, MR, and CR (formulas given in **Figure 5.1**) used in the textile industry and major contaminants in wastewater.^{279,280}

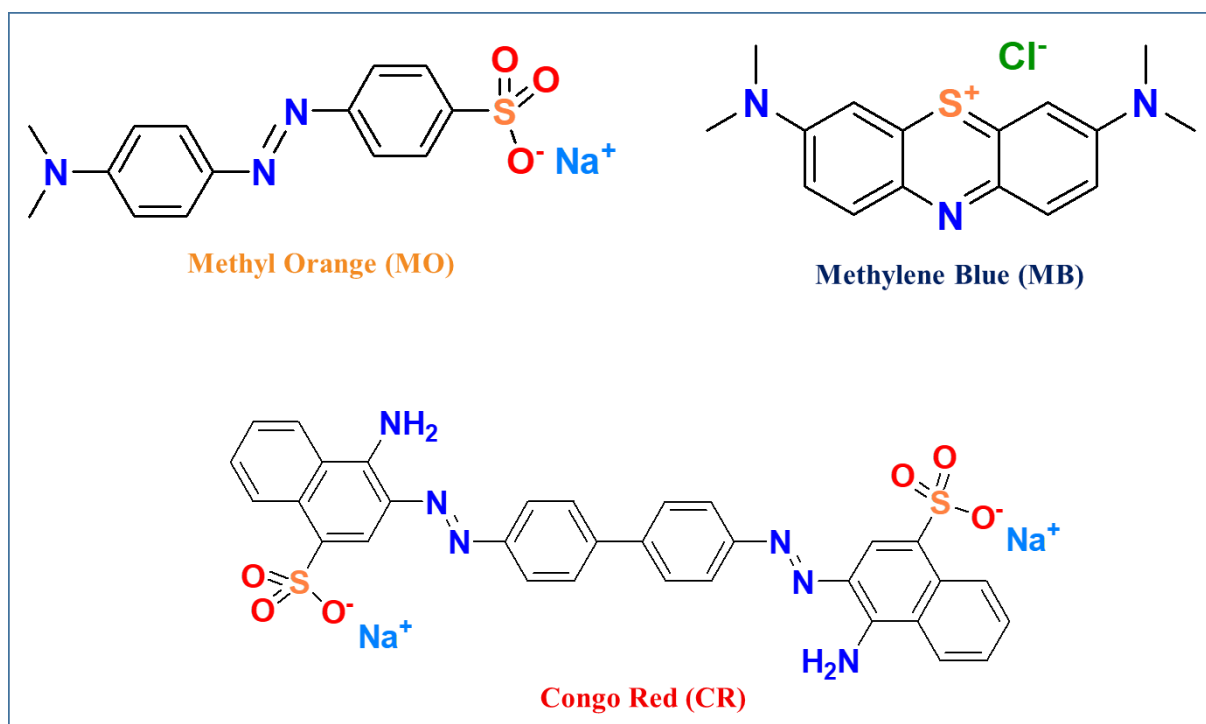


Figure 5.1 The molecular structures of the dyes used for the photocatalytic activity of pure and doped ZnO nanostructures

5.3 Existing issues

TiO_2 and ZnO have commonly investigated photocatalysts for dye degradation due to their non-toxic nature, abundant availability, high reactivity. However, they absorb light in the near UV region, only <5% of our solar spectrum.

Consideration of dopant's chemical nature to improve the photocatalytic properties are less explored especially correlating between photocatalytic activity and dopant chemical hardness.

5.4 Objectives & Action plan

Hydrothermal synthesis of divalent (Ni^{2+} and Sr^{2+}) and trivalent (Cr^{3+} and Al^{3+}) doped ZnO nanostructures were studied.

Study the photocatalytic activity of undoped and doped ZnO nanostructures towards the degradation of methyl orange (MO), methylene blue (MB), and congo red (CR).

We compared their results to derive the trend of chemical hardness with respect to photocatalytic dye degradation.

5.5 Materials and Methods

5.5.1 Materials used

$\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, $\text{Cr}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, $\text{Sr}(\text{NO}_3)_2 \cdot 9\text{H}_2\text{O}$, NaOH (Merck, South Africa), of analytical grades were used as received without any further treatment. Also, the methyl orange (MO), methylene blue (MB), and congo red (CR) were procured from Merck and used without any further treatment.

5.5.2 Synthesis of ZnO nanoparticles

ZnO nanoparticles are synthesized by a hydrothermal method. In a typical synthesis of undoped ZnO nanoparticles, 1.25 mmol of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (374.30 mg) and 0.5 g of NaOH were dissolved in 20 mL Milli-Q water. First, the above solution is transferred into a 25 mL Teflon container. Then, the Teflon container is sealed inside a stainless-steel autoclave and placed in a hot air oven at 180 °C for 18 h. After completing the reaction, the reactor was allowed to cool to room temperature, and then the precipitate was then separated and washed thoroughly with water and ethanol. The final compound was dried at 70 °C for overnight²⁸¹.

To synthesize doped ZnO nanoparticles, 5 mol% of corresponding metal nitrates are used, keeping everything else similar as per the above procedure.

5.5.3 Dye-degradation test

In a typical photocatalytic kinetic study, 3.0 mg of photocatalyst (ZnO, Ni-ZnO, Sr-ZnO, Cr-ZnO, and Al-ZnO) was added in 3.0 mL of 36.0 $\mu\text{mol/L}$ aqueous solutions of MO. The formed suspension was well dispersed by sonication for 5 minutes and allowed to stir to mix catalyst and dye molecules. Then, the mixture of MO and catalyst was irradiated with 360 nm UV radiation inside a UV fluorescence cabinet of 220 volts and 50 Hz under constant stirring conditions. The distance of the quartz vessel from the UV lamp was 15 cm. Samples were collected in the regular interval of 10 minutes, and the change in the concentration of organic dye molecules was examined by recording the UV-Vis spectra in Cary 100 Bio UV-Vis spectrophotometer. Before measuring UV-Vis spectra, the collected samples were centrifuged at 5000 rpm for 2 minutes to separate the catalyst from the dye solution.^{234,282,283} We have also carried out a similar dye-degradation test with MB and CR dyes.

5.5.4 Characterization

All the samples were characterized by powder x-ray diffraction (PANalytical Empyrean) with Cu-K α radiation for phase identification. Field emission scanning electron microscopy (FESEM) of the Carl-Zeiss model (Zeiss ultra 55) was used for the morphology study. The absorption studies were conducted using (Cary 100 Bio UV-Vis spectrophotometer). Dopant concentration was characterized by an inductively coupled plasma atomic emission spectrometer (ICP-AES) using ARCOS, a simultaneous ICP spectrometer.

5.6 Results and Discussion

Hydrothermal synthesis is employed in basic conditions to prepare pure as well as doped ZnO nanostructures. We have characterized as-prepared samples by scanning electron microscope (SEM) and powder X-ray diffraction (XRD). **Figure 5.2** shows powder XRD patterns of pure ZnO and Sr²⁺, Cr³⁺, Ni²⁺, and Al³⁺ doped ZnO nanostructures. All reflections of pure and doped ZnO nanostructures are nearly identical and characteristic of the hexagonal wurtzite structure of ZnO with the space group of P6₃mc.

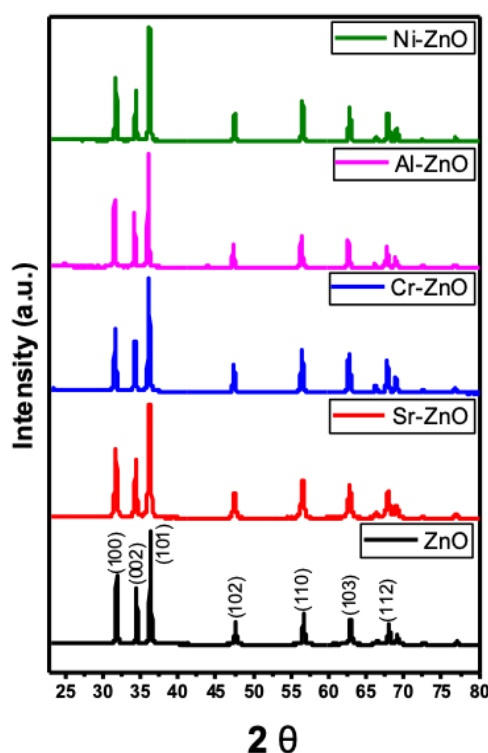


Figure 5.2 XRD patterns of undoped as well as Sr²⁺, Cr³⁺, Al³⁺, and Ni²⁺ doped ZnO nanostructure

SEM images of undoped, as well as Sr^{2+} and Ni^{2+} doped ZnO, are shown in **Figure 5.3(a, b, and c)**, respectively, revealing nanorods morphologies and lengths in the range of 0.48-3.21, 0.43-3.49, and 0.15-1.06 microns and diameters in the range of 0.073-0.293, 0.093-0.528, and 0.025-0.109 microns. **Figure 5.3(d and e)** show SEM images of Cr^{3+} and Al^{3+} doped ZnO nanostructures with nanoparticle/small nanosheet morphologies having sizes in the range of 0.035-0.353 and 0.049-0.474 microns, respectively.

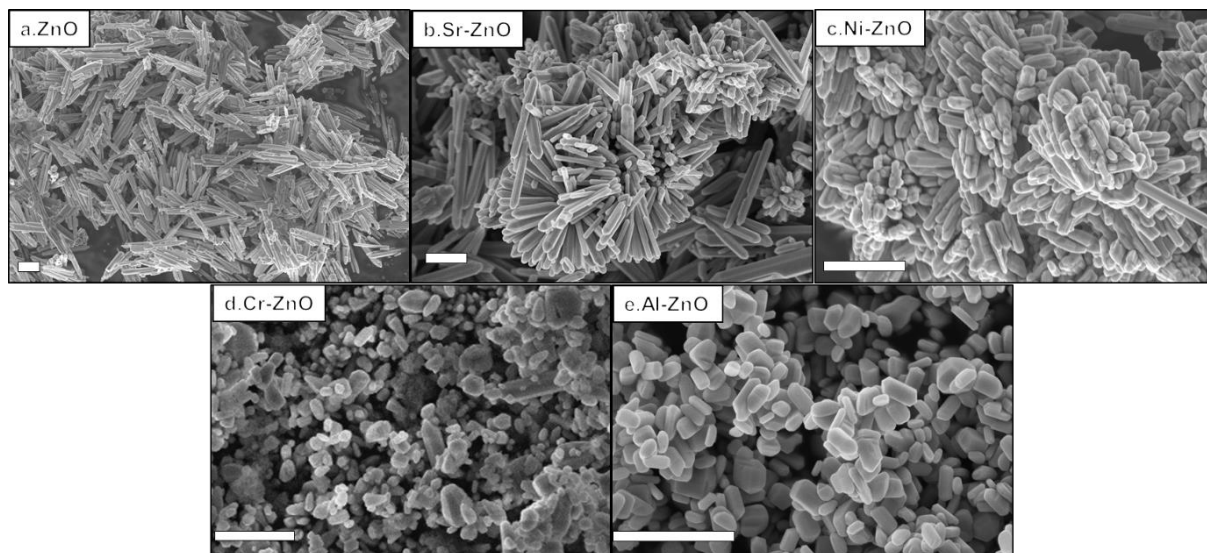


Figure 5.3 FE-SEM images of undoped as well as Sr^{2+} , Ni^{2+} , Cr^{3+} , and Al^{3+} doped ZnO nanostructures. The scale bars of the respective images indicate 1 μm

SEM images of Al and Cr-doped ZnO nanostructures show less agglomeration behavior than undoped and Sr and Ni-doped ZnO nanostructures. Between Al and Cr-doped ZnO nanostructures, Al-doped ZnO shows a less agglomeration tendency. The Al-doped ZnO nanoparticles have also shown better dispersion properties in water than pristine, Sr, and Cr doped ZnO nanostructures, which show sedimentation after 12 h, whereas part of the Al-doped ZnO nanoparticles remains in the suspension. Fewer agglomeration properties of Al-doped ZnO nanostructures could be attributed to surface charges in n-type Al-doped ZnO nanostructures.²⁸⁴ The charge carriers would remain at the surface of the nanoparticles eliciting mutual repulsion. Similar experiments of Re-doped MoS_2 have shown fewer agglomeration properties, leading to a notable effect on their rheological properties.²⁸⁵ Inductive coupled plasma-atomic emission spectroscopy (ICP-AES) estimates the dopant concentration in ZnO nanostructures. The ICP-AES data reveals 4.65 and 4.16 at % doping of Sr and Ni in ZnO nanorods. In contrast, the amount of incorporation of trivalent dopants in ZnO nanostructures is minor (0.34 at.% of Al^{3+} and 0.36 at.% of Cr^{3+}).

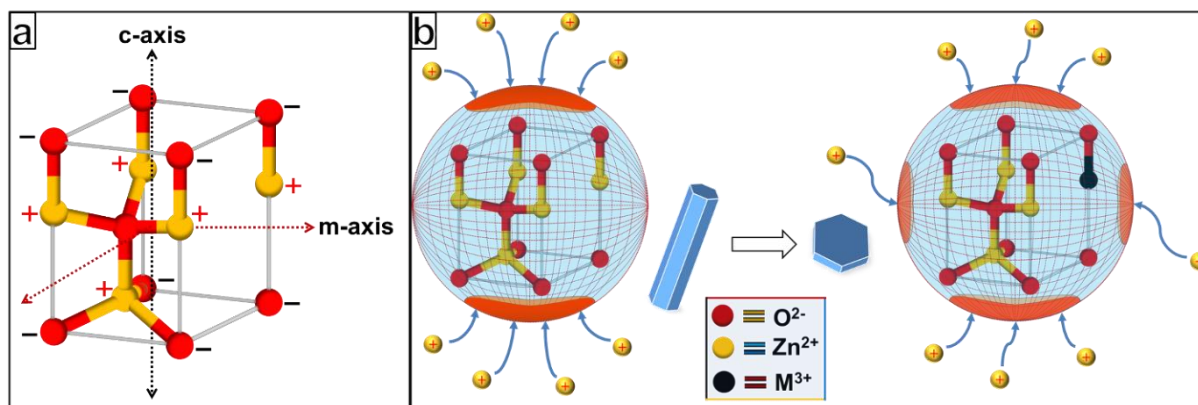


Figure 5.4 (a) Schematic of wurtzite structure, (b) schematic of polarity-cloud dispersion by doping (increasing intensity of the red color indicates an increase in polarity)

As presented above, pristine ZnO and Sr and Ni-doped ZnO form nanorods. On the contrary, ZnO crystallizes into small nanosheets in the presence of 3+ dopants. Several methods have been reported to change the morphology of ZnO nanostructures, including pH and capping agents, etc.^{286,287}

Table 5.1 Some of the relevant photocatalytic dye degradation of doped-ZnO nanostructures reported in the literature

SL No.	System	Dye	Morphology	Size	Optical region	Rate (min ⁻¹)	Ref.
1	Al-ZnO	MO	Films	1.95-2.24 μm thickness	UV	0.0117	244
2	Eu-ZnO	MO	NPs	5.67 nm	UV	0.01514	246
3	Mg-ZnO	RhB	NPs	30-110 nm	UV	0.0126	247
4	Mn-ZnO	MB	NPs	6-7 nm	Vis	0.0197	249
5	Fe-ZnO	2-chlorophenol	NPs	16-14 nm	UV	0.0263	251
6	Ce-ZnO	MB	Nanorods	900nm-3 μm in L, 70-85 nm in D	UV	0.044	252
7	Sr-ZnO	MB	NPs	20-60 nm	Vis	0.1	256

8	Mn,Co-ZnO	MO	NPs	50-70 nm	UV	0.0139, 0.0087	288
9	Ce-ZnO	MB	Bottle-such as	~1 μm in D	UV	0.04	253
10	La-ZnO	Monocrotophos MCP	NPs	100-500 nm	UV	0.1	257
11	Nd-ZnO	MO	Powders	100-200 nm	UV	0.15	260
12	Bi-ZnO	MO	irregular villus-such as	0.5 μm	UV	0.01	261
13	Mg-ZnO	MO	NPs	60-70 nm	UV	0.02	248
14	Hf-ZnO	MB	NPs	17-14 nm	Vis	0.015	262
15	Ni-Co-ZnO	RhB	NPs	16-30 nm	Vis	0.001	268
16	Sm-ZnO	Phenol	Nanorod	43–103 nm	Vis	0.004	264
17	Pd-ZnO	MO	NPs	25-33 nm	Vis	0.01	265
18	Se-ZnO	Trypan blue	NPs	9.5 nm	UV	0.0046	266

*NPs is nanoparticles, *L: Length, and *D: Diameter

However, in the present case, the type of dopant seems to affect the morphology while keeping the rest of the same reaction conditions. Therefore, we propose the following mechanism for morphology change in trivalent doped ZnO nanostructures. ZnO crystallizes in Wurtzite structure, having polar *c*-axis and non-polar *ab*-plane (see **Figure 5.4(a)**). In the growth stage (in the solution-based nucleation and growth mechanism), the newly formed $\text{Zn}(\text{OH})_4^{2-}$ nuclei predominantly attach to the *c*-axis of the already existing ZnO cluster in the solution because of the high polarity along this direction. This is one of the main reasons for ZnO to form nanorod morphology, even without using any surfactants.²⁸⁹

In **Figure 5.4(b)**, we show a schematic relative polarity index around the wurtzite structure. Here, the intensity of the red color represents the magnitude of polarity. As the schematic indicates, ZnO has a polar *c*-axis (indicated with red color) and a non-polar *ab*-plane. When Zn^{2+} is replaced with a 2+ cation, there is no change in polarity in the structure. On the contrary,

when Zn^{2+} is replaced with a $3+$ cation (such as Al^{3+} and Cr^{3+}), polarity is also induced in the *ab*-plane. Thus, newly formed $\text{Zn}(\text{OH})_4^{2-}$ nuclei can approach already existing ZnO clusters in the solution from any direction, leading to the formation of more isotropic morphology. Hence, the formation of small nanosheets or spherical nanoparticles is formed in the case of trivalent doping. Choice of counter ion of metal precursors can also affect the morphology of ZnO nanostructures.²³² In this case, all the metal precursors are nitrates; thus, the possibility of a counter ion effect can be eliminated. Another possibility could be that $3+$ dopant chemical species can adsorb on the polar (001) plane of ZnO and inhibit adsorption of $\text{Zn}(\text{OH})_4^{2-}$ on the (001) plane and slow down the growth of ZnO in this direction. As the percentage of dopant is significantly less, we believe the contribution of this effect is minimal.

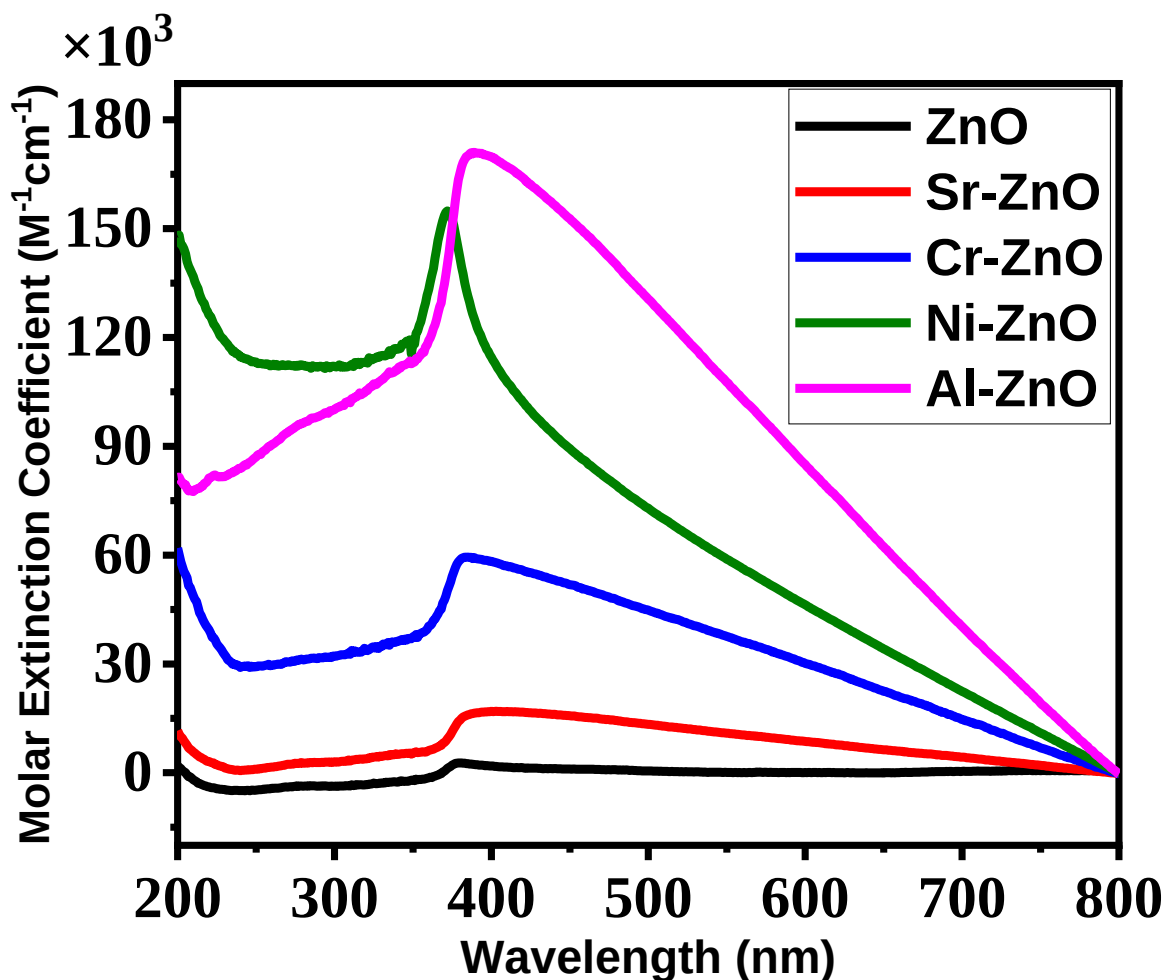


Figure 5.5 Absorption spectra of the pure and doped ZnO nanostructures

Figure 5.5 shows UV-Vis absorption spectra of pristine and doped ZnO nanostructures. The calculated molar absorption coefficients of these nanostructures are 2.7, 16.9, 61.9, 154, 171 $\text{M}^{-1}\text{cm}^{-1}$ for ZnO, Sr^{2+} , Cr^{3+} , Ni^{2+} , and Al^{3+} -doped ZnO nanostructures, respectively. The

higher absorption coefficient of Al^{3+} -doped ZnO nanostructures could be associated with the higher dispersion ability of these nanostructures in water, as described earlier. In addition, the small size of Ni-ZnO nanostructures also shows a high absorption coefficient with an excitonic- such as peak (**Figure 5.5**).

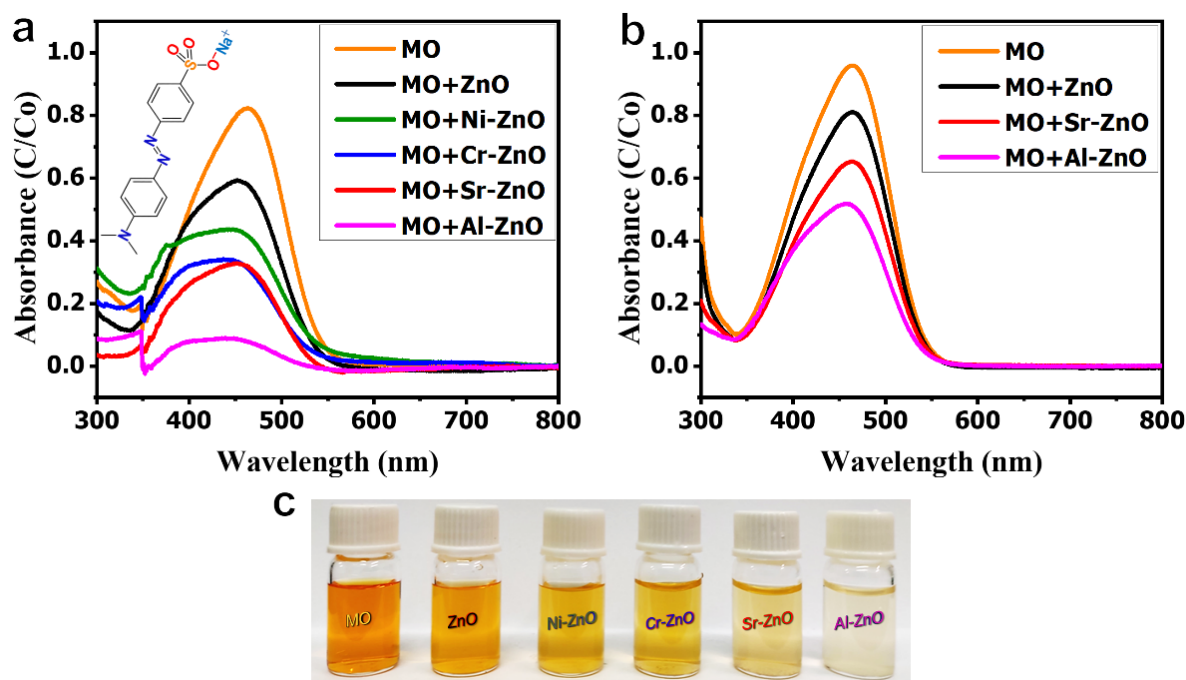


Figure 5.6 Changes in absorption spectra of MO with and without photocatalyst after 60 minutes (a) under UV-light irradiation and (b) under visible-light irradiation. (c) Optical images of MO-degradation with and without photocatalyst under UV-light

Higher absorption coefficient and dispersibility are important properties for a photocatalyst to degrade the dyes in the solution. Photocatalytic activities of pristine ZnO and doped ZnO nanostructures were studied in the case of MO by utilizing the changes in the electronic absorption spectra of the dye. In **Figure 5.6(a)**, we present the typical changes in absorbance of MO on irradiation with and without a photocatalyst. The change in the concentration (C/C_0) of MO with reaction time has been examined and presented in **Figure 5.7**.

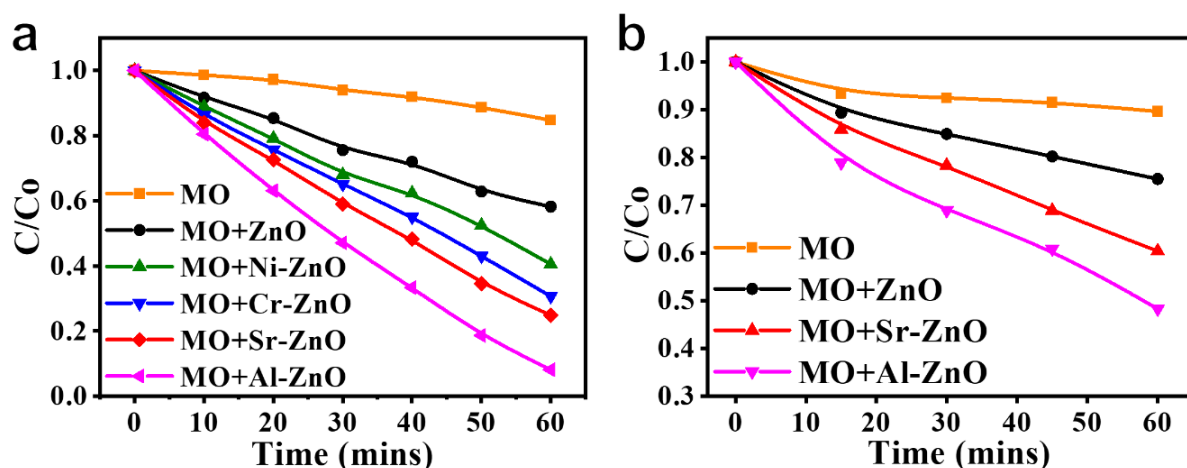


Figure 5.7 The change of normalized concentration with the exposure time for the photocatalytic degradation of MO with and without catalyst (a) under UV-light irradiation and (b) under visible-light irradiation

We observe that the degradation rate varies as $\text{Al-ZnO} > \text{Sr-ZnO} > \text{Cr-ZnO} > \text{Ni-ZnO} > \text{ZnO}$. The value of the rate constant, k_a from the slope of $\ln(C_0/C)$ vs time for ZnO, Ni-ZnO, Cr-ZnO, Sr-ZnO, Al-ZnO are 0.0091, 0.0143, 0.0187, 0.0227, 0.0396 min^{-1} , respectively. The rate constant values obtained here are comparable to the literature values.^{290,291} In **Figure 5.6(c)**, we present photographs showing color variation of dye solutions after 60 min under UV light. The photographs exhibit the trend and show better efficiency of Al-ZnO nanostructures. In **Figure 5.6(b)**, we present typical changes in the absorbance of MO on irradiation of artificial sunlight for 60 min with ZnO, Sr-ZnO, and Al-ZnO nanostructures. We observe a similar degradation trend that varies as $\text{Al-ZnO} > \text{Sr-ZnO} > \text{ZnO}$. However, photocatalyst under UV light is slightly better than under artificial sun. This could be due to bandgap excitation of photocatalysts (~ 3.37 eV) under UV-light that generate more electrons and holes, which are further responsible for dye degradation.

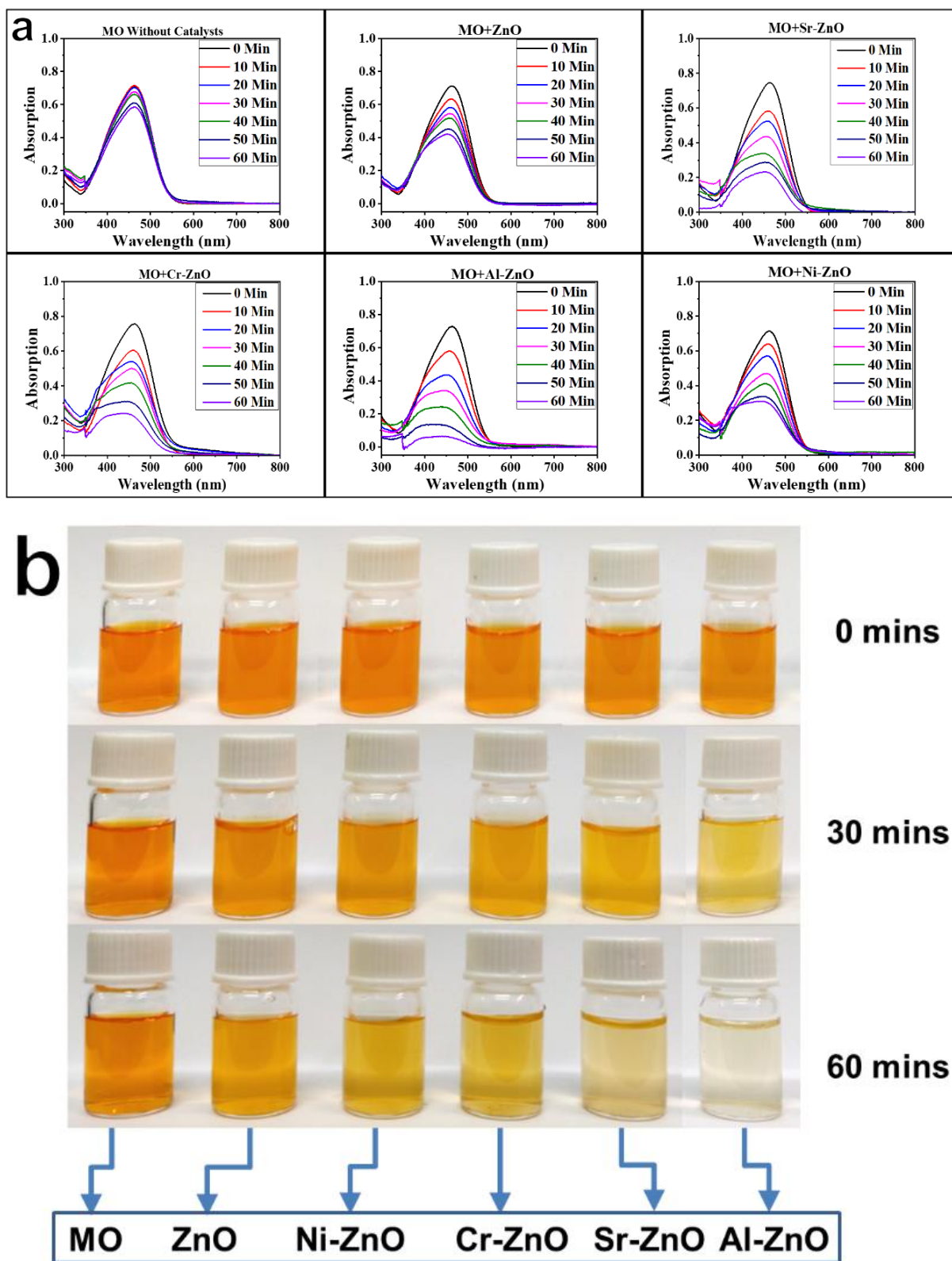


Figure 5.8 (a) Absorption plots of MO dye degradation at different time intervals under UV light, in the presence and absence of photocatalyst. (b) Optical images of MO dye degradation at (0 mins, 30 mins, and 60 mins) under UV light, in the presence and absence of photocatalyst

Figure 5.8(a and b) show absorption spectra and optical images of MO degradation using ZnO, Sr-ZnO, and Al-ZnO nanostructures at different time intervals. The MB and CR degradation studies have also shown similar photocatalytic activity trends, which can be seen from the absorption spectra (**Figure 5.9(a and b)**) and the optical images **Figure 5.9(c and d)**. These catalysts were very robust to be used for multiple cycles of dye degradation studies. After every cycle, a couple of ultrasonication cleaning followed by Milli-Q water washing removes the surface deposited dyes, and the activity was fully re-established. We have re-used all the catalysts for seven cycles without reduction in photocatalytic activity.

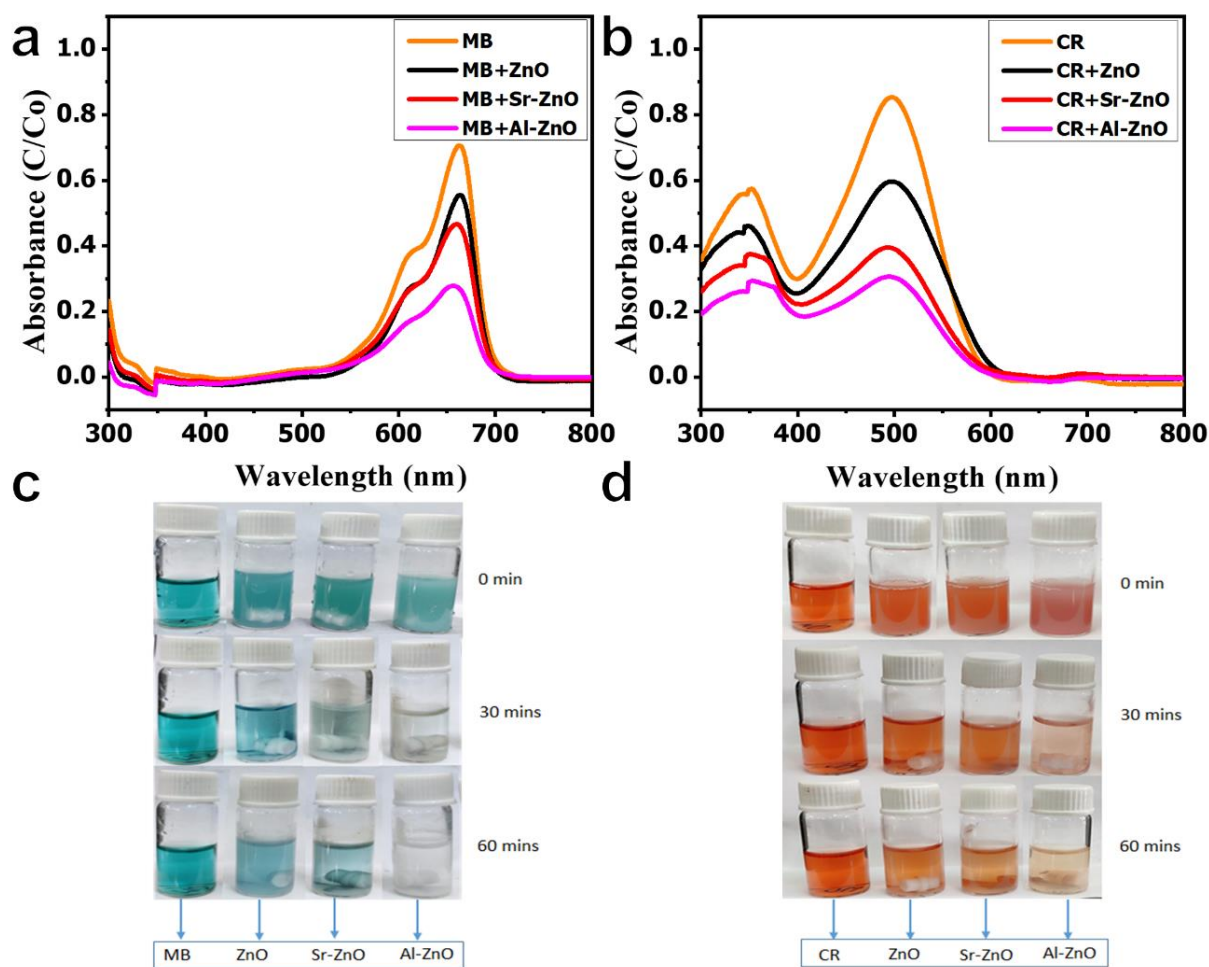


Figure 5.9 (a) and (b) are the absorption spectrum of MB and CR with and without photocatalyst treated under UV-light for 60 mins, respectively. (c) and (d) are the optical images of MB and CR degradation with and without catalyst under UV-light with time, respectively

Dispersion of a photocatalyst in the water and absorption coefficient are important to improve the dye degradation process. Nevertheless, it is interesting to note that the catalytic activity trend does not match the absorption coefficient trend presented in **Figure 5.5**. For

instance, Ni-ZnO and Cr-ZnO nanostructures show better absorption coefficients than Sr-ZnO, but the photocatalytic activity of Sr-ZnO is superior to Ni-ZnO and Cr-ZnO nanostructures. We propose that besides dispersibility and absorption coefficient, the adsorption capability of O₂, H₂O, and dye onto the photocatalyst is an important property so that electron transfer between photocatalyst and adsorbed molecules is efficient. As Sr²⁺ is harder than Ni²⁺ and Cr³⁺, it has a higher tendency to adsorb chemically hard molecules such as H₂O and O₂, subsequently dye degradation process.

The Pearson hardness values of Al > Sr > Cr > Ni > Zn.²⁹² The trend is similar to the trend of photocatalytic activity of different photocatalysts (Al-ZnO > Sr-ZnO > Cr-ZnO > Ni-ZnO > ZnO). Isovalent doping (Sr²⁺ and Ni²⁺) has shown similar morphology and doping concentration in ZnO nanostructures. Similarly, aliovalent doping (Al³⁺ and Cr³⁺) has also shown similar morphology and doping concentrations in ZnO nanostructures. Thus, we believe that comparing the isovalent and aliovalent doped ZnO nanostructures is more reasonable. Even in this case, dye degradation trends are following the chemical hardness trend (Al-ZnO > Cr-ZnO > ZnO and Sr-ZnO > Ni-ZnO > ZnO).

5.7 Conclusions

We have reported isovalent (Sr²⁺ and Ni²⁺) and aliovalent doping (Al³⁺ and Cr³⁺) in ZnO nanostructures using hydrothermal synthesis. The incorporation of isovalent ions is found to be high (~at 4%) compared to the aliovalent dopants (0.3-0.4 at%). Aliovalent doping also tunes the morphology of ZnO nanostructure from nanorods to nanoparticles. The polarity-induced morphology change mechanism is proposed in the case of aliovalent doping. The photocatalytic activity studies of undoped and doped ZnO nanostructures towards MO, MB, and CR demonstrate that catalytic activity is higher in the case of ZnO nanostructures doped with high chemicals hardness elements such as Al³⁺ and Sr²⁺. Further, Al-doped ZnO nanostructures show less agglomeration and high dispersion properties in the water, which is an asset for using these structures for making better composites. We believe the present study contributes to the development of various dye degradation systems and could help in designing materials with a more rational way of improving photocatalytic properties in the future.

CHAPTER-6:

Conclusion and Future Scope

Conclusion

Our experiments with low-temperature synthesis and morphology control of LPOs yielded promising results. We discovered the optimal conditions for synthesising LCrO, LKMO, and LFO phases in the SH method. We discovered that diffusion and byproduct dissolution limit the kinetics of the formation of these materials, particularly LKMO. The addition of precursor seeds was found to allow the limited-simultaneous available reagents to be distributed uniformly throughout the reaction volume, which as a result, regulated the rapid growth of primary nuclei leading to the formation of secondary nuclei and, ultimately, nanocubes or submicron cubes. The same phenomenon allowed for the rapid dissolution of by-products according to Le-Chatelier's principle, resulting in nearly 80 per cent reaction completion in just two days, which would otherwise have taken seven days. Variation in KOH concentration, particularly at 9 M, 12 M, and 18 M, had a significant impact on the surface morphology and yield of the LKMO system. The discussion of Tc being inversely proportional to K per cent demonstrated the significance of potassium substitution in the LMO system.

Another way to improve the kinetics of the system was found to be an incorporation of stirring into the synthesis, commonly known as the dynamic hydrothermal (DH) approach. Thanks to stirring, we could obtain LKMO cubes in less than 6 hours. Using a custom-made reactor to implement a new synthesis technique yielded exciting results in this class of materials were demonstrated using this DH method to obtain LKMO nanorods. We created hollow LKMO cubes by DH-treating pre-synthesized LKMO microcubes under NaOH conditions. The precursor concentration and KOH concentration variation resulted in 111-facet truncated LKMO cubes. Further, we also investigated the electrical properties of single rod two-probe devices made of LKMO nanorods. The profile fitting of XRD data from 12 M LKMO microcubes revealed that the phase is hexagonal (I4/mmm). This controlled method of synthesis of higher symmetry phases in oxide materials; a field that had received only a little attention to date; could be explored through this. The high-temperature continuous XRD measurement revealed a gradual shift in phase from hexagonal to pseudocubic.

The presence of potassium in the perovskite structure resulted in improved catalytic activity and superior stability for oxygen reduction and evolution reactions when compared to many previously reported perovskites and transition metal-based electrocatalysts. Because of Mn^{3+} - Mn^{4+} Zener pairs and abundant oxygen vacancies, the zinc-air battery with LKMO catalyst demonstrated lower round-trip peak potential and excellent stability over 1000 charge-

discharge cycles of continuous operation. When compared to LMO and LSMO perovskite-based catalysts, the robust structure and chemical inertness of synthesised LKMO electrocatalysts exhibited excellent catalytic activity. Apart from electrochemical activity, LKMO microcubes were demonstrated to be a promising candidate for replacing noble metals in organic catalysis. This remarkable catalytic efficiency was witnessed in the reduction of α,β -unsaturated compounds and nitroarenes where a conversion rate of 98% was obtained, which is significantly higher than that of existing materials.

Finally, isovalent (Sr^{2+} and Ni^{2+}) and aliovalent doped (Al^{3+} and Cr^{3+}) ZnO nanostructures were hydrothermally synthesized for photocatalytic degradation of MO, MB, and CR organic dyes under UV. We found that doping ZnO nanostructures with high chemical hardness elements like Al^{3+} and Sr^{2+} improved their photocatalytic activity. Aliovalent doping also influenced the morphology of ZnO nanostructures, ranging from nanorods to nanoparticles. Further, the mechanism of polarity-induced morphology change has also been proposed subsequently.

Future scope

This thesis covers a wide range of topics, including the low-temperature hydrothermal synthesis of potassium substituted perovskite manganite LKMO, chromite LCrO, ferrite LFO. To briefly summarize, the key outcomes of this thesis included; (a) The effect of mineralizer (KOH) concentration on the stabilization of the critical phase in addition to time variation, (b) The importance of pre-synthesized microcubes and STO nanoparticles as templates in improving secondary nucleation through the dynamic hydrothermal approach substantiated by the proposal of the shear jamming concept, (c) Demonstration of electrical conductivity through the fabrication of two-probe LKMO nanorod devices, (d) Estimation of catalytic efficiency of hydrothermally synthesized oxides in zinc-air battery and pharmaceutically important processes, (e) Hydrothermal synthesis of ZnO nanoparticles doped with chemically hard elements and demonstration of photocatalytic activity in addition to the proposal of a polarity-induced morphology tuning mechanism.

Even though all these topics have been discussed in the previous sections, there are a few directions that have not been covered under the scope of this thesis and could be investigated in future:

In **Chapter 2**, the effect of KOH concentration variation on LKMO formation was investigated. The XRD refinement showed that the 12 m case led to a higher symmetry phase

of LKMO than the 18 M case. The crystal structure information of this 12 M phase promises intriguing insights. This could explain how 12 M could stabilize high symmetry, and one can further understand the energetics of this system by doing DFT calculations. In the 12 M cases, the Rietveld refinement analysis of XRD could not provide an exact location of K. Thus, one could do neutron diffraction (ND) to precisely determine atomic positions and thus the crystal structure information. This high symmetry LKMO phase may have magnetocaloric activity, which can be investigated by performing M vs. H magnetic measurements in the VSM-PPMS facility. Because of their high crystallinity, these hydrothermally synthesized LKMO microcubes may exhibit improved magnetoresistance activity. Also, by measuring the I-V sweep for 12 M and 18 M cubes before and after annealing at 850 °C, one could investigate the effect of oxygen vacancies on the mechanism of change of electrical resistance of the LKMO. The multiferroic nature of LKMO may be investigated by measuring ferroelectric and dielectric properties using broadband dielectric spectroscopy (BDS). However, this may involve the need to compress the sample into a pellet; thus making it mandatory to have a good quantitative yield of the material.

Additionally, demonstrates that SH can achieve monovalent substitution of perovskites. This has been demonstrated successfully by the synthesis of K-substituted lanthanum manganite, LKMO. However, a similar monovalent substitution in other transition metal perovskites, such as LaCrO_3 and LaFeO_3 , has not been investigated in this chapter. So, for future scientists who take over, this could be a very good future direction. As a result, the effect of monovalent substitution on crystal structure, electrical-magnetic properties, and catalytic activity could be investigated. This could be accomplished by further optimising the appropriate KOH environment required for individual phases, as KOH dictates individual phase stability.

Further, manganite, chromite, and ferrite perovskite oxide phases were successfully synthesised. It was also demonstrated that 9 M KOH is suitable for stabilising LaCrO_3 , while 18 M KOH is required for LaMnO_3 , and 12 M KOH is required for LaFeO_3 phase. However, the low-temperature hydrothermal synthesis of LaTiO_3 , LaVO_3 , LaCoO_3 , and LaNiO_3 systems was not addressed in this chapter. Obtaining these materials is critical because these have a variety of important functions such as titanate for transparent electrodes, vanadate for semiconductivity, and cobaltate and nickelate for catalytic activities. With the knowledge gained in this chapter, one could consider optimising the KOH concentration, the sequence of precursor addition, exposure to environmental oxygen, and mixing duration to achieve each of these phases in the SH method. In addition to the simple phases, achieving K-substituted

systems will provide enhanced features, opening a new horizon of applications. LaKCoO_3 , for example, may very well have higher electrocatalytic activity. Also, the 12 m case of LKMO resulted in nanorods. However, the synthesis of nanorods of other LPO materials is not covered in this chapter. By adjusting the reaction parameters, particularly the KOH concentration, chromite and ferrite phase nanorods can be obtained. With the availability of this morphology, researchers will be able to investigate the transport properties of these oxide perovskite systems. In the same chapter, the use of seed materials reduced reaction time and energy. Thus, one can think in the same direction for the synthesis of new phases. The nanopowders of the required phase can be obtained using the TMSS method, as described in the auxiliary section, and can be used as seeds in SH. This increases the likelihood of the new phases stabilising in the SH method. This will also result in higher crystallinity and yield.

In **chapter 3**, it was demonstrated how stirring enhanced the reaction kinetics of LKMO-like LPO synthesis. With this knowledge that constant stirring could affect nucleation and crystal growth, further investigations could be done on how the rate of stirring and the duration of stirring in intervals may make a difference in the morphology of the materials, thus affecting nucleation and growth. As a step ahead, since DH has been shown to improve reaction kinetics, this approach could be used to stabilise challenging phases like nickelate and cobaltate which may require long reaction hours. Perhaps using the TMSS method to obtain these phases and using them as seeds to support the reaction further can be considered. This could lead to the discovery of new morphologies, such as useful nanorods. Also, it is evident from this work that La(OH)_3 rods, which are the by-products of the formation of lanthanum-based perovskites, have long and distinctly separated grooves in DH. This was not explored further in this chapter. One could use them as a template to synthesize assembled nanoparticles.

Several perovskite oxides, including SrTiO_3 (STO), have been discovered in the literature using the hydrothermal method. It is popular for a variety of reasons. The lower lattice mismatch with the other oxide perovskites is one of them. It is well known that STO creates functionally interesting interfaces with LPOs. These interfaces are known to exhibit novel properties and to be the source of several new features. Thus, one could consider synthesising STO on top of these LKMO nanorods to achieve a core/shell morphology. HRTEM can be used to characterise this and use it to study interface transport properties by fabricating two- and four-probe devices.

From **Chapter 4**, it is clear that the 18 m LKMO was a stable and bifunctional electrocatalyst. However, the electrocatalytic activity of the 12 m case has not been investigated. Thus, in a similar manner, one could investigate the electrocatalytic activity of

these 12 m LKMO cubes as well as the nanorods obtained from the DH method. Because this phase has a lower K percent, it may be a better catalyst than the 18 m case. In addition, the bifunctional activity of a series of double perovskites synthesised using the DH method, which was not discussed in this chapter, can be investigated. To begin, the electrocatalytic activity of NiMn double perovskite in an electrolyser can be investigated. In the same **Chapter 4**, LKMO was proven to be an excellent reduction catalyst. However, different facet specific activity in similar organic reduction can be investigated further, which has not been done in this chapter.

Chapter 5 investigated the effect of dopant chemical hardness on the photocatalytic activity and morphology of wurtzite structured metal oxide nanoparticles. This method could be further extended to other hexagonally structured functional materials, such as GaN. This method allows one to synthesize important materials like GaN in specific morphologies with controlled shape and sizes.

In general, perovskite oxides are cost-effective and abundant. If synthesised in a controlled manner, they have the potential to replace noble metals in the field of catalysis. Understanding their formation mechanism reveals new information. Because solution-based synthesis of these materials has not been extensively studied, there is a large scope for future research.

Appendix

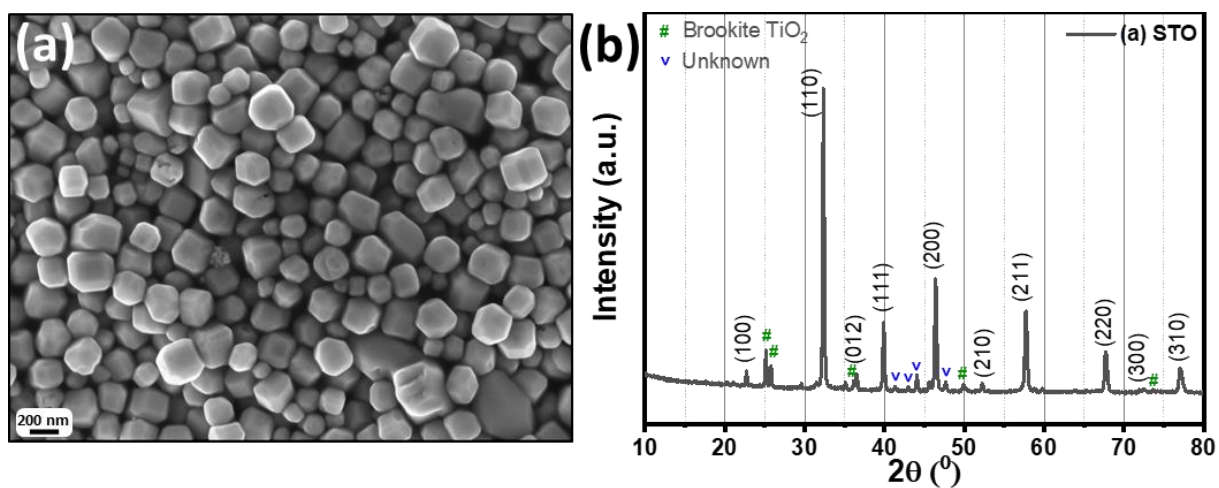


Figure 0.1 (a) SEM image and (b) XRD pattern of SrTiO₃ nanoparticles.

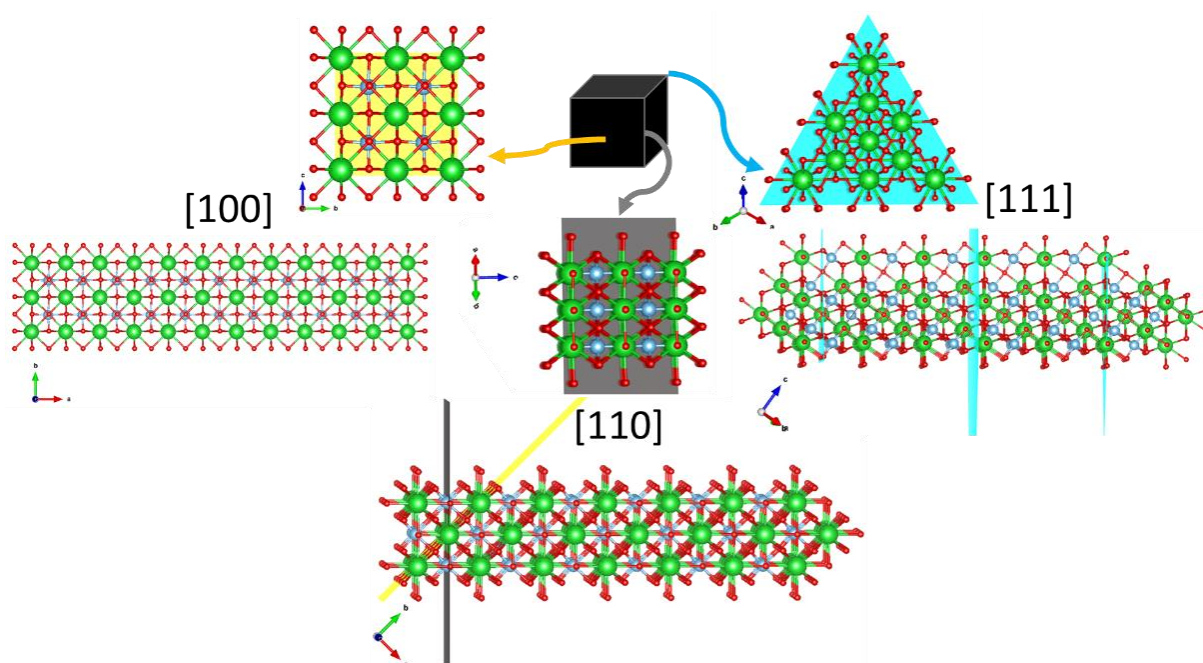
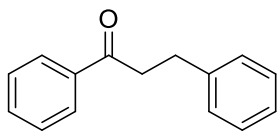
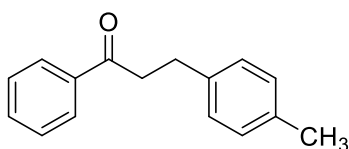


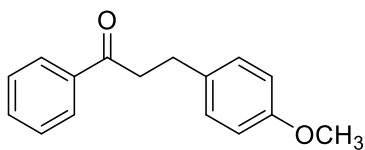
Figure 0.2 Top and lateral view of atomic arrangement along face [1 0 0], edge [1 1 0], and corner [1 1 1] of LKMO (R- $\bar{3}c$ -167) microcube.

Spectral data:

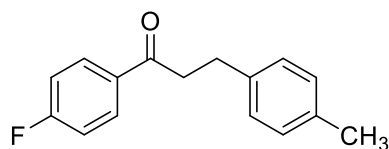
1,3-diphenylpropan-1-one (1a): Colorless liquid; ^1H NMR (400 MHz, CDCl_3): δ = 7.98 (d, J = 7.45 Hz, 2H), 7.57 (t, J = 7.39 Hz, 1H), 7.46 (t, J = 7.39 Hz, 2H), 7.32-7.20 (m, 5H), 3.31 (t, J = 8.01 Hz, 2H) and 3.08 (t, J = 8.01 Hz, 2H); ^{13}C NMR (100 MHz, CDCl_3): δ = 199.31, 141.40, 136.97, 133.15, 128.70, 128.63, 128.53, 128.14, 126.24, 40.54 and 30.23.



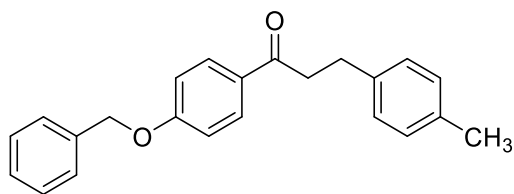
1-phenyl-3-(p-tolyl)propan-1-one (1b): Colorless liquid; ^1H NMR (400 MHz, CDCl_3): δ = 7.96 (d, J = 7.49 Hz, 2H), 7.55 (t, J = 7.49 Hz, 1H), 7.45 (t, J = 7.67 Hz, 2H), 7.13 (q, J = 16 Hz, 8.15 Hz, 4H), 3.28 (t, J = 7.70 Hz, 2H), 3.30 (t, J = 7.70 Hz, 2H), 2.32 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3): δ = 199.48, 138.32, 137.01, 135.76, 133.16, 129.33, 128.72, 128.42, 128.17, 40.75, 29.84, 21.13.



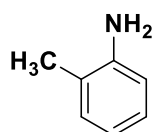
3-(4-methoxyphenyl)-1-phenylpropan-1-one (1c): Colorless liquid; ^1H NMR (400 MHz, CDCl_3): δ = 7.95 (d, J = 7.5 Hz, 2H), 7.55 (t, J = 7.38 Hz, 1H), 7.45 (t, J = 7.72, 2H), 7.17 (d, J = 8.56 Hz, 2H), 6.84 (d, J = 8.56 Hz, 2H), 3.78 (s, 3H), 3.27 (t, J = 7.85 Hz, 2H), 3.01 (t, J = 7.85 Hz, 2H); ^{13}C NMR (100 MHz, CDCl_3): δ = 199.55, 158.10, 137.01, 133.44, 133.17, 129.48, 128.72, 128.17, 114.06, 55.40, 40.85, 29.40.



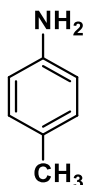
1-(4-fluorophenyl)-3-(p-tolyl)propan-1-one (1d): Pale yellow color liquid; ^1H NMR (400 MHz, CDCl_3): δ = 7.98 (J = 8.88, 5.48, 2H), 7.16-7.09 (m, 6H), 3.25 (t, J = 8.09 Hz, 2H), 3.03 (t, J = 7.3 Hz, 2H), 2.33 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3): δ = 197.83, 167.10, 164.57, 138.16, 135.82, 133.48, 133.46, 130.82, 130.73, 129.35, 128.40, 115.90, 115.68, 40.64, 29.82, 21.11.



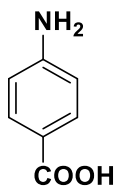
1-(4-(benzyloxy)phenyl)-3-(p-tolyl)propan-1-one (1e): Colourless solid; ^1H NMR (400 MHz, CDCl_3): δ = 7.94 (d, J = 8.5 Hz, 2H), 7.49-7.33 (m, 5H), 7.13 (q, J = 15.0 and 7.5 Hz, 4H), 7.00 (d, J = 8.5 Hz, 2H), 5.13 (s, 2H), 3.23 (t, J = 7.6 Hz, 2H), 3.02 (t, J = 7.5 Hz, 2H) and 2.32 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3): δ = 198.06, 162.71, 138.48, 136.33, 135.70, 130.45, 129.31, 128.83, 128.42, 128.37, 127.60, 114.71, 70.27, 40.42, 30.04, 21.13.



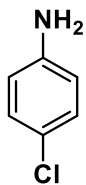
***o*-Toluidine (2a):** ^1H NMR (400 MHz, CDCl_3): δ = 7.23-7.11 (m, 2H), 6.55 (m, 1H), 6.78 (m, 1H), 3.69 (s, 2H, NH_2), 2.28 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3): δ = 144.72, 130.55, 127.09, 122.47, 118.71, 115.08, 17.5



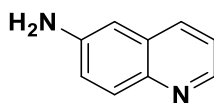
***p*-Toluidine (2b):** ^1H NMR (400 MHz, CDCl_3): δ = 7.00 (d, J = 8.02 Hz), 6.64 (d, J = 8.20 Hz), 3.55 (s, 2H, NH_2), 2.27 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3): δ = 143.81, 129.77, 127.82, 115.29, 20.47



4-Aminobenzoic acid (2c): ^1H NMR (400 MHz, CDCl_3): δ = 11.95 (d, J = 8.76 Hz), 6.54 (d, J = 8.69 Hz), 5.86 (s, 2H, NH_2); ^{13}C NMR (100 MHz, CDCl_3): δ = 167.97, 153.61, 131.70, 117.33, 113.25.

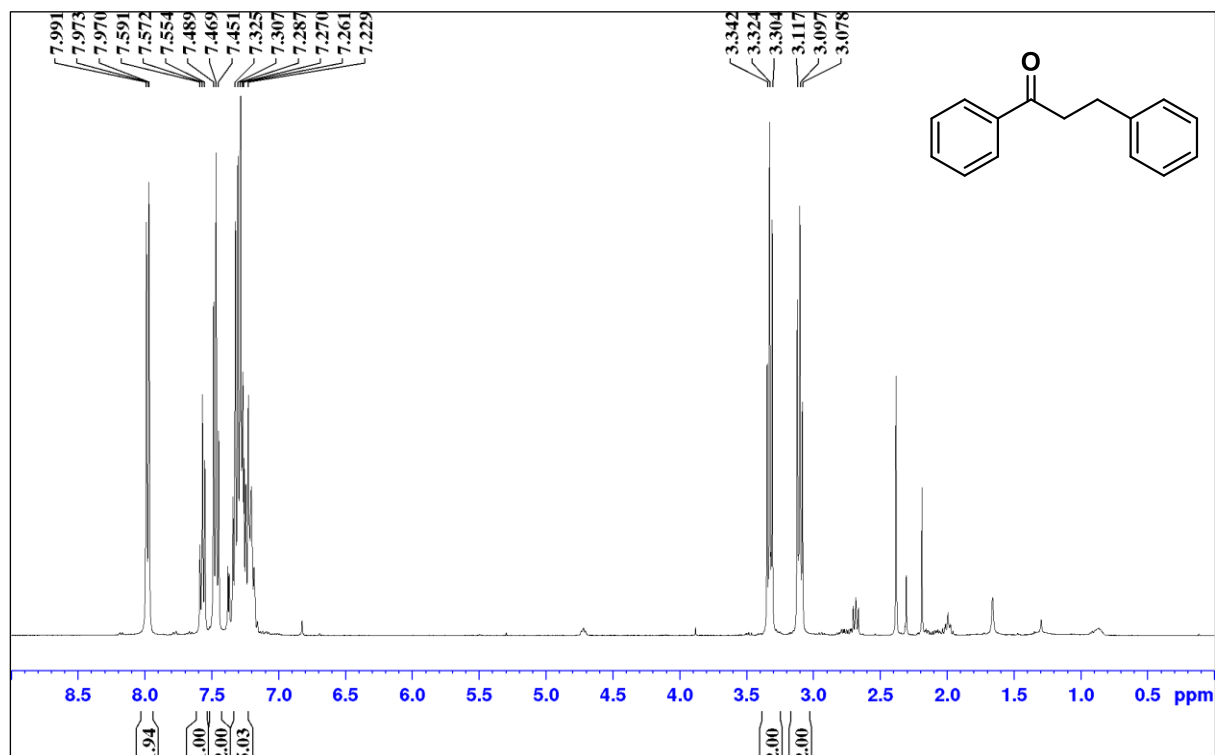


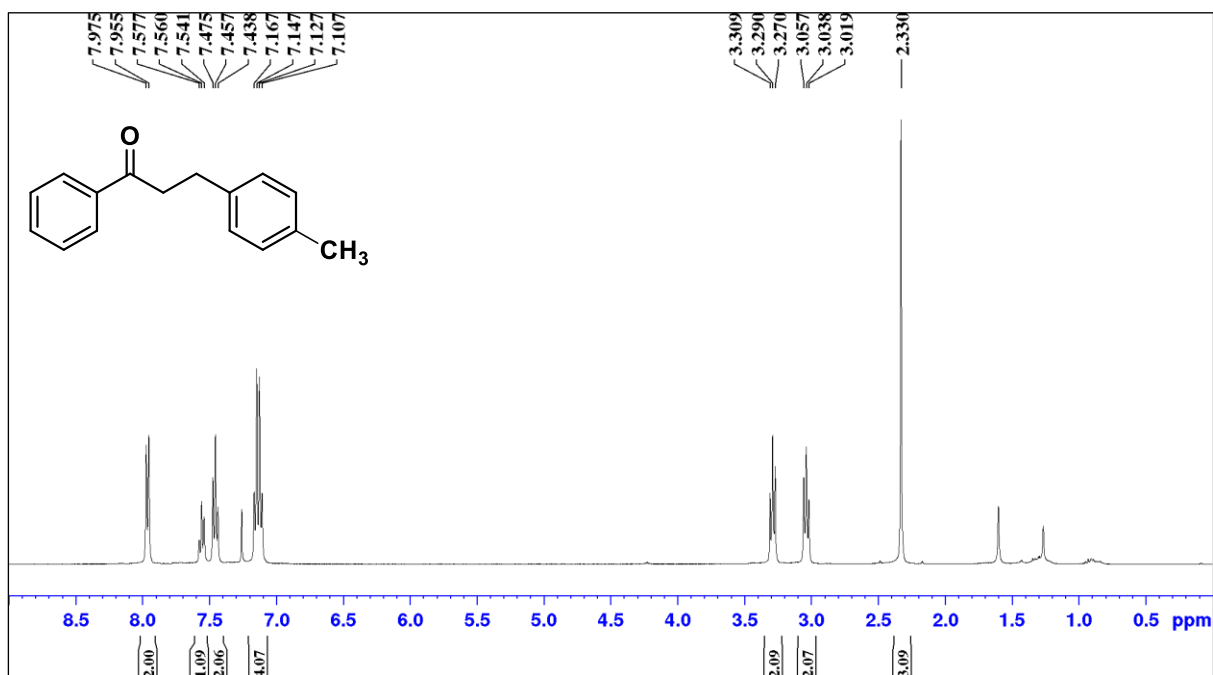
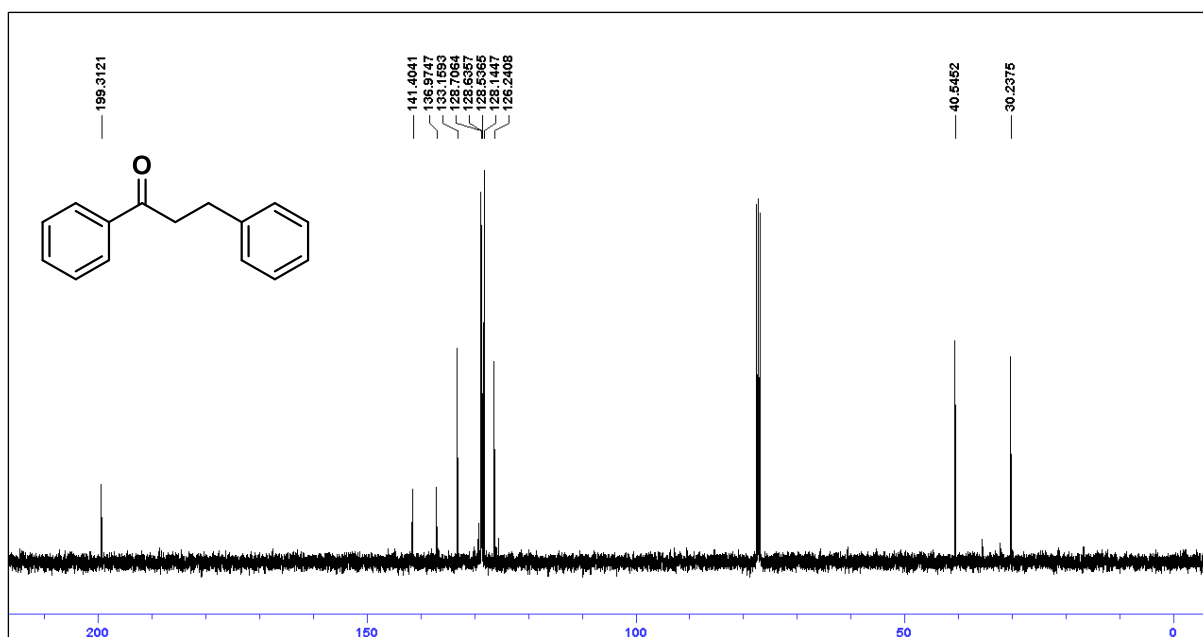
4-Chloroaniline (2d): ^1H NMR (400 MHz, CDCl_3): δ = 7.12 (d, J = 8.71 Hz), 6.62 (d, J = 8.74), 3.68 (s, 2H, NH_2); ^{13}C NMR (100 MHz, CDCl_3): δ = 144.97, 129.13, 123.17, 116.28

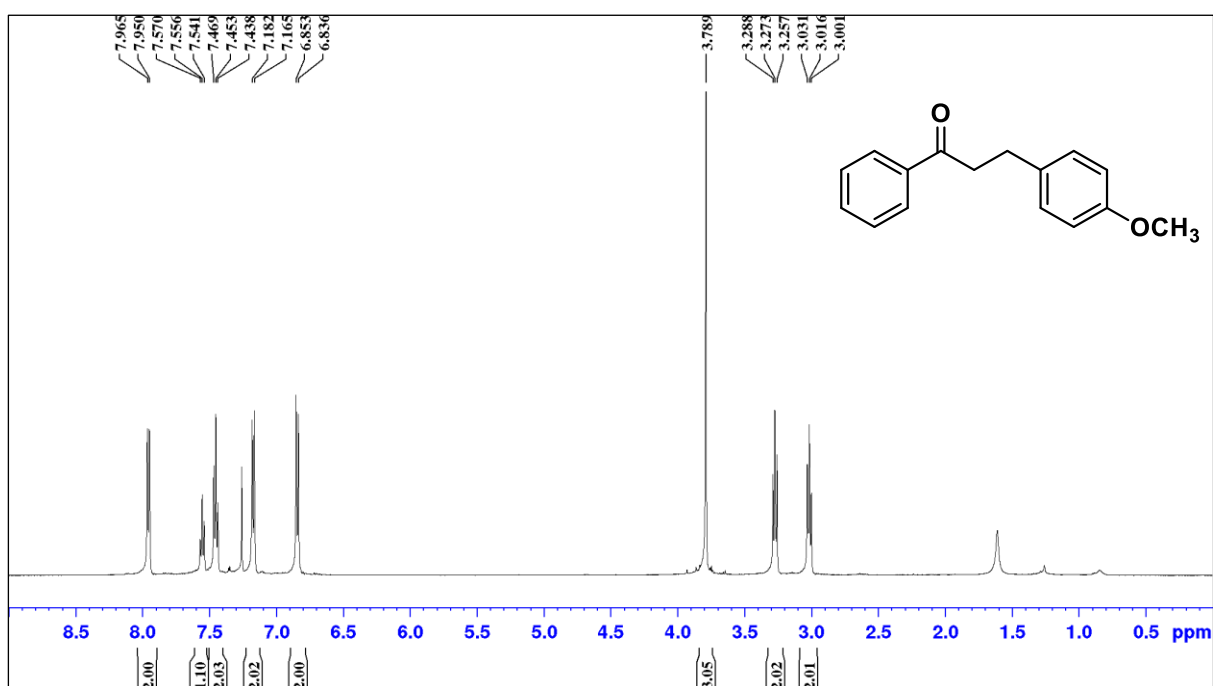
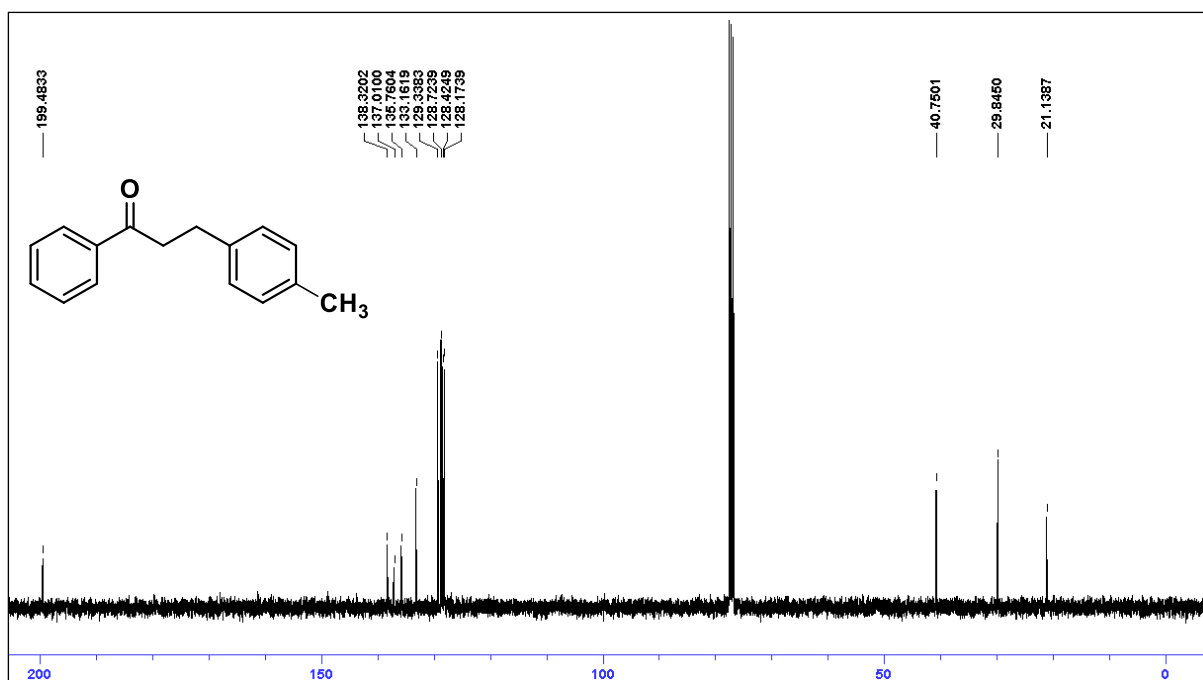


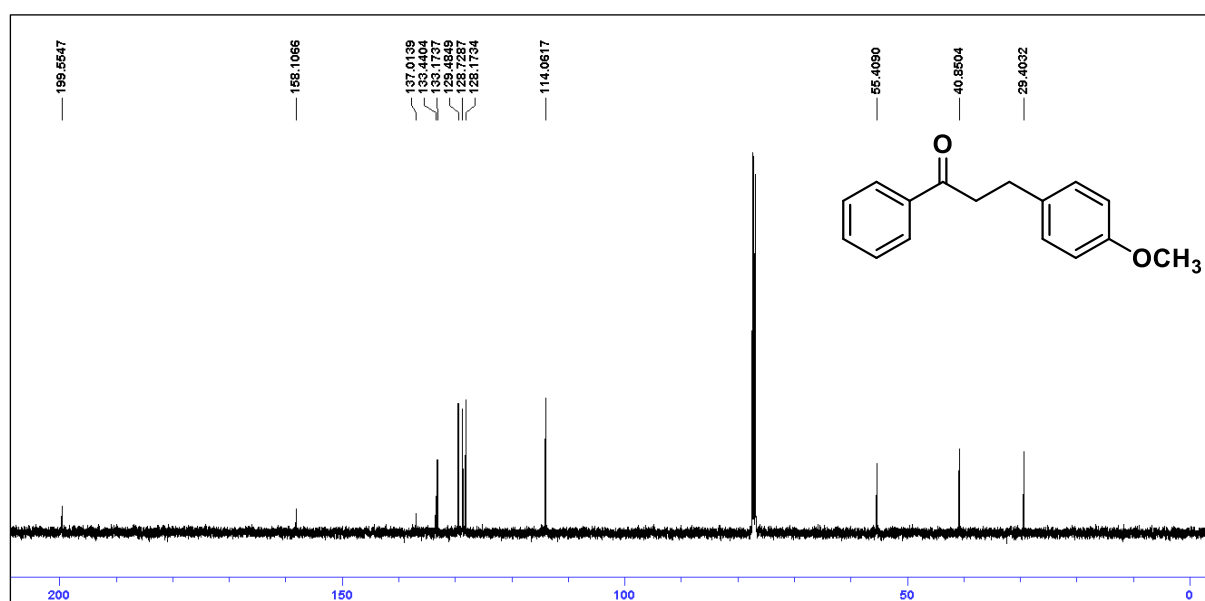
Quinolin-6-amine (2e): ^1H NMR (400 MHz, CDCl_3): δ = 8.64 (s, 1H), 7.88 (m, 2H), 7.24 (m, 1H), 7.12 (m, 1H), 6.06 (m, 1H), 4.05 (s, 2H, NH_2); ^{13}C NMR (100 MHz, CDCl_3): δ = 146.75, 144.76, 143.38, 133.80, 130.45, 129.82, 121.62, 121.41, 107.38.

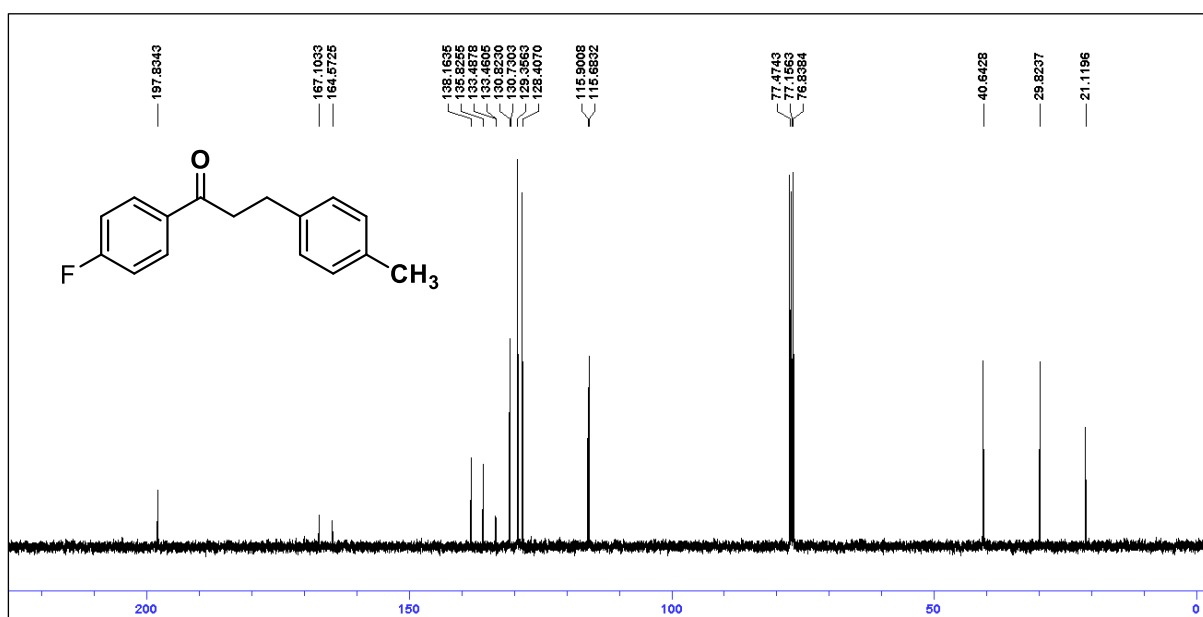
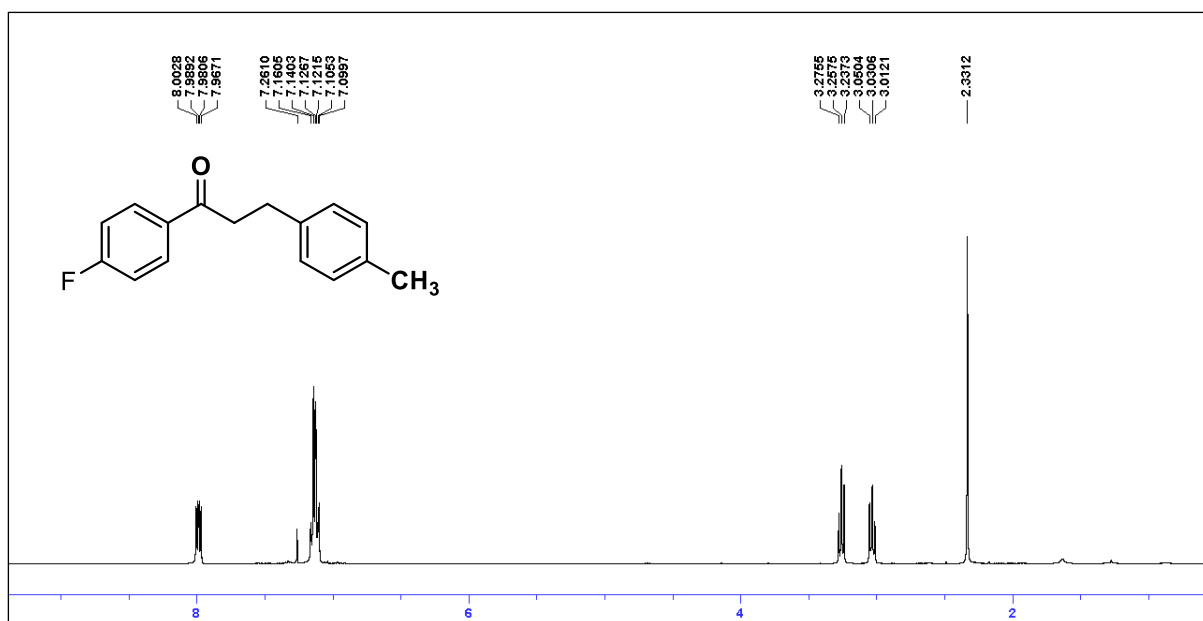
^1H and ^{13}C spectrums of hydrogenated $\alpha\beta$ -unsaturated carbonyl compounds:

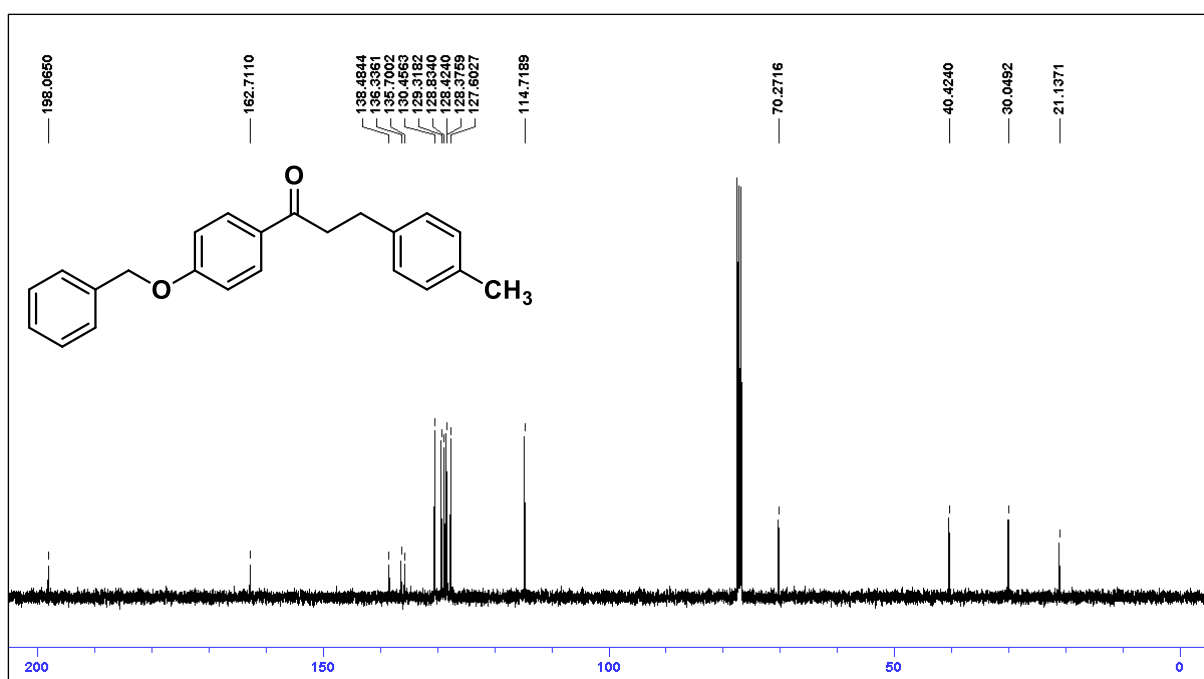
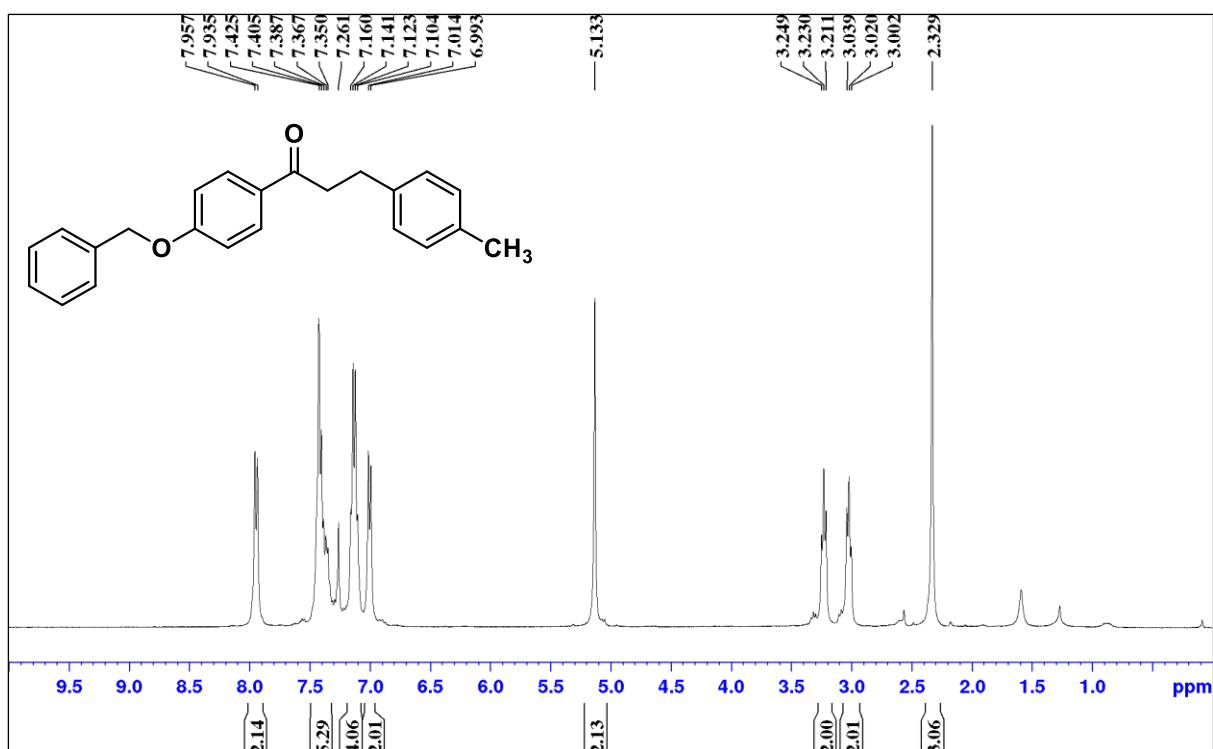


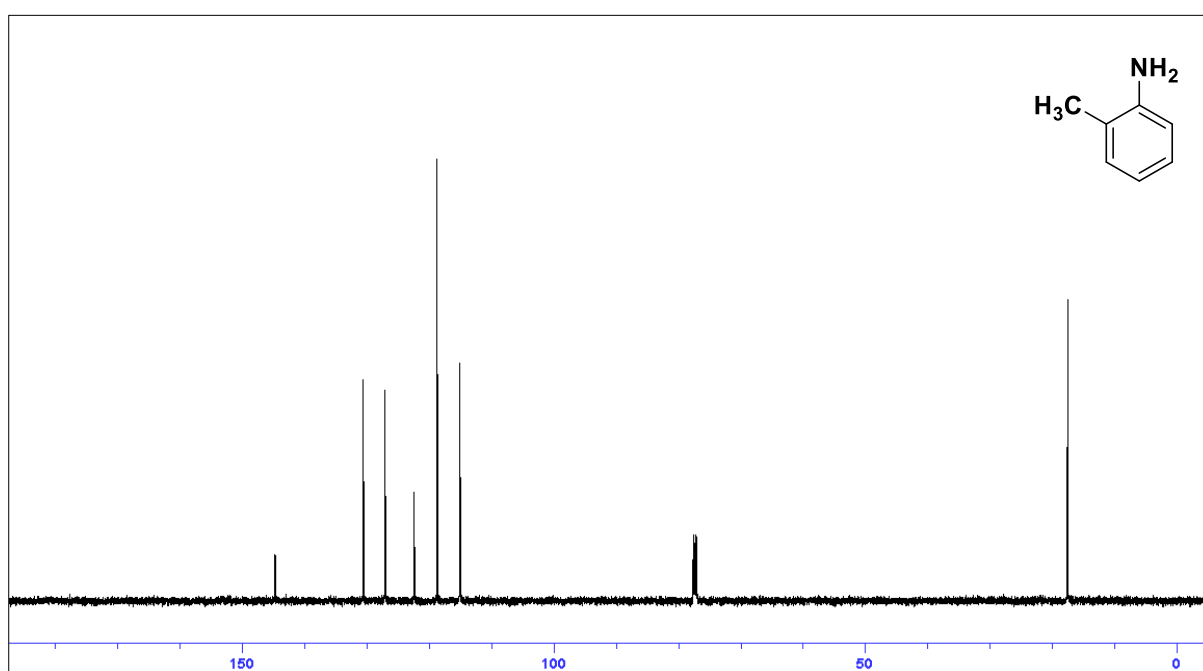
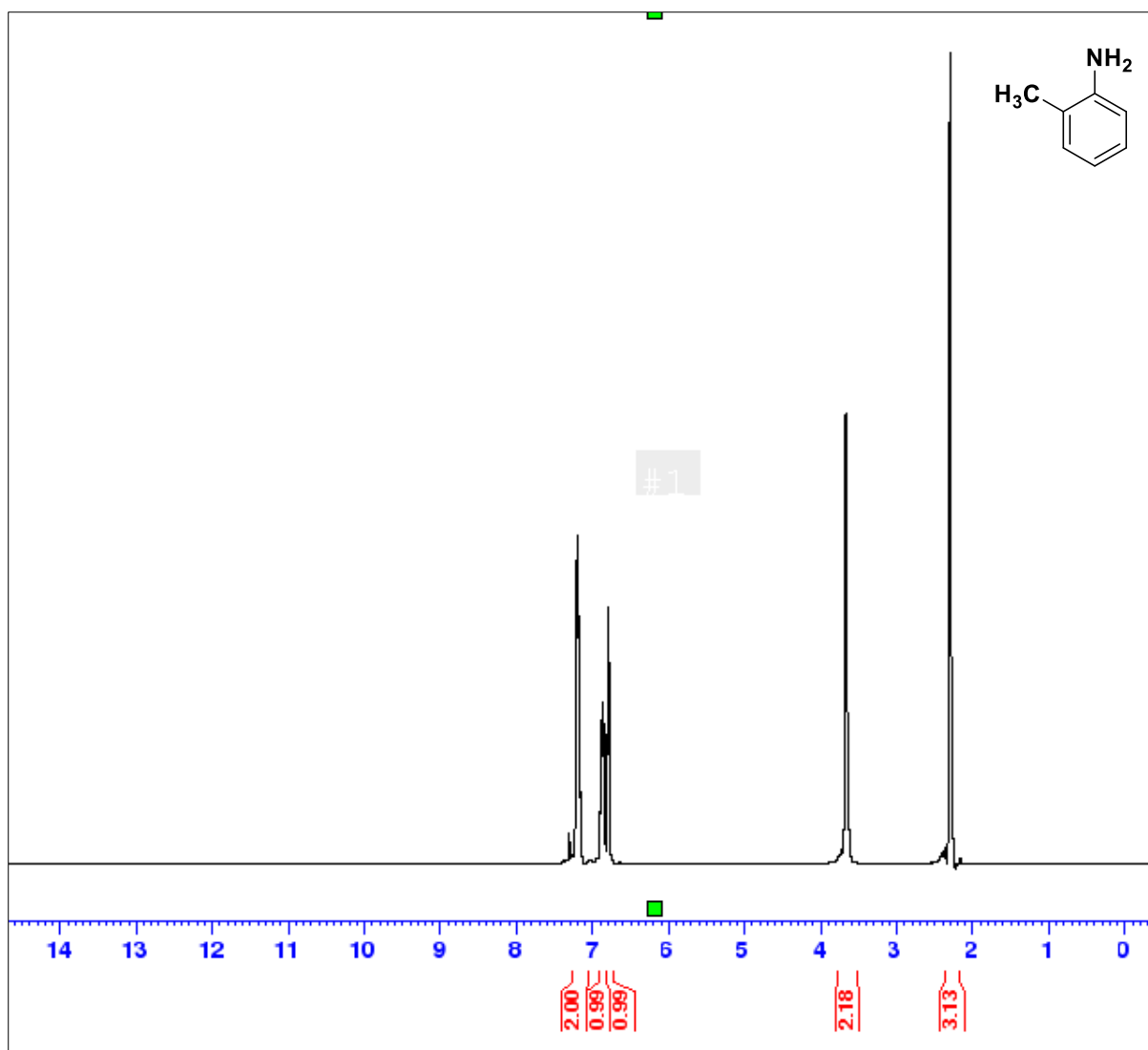


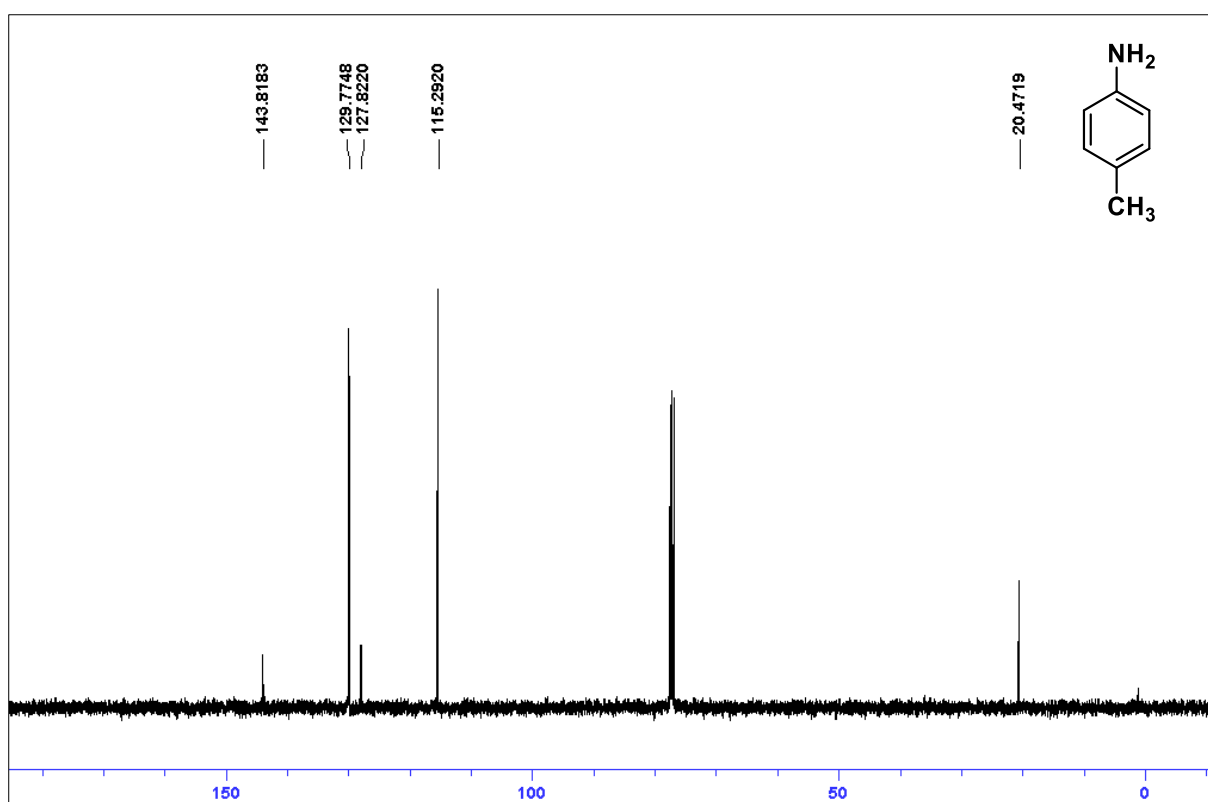
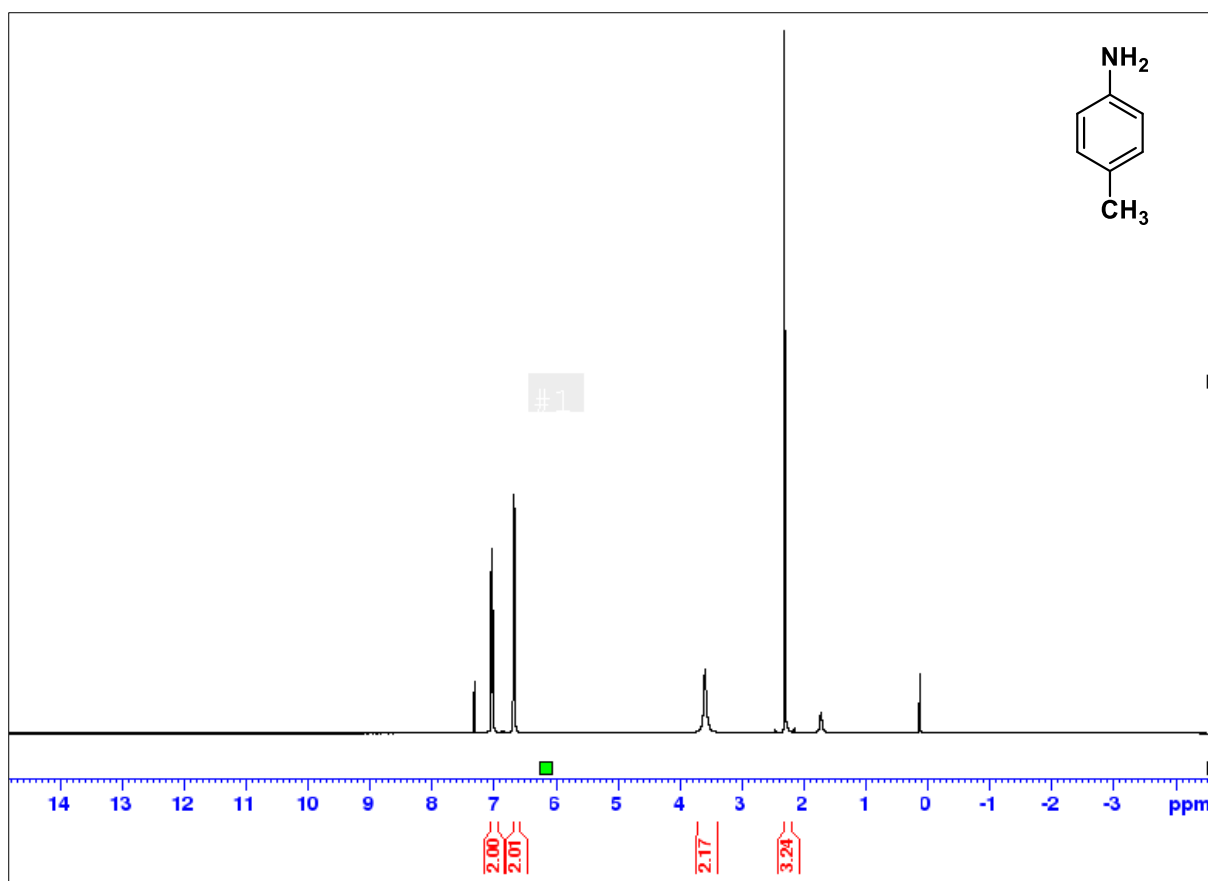


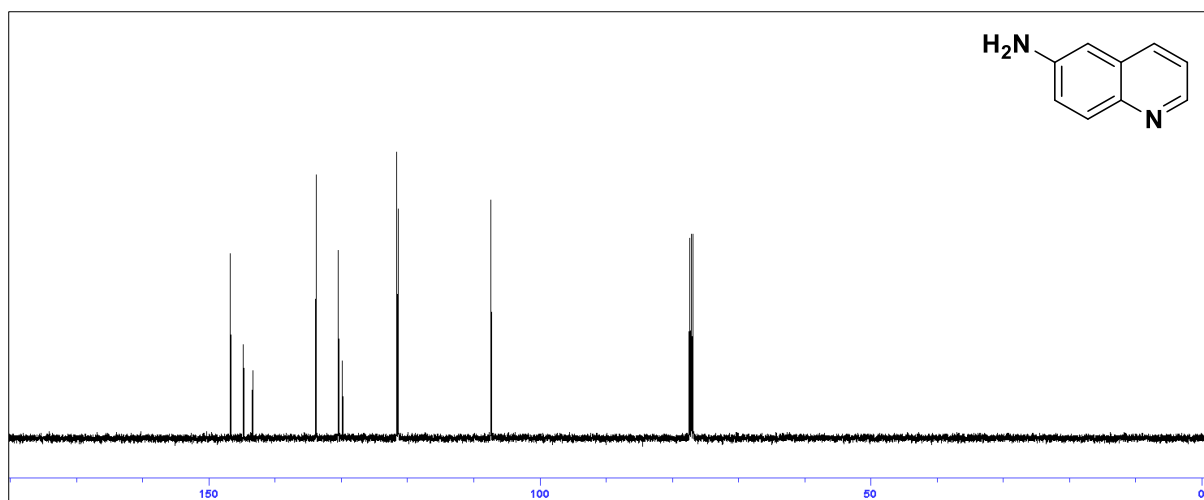
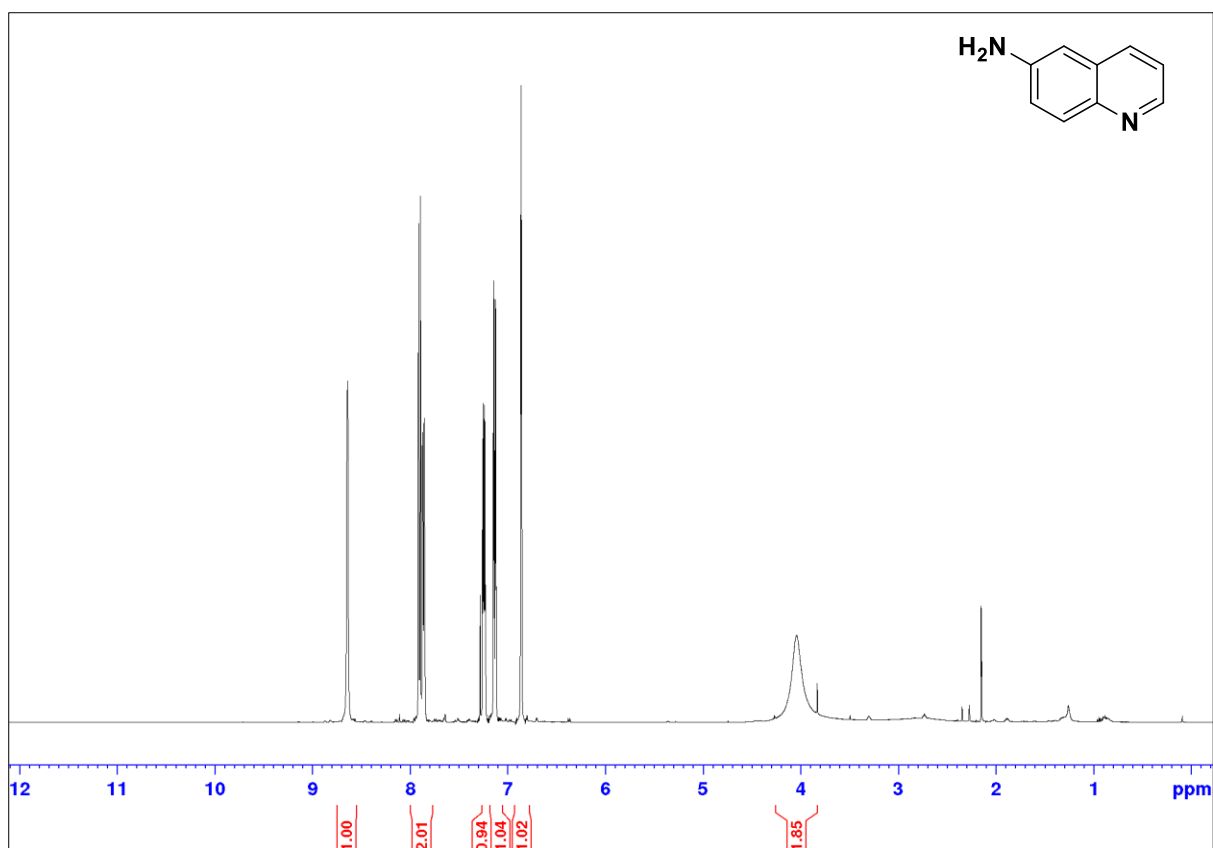


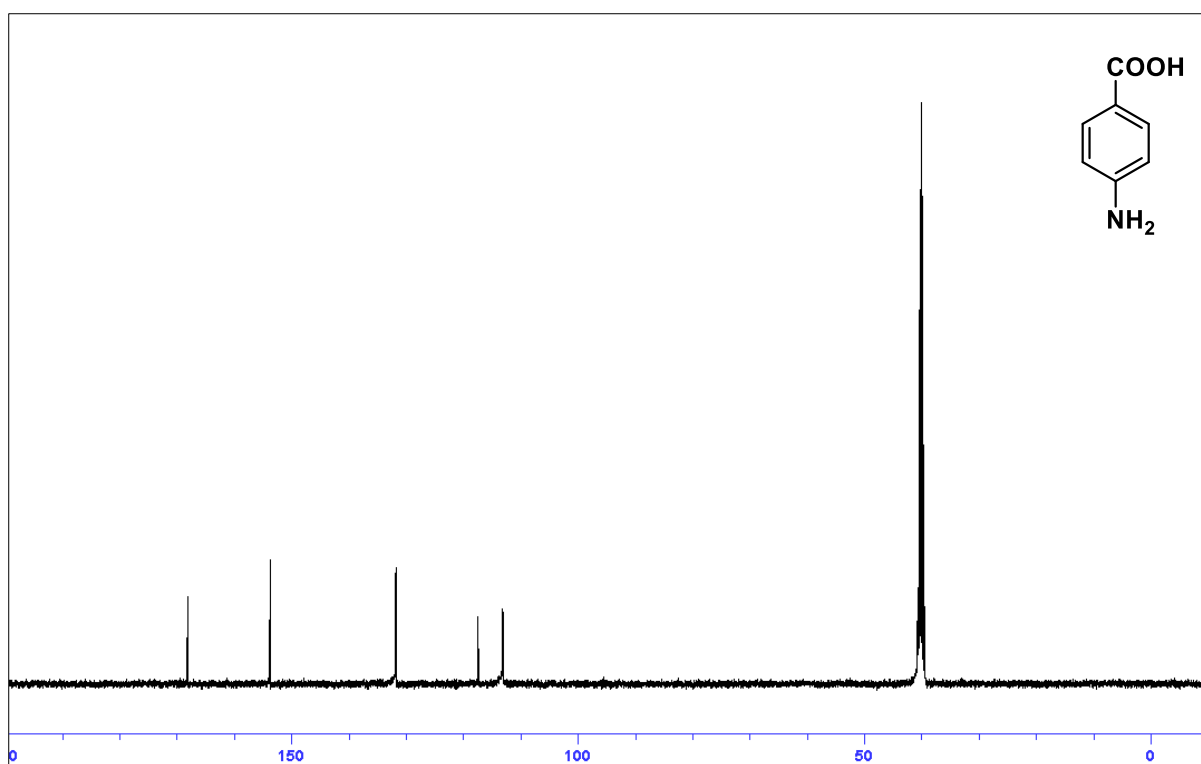
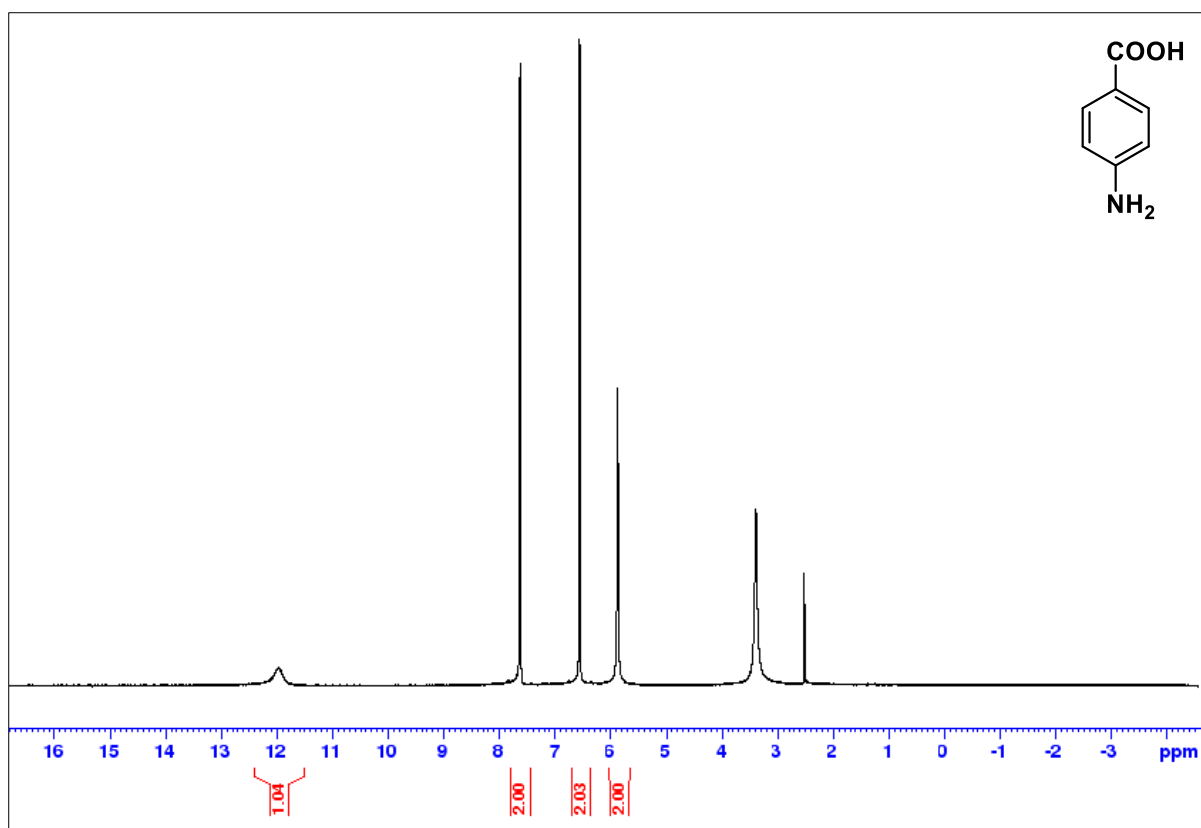


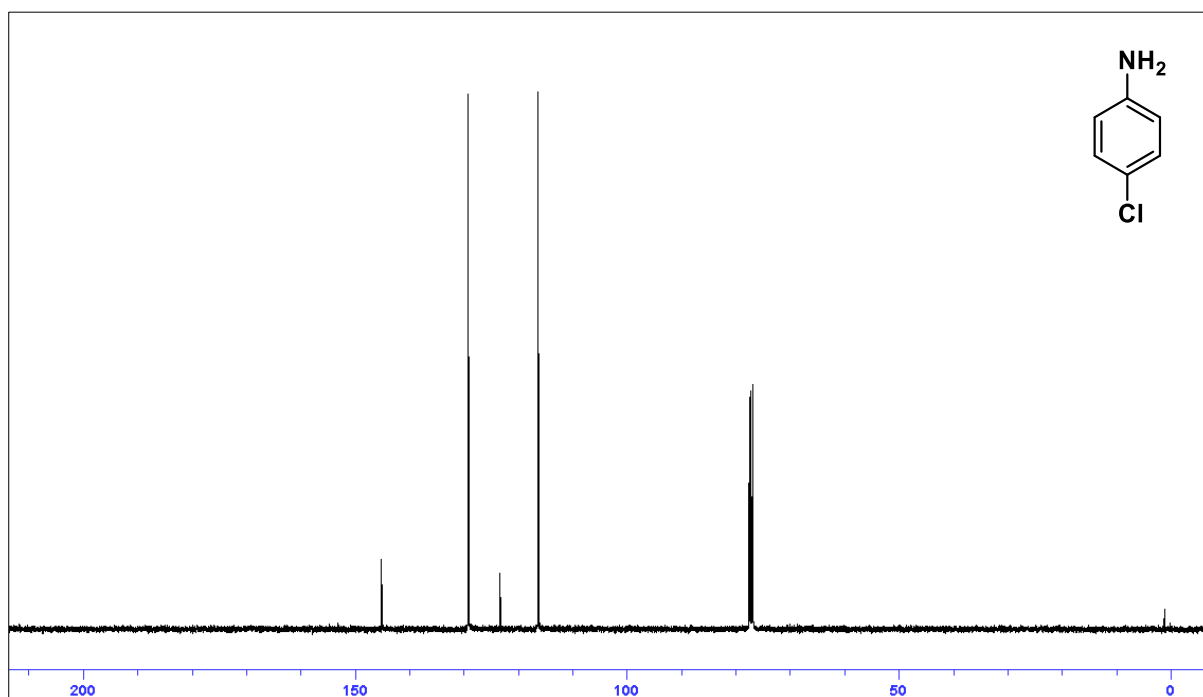
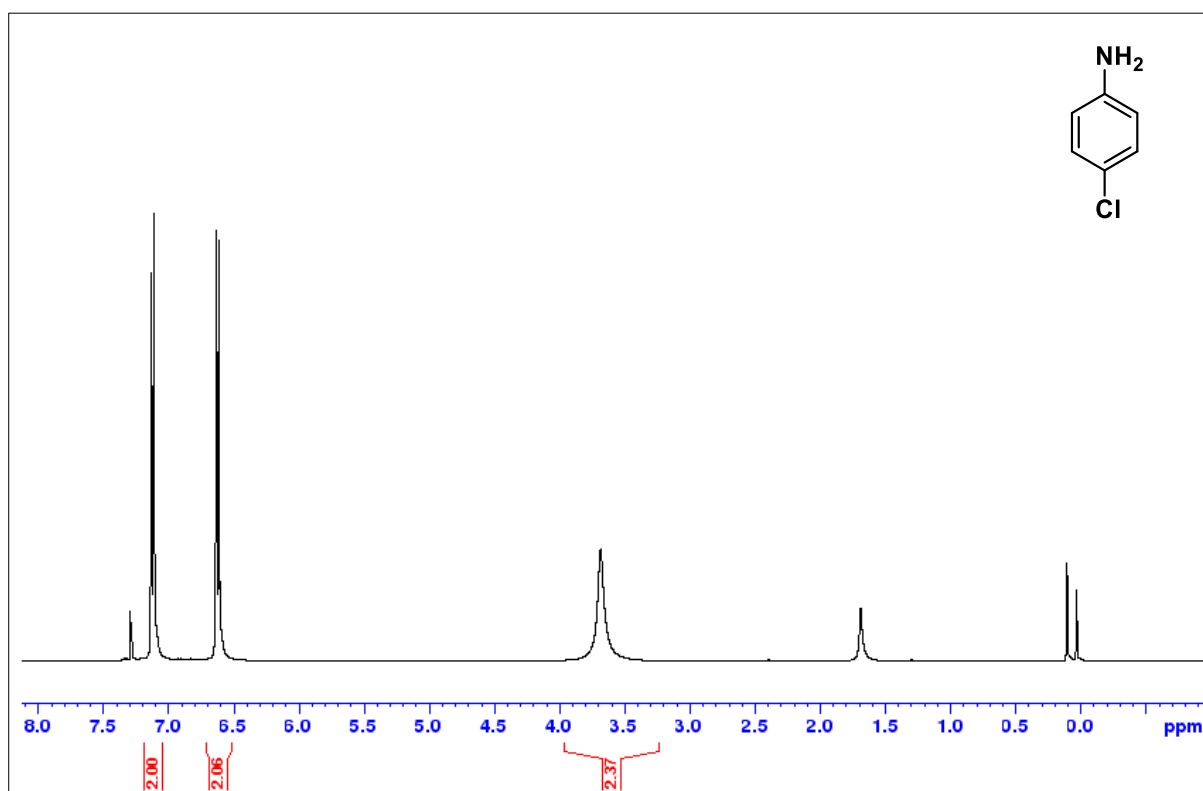










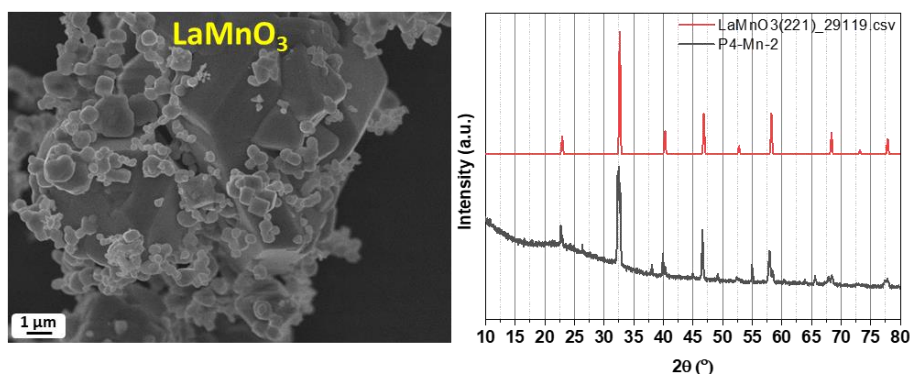


Synthesis of perovskite oxides by TMSS method :-

There is no standard procedure to synthesize the lanthanum-based perovskite oxide nanorods in the literature. One can think of clubbing the seed addition to this dynamic hydrothermal, which leads to the formation of the nanorods the LPOs like that of LKMO. To synthesize the other materials in a similar reaction, we synthesized the precursor materials using the molten salt method (MSS). Compared to solid-state reactions, the TMSS method lowers the reaction temperature by allowing faster mass transport in the liquid environment of the molten salts through convection and diffusion.²⁹³ Thus, from understanding the growth mechanism, we can choose the TMSS method, which exactly or nearly provides similar ionic mobility and high energies. We used molten 1:1 NaCl+KCl eutectic mixture as the reaction medium. Due to short diffusion distance and high mobilities of reactant oxides in this medium reduces the temperature and time of the reaction.

In an as typical reaction, pre-synthesized $\text{La}(\text{OH})_3$ nanorods used as the starting material as a source of La. Adding few drops of the aqueous eutectic mixture along with the respective transition metal salt to it forms the reaction mixture. Till date, LMO, LFO, LCoO, LNO, & LAO synthesize by this method at 800 °C in 12hrs. As the used salts are very much soluble in water, thus the washing got easier. **Figure 0.3**, shows the SEM and XRD of these compounds.

The fundamental reaction mechanism is that in the molten salt medium, the more soluble reactants diffuse onto the surface of less soluble reagent and react with them to give the final product. If both fully dissolve then they diffuse to react and results in an irregular shape. Except LFO others are deviating from the one-dimensional shape, indicating that the $\text{La}(\text{OH})_3$ nanorods are partly getting soluble in this particular eutectic mixture. In future one can use these as seeds and try to obtain the phase as well as nanorods of other materials. 5 mM precursor concentration, 12 M KOH at 850 rpm in 2 days.



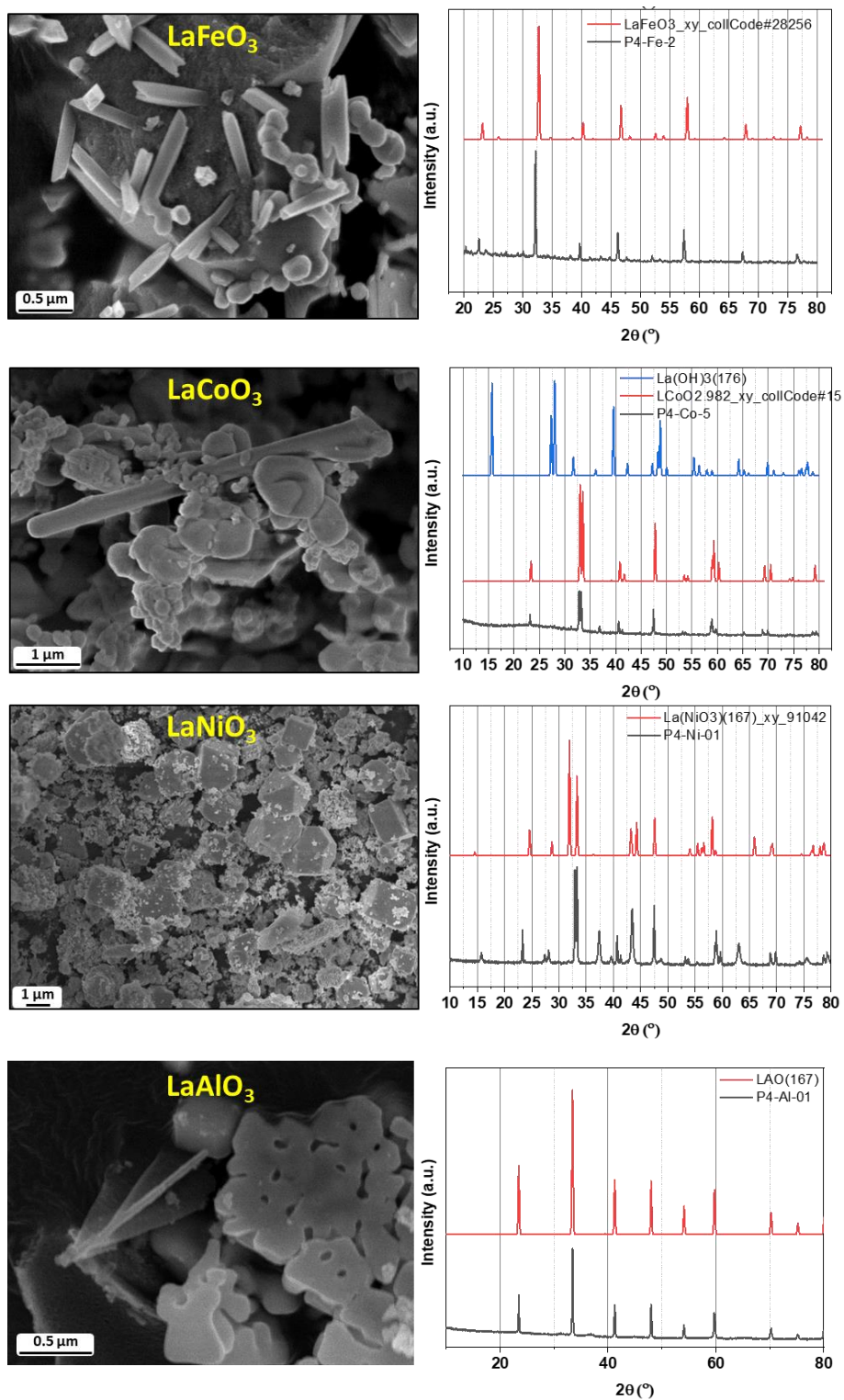


Figure 0.3 XRD and SEM of LMO, LFO, LCoO, LNO, & LAO obtained from TMSS

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